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Zigbee IEEE 802.15.4 PHY/MAC Test Plan

Version 2.0

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Abstract:	This document describes the tests for the Zigbee PRO/RF4CE PHY/MAC Level certification programs.

Keywords:	Qualification, Certification, 802.15.4, PHY, MAC, 2.4 GHz, Sub-GHz, GB SE Sub-GHz

Notice of Disclosure

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Version History

This table shows the version history for this specification.

Document Version History

Version	Description
0.0	Supporting ZigBee IP, ZigBee/ZigBee PRO and RF4CE network stacks.
1.0	Update to support Japan HEMS extension to ZigBee IP network stack.
2.0	Editorial restructuring, incorporating GB SE Sub-GHz PHY/MAC Test Plan, and removing PHY & MAC test cases supporting Zigbee IP and Japan HEMS.

1 Introduction

1.1 Scope

This document covers the Test Cases used to certify conformance to the Zigbee PRO/RF4CE IEEE 802.15.4 PHY and MAC layers.

Certification of a PHY/MAC as just an RFD is not premitted. Many of the test cases require an FFD as the DUT in order to verify the behavior correctly.

1.2 Purpose

The purpose of the Zigbee PRO/RF4CE IEEE 802.15.4 PHY/MAC Test Plan is to define a set of Test Cases to certify the Zigbee IEEE 802.15.4 PHY and MAC layers. Test specifications for other levels of certification are covered in other documents.

2 Contributors

The following was the chair of the Zigbee Work Group, PRO Core Balloting Group when this document was released:

Rob Alexander: Zigbee PRO Core Balloting Group *Chair*

The editor of this document was the following:

Clint Powell: Certification Advisory Group *Chair*, PHY/MAC Advisory Group *Chair*

3 References

The following standards and specifications contain provisions, which through reference in this document constitute provisions of this specification. All the standards and specifications listed are normative references. At the time of publication, the editions indicated were valid. All standards and specifications are subject to revision, and parties to agreements based on this specification are encouraged to investigate the possibility of applying the most recent editions of the standards and specifications indicated below.

3.1 Reference Documents

[R1] IEEE Std 802.15.4™-2003: Wireless Medium Access Control (MAC) and Physical Layer (PHY) Specifications for Low Rate Wireless Personal Area Networks (LR-WPANs)

[R2] Zigbee document 05-3474 Zigbee Specification, Zigbee Core Stack Working Group

[R3] [EN 300-220] ETSI EN 300 220-1 V2.4.1 (2012-01)….. Electromagnetic compatibility and Radio spectrum Matters (ERM); Short Range Devices (SRD); Radio equipment to be used in the 25 MHz to 1000 MHz frequency range with power levels ranging up to 500 mW; Part 1: Technical characteristics and test methods

[R4] [EN 304-204] ETSI EN 303 204-1 V1.1.0 (2014-06)….. Electromagnetic compatibility and Radio spectrum Matters (ERM); Network Based Short Range Devices (SRD); Radio equipment to be used in the 870 MHz to 876 MHz frequency range with power levels ranging up to 500 mW; Part 1:Technical characteristics and test methods

[R5] Zigbee document 14-0333-02 Zigbee IEEE 802.15.4 PICS

[R6] Zigbee document 16-04012-002 LBT Test Parameters

[R7] IEEE Std 802.15.4™-2020: Low-Rate Wireless Personal Area Networks (WPANs)

3.2 Zigbee PRO/RF4CE Supported PHYS and 802.15.4 Clauses

Below is a table mapping the relationship between each of the specific PHYs supported by Zigbee PRO/RF4CE and the related 802.15.4 Clause. The Band Descriptor is used throughout the test plan sections to identify which test cases are applicale to to a specific PHY

Band Descriptor	**PHY**	**IEEE Std 802.15.4™-2020 Associated Clause**
2003 2.4 GHz	2.4 GHz, 250kbps, O-QPSK	Clause 12
2003 Sub-GHz	868 MHz, 20kbps, BPSK 915 MHz, 40kbps, BPSK	Clause 13
GB SE Sub-GHz	863-876 MHz, 100kbps, Filtered 2FSK 915-921 MHz, 100kbps, Filtered 2FSK	Clause 19 based*

*several 802.15.4 modulation parameters were deviated from during the definition of this PHY for GB SE Sub-GHz

4 Acronyms, Abbreviations, and Definitions

4.1 Acronyms and Abbreviations

This section provides a list of acronyms and abbreviations used within this document.

BO....................Beacon Order
CSMA ....Carrier Sense Multiple Access
DUT ....Device Under Test
FFD ....IEEE 802.15.4 Full Function Device
EB ....Enhanced Beacon
EBR..................Enhanced Beacon Request
GB....................Great Britain
IE......................Information Element
IEEE.................Institute of Electrical and Electronic Engineers
MAC ....Medium Access Control
PAN ....Personal Area Network
PHY ....Physical Layer
PICS ....Protocol Implementation Conformance Statement
RFD..................IEEE 802.15.4 Reduced Function Device
SE.....................Smart Energy
SO ....Superframe Order
ZH ....Zigbee Host
ZR ....Zigbee Router

4.2 Definitions

5 Test Specification Overview

5.1 Certification Scheme

Figure 1 illustrates the different layers of certification. This document covers the (IEEE 802.15.4) MAC/PHY portion of certification testing.

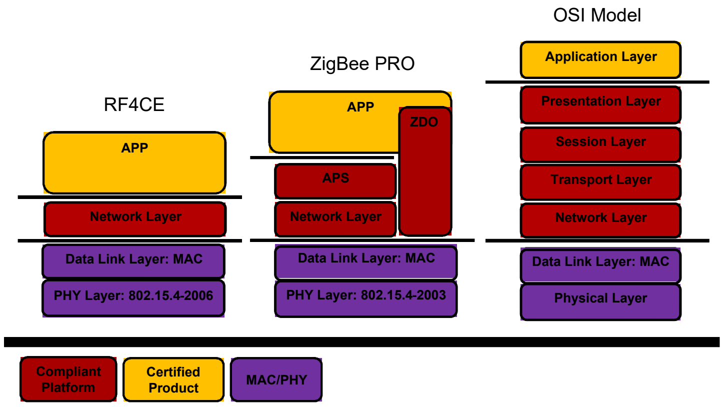


Figure 1 - Certification Layers

5.2 Test Specification Structure

This Test Specification separates the testing into individual Test Cases (TC). Each one of them is structured into the following parts:

- Setup Conditions: Pre-conditions necessary for testing.
- Test Procedure: Actual test method using Device Under Test (DUT).
- Expected Outcome: Pass/Fail conditions, expected observables.
- Notes: General, descriptive and explanatory text as necessary.

The TCs are grouped by Certification Level (CL). Each of the CL is subdivided into Categories (CTG) that starts at the lowest level and builds upward. Each CTG has an ordered list of Services (SRV) or Features (FTR) to be tested, as shown in Table 1.

Table 1 - Test Cases Classification

CL	CTG	SRV or FTR
IEEE 802.15.4 (154)	PHY (PHY)	Transmit Receive Turnaround time
	MAC (MAC)	Scanning Beacon management Channel access Frame validation Ack frame delivery Retries Association Data Warm start

Test Case Naming Convention

Taking off from Table 1 the names of the test cases are formed by:

< Test Name> => **TP**/<CL>/<CTG>/<SRV or FTR><#>

Where TP is for Test Procedure, CL is for Certification Level, CTG is for Category, SRV or FTR is for Service or Feature, and the # is the number from 01 to N since there can be several tests for a specific feature.

Conformance Level

Expected: A key word used to describe the behavior of the hardware or software in the design models *assumed* by the Draft. Other Hardware and software design models may also be implemented.

May: A key word that indicates flexibility of choice with *no preference implied*.

Shall: A key word indicating that a requirement is mandatory. Designers are required to implement all such mandatory requirements.

Should: A key word indicating flexibility of choice with a strongly preferred alternative. Equivalent to the phrase *is recommended.*

5.3 Tests Cases Format

Test Configuration

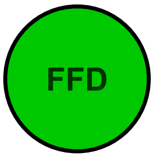
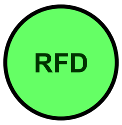
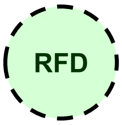
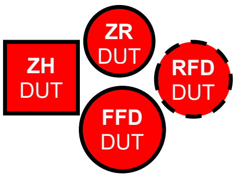
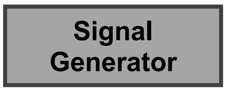
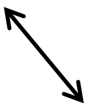
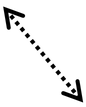
A minimum possible number of devices are employed in performing each of the test cases. The focus of each test is to verify the function and performance of the Zigbee platform, or node. As such, all test cases form only a very limited network, and do not attempt to construct arbitrarily large networks. The only exception to this rule is the test cases examining the maximum parameters of network formation.

Each test case involves the DUT in the role of either of the logical device types FFD (Full Function Device), RFD (Reduced Function Device).

Diagram

Each of the certification levels has a set of figures to represent the devices needed to perform the test as well as the device under test. An explanation of the figures is given below:

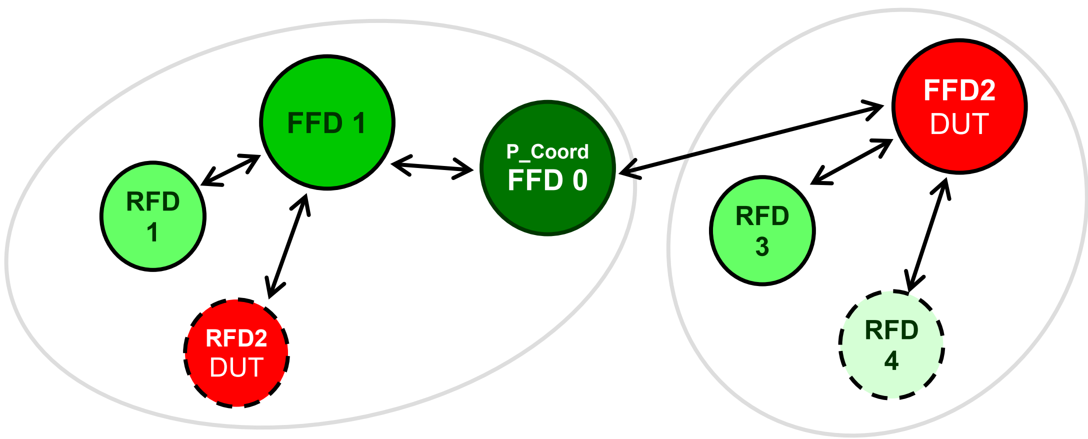
Figure 2 - Figure Description

Figure	Description
	Full Function Device acting as a PAN Coordinator Preferred use: IEEE 802.15.4 test plan.
	Full Function Device that is not a PAN Coordinator. Preferred use: IEEE 802.15.4 test plan.
	Reduced Function Device. Preferred use: IEEE 802.15.4 test plan.
	Sleepy Reduced Function Device, dotted outline means sleepy (polling device). Preferred use: IEEE 802.15.4 test plan.
	Device Under Test (DUT). Any device described above that appears in color red represents a DUT, the shape does not change only the color. Preferred use: any of the test plans, respecting the right figures.
	Signal Generator - Equipment used for 802.15.4 testing. Preferred use: IEEE 802.15.4 test plan.
	Signal Analyzer - Equipment used for 802.15.4 testing. Preferred use: IEEE 802.15.4 test plan.
	Communication between nodes or devices. Devices that are expected to communicate may be connected by arrows. Preferred use: any of the test plans.
	Communication in question, since it is between a secure and an un-secure network or device. Preferred use: any of the test plans.
	Communication Range. The devices that are expected to be within communication range should be surrounded by a transparent oval. Preferred use: in any of the test plans.

5.3.1 Zigbee PHY/MAC Test Case Format Example

Add here a brief explanation on the test case; explain how the service or feature is being tested and what the purpose of this particular test case is.

Setup Conditions:

Diagram
. 

The Diagram is the image that represents the configuration that is deployed for the Test Procedure.

Initial Conditions:

Device name	**Description**
All the devices	*Security: no for all the devices* *Channel: 0x14 for all the devices* *PAN id: 0x1AAA for all the devices* *MAC Key: 0xCFCECDCCCBCAC9C8C7C6C5C4C3C2C1C0 for all the devices*
P_Coord FFD0	*PAN Coordinator* *MAC: 0xACDE480000000001*
FFD1	*MAC: 0xACDE480000000002*
FFD2 (DUT)	*MAC: 0xACDE480000000003*
RFD1	*MAC: 0xACDE480000000004*
RFD2 (DUT)	*Polling device* *MAC: 0xACDE480000000005*
RFD3	*MAC: 0xACDE480000000006*
RFD4	*Polling Device* *MAC: 0xACDE480000000007*

Test Procedure:

Step No.	Step description	Expected outcome
#	Step # description	**Pass:** ***Describe here the expected behavior according to the specification, so that this step has considered successfully completed, and we can continue with the next one.*** **Fail:** ***Describe here the behavior in this step that would indicate that the step has not completed successfully and may need the test to restart, or just repeat the step.***
1	FFD0 forms the network.	**Pass: The FFD0 forms the network and becomes the PAN Coordinator.** **Fail: The FFD0 does not form the network or Joins a previous network.**
2	**FFD1 and FFD2 join FFD0**	**Pass: FFD1 and FFD2 join to FFD0.** **Fail: FFD1 or FFD2 do not join to FFD0**

Notes:

If required, add notes and comments here.

Number and Field Representation

Throughout the document, addresses and PAN IDs are shown in numerical format.

Numerical Format

All numbers shown as decimal or hexadecimal (preceded with ‘0x’) follow the convention for numerical representation, i.e. most significant digit on the left. Examples:

- 0xACDE480000000001
- 0xCAFE
- 78

With regard to octets, the least significant byte is shown on the right and the most significant byte is shown on the left.

Octet Format

An octet is always shown in numerical format (usually hexadecimal) with the least significant bit shown on the right and the most significant bit shown on the left. Example:

- 0x0D

Octet String

An octet string is represented with the first octet shown on the left. ‘First’ can mean first octet over the air or first octet in memory. An example:

- 0x32 0x45 0x65 0x78 0x23

6 Zigbee PHY Testing

Where IEEE 802.15.4 testing involves the use a of radio/platform as part of the test harness the radio/platform utilized in/by the test harness shall be different from that of the DUT being tested.

PHY Testing Approach

All 802.15.4 PHY tests shall be done conducted at the antenna termination point (without the antenna connected) with all path loss for cabling, attenuators, couplers, etc. accounted for.

Testing of the effects of the antenna, including matching and its radiation pattern shall be reserved for end product certification testing.

While it is acknowledged that aspects of the PHY may shift (or drift) over the working temperature range of the DUT, the PHY tests described here are to be implemented at or near room temperature (~ 25° C) and typical humidity. The effects on the PHY due to temperature and humidity variation shall be reserved for end product certification testing.

All PHY test specifications/limits already take into account any allowable variation and/or tolerance. Any test result beyond the test specification limits is to be considered a failure for that test case.

It is also acknowledged that some of these tests may be run simultaneously
(i.e. Transmit Test 1 and Transmit Test 2).

Interval Between Packet Transmissions for RX and TX Testing

To allow sufficient time for SPI transfers between the DUT and the tester interface, the interval between packet transmissions shall be no less than 100 mSec.

6.1 PHY Testing Specifics

6.1.1 2003 (2.4 GHz & Sub-GHz) PHY Testing Specifics

6.1.1.1 802.15.4 Field Representation for 2003 (2.4 GHz & Sub-GHz) PHY

802.15.4 fields are always represented in the specification in network bit and network byte order, i.e. the order they are transmitted in. This means fields are shown with the least significant bit and least significant byte on the left, which is the opposite of numerical representation. For example, the frame control field is represented as follows:

Zigbee Network Stack	**Bits 0-2**	**3**	**4**	**5**	**6**	**7-9**	**10-11**	**12-13***	**14-15**
Zigbee PRO/ RF4CE	Frame type	Security Enabled	Frame Pending	Ack Request	PAN ID compression	Reserved	Dest Addressing Mode	Reserved	Source Mode Addressing

In this document, 802.15.4 fields are shown in numerical order. For example, the frame control field is represented as follows:

Frame control: 0x9849			
Bitfield	**Field name**	**Hex. value**	**Field value**
.... .001	Frame Type	0x1	Data
.... 0...	Security Enabled	0x0	Disabled
....0	Frame Pending	0x0	No more data
....0.	Acknowledgment request	0x0	Acknowledgement not required
.... .1..	PAN ID compression	0x1	PAN ID compressed
.... ..00 0...	(reserved)	0x0	(reserved)
.... 10..	Destination Addressing Mode	0x2	16-bit short address
..01	Frame Version	0x1	802.15.4-2006 frame
10..	Source Addressing Mode	0x2	16-bit short address

However, this will be transmitted bitwise as follows, with the first bit shown on the left:

0b1 0b0 0b0 0b0 0b0 0b0 0b1 0b0 0b0 0b0 0b0 0b1 0b1 0b0 0b0 0b1

This will be transmitted octetwise as follows, with the first octet shown on the left:

0x41 0x98

6.1.1.2 PSDU MAC Payload for 2003 (2.4 GHz & Sub-GHz) PHY RX Testing

A standard approach is defined here to ensure that all symbol transitions are used in the generation of the PSDU data.

There are a total of 16 symbols and each symbol can be preceded by 16 different symbols. This gives a total of 256 combinations of 2 symbols - i.e. bytes. A maximum of 16x16 = 256 bytes are needed to test for all possible symbol transitions. Removing redundant symbol transitions results in all symbol transitions being represented in less than 16 MAC compliant (20 byte PHY Payload) packets, where the remaining symbols of the sixteenth payload are zero padded. The 16 packet payloads to be used are shown in Figure 3.

The xx in Figure 3 is the mac sequence number which can be any value and the FCS (CRC1 and CRC2) is computed normally.

This 16 packet sequence will be repeated 63 or 625 times for the purposes of a 1000 or 10,000 packet PER test respectively. Using this approach insures that all symbols (from 0 to F) are used and that all transitions between symbols are used as well.

packet	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	CRC1	CRC2
0	41	88	xx	AA	1A	FF	FF	44	33	00	10	20	30	40	50	60	70	80		
1	41	88	xx	AA	1A	FF	FF	44	33	09	0A	0B	0C	0D	0E	0F	11	21		
2	41	88	xx	AA	1A	FF	FF	44	33	13	14	15	16	17	18	19	1A	1B		
3	41	88	xx	AA	1A	FF	FF	44	33	1C	1D	1E	1F	22	32	42	52	62		
4	41	88	xx	AA	1A	FF	FF	44	33	27	28	29	2A	2B	2C	2D	2E	2F		
5	41	88	xx	AA	1A	FF	FF	44	33	33	43	53	63	73	83	93	A3	B3		
6	41	88	xx	AA	1A	FF	FF	44	33	3C	3D	3E	3F	44	54	64	74	84		
7	41	88	xx	AA	1A	FF	FF	44	33	49	4A	4B	4C	4D	4E	4F	55	65		
8	41	88	xx	AA	1A	FF	FF	44	33	57	58	59	5A	5B	5C	5D	5E	5F		
9	41	88	xx	AA	1A	FF	FF	44	33	66	76	86	96	A6	B6	C6	D6	E6		
10	41	88	xx	AA	1A	FF	FF	44	33	6F	77	87	97	A7	B7	C7	D7	E7		
11	41	88	xx	AA	1A	FF	FF	44	33	7F	88	98	A8	B8	C8	D8	E8	F9		
12	41	88	xx	AA	1A	FF	FF	44	33	99	A9	B9	C9	D9	E9	FA	AB	AC		
13	41	88	xx	AA	1A	FF	FF	44	33	AD	AE	AF	B1	BB	CB	DB	EB	FC		
14	41	88	xx	AA	1A	FF	FF	44	33	CA	CC	DC	EC	FD	DE	DF	EE	FF		
15	41	88	xx	AA	1A	FF	FF	44	33	F3	F6	00	00	00	00	00	00	00		

Figure 3 - 16 Packet Payloads Used for 2003 (2.4 GHz & Sub-GHz) PHY RX Testing

6.1.1.3 PER Measurement for 2003 (2.4 GHz & Sub-GHz) PHY RX Testing

PER measurements may be carried out via more than one approach:

- One approach is that the DUT may have PER software running to measure the error rate. This approach makes use of test mode (utility) software provided by the platform vendor that is run on the DUT. The software running on the DUT is used in conjunction with software running on the test equipment, including the PC controlling the test equipment.
- Other approaches exist and may be suitable in lieu of the above approach.

All receive tests involving PER use < 1% as the pass criteria, per the 802.15.4 standard. The standard, however does not say how many packets to use for the tests involving PER. The steps below outline an approach to PER testing that strikes a balance between time (and therefore cost) and the confidence of the results obtained.

- In computing the PER it must be assured that the information used to compute the error rate, both the number of samples used and the number of errors detected, comprise a statistically valid result.
    - Testing 1000 packets for a <1% PER, where the fail criteria is more than 9 packets in error, results in a probability of a wrong FAIL verdict of 54.3%. To reduce this wrong FAIL verdict to an acceptable amount (<1e-3, an order of magnitude less than the error rate being tested for) the number of packets in error should be increased to 21. This leaves a grey area between 9 and 21 packets in error when testing with 1000 packets. For 10,000 packets the number goes from 21 to 132.
    - Conversely, and more importantly, to achieve a 95% or better correct PASS verdict confidence requires 4 or fewer packets received in error.
    - Given both of the above the following approach shall be used in all RX PER testing.
        - Step 1: Generate N=1000 packets with signal generator and determine # of packet errors
        - Step 1 PASS Criteria: less than or equal to 4 packet errors
        - Step 1 FAIL Criteria: more than 21 packet errors
        - Step 1 UNDECIDED: more than 4 but less than or equal to 21 packet errors, proceed to Step 2 (grey area)
        - Step 2: Generate N=10, 000 packets with signal generator and determine # of packet errors
        - Step 2 PASS Criteria: less than or equal to 83 packet errors
        - Step 2 FAIL Criteria (a): more than 132 packet errors
        - Step 2 FAIL Criteria (b): more than 83 packet errors and less than or equal to 132 packet errors (a device falling into the grey area twice is not doing a good job)

6.1.1.4 Number of Channels Defined for the 2003 (2.4 GHz & Sub-GHz) PHY

- The IEEE 802.15.4 Standard [R1] specifies the use of 16 channels, for the O-QPSK PHY, in the 2.4 GHz band as follows:
    - Channels 11 to 26, where fc [MHz] = 2405+5*(k-11), where k = 11 (0x0B) to 26 (0x1A)
- The IEEE 802.15.4 Standard [R1] specifies the use of 11 channels, for the BPSK PHY, in the 868 and 915 MHz bands as follows:
    - 868 MHz band: Channel 0, where fc [MHz] = 868.3,

- 915 MHz band: Channels 1 to 10, where fc [MHz] = 906 + 2* (k-1), where k = 1, 2, …, 10

6.1.2 GB SE Sub-GHz PHY Testing Specifics

6.1.2.1 802.15.4 Field Representation for GB SE Sub-GHz PHY

802.15.4 fields are always represented in the specification in network bit and network byte order, i.e. the order in which they are transmitted in. This means fields are shown with the least significant bit and least significant byte on the left, which is the opposite of numerical representation. For example, the frame control field is represented as follows:

Bits 0-2	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**10-11**	**12-13**	**14-15**
Frame type	Security Enabled	Frame Pending	Ack Request	PAN ID compression	Reserved	Sequence Nmber Suppression	IE List Present	Dest Addressing Mode	Frame Version	Source Addressing Mode

In this document, 802.15.4 fields are shown in numerical order. For example, the frame control field is represented as follows:

Frame control: 0x9849			
Bitfield	**Field name**	**Hex. value**	**Field value**
`.... .... .... .001`	Frame Type	0x1	Data
`.... .... .... 0...`	Security Enabled	0x0	Disabled
`.... .... ...0 ....`	Frame Pending	0x0	No more data
`.... .... ..0. ....`	Acknowledgment request	0x0	Acknowledgement not required
`.... .... .1.. ....`	PAN ID compression	0x1	PAN ID compressed
`.... .... 0... ....`	(reserved)	0x0	(reserved)
`.... ...0 .... ....`	Sequence Number Suppression	0x0	Sequence number present in frame
`.... ..0. .... ....`	IE List Present	0x0	No IEs are contained in frame
`.... 10.. .... ....`	Destination Addressing Mode	0x2	16-bit short address
`..01 .... .... ....`	Frame Version	0x1	802.15.4-2006 frame
`10.. .... .... ....`	Source Addressing Mode	0x2	16-bit short address

However, this will be transmitted bitwise as follows, with the first bit shown on the left:

0b1 0b0 0b0 0b0 0b0 0b0 0b1 0b0 0b0 0b0 0b0 0b1 0b1 0b0 0b0 0b1

This will be transmitted octetwise as follows, with the first octet shown on the left:

0x41 0x98

6.1.2.2 PSDU MAC Payload for GB SE Sub-GHz PHY RX Testing

A standard approach is defined here to ensure that all symbol transitions are used in the generation of the PSDU MAC Payload data.

There are a total of 16 symbols and each symbol can be preceded by 16 different symbols, where symbol in this case refers to the hex mac symbol (or half octet). This gives a total of 256 combinations of 2 symbols - i.e. bytes. A maximum of 16x16 = 256 bytes are needed to test for all possible symbol transitions. Removing redundant symbol transitions results in all symbol transitions being represented in the 128 byte sequence given below in Figure 4:

0	0	1	0	2	0	3	0	4	0	5	0	6	0	7	0	8	0	9	0	A	0	B	0	C	0	D	0	E	0	F	1
1	2	1	3	1	4	1	5	1	6	1	7	1	8	1	9	1	A	1	B	1	C	1	D	1	E	1	F	2	2	3	2
4	2	5	2	6	2	7	2	8	2	9	2	A	2	B	2	C	2	D	2	E	2	F	3	3	4	3	5	3	6	3	7
3	8	3	9	3	A	3	B	3	C	3	D	3	E	3	F	4	4	5	4	6	4	7	4	8	4	9	4	A	4	B	4
C	4	D	4	E	4	F	5	5	6	5	7	5	8	5	9	5	A	5	B	5	C	5	D	5	E	5	F	6	6	7	6
8	6	9	6	A	6	B	6	C	6	D	6	E	6	F	7	7	8	7	9	7	A	7	B	7	C	7	D	7	E	7	F
8	8	9	8	A	8	B	8	C	8	D	8	E	8	F	9	9	A	9	B	9	C	9	D	9	E	9	F	A	A	B	A
C	A	D	A	E	A	F	B	B	C	B	D	B	E	B	F	C	C	D	C	E	C	F	D	D	E	D	F	E	E	F	F

Figure 4 - Octet (byte) Sequence Used for GB SE Sub-GHz PHY RX Testing

For GB SE Sub-GHz PHY RX Testing the MAC Payload of the 1st packet sent in an RX test shall use the 1st 116 octets (bytes) of the sequence shown in Figure 4. The MAC Payload of the 2nd packet sent in an RX test shall use the last 12 octets (bytes) of the sequence shown in Figure 4, followed by 1st 104 octets (bytes) of the sequence shown in Figure 4. The remaining odd packets sent in an RX test shall use the same MAC Payload as the 1st packet sent in an RX test and the remaing even packets sent in an RX test shall use the same MAC Payload as the 2nd packet sent in an RX test.

The 9 octets (bytes) of MAC Header (MHR) shall be set appropriately and the 2 octet (byte) CRC shall be computed normally for each of the packets sent in an RX test.

6.1.2.3 PER Measurement for GB SE Sub-GHz PHY RX Testing

- See 6.1.1.3.

6.1.2.4 Number of Channels Defined for the GB SE Sub-GHz PHY

There are a total of 90 channels specified for use in the GB SE Sub-GHz PHY. The full PHY specification, including the channel numbers (also included below), used by ZSE 1.4 (GB SE Sub-GHz) is contained in Annex D of the Zigbee Specification [R2].

Channel numbers for the GB SE Sub-GHz PHY are specified as follows:

$$ChanCenterFreq = ChanCenterFreq0 + NumChan * ChanSpacing$$

where *ChanCenterFreq0* is the first channel center frequency in MHz, *ChanSpacing* is the separation between adjacent channels in MHz, and *NumChan* is the channel number from 0 to *TotalNumChan–1*.

For the 863 MHz - 876 MHz band

TotalNumChan = 63

ChanSpacing = 0.2

NumChan goes from 0 to 62

ChanCenterFreq0 = 863.25

Resulting in the following for the 863 MHz - 876 MHz band:

Chan. #	Fc (MHz)
0	863.25
1	863.45
…	…
61	875.45
62	875.65

For the 915 MHz - 921 MHz band

TotalNumChan = 27

ChanSpacing = 0.2

NumChan goes from 0 to 26

ChanCenterFreq0 = 915.35

Resulting in the following for the 915 MHz - 921 MHz band:

Chan. #	Fc (MHz)
0	915.35
1	915.55
…	…
25	920.35
26	920.55

Four channel pages are used for the GB SE Sub-GHz PHY, using the existing 32-bit mechanism (5-bits of channel-page, 27-bits of channel-mask), spreading the channels across the 4 pages as follows:

Channel Page	Description
28	863 MHz Band Channels 0-26
29	863 MHz Band Channels 27-34, 62
30	863 MHz Band Channels 35-61 -- the "high-power band"
31	915 MHz Band Channels 0-26

6.1.2.5 Required Testing Channels for the GB SE Sub-GHz PHY Certification Testing

For certification testing it is realized that conducting PHY Transmit and Receive testing on all of the channels above represents an extraordinary effort. Therefore, while devices submitted for certification testing shall operate on all 90 of the above channels, for the purposes of certification testing only the following representative set of required testing channels from each of the 4 channel pages is required to be tested, reducing the total number of channels required to be tested from 90 to 20.

Channel Page	Required Testing Channels
28	0, 6, 13, 20, 26
29	27, 30, 32, 34, 62
30	35, 41, 48, 55, 61
31	0, 6, 13, 20, 26

6.1.2.6 Mandatory Use of Scrambling in the GB SE Sub-GHz PHY

Scrambling (data whitening) is required for the GB SE Sub-GHz PHY. Therefeore all PHY RX and TX testing will be conducted with scrambling turned on (i.e. all packets are to be sent and received scrambled).

6.2 PHY Test Case Quick Reference

Below is a summary of all test cases for all PHY's covered in the this test plan. A mapping of the previous PHY test case number to the current test plans test case number is also provided. The latter will alleviate having to update the PICs for all PHY test plans existing prior to the generation of the consolidated test plan.

Consolidated Test Case #	Description	2003 2.4 GHz	2003 Sub-GHz	GB SE Sub-GHz
TX-01	TX-Mod, Spread, Bits/Sym	TX-01	TX-01	TX-01
TX-02	TX-EVM	TX-02	TX-02	
TX-03	TX-Frequency Deviation/Zero Crossing			TX-02
TX-04	TX-Center Freq. Offset	TX-03	TX-03	TX-03
TX-05	TX-Min. Tx Power	TX-04	TX-04	
TX-06	TX-Power Adjustment Range			TX-04
TX-07	TX-Energy Emissions Mask	TX-05	TX-05	TX-05
RX-01	RX-PSR at Sensitivity	RX-01	RX-01	RX-01
RX-02	RX-Co-Channel Interferer			RX-02
RX-03	RX-Adj., Alt., Other Channel Interferer	RX-02/03	RX-02/03	RX-03
RX-04	RX-Blocker			RX-04
RX-05	RX-Max. Input Signal w/100% PSR	RX-04	RX-04	RX-05
RX-06	RX-ED (Energy Detect)	RX-05	RX-05	RX-06
RX-07	RX-LQI (Link Quality Indicator)	RX-06	RX-06	RX-07
RX-08	RX-CCA (Clear Channel Assesment)	RX-07	RX-07	RX-08
RX-09	RX-RSSI Reporting			RX-09
TAT-01	TAT-RX-to-TX Turnaround Time	TAT-01	TAT-01	TAT-01
TAT-02	TAT-TX-to-RX Turnaround Time	TAT-02	TAT-02	TAT-02

6.3 Zigbee IEEE 802.15.4 PHY Test Cases

TP/154/PHY/TRANSMIT-01

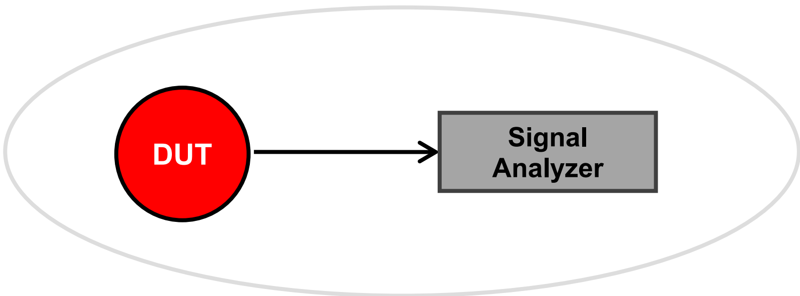
Tests that for the PHY being tested, the DUT supports transmission of the correct modulation for the PHY being tested. At the Output Power Level specified in the table below the DUT shall send a standards conforming signal and packet with the Preamble Length, PHY Payload (PSDU) and FCS/CRC specified in the table below.

The data sequence is demodulated with the Signal Analyzer to verify proper modulation, spreading, and bits per symbol. This test shall also check that the phy symbol to mac hex symbol mapping is done correctly as well.

	PHY	**Output Power Level**	**Preamble Length**	**PHY Payload (PSDU) and FCS/CRC**
2003 2.4 GHz	2.4 GHz, 250kbps, O-QPSK	≥ -3 dBm	4 bytes	"41 88 xx AA 1A FF FF 44 33 01 23 45 67 89 AB CD EF FE DC BA 98 76 54 32 10 CRC1 CRC2", where xx is the mac sequence number which can be any value and the FCS (CRC1 and CRC2) is computed normally.
2003 Sub-GHz	868 MHz, 20kbps, BPSK 915 MHz, 40kbps, BPSK	≥ -3 dBm	4 bytes	"41 88 xx AA 1A FF FF 44 33 01 23 45 67 89 AB CD EF FE DC BA 98 76 54 32 10 CRC1 CRC2", where xx is the mac sequence number which can be any value and the FCS (CRC1 and CRC2) is computed normally.
GB SE Sub-GHz	863-876 MHz, 100kbps, Filtered 2FSK 915-921 MHz, 100kbps, Filtered 2FSK	≥ -3 dBm	8 bytes	Arbitrary bit pattern for the payload (PSDU), and correct CRC, where the payload (PSDU) consists of 127 bytes.

Setup Conditions:

Diagram



The Diagram is the image that represents the configuration deployed for the Test Procedure.

482 ***Initial Conditions:***

Device name	Description
DUT	*Channel: for this test case, all of the required testing channels for the specific PHY being tested are specified in the table at the bottom of the Test Procedure.*
Signal Analyzer	*Signal Analyzer shall receive output from the DUT.* *Signal Analyzer shall be configured in the following manner:* *Reference Level is set to an appropriate value according to the stated DUT maximum output power level. If the DUT output power level exceeds analyzer's allowed maximum RF input level, use additional attenuators.* *All other parameters of the Signal Analyzer shall be set as defined in the table below.*

483

	Band	Span	Resolution BW (RBW)	Video BW (VBW)	Attenuation
2003 2.4 GHz	2.4 GHz	5 MHz	3 MHz	≥ RBW	Auto
2003 Sub-GHz	868 MHz 915 MHz	2 MHz 3 MHz	≤ 100 kHz	≥ RBW	Auto
GB SE Sub-GHz	863-876 MHz 915-921 MHz	200 kHz	100 kHz	≥ RBW	Auto

484 ***Test Procedure:***

Step No.	Step description	Expected outcome
1	DUT shall transmit a standards conforming signal and packet with the specified PHY Payload (PSDU) and FCS/CRC. The signal level shall be at the specified Output Power Level. If the DUT is capable of properly transmitting at a maximum output power greater than minimum specified Output Power Level then the output power of the DUT shall be set to its maximum output power level. Signal Analyzer shall receive and detect the exact same bit and symbol pattern transmitted by the DUT for 100 packets.	Pass: For each of the required testing channels none of the bits and symbols of the transmitted packets are received in error and that the specified Preamble Length was sent. Fail: For any of the required testing channels 1 or more of the bits and/or symbols of the transmitted packets are received in error and/or that the specified Preamble Length was not sent.
2	For the PHY being tested perform test for each of the following required Channels to Test at the corresponding center frequencies specified in the table below.	

	PHY	Channels to Test	Center Frequency Calculation
2003 2.4 GHz	2.4 GHz, 250kbps, O-QPSK	11 to 26	fc [MHz] = 2405+5*(k-11), where k = 11 (0x0B) to 26 (0x1A)
2003 Sub-GHz	868 MHz, 20kbps, BPSK 915 MHz, 40kbps, BPSK	0 1 to 10	fc [MHz] = 868.3 fc [MHz] = 906 + 2* (k-1), where k = 1, 2, …, 10
GB SE Sub-GHz	863-876 MHz, 100kbps, Filtered 2FSK 915-921 MHz, 100kbps, Filtered 2FSK	Ch Page [Ch #'s] 28 [0, 6, 13, 20, 26] 29 [27, 30, 32, 34, 62] 30 [35, 41, 48, 55, 61] 31 [0, 6, 13, 20, 26]	fc [MHz] = ChanCenterFreq0 + Ch # * ChanSpacing where ChanSpacing = 0.200 MHz For 863-876 MHz, Ch Page 28-30, ChanCenterFreq0 = 863.25 MHz For 915-921 MHz, Ch Page 31, ChanCenterFreq0 = 915.35 MHz

Notes:

None.

TP/154/PHY/TRANSMIT-02

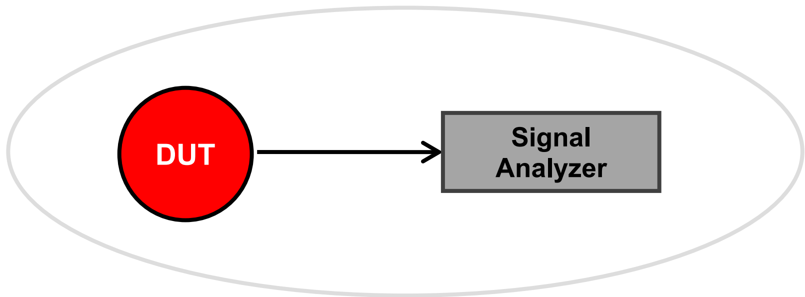
Tests that for the PHY being tested, during PHY operation of the DUT the Error Vector Magnitude (EVM) meets the EVM specified in the table below.

The EVM measurement shall be done at the Output Power Level specified below, using a standards conforming signal and packet with an arbitrary bit pattern with the payload (PSDU) consisting of the number of bytes specified in the table below, and correct CRC.

	PHY	Output Power Level	Payload (PSDU) Length	EVM	Measurement Duration
2003 2.4 GHz	2.4 GHz, 250kbps, O-QPSK	≥ -3 dBm	≥ 20 bytes	< 35%	1000 chips
2003 Sub-GHz	868 MHz, 20kbps, BPSK 915 MHz, 40kbps, BPSK	≥ -3 dBm	≥ 20 bytes	< 35%	1000 chips

Setup Conditions:

Diagram



The Diagram is the image that represents the configuration deployed for the Test Procedure.

Initial Conditions:

Device name	Description
DUT	*Channel: for this test case, all of the required testing channels for the specific PHY being tested are specified in the table at the bottom of the Test Procedure.*
Signal Analyzer	*Signal Analyzer shall receive output from the DUT.* *Reference modulation shall be available through a vector signal analyzer (VSA) which is capable of digital demodulation while performing carrier lock, symbol timing recovery, and amplitude adjustment:* *Where the Reference Level is set to an appropriate value according to the stated DUT maximum output power level. If the DUT output power level exceeds analyzer's allowed maximum RF input level, use additional attenuators* *Set the VSA to digital demodulation mode, display as Polar/vector and configured for the signal in the manner specified in the table below*

498

	Band	Span	Resolution BW (RBW)	Time Duration	Modulation Scheme	Pulse Shape	Samples per Symbol
2003 2.4 GHz	2.4 GHz	5 MHz	3 MHz	1000 chips	MSK Type 2 symbol rate = 2 MHz	Half Sine	≥ 4
2003 Sub-GHz	868 MHz 915 MHz	2 MHz 3 MHz	Auto	1000 chips	BPSK, chip rate = 300 kchips/s BPSK, chip rate = 600 kchips/s	Raised Cosine roll-off=1.0	≥ 4

499 ***Test Procedure:***

Step No.	Step description	Expected outcome
1	DUT shall transmit a standards conforming signal and packet with the specified payload (PSDU), and correct CRC, with a payload (PSDU) length of Payload (PSDU) Length. The signal level shall be at the specified Output Power Level. If the DUT is capable of properly transmitting at a maximum output power greater than minimum specified Output Power Level then the output power of the DUT shall be set to its maximum output power level. Using the Signal Analyzer initial conditions configuration settings measure the error vector magnitude (EVM) of the DUT.	Pass: For each of the required testing channels the specified EVM is met. Fail: For any of the required testing channels the specified EVM is not met.
2	For the PHY being tested perform test for each of the following required Channels to Test at the corresponding center frequencies specified in the table below.	

500

	PHY	Channels to Test	Center Frequency Calculation
2003 2.4 GHz	2.4 GHz, 250kbps, O-QPSK	11 to 26	fc [MHz] = 2405+5*(k-11), where k = 11 (0x0B) to 26 (0x1A)
2003 Sub-GHz	868 MHz, 20kbps, BPSK 915 MHz, 40kbps, BPSK	0 1 to 10	fc [MHz] = 868.3 fc [MHz] = 906 + 2* (k-1), where k = 1, 2, …, 10

501 ***Notes:***

502 *None.*

503

TP/154/PHY/TRANSMIT-03

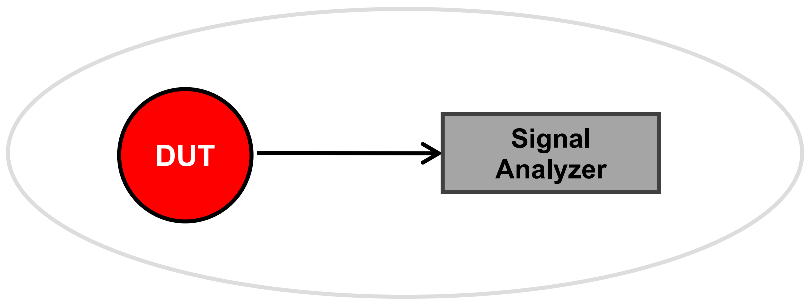
Tests that for the PHY being tested, during PHY operation of the DUT the frequency deviations and demodulated eye diagram meet the Frequency Deviation Tolerance and Zero Crossing Tolerance specified in the table below.

The eye diagram and frequency deviation measurements shall encompass all portions of a transmitted frame, and be done at the Output Power Level specified in the table below, using a standards conforming signal and packet with an arbitrary bit pattern with the Payload (PSDU) Length number of bytes specified in the table below, and correct CRC.

	PHY	Output Power Level	Payload (PSDU) Length	Frequency Deviation Tolerance	Zero Crossing Tolerance
GB SE Sub-GHz	863-876 MHz, 100kbps, Filtered 2FSK 915-921 MHz, 100kbps, Filtered 2FSK	≥ -3 dBm	127 bytes	–fdev +/- 30%, +fdev +/- 30% where fdev = 35 kHz	Ts ≤ +/- 12.5%

Setup Conditions:

Diagram



The Diagram is the image that represents the configuration deployed for the Test Procedure.

Initial Conditions:

Device name	Description
DUT	*Channel: for this test case, all of the required testing channels for the specific PHY being tested are specified in the table at the bottom of the Test Procedure.*
Signal Analyzer	*Signal Analyzer shall receive output from the DUT.* *Signal Analyzer shall be configured to generate an eye diagram:* *Where the Reference Level is set to an appropriate value according to the stated DUT maximum output power level. If the DUT output power level exceeds analyzer's allowed maximum RF input level, use additional attenuators* *For the signal specified in the table below*

517

	Band	Signal Demodulation	Symbol Rate	Modulation Index	# of Symbol Periods	Persistence
GB SE Sub-GHz	863-876 MHz 915-921 MHz	2 Level FSK	100 kHz	0.7	2	≥ 16

518 ***Test Procedure:***

Step No.	Step description	Expected outcome
1	DUT shall transmit a standards conforming signal and packet with the specified payload (PSDU), and correct CRC, with a payload (PSDU) length of Payload (PSDU) Length. The signal level shall be at the specified Output Power Level. If the DUT is capable of properly transmitting at a maximum output power greater than minimum specified Output Power Level then the output power of the DUT shall be set to its maximum output power level. Using the Signal Analyzer initial conditions configuration settings, and employing the frequemncy deviation and eye diagram capability of the Signal Analyzer, measure the Frequency Deviation Tolerance and Zero Crossing Tolerance of the DUT.	Pass: For each of the required testing channels the specified Frequency Deviation Tolerance and Zero Crossing Tolerance are met. Fail: For any of the required testing channels the specified Frequency Deviation Tolerance and/or Zero Crossing Tolerance are not met.
2	For the PHY being tested perform test for each of the following required Channels to Test at the corresponding center frequencies specified in the table below.	

519

	PHY	Channels to Test	Center Frequency Calculation
GB SE Sub-GHz	863-876 MHz, 100kbps, Filtered 2FSK 915-921 MHz, 100kbps, Filtered 2FSK	Ch Page [Ch #'s] 28 [0, 6, 13, 20, 26] 29 [27, 30, 32, 34, 62] 30 [35, 41, 48, 55, 61] 31 [0, 6, 13, 20, 26]	fc [MHz] = ChanCenterFreq0 + Ch # * ChanSpacing where ChanSpacing = 0.200 MHz For 863-876 MHz, Ch Page 28-30, ChanCenterFreq0 = 863.25 MHz For 915-921 MHz, Ch Page 31, ChanCenterFreq0 = 915.35 MHz

520 ***Notes:***

521 *None.*

522

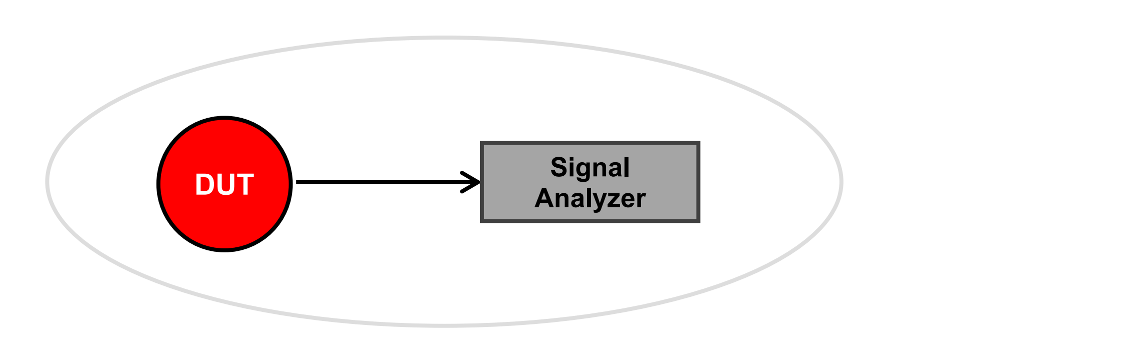
TP/154/PHY/TRANSMIT-04

Tests that for the PHY being tested, the transmit center frequency tolerance for each channel of the DUT meets the Maximum Center Frequency Offset specified in the table below. At the Output Power Level specified in the table below the DUT shall send a standards conforming signal and packet with the Preamble Length, PHY Payload (PSDU) and FCS/CRC specified in the table below.

	PHY	Output Power Level	Maximum Center Frequency Offset	PHY Payload (PSDU) and FCS/CRC
2003 2.4 GHz	2.4 GHz, 250kbps, O-QPSK	≥ -3 dBm	≤ +/- 40ppm or ≤ +/- 96.2 kHz at 2405 MHz to ≤ +/- 99.2 kHz at 2480 MHz	"41 88 xx AA 1A FF FF 44 33 01 23 45 67 89 AB CD EF FE DC BA 98 76 54 32 10 CRC1 CRC2", where xx is the mac sequence number which can be any value and the FCS (CRC1 and CRC2) is computed normally.
2003 Sub-GHz	868 MHz, 20kbps, BPSK 915 MHz, 40kbps, BPSK	≥ -3 dBm	≤ +/- 40ppm or ≤ +/- 34.7 kHz at 868.3 MHz ≤ +/- 36.2 kHz at 906 MHz to ≤ +/- 37.0 kHz at 924 MHz	"41 88 xx AA 1A FF FF 44 33 01 23 45 67 89 AB CD EF FE DC BA 98 76 54 32 10 CRC1 CRC2", where xx is the mac sequence number which can be any value and the FCS (CRC1 and CRC2) is computed normally.
GB SE Sub-GHz	863-876 MHz, 100kbps, Filtered 2FSK 915-921 MHz, 100kbps, Filtered 2FSK	≥ -3 dBm	≤ +/- 20ppm or ≤ +/- 17.26 kHz at 863.25 MHz to ≤ +/- 18.41 kHz at 920.55 MHz	Arbitrary bit pattern for the payload (PSDU), and correct CRC, where the payload (PSDU) consists of 127 bytes.

Setup Conditions:

Diagram



The Diagram is the image that represents the configuration deployed for the Test Procedure.

531 ***Initial Conditions:***

Device name	Description
DUT	*Channel: for this test case, all of the required testing channels for the specific PHY being tested are specified in the table at the bottom of the Test Procedure.*
Signal Analyzer	*Signal Analyzer shall receive output from the DUT.* *Signal Analyzer shall be configured in the following manner:* *Reference Level is set to an appropriate value according to the stated DUT maximum output power level. If the DUT output power level exceeds analyzer's allowed maximum RF input level, use additional attenuators.* *All other parameters of the Signal Analyzer shall be set as defined in the table below.*

532

	Band	Span	Resolution BW (RBW)	Video BW (VBW)	Attenuation
2003 2.4 GHz	2.4 GHz	5 MHz	3 MHz	≥ RBW	Auto
2003 Sub-GHz	868 MHz 915 MHz	2 MHz 3 MHz	≤ 100 kHz	≥ RBW	Auto
GB SE Sub-GHz	863-876 MHz 915-921 MHz	300 kHz	100 kHz	≥ RBW	Auto

533 ***Test Procedure:***

Step No.	Step description	Expected outcome
1	DUT shall transmit a standards conforming signal and packet with the specified PHY Payload (PSDU) and FCS/CRC. The signal level shall be at the specified Output Power Level. If the DUT is capable of properly transmitting at a maximum output power greater than minimum specified Output Power Level then the output power of the DUT shall be set to its maximum output power level. Measure, over the specified Span, the frequency of the peak spectral value of the DUT for the channel being tested.	Pass: For each sample of each of the required testing channels the specified Maximum Center Frequency Offset is met. Fail: For any sample of any of the required testing channels the specified Maximum Center Frequency Offset is not met.
2	Calculate the deviation of the measured peak value from the ideal peak value at the Center frequency for the channel being tested. Repeat Measurement and Calculation and record for 10 samples.	

3	For the PHY being tested perform test for each of the following required Channels to Test at the corresponding center frequencies specified in the table below.	

	PHY	Channels to Test	Center Frequency Calculation
2003 2.4 GHz	2.4 GHz, 250kbps, O-QPSK	11 to 26	fc [MHz] = 2405+5*(k-11), where k = 11 (0x0B) to 26 (0x1A)
2003 Sub-GHz	868 MHz, 20kbps, BPSK 915 MHz, 40kbps, BPSK	0 1 to 10	fc [MHz] = 868.3 fc [MHz] = 906 + 2* (k-1), where k = 1, 2, …, 10
GB SE Sub-GHz	863-876 MHz, 100kbps, Filtered 2FSK 915-921 MHz, 100kbps, Filtered 2FSK	Ch Page [Ch #'s] 28 [0, 6, 13, 20, 26] 29 [27, 30, 32, 34, 62] 30 [35, 41, 48, 55, 61] 31 [0, 6, 13, 20, 26]	fc [MHz] = ChanCenterFreq0 + Ch # * ChanSpacing where ChanSpacing = 0.200 MHz For 863-876 MHz, Ch Page 28-30, ChanCenterFreq0 = 863.25 MHz For 915-921 MHz, Ch Page 31, ChanCenterFreq0 = 915.35 MHz

Notes:

None.

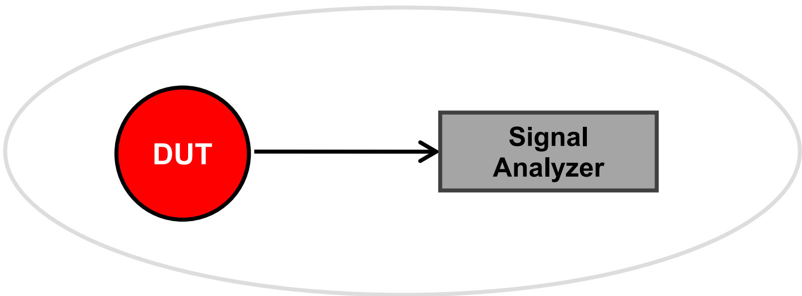
TP/154/PHY/TRANSMIT-05

Tests that for the PHY being tested, DUT is capable of transmitting at the Minimum Tx Power Level and Maximum Tx Power Level specified in the table below, using a standards conforming signal and packet with an arbitrary bit pattern with the Payload (PSDU) Length number of bytes specified in the table below, and correct CRC.

	PHY	Minimum Tx Power Level	Maximum Tx Power Level	Payload (PSDU) Length
2003 2.4 GHz	2.4 GHz, 250kbps, O-QPSK	≥ -3 dBm	DUT specific	≥ 20 bytes
2003 Sub-GHz	868 MHz, 20kbps, BPSK 915 MHz, 40kbps, BPSK	≥ -3 dBm	DUT specific	≥ 20 bytes

Setup Conditions:

Diagram



The Diagram is the image that represents the configuration deployed for the Test Procedure.

Initial Conditions:

Device name	Description
DUT	*Channel: for this test case, all of the required testing channels for the specific PHY being tested are specified in the table at the bottom of the Test Procedure.*
Signal Analyzer	*Signal Analyzer shall receive output from the DUT.* *Signal Analyzer shall be configured in the following manner:* *Reference Level is set to an appropriate value according to the stated DUT maximum output power level. If the DUT output power level exceeds analyzer's allowed maximum RF input level, use additional attenuators.* *All other parameters of the Signal Analyzer shall be set as defined in the table below.*

	Band	Span	Resolution BW (RBW)	Video BW (VBW)	Attenuation	Detector Type
2003 2.4 GHz	2.4 GHz	5 MHz	3 MHz	≥ RBW	Auto	RMS
2003 Sub-GHz	868 MHz 915 MHz	2 MHz 3 MHz	3 MHz	≥ RBW	Auto	RMS

Test Procedure:

Step No.	**Step description**	**Expected outcome**
1	DUT shall transmit a standards conforming signal and packet with the specified payload (PSDU), and correct CRC, with a payload (PSDU) length of Payload (PSDU) Length. The output power of the DUT shall be set to its maximum output power level. Set the signal/spectrum analyzer to trigger on the detected RF/IF power. If a power meter with averaging detector is used, correct the measurement results by the stated duty cycle.	Pass: For each of the required testing channels the specified Minimum Tx Power Level and Maximum Tx Power Level is met. Fail: For any of the required testing channels the specified Minimum Tx Power Level or Maximum Tx Power Level is not met.
2	For the PHY being tested perform test for each of the following required Channels to Test at the corresponding center frequencies specified in the table below.	

	PHY	**Channels to Test**	**Center Frequency Calculation**
2003 2.4 GHz	2.4 GHz, 250kbps, O-QPSK	11 to 26	fc [MHz] = 2405+5*(k-11), where k = 11 (0x0B) to 26 (0x1A)
2003 Sub-GHz	868 MHz, 20kbps, BPSK 915 MHz, 40kbps, BPSK	0 1 to 10	fc [MHz] = 868.3 fc [MHz] = 906 + 2* (k-1), where k = 1, 2, …, 10

Notes:

None.

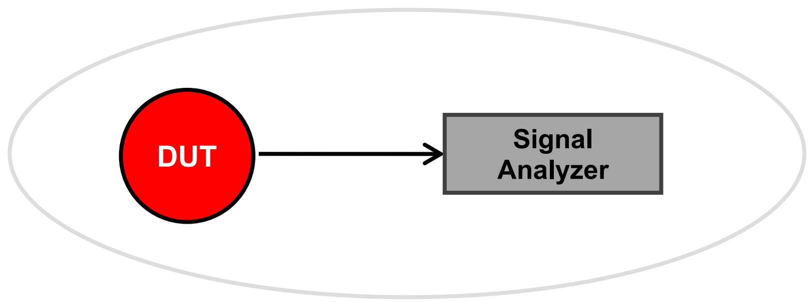
TP/154/PHY/TRANSMIT-06

Tests that for the PHY being tested, DUT is capable of transmitting a standards conforming signal and packet with an arbitrary payload and correct CRC and varying its transmit power on a packet basis over a defined range (from the Minimum Tx Power Level to the Maximum Tx Power Level), with a step size ≤ 2 dB and an accuracy of 3 dB or better. This test case, unlike the preceding Tx power test case, ensures interoperable Adpative Transmit Power Control for the PHY's utilizing it.

	PHY	Payload (PSDU) Length	Minimum Tx Power Level	Maximum Tx Power Level	Step Size	Accuracy
GB SE Sub-GHz	863-876 MHz, 100kbps, Filtered 2FSK 915-921 MHz, 100kbps, Filtered 2FSK	127 bytes	-15 dBm (+/- 1dB)	14 dBm (+ 0dB/ - 3dB)	≤ 2 dB	≤ 3 dB

Setup Conditions:

Diagram



The Diagram is the image that represents the configuration deployed for the Test Procedure.

Initial Conditions:

Device name	Description
DUT	*Channel: for this test case, all of the required testing channels for the specific PHY being tested are specified in the table at the bottom of the Test Procedure.*
Signal Analyzer	*Signal Analyzer shall receive output from the DUT.* *Signal Analyzer shall be configured in the following manner:* *Reference Level is set to an appropriate value according to the stated DUT maximum output power level. If the DUT output power level exceeds analyzer's allowed maximum RF input level, use additional attenuators.* *All other parameters of the Signal Analyzer shall be set as defined in the table below.*

	Band	Span	Resolution BW (RBW)	Video BW (VBW)	Attenuation	Detector Type
GB SE Sub-GHz	863-876 MHz 915-921 MHz	200 kHz	100 kHz	≥ RBW	Auto	RMS

Test Procedure:

Step No.	Step description	Expected outcome
1	DUT shall transmit a standards conforming signal and packet with the specified payload (PSDU), and correct CRC, with a payload (PSDU) length of Payload (PSDU) Length. The output power of the DUT shall initially be set to the specified Minimum Tx Power Level. The test shall be repeated, with the Tx power increased by not more than the specified Step Size each time, until the Tx power is equal to the specified Maximum Tx Power Level. Set the signal/spectrum analyzer to trigger on the detected RF/IF power. If a power meter with averaging detector is used, correct the measurement results by the stated duty cycle.	Pass: For each of the required testing channels the DUT is capable of transmitting packets at power levels over the defined range, and meeting the specified Step Size and Accuracy. Fail: For any of the required testing channels the DUT is not capable of transmitting packets at power levels over the defined range, and/or not meeting the specified Step Size and/or Accuracy.
2	For the PHY being tested perform test for each of the following required Channels to Test at the corresponding center frequencies specified in the table below.	

	PHY	Channels to Test	Center Frequency Calculation
GB SE Sub-GHz	863-876 MHz, 100kbps, Filtered 2FSK 915-921 MHz, 100kbps, Filtered 2FSK	Ch Page [Ch #'s] 28 [0, 6, 13, 20, 26] 29 [27, 30, 32, 34, 62] 30 [35, 41, 48, 55, 61] 31 [0, 6, 13, 20, 26]	fc [MHz] = ChanCenterFreq0 + Ch # * ChanSpacing where ChanSpacing = 0.200 MHz For 863-876 MHz, Ch Page 28-30, ChanCenterFreq0 = 863.25 MHz For 915-921 MHz, Ch Page 31, ChanCenterFreq0 = 915.35 MHz

Notes:

None.

TP/154/PHY/TRANSMIT-07

Tests that for the PHY being tested, the emissions of the DUT are attenuated by at least the Emission Limit specified in the tables below, over the Frequency Offset specified in the tables below.

The emissions measurement shall be done at the TX Power specified in the tables below, and the measurement interval shall encompass all portions of a transmitted frame consisting of a standards conforming signal and packet with an arbitrary payload and correct CRC for the Payload (PSDU) Length specified in the tables below.

	PHY	**Tx Power**	**Frequency Offset**	**Relative Emission Limit**	**Absolute Emission Limit**	**Payload (PSDU) Length**
2003 2.4 GHz	2.4 GHz, 250kbps, O-QPSK 2405 – 2480 MHz (Channels 11-26)	Max.	$\lvert f - f_c \rvert > 3.5$ MHz	-20 dB	-30 dBm	≥ 20 bytes
2003 Sub-GHz	868 MHz, 20kbps, BPSK 868.3 MHz (Channel 0) 915 MHz, 40kbps, BPSK 906 – 924 MHz (Channels 1-10)	Max.	$\lvert f - f_c \rvert > 1.2$ MHz	-20 dB	-20 dBm	≥ 20 bytes

	PHY	**Tx Power**	**Frequency Offset**	**Absolute Emission Limit[1]**	**Payload (PSDU) Length**
GB SE Sub-GHz	863 – 870 MHz (Channels 0-33)[2]	+ 14 dBm	$\leq f_{el}, \geq f_{eu}$ $\leq (f_{el}$ - 200 kHz), $\geq (f_{eu}$ + 200 kHz) $\leq (f_{el}$ - 400 kHz), $\geq (f_{eu}$ + 400 kHz) $\leq (f_{el}$ - 1000 kHz), $\geq (f_{eu}$ + 1000 kHz)	-30 dBm -36 dBm -46 dBm -56 dBm	127 bytes
	870 – 876 MHz (Channel 34-48)[2,3]	+ 14 dBm	$\leq (f_c$ - 100 kHz), $\geq (f_c$ + 100 kHz) $\leq (f_c$ - 500 kHz), $\geq (f_c$ + 500 kHz) $\leq f_{el}, \geq f_{eu}$ $\leq (f_{el}$ - 200 kHz), $\geq (f_{eu}$ + 200 kHz) $\leq (f_{el}$ - 400 kHz), $\geq (f_{eu}$ + 400 kHz) $\leq (f_{el}$ - 1000 kHz), $\geq (f_{eu}$ + 1000 kHz)	0 dBm -36 dBm -30 dBm -36 dBm -46 dBm -56 dBm	127 bytes
	915 – 921 MHz (Channels 0-12)[2]	+ 14 dBm	$\leq f_{el}, \geq f_{eu}$ $\leq (f_{el}$ - 200 kHz), $\geq (f_{eu}$ + 200 kHz) $\leq (f_{el}$ - 400 kHz), $\geq (f_{eu}$ + 400 kHz) $\leq (f_{el}$ - 1000 kHz), $\geq (f_{eu}$ + 1000 kHz)	-30 dBm -36 dBm -46 dBm -56 dBm	127 bytes

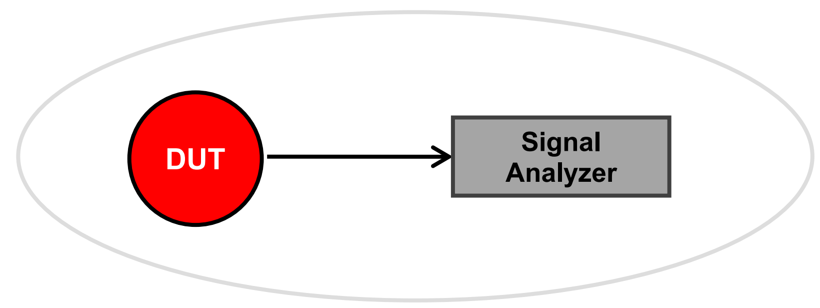
[1]Measured using a RBW = 1 kHz

[2]f_{el} and f_{eu} are the lower and upper band edges

[3] f_c is the center of the channel

579 ***Setup Conditions:***

Diagram



580 *The Diagram is the image that represents the configuration deployed for the Test Procedure.*

581 ***Initial Conditions:***

Device name	Description
DUT	*Channel: for this test case, all of the required testing channels for the specific PHY being tested are specified in the table at the bottom of the Test Procedure.*
Signal Analyzer	*Signal Analyzer shall receive output from the DUT.* *Signal Analyzer shall be configured in the following manner:* *Reference Level is set to an appropriate value according to the stated DUT maximum output power level. If the DUT output power level exceeds analyzer's allowed maximum RF input level, use additional attenuators.* *All other parameters of the Signal Analyzer shall be set as defined in the table below.*

582

	Band	Span	Resolution BW (RBW)	Video BW (VBW)	Attenuation	Detector Type
2003 2.4 GHz	2.4 GHz	5 MHz	100 kHz	≥ RBW	Auto	RMS
2003 Sub-GHz	868 MHz 915 MHz	2 MHz 3 MHz	100 kHz	≥ RBW	Auto	RMS
GB SE Sub-GHz	863-876 MHz 915-921 MHz	$f_{eu} - f_{el} + 4$ MHz, for the band in which the channel resides	1 kHz	≥ RBW	Auto	RMS

583 ***Test Procedure:***

Step No.	Step description	Expected outcome
1	DUT shall transmit a standards conforming signal and packet with an arbitrary bit pattern for the payload (PSDU), and correct CRC, with the specified Payload (PSDU)	Pass: All the power measurments at the frequency offsets indicated in the table are at or below the specified Emission Limit value for each* of the required testing channels.

	Length at the specified Tx Power for for each channel in the table below. Set the signal/spectrum analyzer to trigger on the detected RF/IF power. Using the Signal Analyzer initial conditions configuration settings make the power measurements as specified in the table.	**In the case of Channel 26 (for 2003 2.4 GHz), for the Absolute Emission Limit, this test may be performed at a Tx power level less than the specified Tx Power, provided that the Tx power level used is at least -3dBm. In the case where the specified Tx Power is not used in the testing of Channel 26, the Tx power level used to successfully pass the Absolute Emission Limit is to be recorded as part of the Pass/Fail test results.* Fail: At least one of the power measurements at the frequency offsets indicated in the table are not at or below the specified Emission Limit value for at least one of the required testing channels. ***See exception above in Pass Criteria for testing on Channel 26 (for 2003 2.4 GHz).*
2	For tests with relative emissions levels, measure the relative emission limit vs. the peak of the PSD centered at the channels center frequency.	
3	For the PHY being tested perform test for each of the following required Channels to Test at the corresponding center frequencies specified in the table below.	

584

	PHY	**Channels to Test**	**Center Frequency Calculation**
2003 2.4 GHz	2.4 GHz, 250kbps, O-QPSK	11 to 26	fc [MHz] = 2405+5*(k-11), where k = 11 (0x0B) to 26 (0x1A)
2003 Sub-GHz	868 MHz, 20kbps, BPSK 915 MHz, 40kbps, BPSK	0 1 to 10	fc [MHz] = 868.3 fc [MHz] = 906 + 2* (k-1), where k = 1, 2, …, 10
GB SE Sub-GHz	863-876 MHz, 100kbps, Filtered 2FSK 915-921 MHz, 100kbps, Filtered 2FSK	Ch Page [Ch #'s] 28 [0, 6, 13, 20, 26] 29 [27, 30, 32, 34, 62] 30 [35, 41, 48, 55, 61] 31 [0, 6, 13, 20, 26]	fc [MHz] = ChanCenterFreq0 + Ch # * ChanSpacing where ChanSpacing = 0.200 MHz For 863-876 MHz, Ch Page 28-30, ChanCenterFreq0 = 863.25 MHz For 915-921 MHz, Ch Page 31, ChanCenterFreq0 = 915.35 MHz

585 ***Notes:***

586 *None.*

587

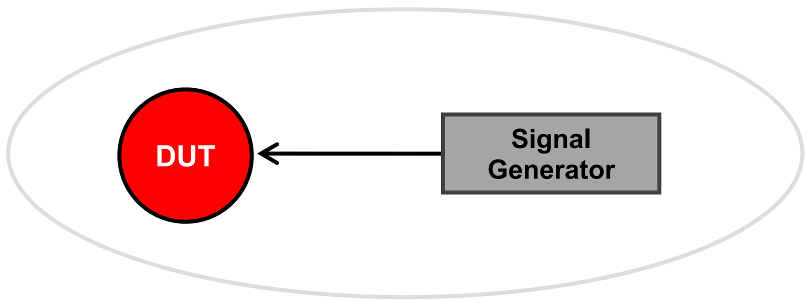
TP/154/PHY/RECEIVE-01

Tests that for the PHY being tested, the DUT receiver is capable of receiving packets at the Sensitivity Level specified in the table below, with the PER specified in the table below, for a packet with a Payload PSDU length in bytes (octets) in the data payload specified in the table below.

	PHY	PER	Payload PSDU (bytes)	Sensitivity Level
2003 2.4 GHz	2.4 GHz, 250kbps, O-QPSK	≤ 1%	20	-85 dBm
2003 Sub-GHz	868 MHz, 20kbps, BPSK 915 MHz, 40kbps, BPSK	≤ 1%	20	-92 dBm
GB SE Sub-GHz	863-876 MHz, 100kbps, Filtered 2FSK 915-921 MHz, 100kbps, Filtered 2FSK	≤ 1%	127	-99 dBm (+/- 1 dB)

Setup Conditions:

Diagram



The Diagram is the image that represents the configuration deployed for the Test Procedure.

Initial Conditions:

Device name	Description
DUT	*Channel: for this test case, all of the required testing channels for the specific PHY being tested are specified in the table at the bottom of the Test Procedure.*
Signal Generator	*DUT shall receive output from a Signal Generator capable of generating digitally modulated radio frequency signals for the band of operation.* *Signal Generator shall be configured to transmit a standards conforming signal and packet with an arbitrary payload and correct CRC with the following parameters:* *Using the PSDU generation approach for the specific PHY(s) being tested, as described in 6.1.* *Using the odd and even packet sequence.* *Configuring the Signal Generator as defined in the table below.*

	Band	PSDU # Bits	Bit Pattern	Mod.	Symbol Rate (kHz)	Mod. Index	Pulse Shape	PSDU Data Whitening
2003 2.4 GHz	2.4 GHz	20 x 8	Phase Continous	O-QPSK	250	n/a	Half Sine	n/a
2003 Sub-GHz	868 MHz 915 MHz	20 x 8	Phase Continous	BPSK	20 40	n/a	Raised Cosine roll-off=1.0	n/a
GB SE Sub-GHz	863-876 MHz 915-921 MHz	127 x 8	Phase Continous	2FSK	100	0.7	Gaussian BT=0.5	PN9 in *16.2.3 of [R7]*

596 ***Test Procedure:***

Step No.	Step description	Expected outcome
1	Adjust the output power of Signal Generator to generate an input level equal to the specified Sensitivity Level at DUT receiver input. Measure PER while the Signal Generator transmits packets and DUT receiver receives packets on the channel being tested. Signal Generator shall generate at least 1000 packets at each power level being tested.	Pass: For each of the required testing channels the PER of the DUT meets the specified PER. Fail: For any of the required testing channels the PER of the DUT does not meet the specified PER.
2	For the PHY being tested perform test for each of the following required Channels to Test at the corresponding center frequencies specified in the table below.	

597

	PHY	Channels to Test	Center Frequency Calculation
2003 2.4 GHz	2.4 GHz, 250kbps, O-QPSK	11 to 26	fc [MHz] = 2405+5*(k-11), where k = 11 (0x0B) to 26 (0x1A)
2003 Sub-GHz	868 MHz, 20kbps, BPSK 915 MHz, 40kbps, BPSK	0 1 to 10	fc [MHz] = 868.3 fc [MHz] = 906 + 2* (k-1), where k = 1, 2, …, 10
GB SE Sub-GHz	863-876 MHz, 100kbps, Filtered 2FSK 915-921 MHz, 100kbps, Filtered 2FSK	Ch Page [Ch #'s] 28 [0, 6, 13, 20, 26] 29 [27, 30, 32, 34, 62] 30 [35, 41, 48, 55, 61] 31 [0, 6, 13, 20, 26]	fc [MHz] = ChanCenterFreq0 + Ch # * ChanSpacing where ChanSpacing = 0.200 MHz For 863-876 MHz, Ch Page 28-30, ChanCenterFreq0 = 863.25 MHz For 915-921 MHz, Ch Page 31, ChanCenterFreq0 = 915.35 MHz

598 ***Notes:***

599 *None.*

TP/154/PHY/RECEIVE-02

Tests that for the PHY being tested, the DUT receiver is capable of receiving packets at the Desired Channel Test Input Level specified in the table below, with the PER specified in the table below, for a packet with a Payload PSDU length in bytes (octets) in the data payload specified in the table below,

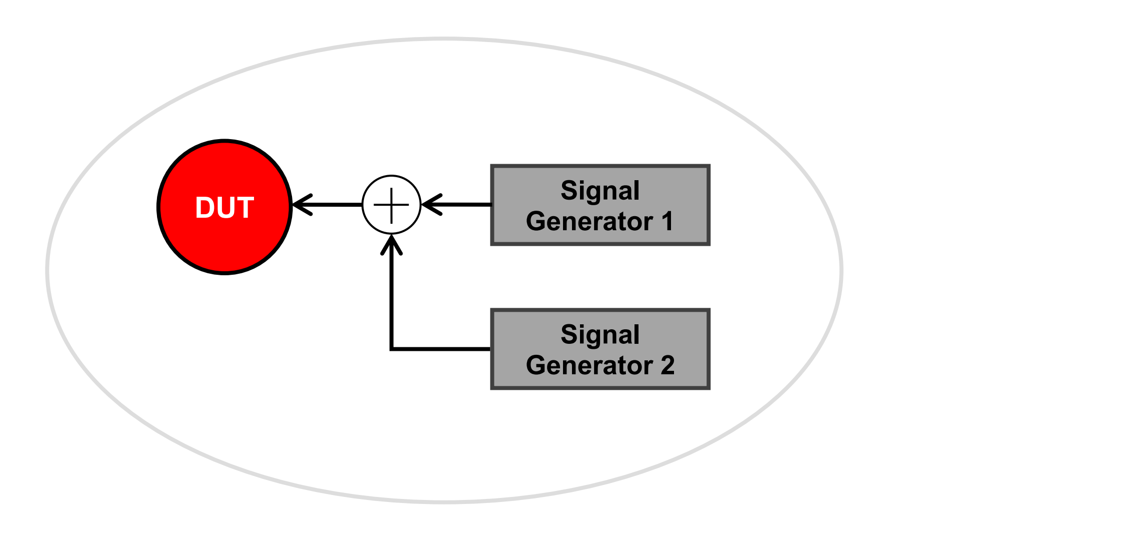
	PHY	PER	Payload PSDU (bytes)	Desired Channel Test Input Level
GB SE Sub-GHz	863-876 MHz, 100kbps, Filtered 2FSK 915-921 MHz, 100kbps, Filtered 2FSK	$\leq$ 1%	127	-79 dBm

given a jamming signal at the specified Co-Channel Relative Jamming Level below on the same channel, for a packet with a Payload PSDU length in bytes (octets) in the data payload specified in the table below.

	PHY	Payload PSDU (bytes)	Co-Channel Relative Jamming Level
GB SE Sub-GHz	863-876 MHz, 100kbps, Filtered 2FSK 915-921 MHz, 100kbps, Filtered 2FSK	127	-15 dB

Setup Conditions:

Diagram



The Diagram is the image that represents the configuration deployed for the Test Procedure.

Initial Conditions:

Device name	Description
DUT	*Channel: for this test case, all of the required testing channels for the specific PHY being tested are specified in the table at the bottom of the Test Procedure.*
Signal Generator 1/2	*DUT shall receive output from Signal Generator 1 capable of generating digitally modulated radio frequency signals for the band of operation.* *Signal Generator 2 may either be set to transmit synchronously with Signal Generator 1 or Signal Generator 2 may continuously stream packets during the test.*

	Signal Generator 1/2 shall be configured to transmit a standards conforming signal and packet with an arbitrary payload and correct CRC with the following parameters: *Using the PSDU generation approach for the specific PHY(s) being tested, as described in 6.1.* *Using the odd and even packet sequence.* *Center frequency: see Step 2 below.* *Configuring Signal Generator 1/2 as defined in the table below.*

609

	Band	PSDU # Bits	Bit Pattern	Mod.	Symbol Rate (kHz)	Mod. Index	Pulse Shape	PSDU Data Whitening
GB SE Sub-GHz	863-876 MHz 915-921 MHz	127 x 8	Phase Continuous	2FSK	100	0.7	Gaussian BT=0.5	PN9 in *16.2.3 of [R7]*

610 ***Test Procedure:***

Step No.	Step description	Expected outcome
1	Adjust the output power of Signal Generator 1 to generate an input level equal to the Desired Channel Test Input Level at DUT receiver input. Adjust the output power of Signal Generator 2 to generate an input level equal to the Co-Channel Relative Jamming Level at DUT receiver input. Signal Generator 1 shall be transmitting on the DUT receiving channel, and Signal Generator 2 shall be transmitting on the same channel that Signal Generator 1 is transmitting in. Measure PER while the Signal Generator transmits packets and DUT receiver receives packets on the channel being tested. Signal Generator 1 and Signal Generator 2 shall generate at least 1000 packets at each power being level tested. For each channel that the DUT is tested on a jamming signal on the same channel shall be tested.	Pass: For each of the required testing channels at the Relative Jamming Level, the PER of the DUT meets the specified PER. Fail: For any of the required testing channels at the Relative Jamming Level, the PER of the DUT does not meet the specified PER.
2	For the PHY being tested perform test for each of the following required Channels to Test at the corresponding center frequencies specified in the table below.	

611

612

	PHY	Channels to Test	Center Frequency Calculation
GB SE Sub-GHz	863-876 MHz, 100kbps, Filtered 2FSK 915-921 MHz, 100kbps, Filtered 2FSK	Ch Page [Ch #'s] 28 [0, 6, 13, 20, 26] 29 [27, 30, 32, 34, 62] 30 [35, 41, 48, 55, 61] 31 [0, 6, 13, 20, 26]	fc [MHz] = ChanCenterFreq0 + Ch # * ChanSpacing where ChanSpacing = 0.200 MHz For 863-876 MHz, Ch Page 28-30, ChanCenterFreq0 = 863.25 MHz For 915-921 MHz, Ch Page 31, ChanCenterFreq0 = 915.35 MHz

Notes:

None.

TP/154/PHY/RECEIVE-03

Tests that for the PHY being tested, the DUT receiver is capable of receiving packets at the Desired Channel Test Input Level specified in the table below, with the PER specified in the table below, for a packet with a Payload PSDU length in bytes (octets) in the data payload specified in the table below,

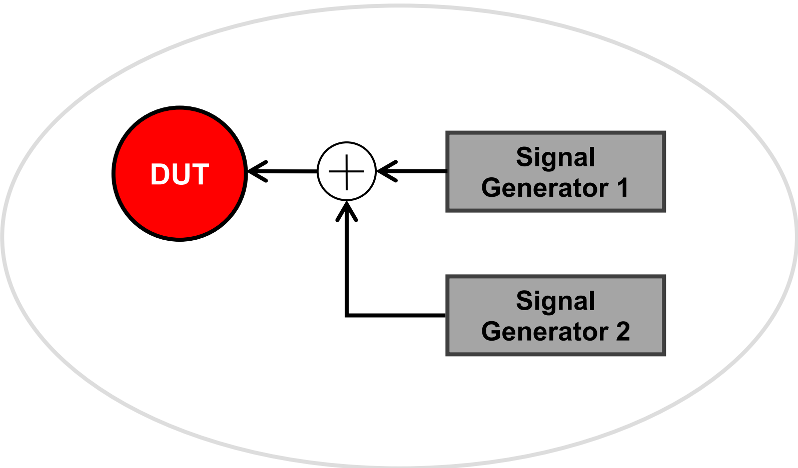
	PHY	PER	Payload PSDU (bytes)	Desired Channel Test Input Level
2003 2.4 GHz	2.4 GHz, 250kbps, O-QPSK	≤ 1%	20	-82 dBm
2003 Sub-GHz	868 MHz, 20kbps, BPSK 915 MHz, 40kbps, BPSK	≤ 1%	20	-89 dBm
GB SE Sub-GHz	863-876 MHz, 100kbps, Filtered 2FSK 915-921 MHz, 100kbps, Filtered 2FSK	≤ 1%	127	-96 dBm

given a jamming signal at the specific Relative Jamming Levels for the specific Relative Jamming Channels in the table below.

	PHY	Payload PSDU (bytes)	Adjacent Channel Relative Jamming Level/Channel	Alternate Channel Relative Jamming Level/Channel	5th Channel Relative Jamming Level/Channel
2003 2.4 GHz	2.4 GHz, 250kbps, O-QPSK	20	0 dB @ +/- 5 MHz	+30 dB @ +/- 10 MHz	n/a
2003 Sub-GHz	868 MHz, 20kbps, BPSK 915 MHz, 40kbps, BPSK	20	0 dB @ +/- 2 MHz	+30 dB @ +/- 4 MHz	n/a
GB SE Sub-GHz	863-876 MHz, 100kbps, Filtered 2FSK 915-921 MHz, 100kbps, Filtered 2FSK	127	+17 dB @ +/- 200 kHz	+30 dB @ +/- 400 kHz	+35 dB @ +/- 1000 kHz

Setup Conditions:

Diagram



The Diagram is the image that represents the configuration deployed for the Test Procedure.

626 ***Initial Conditions:***

Device name	Description
DUT	*Channel: for this test case, all of the required testing channels for the specific PHY being tested are specified in the table at the bottom of the Test Procedure.*
Signal Generator 1/2	*DUT shall receive output from Signal Generator 1 capable of generating digitally modulated radio frequency signals for the band of operation.* *Signal Generator 2 may either be set to transmit synchronously with Signal Generator 1 or Signal Generator 2 may continuously stream packets during the test.* *Signal Generator 1/2 shall be configured to transmit a standards conforming signal and packet with an arbitrary payload and correct CRC with the following parameters:* *Using the PSDU generation approach for the specific PHY(s) being tested, as described in 6.1.* *Using the odd and even packet sequence.* *Center frequency: see Step 2 below.* *Configuring Signal Generator 1/2 as defined in the table below.*

627

	Band	**PSDU # Bits**	**Bit Pattern**	**Mod.**	**Symbol Rate (kHz)**	**Mod. Index**	**Pulse Shape**	**PSDU Data Whitening**
2003 2.4 GHz	2.4 GHz	20 x 8	Phase Continuous	O-QPSK	250	n/a	Half Sine	n/a
2003 Sub-GHz	868 MHz 915 MHz	20 x 8	Phase Continuous	BPSK	20 40	n/a	Raised Cosine roll-off=1.0	n/a
GB SE Sub-GHz	863-876 MHz 915-921 MHz	127 x 8	Phase Continuous	2FSK	100	0.7	Gaussian BT=0.5	PN9 in *16.2.3 of [R7]*

628 ***Test Procedure:***

Step No.	Step description	Expected outcome
1	Adjust the output power of Signal Generator 1 to generate an input level equal to the Desired Channel Test Input Level at DUT receiver input. Adjust the output power of Signal Generator 2 to generate an input level equal to the Relative Jamming Level at DUT receiver input. Signal Generator 1 shall be transmitting in the DUT receiving channel, and Signal Generator 2 shall be transmitting in either the high or low jamming channel relative to the channel Signal Generator 1 is transmitting in.	Pass: For each of the required testing channels at the Relative Jamming Level for both the high and low Relative Jamming Channels, the PER of the DUT meets the specified PER. Fail: For any of the required testing channels at any of the Relative Jamming Levels for either the high and low Relative Jamming Channels, the PER of the DUT does not meet the specified PER.

	Measure PER while the Signal Generator transmits packets and DUT receiver receives packets on the channel being tested. Signal Generator 1 and Signal Generator 2 shall generate at least 1000 packets at each power being level tested. For each channel that the DUT is tested on a jamming signal on both high and low Relative Jamming Channels for each of the Relative Jamming Levels shall be tested.	
2	For the PHY being tested perform test for each of the following required Channels to Test at the corresponding center frequencies specified in the table below.	

	PHY	**Channels to Test**	**Center Frequency Calculation**
2003 2.4 GHz	2.4 GHz, 250kbps, O-QPSK	11 to 26	fc [MHz] = 2405+5*(k-11), where k = 11 (0x0B) to 26 (0x1A)
2003 Sub-GHz	868 MHz, 20kbps, BPSK 915 MHz, 40kbps, BPSK	0 1 to 10	fc [MHz] = 868.3 fc [MHz] = 906 + 2* (k-1), where k = 1, 2, …, 10
GB SE Sub-GHz	863-876 MHz, 100kbps, Filtered 2FSK 915-921 MHz, 100kbps, Filtered 2FSK	Ch Page [Ch #'s] 28 [0, 6, 13, 20, 26] 29 [27, 30, 32, 34, 62] 30 [35, 41, 48, 55, 61] 31 [0, 6, 13, 20, 26]	fc [MHz] = ChanCenterFreq0 + Ch # * ChanSpacing where ChanSpacing = 0.200 MHz For 863-876 MHz, Ch Page 28-30, ChanCenterFreq0 = 863.25 MHz For 915-921 MHz, Ch Page 31, ChanCenterFreq0 = 915.35 MHz

Notes:

None.

TP/154/PHY/RECEIVE-04

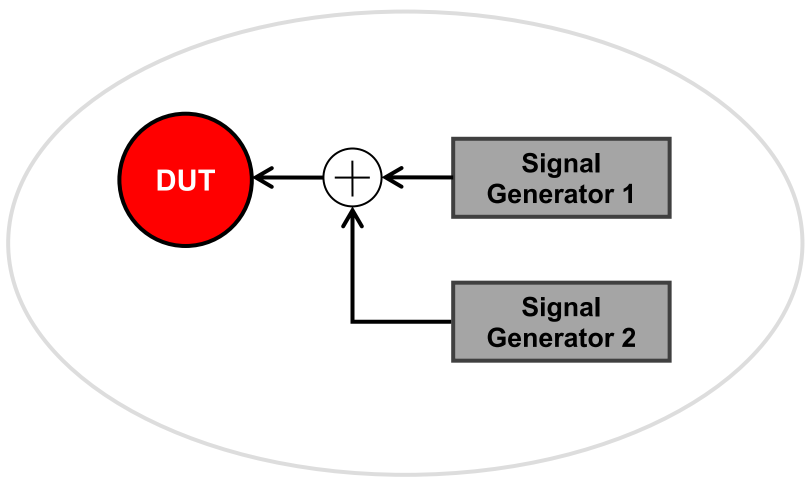
Tests that for the PHY being tested, the DUT receiver is capable of receiving packets at the Desired Channel Test Input Level specified in the table below, with the PER specified in the table below, for a packet with a Payload PSDU length in bytes (octets) in the data payload specified in the table below.

	PHY	PER	Payload PSDU (bytes)	Desired Channel Test Input Level
GB SE Sub-GHz	868.05 MHz, 100kbps, Filtered 2FSK	≤ 1%	127	-96 dBm

given a jamming signal at the specified Relative Blocking Levels for the specified Relative Blocking Channels in the table below.

	PHY	Payload PSDU (bytes)	Blocker 1 Relative Blocking Level/Channel	Blocker 2 Relative Blocking Level/Channel	Blocker 3 Relative Blocking Level/Channel
GB SE Sub-GHz	868.05 MHz, 100kbps, Filtered 2FSK	127	+40 dB @ +/- 2 MHz	+45 dB @ +/- 6 MHz	+50 dB @ +/- 10 MHz

Setup Conditions:

Diagram


The Diagram is the image that represents the configuration deployed for the Test Procedure.

Initial Conditions:

Device name	**Description**
DUT	*Channel: for this test case, all of the required testing channels for the specific PHY being tested are specified in the table at the bottom of the Test Procedure.*
Signal Generator 1/2	*DUT shall receive output from Signal Generator 1 capable of generating digitally modulated radio frequency signals for the band of operation.* *Signal Generator 2 may either be set to transmit synchronously with Signal Generator 1 or Signal Generator 2 may continuously stream packets during the test.*

	Signal Generator 1/2 shall be configured to transmit a standards conforming signal and packet with an arbitrary payload and correct CRC with the following parameters: *Using the PSDU generation approach for the specific PHY(s) being tested, as described in 6.1.* *Using the odd and even packet sequence.* *Center frequency: see Step 2 below.* *Configuring Signal Generator 1/2 as defined in the table below.*

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	Band	PSDU # Bits	Bit Pattern	Mod.	Symbol Rate (kHz)	Mod. Index	Pulse Shape	PSDU Data Whitening
GB SE Sub-GHz	863-876 MHz 915-921 MHz	127 x 8	Phase Continuous	2FSK	100	0.7	Gaussian BT=0.5	PN9 in *16.2.3 of [R7]*

643 ***Test Procedure:***

Step No.	Step description	Expected outcome
1	Adjust the output power of Signal Generator 1 to generate an input level equal to the Desired Channel Test Input Level at DUT receiver input. Adjust the output power of Signal Generator 2 to generate an input level equal to the Relative Blocking Level at DUT receiver input. Signal Generator 1 shall be transmitting in the DUT receiving channel, and Signal Generator 2 shall be transmitting in either the high or low blocking channel relative to the channel Signal Generator 1 is transmitting in. Measure PER while the Signal Generator transmits packets and DUT receiver receives packets on the channel being tested. Signal Generator 1 and Signal Generator 2 shall generate at least 1000 packets at each power being level tested. For the specified channel that the DUT is tested on a blocking signal on both high and low Relative Blocking Channels for each of the Relative Blocking Levels shall be tested.	Pass: For each of the required testing channels at all the Relative Blocking Levels, the Relative Blocking Levels for both the high and low Relative Blocking Channels results in a PER that meets the specified PER. Fail: For any of the required testing channels at any of the Relative Bocking Levels, the Relative Blocking Levels for either the high or low Relative Blocking Channels results in a PER that does not meet the specified PER.
2	For the PHY being tested perform test for each of the following required Channels to Test at the corresponding center frequencies specified in the table below.	

	PHY	Channels to Test	Center Frequency Calculation
GB SE Sub-GHz	868.05 MHz, 100kbps, Filtered 2FSK	Ch Page [Ch #'s] 28 [24]	fc [MHz] = ChanCenterFreq0 + Ch # * ChanSpacing where ChanSpacing = 0.200 MHz For 863-876 MHz, Ch Page 28-30, ChanCenterFreq0 = 863.25 MHz

Notes:

None.

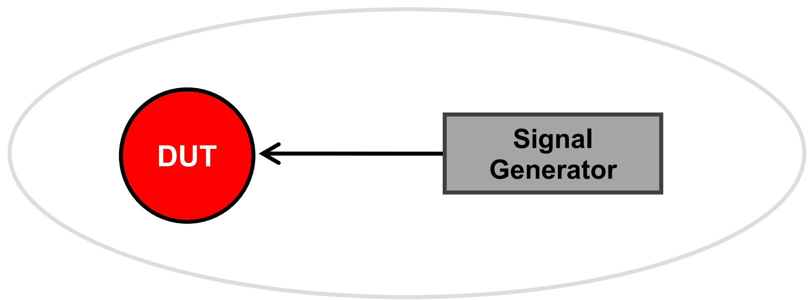
TP/154/PHY/RECEIVE-05

Tests that for the PHY being tested, the DUT receiver is capable of receiving packets at the Maximum Input Power Level specified in the table below, with the PER specified in the table below, for a packet with a Payload PSDU length in bytes (octets) in the data payload specified in the table below.

	PHY	PER	Payload PSDU (bytes)	Maximum Input Power Level
2003 2.4 GHz	2.4 GHz, 250kbps, O-QPSK	≤ 1%	20	-20 dBm
2003 Sub-GHz	868 MHz, 20kbps, BPSK 915 MHz, 40kbps, BPSK	≤ 1%	20	-20 dBm
GB SE Sub-GHz	863-876 MHz, 100kbps, Filtered 2FSK 915-921 MHz, 100kbps, Filtered 2FSK	≤ 1%	127	-20 dBm

Setup Conditions:

Diagram



The Diagram is the image that represents the configuration deployed for the Test Procedure.

Initial Conditions:

Device name	Description
DUT	*Channel: for this test case, all of the required testing channels for the specific PHY being tested are specified in the table at the bottom of the Test Procedure.*
Signal Generator	*DUT shall receive output from a Signal Generator capable of generating digitally modulated radio frequency signals for the band of operation.* *Signal Generator shall be configured to transmit a standards conforming signal and packet with an arbitrary payload and correct CRC with the following parameters:* *Using the PSDU generation approach for the specific PHY(s) being tested, as described in 6.1.* *Using the odd and even packet sequence.* *Configuring the Signal Generator as defined in the table below.*

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	Band	PSDU # Bits	Bit Pattern	Mod.	Symbol Rate (kHz)	Mod. Index	Pulse Shape	PSDU Data Whitening
2003 2.4 GHz	2.4 GHz	20 x 8	Phase Continuous	O-QPSK	250	n/a	Half Sine	n/a
2003 Sub-GHz	868 MHz 915 MHz	20 x 8	Phase Continuous	BPSK	20 40	n/a	Raised Cosine roll-off=1.0	n/a
GB SE Sub-GHz	863-876 MHz 915-921 MHz	127 x 8	Phase Continuous	2FSK	100	0.7	Gaussian BT=0.5	PN9 in *16.2.3 of [R7]*

658 ***Test Procedure:***

Step No.	Step description	Expected outcome
1	Adjust the output power of Signal Generator, to generate an input level equal to the specified Maximum Input Power Level at DUT receiver input. Measure PER while the Signal Generator transmits packets and DUT receiver receives packets on the channel being tested. Signal Generator shall generate at least 1000 packets at each power level being tested.	Pass: For each of the required testing channels the PER of the DUT meets the specified PER. Fail: For any of the required testing channels the PER of the DUT does not meet the specified PER.
2	For the PHY being tested perform test for each of the following required Channels to Test at the corresponding center frequencies specified in the table below.	

659

	PHY	Channels to Test	Center Frequency Calculation
2003 2.4 GHz	2.4 GHz, 250kbps, O-QPSK	11 to 26	fc [MHz] = 2405+5*(k-11), where k = 11 (0x0B) to 26 (0x1A)
2003 Sub-GHz	868 MHz, 20kbps, BPSK 915 MHz, 40kbps, BPSK	0 1 to 10	fc [MHz] = 868.3 fc [MHz] = 906 + 2* (k-1), where k = 1, 2, …, 10
GB SE Sub-GHz	863-876 MHz, 100kbps, Filtered 2FSK 915-921 MHz, 100kbps, Filtered 2FSK	Ch Page [Ch #'s] 28 [0, 6, 13, 20, 26] 29 [27, 30, 32, 34, 62] 30 [35, 41, 48, 55, 61] 31 [0, 6, 13, 20, 26]	fc [MHz] = ChanCenterFreq0 + Ch # * ChanSpacing where ChanSpacing = 0.200 MHz For 863-876 MHz, Ch Page 28-30, ChanCenterFreq0 = 863.25 MHz For 915-921 MHz, Ch Page 31, ChanCenterFreq0 = 915.35 MHz

660 ***Notes:***

661 *None.*

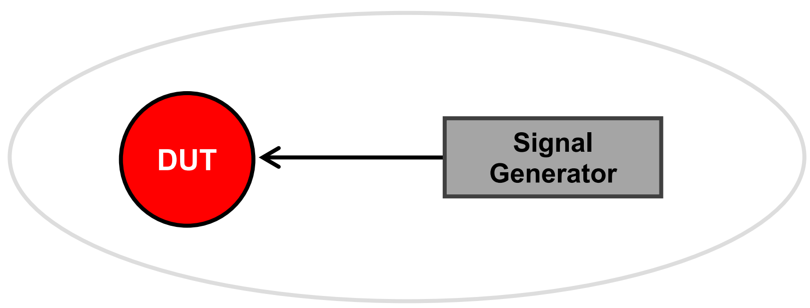
TP/154/PHY/RECEIVE-06

Tests that the DUT is capable of reporting Energy Detection (ED) values with the proper monotonicity and linearity requirements.

Tests that for the PHY being tested, the DUT receiver reports the minimum ED value of zero for the Minimum ED Power Level specified in the table below, spanning the Range of Detection specified in the table below at which the DUT reports an ED value of zero, with the ED Accuracy specified in the table below, in the ED Measurement Time to average over specified in the table below.

	PHY	**Minimum ED Power Level**	**Range of Detection**	**ED Accuracy**	**ED Measurement Time**
2003 2.4 GHz	2.4 GHz, 250kbps, O-QPSK	< -75 dBm	> 40 dB	< +/- 6 dB	8 symbol periods
2003 Sub-GHz	868 MHz, 20kbps, BPSK 915 MHz, 40kbps, BPSK	< -82 dBm	> 40 dB	< +/- 6 dB	8 symbol periods
GB SE Sub-GHz	863-876 MHz, 100kbps, Filtered 2FSK 915-921 MHz, 100kbps, Filtered 2FSK	< -89 dBm	> 40 dB	< +/- 6 dB	16 symbol periods

Setup Conditions:

Diagram


The Diagram is the image that represents the configuration deployed for the Test Procedure.

Initial Conditions:

Device name	**Description**
DUT	*Channel: for this test case, all of the required testing channels for the specific PHY being tested are specified in the table at the bottom of the Test Procedure.*
Signal Generator	*DUT shall receive output from a Signal Generator capable of generating an unmodulated carrier wave (CW).* *Signal Generator shall be configured to transmit an unmodulated CW at the center frequency of the channels being tested.*

Test Procedure:

Step No.	Step description	Expected outcome
1	DUT shall perform an ED measurement 10 times for each 2 dB transmitted power level step over the Range of Detection, starting from the highest input power level corresponding to the minimum ED value (0). Repeat above measurement for each channel in the band of operation. For the channel being measured, map the 10 measurement ED distribution for each received signal power level as a function of the calculated received signal power level, which is determined using the transmitted power level and accounting for all associated attenuation.	Pass: For each of the required testing channels the ED measurement is linear (within the specified ED Accuracy) over the specified Range of Detection starting from the highest input power level corresponding to the minimum ED value (0), where the ED value is zero for input power levels below the highest power level corresponding to the minimum ED value (0), and the ED value is greater than zero for input power levels above the highest power level corresponding to the minimum ED value (0), and the highest input power level corresponding to the minimum ED value (0) meets the specified Minimum ED Power Level. Fail: For any of the required testing channels the ED measurement is not linear (within the specified ED Accuracy) over the specified Range of Detection starting from the highest input power level corresponding to the minimum ED value (0), or the ED value is not zero for input power levels below the highest power level corresponding to the minimum ED value (0), or the ED value is zero for input power levels above the highest power level corresponding to the minimum ED value (0), or the highest input power level corresponding to the minimum ED value (0) does not meet the specified Minimum ED Power Level.
2	For the PHY being tested perform test for each of the following required Channels to Test at the corresponding center frequencies specified in the table below.	

	PHY	Channels to Test	Center Frequency Calculation
2003 2.4 GHz	2.4 GHz, 250kbps, O-QPSK	11 to 26	fc [MHz] = 2405+5*(k-11), where k = 11 (0x0B) to 26 (0x1A)
2003 Sub-GHz	868 MHz, 20kbps, BPSK 915 MHz, 40kbps, BPSK	0 1 to 10	fc [MHz] = 868.3 fc [MHz] = 906 + 2* (k-1), where k = 1, 2, …, 10
GB SE Sub-GHz	863-876 MHz, 100kbps, Filtered 2FSK 915-921 MHz, 100kbps, Filtered 2FSK	Ch Page [Ch #'s] 28 [0, 6, 13, 20, 26] 29 [27, 30, 32, 34, 62] 30 [35, 41, 48, 55, 61] 31 [0, 6, 13, 20, 26]	fc [MHz] = ChanCenterFreq0 + Ch # * ChanSpacing where ChanSpacing = 0.200 MHz For 863-876 MHz, Ch Page 28-30, ChanCenterFreq0 = 863.25 MHz For 915-921 MHz, Ch Page 31, ChanCenterFreq0 = 915.35 MHz

Notes:

Figure 5 *shows an example of the +/- 6dB accuracy requirement.*

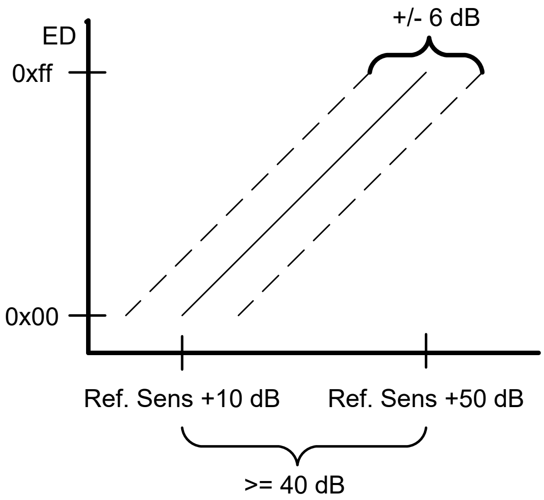


Figure 5 - Example of ED +/- 6 dB Accuracy

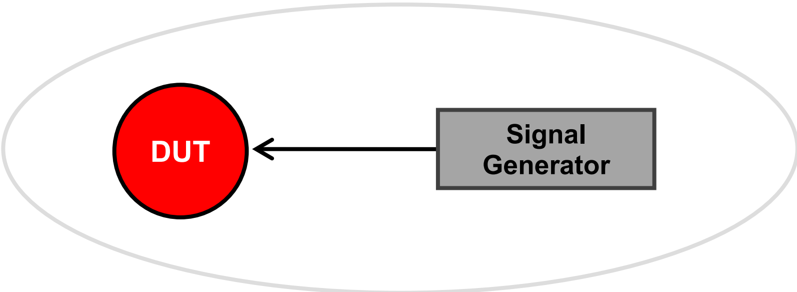
TP/154/PHY/RECEIVE-07

Tests that the DUT receiver is capable of reporting Link Quality Indicator (LQI).

Tests that for the PHY being tested, the minimum (0x00) and maximum (0xff) LQI values should be associated with the lowest and highest quality compliant signals detectable by the DUT receiver, and at least eight unique values of LQI are used.

Setup Conditions:

Diagram



The Diagram is the image that represents the configuration deployed for the Test Procedure.

Initial Conditions:

Device name	Description
DUT	*Channel: for this test case, all of the required testing channels for the specific PHY being tested are specified in the table at the bottom of the Test Procedure.*
Signal Generator	*DUT shall receive output from a Signal Generator capable of generating digitally modulated radio frequency signals for the band of operation.* *Signal Generator shall be configured to transmit a standards conforming signal and packet with an arbitrary payload and correct CRC with the following parameters:* *Using the PSDU generation approach for the specific PHY(s) being tested, as described in 6.1.* *Using the odd and even packet sequence.* *Configuring the Signal Generator as defined in the table below.*

	Band	PSDU # Bits	Bit Pattern	Mod.	Symbol Rate (kHz)	Mod. Index	Pulse Shape	PSDU Data Whitening
2003 2.4 GHz	2.4 GHz	20 x 8	Phase Continuous	O-QPSK	250	n/a	Half Sine	n/a
2003 Sub-GHz	868 MHz 915 MHz	20 x 8	Phase Continuous	BPSK	20 40	n/a	Raised Cosine roll-off=1.0	n/a
GB SE Sub-GHz	863-876 MHz 915-921 MHz	127 x 8	Phase Continuous	2FSK	100	0.7	Gaussian BT=0.5	PN9 in *16.2.3 of [R7]*

Test Procedure:

Step No.	Step description	Expected outcome
1	Decrease the quality of the input signal until the lowest value of LQI (0x00) is observed. Gradually increase the quality of the input signal noting unique LQI values until 6 unique values have been observed. Increase the quality of the input signal until the highest value of LQI (0xff) is observed.	Pass: For each of the required testing channels at least 8 unique LQI values including 0x00 and 0xff are observable. Fail: For any of the required testing channels less than 8 unique LQI values are observable or 0x00 is not observable or 0xff is not observable.
2	For the PHY being tested perform test for each of the following required Channels to Test at the corresponding center frequencies specified in the table below.	

	PHY	Channels to Test	Center Frequency Calculation
2003 2.4 GHz	2.4 GHz, 250kbps, O-QPSK	11 to 26	fc [MHz] = 2405+5*(k-11), where k = 11 (0x0B) to 26 (0x1A)
2003 Sub-GHz	868 MHz, 20kbps, BPSK 915 MHz, 40kbps, BPSK	0 1 to 10	fc [MHz] = 868.3 fc [MHz] = 906 + 2* (k-1), where k = 1, 2, …, 10
GB SE Sub-GHz	863-876 MHz, 100kbps, Filtered 2FSK 915-921 MHz, 100kbps, Filtered 2FSK	Ch Page [Ch #'s] 28 [0, 6, 13, 20, 26] 29 [27, 30, 32, 34, 62] 30 [35, 41, 48, 55, 61] 31 [0, 6, 13, 20, 26]	fc [MHz] = ChanCenterFreq0 + Ch # * ChanSpacing where ChanSpacing = 0.200 MHz For 863-876 MHz, Ch Page 28-30, ChanCenterFreq0 = 863.25 MHz For 915-921 MHz, Ch Page 31, ChanCenterFreq0 = 915.35 MHz

Notes:

In most instances this test can be done by replacing quality with signal level in Step No. 1. In those instances where the proper generation of LQI values can not be shown in this manner it is the responsibility of the owner of the device being tested to define the proper stimulus for the test house to use to be able to test for the proper generation of LQI values.

TP/154/PHY/RECEIVE-08

Tests that for the PHY being tested, DUT receiver is capable of reporting on channel traffic by at least 1 of the 3 Clear Channel Assessment (CCA) methods (modes), at the CCA ED Thershold specified in the table below or the Sensitivity Level +1 dB specified in the table below depending on the CCA Mode being tested, over the the CCA Detection Time specified in the table below.

	PHY	CCA ED Threshold	Sensitivity Level +1 dB	CCA DetectionTime
2003 2.4 GHz	2.4 GHz, 250kbps, O-QPSK	> -68 dBm	-84 dBm	8 symbol periods
2003 Sub-GHz	868 MHz, 20kbps, BPSK 915 MHz, 40kbps, BPSK	> -75 dBm	-91 dBm	8 symbol periods
GB SE Sub-GHz	863-876 MHz, 100kbps, Filtered 2FSK 915-921 MHz, 100kbps, Filtered 2FSK	> -87 dBm	-98 dBm	16 symbol periods

All supported modes of CCA shall be tested.

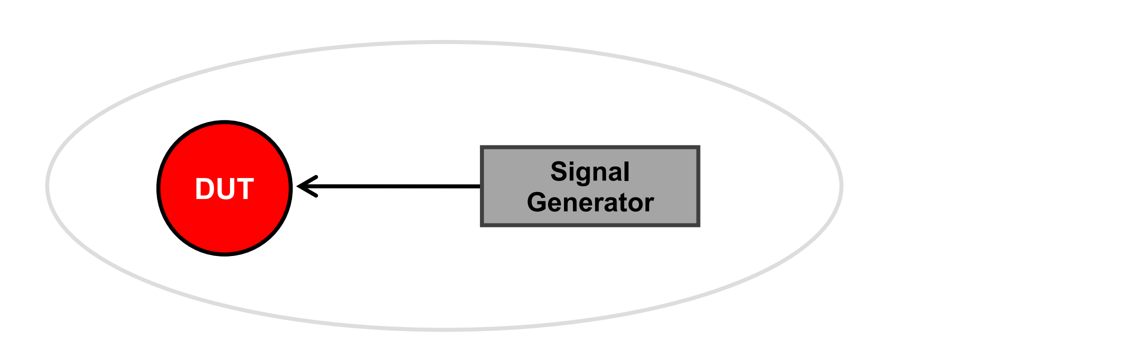
CCA Mode 1 shall report a busy medium upon detecting any energy meeting the specified CCA ED Threshold.

CCA Mode 2 shall report a busy medium only upon the detection of a signal compliant with that described in the initial conditions below for CCA Modes 2 and 3.

CCA Mode 3 shall report a busy medium using a logical combination of CCA Modes 1 and 2.

Setup Conditions:

Diagram



The Diagram is the image that represents the configuration deployed for the Test Procedure.

Initial Conditions:

Device name	Description
DUT	*Channel: for this test case, all of the required testing channels for the specific PHY being tested are specified in the table at the bottom of the Test Procedure.*

Signal Generator	*DUT shall receive output from a Signal Generator capable of generating an unmodulated carrier wave (CW) and capable of generating digitally modulated radio frequency signals for the band of operation.* *For testing CCA Modes 1 and 3 the Signal Generator shall be configured to transmit an unmodulated CW at the center frequency of the channels being tested.* *For testing CCA Modes 2 and 3 the Signal Generator shall be configured to transmit a standards conforming signal and packet with an arbitrary payload and correct CRC with the following parameters:* *Using the PSDU generation approach for the specific PHY(s) being tested, as described in 6.1.* *Using the odd and even packet sequence.* *Configuring the Signal Generator as defined in the table below.*

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	Band	PSDU # Bits	Bit Pattern	Mod.	Symbol Rate (kHz)	Mod. Index	Pulse Shape	PSDU Data Whitening
2003 2.4 GHz	2.4 GHz	20 x 8	Phase Continuous	O-QPSK	250	n/a	Half Sine	n/a
2003 Sub-GHz	868 MHz 915 MHz	20 x 8	Phase Continuous	BPSK	20 40	n/a	Raised Cosine roll-off=1.0	n/a
GB SE Sub-GHz	863-876 MHz 915-921 MHz	127 x 8	Phase Continuous	2FSK	100	0.7	Gaussian BT=0.5	PN9 in *16.2.3 of [R7]*

713 ***Test Procedure:***

Step No.	Step description	Expected outcome
1	CCA Mode 1 Test: DUT shall perform a CCA measurement 10 times with the Signal Generator generating an unmodulated CW with the specified CCA ED Threshold received signal power level. CCA Mode 2 Test: DUT shall perform a CCA measurement 10 times with the Signal Generator generating a compliant signal at the specified Sensitivity Level +1 received signal power level. CCA Mode 3 Test: DUT shall perform a CCA measurement 10 times with the Signal Generator generating a compliant signal at the specified Sensitivity Level +1 received signal power level. DUT shall also perform a CCA measurement 10 times with the Signal Generator generating an unmodulated CW at the specified CCA ED Threshold received signal power level.	Pass: For each of the required testing channels all CCA measurements of the supported modes report a busy medium. Fail: For any of the required testing channels any CCA measurements of the supported modes do not report a busy medium.

2	For the PHY being tested perform test for each of the following required Channels to Test at the corresponding center frequencies specified in the table below.	

	PHY	**Channels to Test**	**Center Frequency Calculation**
2003 2.4 GHz	2.4 GHz, 250kbps, O-QPSK	11 to 26	fc [MHz] = 2405+5*(k-11), where k = 11 (0x0B) to 26 (0x1A)
2003 Sub-GHz	868 MHz, 20kbps, BPSK 915 MHz, 40kbps, BPSK	0 1 to 10	fc [MHz] = 868.3 fc [MHz] = 906 + 2* (k-1), where k = 1, 2, …, 10
GB SE Sub-GHz	863-876 MHz, 100kbps, Filtered 2FSK 915-921 MHz, 100kbps, Filtered 2FSK	Ch Page [Ch #'s] 28 [0, 6, 13, 20, 26] 29 [27, 30, 32, 34, 62] 30 [35, 41, 48, 55, 61] 31 [0, 6, 13, 20, 26]	fc [MHz] = ChanCenterFreq0 + Ch # * ChanSpacing where ChanSpacing = 0.200 MHz For 863-876 MHz, Ch Page 28-30, ChanCenterFreq0 = 863.25 MHz For 915-921 MHz, Ch Page 31, ChanCenterFreq0 = 915.35 MHz

Notes:

None.

TP/154/PHY/RECEIVE-09

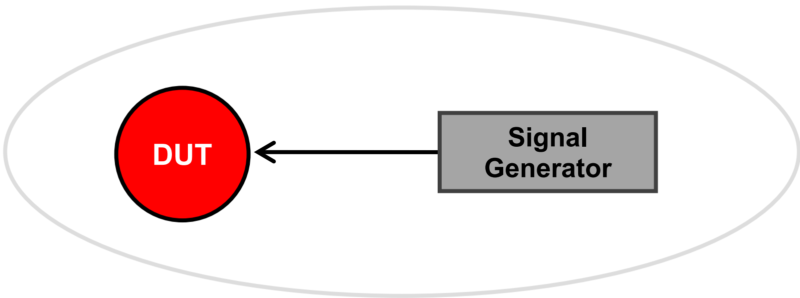
Tests that for the PHY being tested, DUT receiver is capable of measuring and reporting the Signal Strength (Rssi) of a received packet, in dBm, from the Min. Rssi Reporting Signal Level to the Max. Rssi Reporting Signal Level, meeting the Rssi Step Size, and the Rssi Linearity (Accuracy), all specified in the table below.

	PHY	Min. Rssi Reporting Signal Level	Max. Rssi Reporting Signal Level	Rssi Step Size	Rssi Linearity (Accuracy)
GB SE Sub-GHz	863-876 MHz, 100kbps, Filtered 2FSK 915-921 MHz, 100kbps, Filtered 2FSK	≤ -97 dBm	≥ -50 dBm	≤ 2 dB	≤ 3 dB

The Signal Strength measurement may be taken over any portion of the received packet, but shall be representative of the packet.

Setup Conditions:

Diagram



The Diagram is the image that represents the configuration deployed for the Test Procedure.

Initial Conditions:

Device name	Description
DUT	*Channel: for this test case, all of the required testing channels for the specific PHY being tested are specified in the table at the bottom of the Test Procedure.*
Signal Generator	*DUT shall receive output from a Signal Generator capable of generating digitally modulated radio frequency signals for the band of operation.* *Signal Generator shall be configured to transmit a standards conforming signal and packet with an arbitrary payload and correct CRC with the following parameters:* *Using the PSDU generation approach for the specific PHY(s) being tested, as described in 6.1.* *Using the odd and even packet sequence.* *Configuring the Signal Generator as defined in the table below.*

	Band	PSDU # Bits	Bit Pattern	Mod.	Symbol Rate (kHz)	Mod. Index	Pulse Shape	PSDU Data Whitening
GB SE Sub-GHz	863-876 MHz 915-921 MHz	127 x 8	Phase Continuous	2FSK	100	0.7	Gaussian BT=0.5	PN9 in *16.2.3 of [R7]*

Test Procedure:

Step No.	Step description	Expected outcome
1	DUT shall perform a Signal Strength measurement 10 times for each step size meeting the Rssi Step Size from the Min. Rssi Reporting Signal Level to the Max. Rssi Reporting Signal Level. Repeat above measurement for each channel in the band of operation. For the channel being measured, map the 10 measurement Signal Strength distribution for each received signal power level as a function of the calculated received signal power level, which is determined using the transmitted power level and accounting for all associated attenuation.	Pass: For each of the required testing channels each Signal Strength measurement is linear and meeting the Rssi Step Size (within the Rssi Linearity (Accuracy)) from the Min. Rssi Reporting Signal Level to the Max. Rssi Reporting Signal Level. Fail: For any of the required testing channels any of the Signal Strength measurements are not linear or meeting the Rssi Step Size (within the at least Rssi Linearity (Accuracy)) from the Min. Rssi Reporting Signal Level to the Max. Rssi Reporting Signal Level.
2	For the PHY being tested perform test for each of the following required Channels to Test at the corresponding center frequencies specified in the table below1	

	PHY	Channels to Test	Center Frequency Calculation
GB SE Sub-GHz	863-876 MHz, 100kbps, Filtered 2FSK 915-921 MHz, 100kbps, Filtered 2FSK	Ch Page [Ch #'s] 28 [0, 6, 13, 20, 26] 29 [27, 30, 32, 34, 62] 30 [35, 41, 48, 55, 61] 31 [0, 6, 13, 20, 26]	fc [MHz] = ChanCenterFreq0 + Ch # * ChanSpacing where ChanSpacing = 0.200 MHz For 863-876 MHz, Ch Page 28-30, ChanCenterFreq0 = 863.25 MHz For 915-921 MHz, Ch Page 31, ChanCenterFreq0 = 915.35 MHz

Notes:

None.

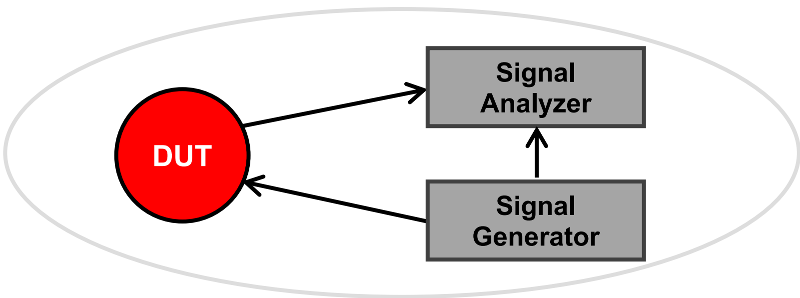
TP/154/PHY/TURNAROUND-TIME-01

Tests that for the PHY being tested, DUT has a RX-to-TX turnaround time meets the RX-to-TX Turnaround Time specified in the table below. The specified RX->TX Turnaround Time is greater than the specified TX->RX Turnaround Time for some PHYs due to the extra time resulting from implementing LBT.

	PHY	**RX-to-TX Turnaround Time**
2003 2.4 GHz	2.4 GHz, 250kbps, O-QPSK	12 Symbols < x < 12.75 symbols (192 µs < x < 204 µs)
2003 Sub-GHz	868 MHz, 20kbps, BPSK 915 MHz, 40kbps, BPSK	12 Symbols < x < 12.75 symbols (600 µs < x < 637.5 µs) 12 Symbols < x < 12.75 symbols (300 µs < x < 318.75 µs)
GB SE Sub-GHz	863-876 MHz, 100kbps, Filtered 2FSK 915-921 MHz, 100kbps, Filtered 2FSK	x < 100 symbols (x < 1000 µs)

Setup Conditions:

Diagram



The Diagram is the image that represents the configuration deployed for the Test Procedure.

Initial Conditions:

Device name	**Description**
DUT	*Channel: for this test case, all of the required testing channels for the specific PHY being tested are specified in the table at the bottom of the Test Procedure.*
Signal Analyzer	*Configure the Signal Analyzer to start acquisition on analog edge triggering.*
Signal Generator	*DUT shall receive output from a Signal Generator capable of generating digitally modulated radio frequency signals for the band of operation.* *The Signal Generator equipment shall be capable of providing tight synchronization with the Signal Analyzer.* *Signal Generator shall be configured to transmit a standards conforming signal and packet with an arbitrary payload and correct CRC with the following parameters:* *Using the PSDU generation approach for the specific PHY(s) being tested, as described in 6.1.*

	Using the odd and even packet sequence. *Configuring the Signal Generator as defined in the table below.* *Set frame to obtain Ack frame from DUT (i.e. set ack. request bit to 1)*

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	Band	PSDU # Bits	Bit Pattern	Mod.	Symbol Rate (kHz)	Mod. Index	Pulse Shape	PSDU Data Whitening
2003 2.4 GHz	2.4 GHz	20 x 8	Phase Continuous	O-QPSK	250	n/a	Half Sine	n/a
2003 Sub-GHz	868 MHz 915 MHz	20 x 8	Phase Continuous	BPSK	20 40	n/a	Raised Cosine roll-off=1.0	n/a
GB SE Sub-GHz	863-876 MHz 915-921 MHz	127 x 8	Phase Continuous	2FSK	100	0.7	Gaussian BT=0.5	PN9 in *16.2.3 of [R7]*

745 ***Test Procedure:***

Step No.	Step description	Expected outcome
1	Signal Generator transmits a packet and DUT receiver receives the packet. Signal Generator generates a Start Trigger at the end of generation on the Signal Analyzer. DUT Acks to signal received from Signal Generator. On the Signal Analyzer measure the delay between the Start Trigger and the start of reception of the Ack frame from the DUT.	Pass: For each of the required testing channels all of the RX-to-TX Turnaround measurements meet the specified RX-to-TX Turnaround Time. Fail: For any of the required testing channels any of the RX-to-TX Turnaround measurements do not meet the specified RX-to-TX Turnaround Time.
2	For the PHY being tested perform test for each of the following required Channels to Test at the corresponding center frequencies specified in the table below.	

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	PHY	Channels to Test	Center Frequency Calculation
2003 2.4 GHz	2.4 GHz, 250kbps, O-QPSK	11 to 26	fc [MHz] = 2405+5*(k-11), where k = 11 (0x0B) to 26 (0x1A)
2003 Sub-GHz	868 MHz, 20kbps, BPSK 915 MHz, 40kbps, BPSK	0 1 to 10	fc [MHz] = 868.3 fc [MHz] = 906 + 2* (k-1), where k = 1, 2, …, 10
GB SE Sub-GHz	863-876 MHz, 100kbps, Filtered 2FSK 915-921 MHz, 100kbps, Filtered 2FSK	Ch Page [Ch #'s] 28 [0, 6, 13, 20, 26] 29 [27, 30, 32, 34, 62] 30 [35, 41, 48, 55, 61] 31 [0, 6, 13, 20, 26]	fc [MHz] = ChanCenterFreq0 + Ch # * ChanSpacing where ChanSpacing = 0.200 MHz For 863-876 MHz, Ch Page 28-30, ChanCenterFreq0 = 863.25 MHz For 915-921 MHz, Ch Page 31, ChanCenterFreq0 = 915.35 MHz

Notes:

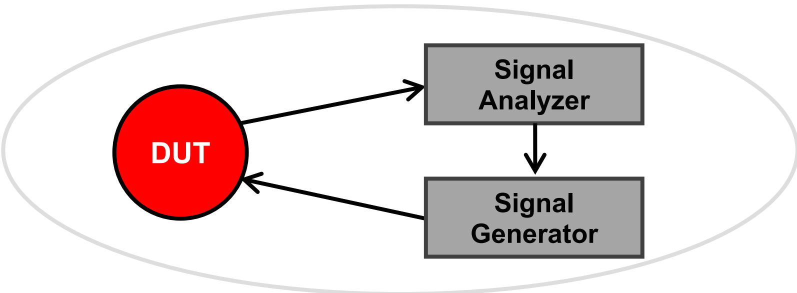
None.

TP/154/PHY/TURNAROUND-TIME-02

Tests that for the PHY being tested, DUT has a TX-to-RX turnaround time less than or equal to the TX-to-RX Turnaround Time specified in the table below.

	PHY	TX-to-RX Turnaround Time
2003 2.4 GHz	2.4 GHz, 250kbps, O-QPSK	x = 12.75 symbols (x = 204 µs)
2003 Sub-GHz	868 MHz, 20kbps, BPSK 915 MHz, 40kbps, BPSK	x = 12.75 symbols (x = 637.5 µs) x = 12.75 symbols (x = 318.75 µs)
GB SE Sub-GHz	863-876 MHz, 100kbps, Filtered 2FSK 915-921 MHz, 100kbps, Filtered 2FSK	x = 46 symbols (x = 460 µs)

Setup Conditions:

Diagram


The Diagram is the image that represents the configuration deployed for the Test Procedure.

Initial Conditions:

Device name	Description
DUT	*Channel: for this test case, all of the required testing channels for the specific PHY being tested are specified in the table at the bottom of the Test Procedure.*
Signal Analyzer	*Configure the Signal Analyzer to start acquisition on analog edge triggering.*
Signal Generator	*DUT shall receive output from a Signal Generator capable of generating digitally modulated radio frequency signals for the band of operation.* *The Signal Generator equipment shall be capable of providing tight synchronization with the Signal Analyzer.* *Signal Generator shall be configured to transmit a standards conforming signal and packet with an arbitrary payload and correct CRC with the following parameters:* *Using the PSDU generation approach for the specific PHY(s) being tested, as described in 6.1.* *Using the odd and even packet sequence.* *Configuring the Signal Generator as defined in the table below.*

	Band	PSDU # Bits	Bit Pattern	Mod.	Symbol Rate (kHz)	Mod. Index	Pulse Shape	PSDU Data Whitening
2003 2.4 GHz	2.4 GHz	20 x 8	Phase Continuous	O-QPSK	250	n/a	Half Sine	n/a
2003 Sub-GHz	868 MHz 915 MHz	20 x 8	Phase Continuous	BPSK	20 40	n/a	Raised Cosine roll-off=1.0	n/a
GB SE Sub-GHz	863-876 MHz 915-921 MHz	127 x 8	Phase Continuous	2FSK	100	0.7	Gaussian BT=0.5	PN9 in *16.2.3 of [R7]*

Test Procedure:

Step No.	Step description	Expected outcome
1	DUT transmits a packet to the Signal Analyzer with a request to Ack. Signal Analyzer generates a Start Trigger at the end of reception of signal on the Signal Generator. Signal Generator waits for the specified TX-to-RX Turnaround Time from the time of the Start Trigger and transmits Ack packet to DUT. Record whether or not DUT re-requests Ack.	Pass: For each of the required testing channels no re-requests for Ack are generated by the DUT. Fail: For any of the required testing channels a re-request for Ack is generated by the DUT.
2	For the PHY being tested perform test for each of the following required Channels to Test at the corresponding center frequencies specified in the table below.	

	PHY	Channels to Test	Center Frequency Calculation
2003 2.4 GHz	2.4 GHz, 250kbps, O-QPSK	11 to 26	fc [MHz] = 2405+5*(k-11), where k = 11 (0x0B) to 26 (0x1A)
2003 Sub-GHz	868 MHz, 20kbps, BPSK 915 MHz, 40kbps, BPSK	0 1 to 10	fc [MHz] = 868.3 fc [MHz] = 906 + 2* (k-1), where k = 1, 2, …, 10
GB SE Sub-GHz	863-876 MHz, 100kbps, Filtered 2FSK 915-921 MHz, 100kbps, Filtered 2FSK	Ch Page [Ch #'s] 28 [0, 6, 13, 20, 26] 29 [27, 30, 32, 34, 62] 30 [35, 41, 48, 55, 61] 31 [0, 6, 13, 20, 26]	fc [MHz] = ChanCenterFreq0 + Ch # * ChanSpacing where ChanSpacing = 0.200 MHz For 863-876 MHz, Ch Page 28-30, ChanCenterFreq0 = 863.25 MHz For 915-921 MHz, Ch Page 31, ChanCenterFreq0 = 915.35 MHz

Notes:

Employing the signal generator testing approach above requires that the next frame seq. # be set using the macDSN PIB attribute. This allows for the frame # expected at the DUT receiver to be set to the same values as the frame # in the ACK packet received from the signal generation.

7 Zigbee MAC Testing

7.1 Zigbee Network Stacks and the MAC Test Cases

Section 7.3 contains the IEEE 802.15.4 MAC tests cases supporting all Zigbee networks stacks. Each of the different Zigbee network stacks utilize a majority of the same test cases. There are however some differences. The table below shows how the the different Zigbee network stacks map to their applicable MAC test cases, based on the test case name and associated superscript number.

Test Case Name Example	Applicable Zigbee Network Stack
Testcasename	Zigbee PRO/RF4CE
Testcasename[1]	Zigbee PRO
Testcasename[2]	RF4CE

7.2 MAC Test Case Considerations

7.2.1 MAC Test Case Considerations for Zigbee PRO/RF4CE

Zigbee PRO/RF4CE network stacks utilize IEEE 802.15.4-2003 for the lower layers, in which case:

- Bit 6 is Intra-PAN
- Bit 12-13 is Reserved

7.2.2 MAC Test Case Considerations for GB SE Sub-GHz

The MAC test cases for ZSE 1.4 (with the GB SE Sub-GHz PHY) certification includes those used by the Zigbee PRO stack including several test cases marked as optional in general for Zigbee PRO but identified as mandatory in the PICS when supporting GB SE Sub-GHz.

A translated symbol time for the PHY, used by GB SE Sub-GHz, of 10 microseconds (representing the actual 100 kbps PHY symbol rate utilized by the GB SE Sub-GHz PHY) is used in the GB SE Sub-GHz PHY test cases. For the purposes of any MAC and PHY timing parameter calculations used in the MAC test cases, the defined symbol duration of 20 microseconds is to be used, per Section 19.1 of [R7].

7.3 Zigbee IEEE 802.15.4 MAC Test Cases

TP/154/MAC/SCANNING-01

Perform an energy detect scan on several channels to check correct EnergyDetectList is generated.

***Setup Conditions*:**

Diagram


The Diagram is the image that represents the configuration deployed for the Test Procedure.

Initial Conditions:

Device name	**Description**
FFD	*MAC: 0xACDE480000000001*

Test Procedure:

Step No.	**Step description**	**Expected outcome**
1	Set MLME-RESET.request(TRUE) to DUT.	Pass: FFD returns SUCCESS. Fail: FFD does not respond.
2	Set to DUT: MLME-SCAN.request (ScanType=0x00, ScanChannels==0x00007FFF800(for 2.4GHz O-QPSK PHY) ScanChannels==0x000000007ff(for 868/915MHz BPSK PHY) ScanChannels==0x00000003fff(for 920MHz FSK PHY), ScanDuration=0x05, ChannelPage=0x00, SecurityLevel=0x00, KeyIDMode=xx, KeySource=NULL, KeyIndex=xx) Energy Scan, all channels, duration=[*aBaseSuperframeDuration* * (2*n* + 1)] symbols per channel	Pass: The DUT shall return a Success result, such that: MLME-SCAN.confirm (Status=0x00=”SUCCESS”, ScanType=0x00=ED Scan, ChannelPage=0, UnscannedChannels=0x00 00 00 00, ResultListSize=16(for 2.4GHz O-QPSK PHY) ResultListSize=11(for 868/915MHz BPSK PHY) ResultListSize=14(for 920MHz FSK PHY), EnergyDetectList=values representing the energy on the different channel, PANDescriptorList=NULL) Fail: DUT does not respond, or EnergyDetectList contains unexpected values

Notes:

Test is done twice: In a shielded environment where the energy value is expected to be zero on all channels and during the second test a designated signal is used on one channel.

TP/154/MAC/SCANNING-02

Perform an active scan on several channels to check correct negative response when coordinator(s) not present.

Setup Conditions:

Diagram

The Diagram is the image that represents the configuration deployed for the Test Procedure.

Initial Conditions:

Device name	Description
FFD	*MAC: 0xACDE480000000001*

Test Procedure:

Step No.	Step description	Expected outcome
1	Set MLME-RESET.request(TRUE) to DUT.	Pass: FFD returns SUCCESS. Fail: FFD does not respond.
2	Set to DUT: MLME-SCAN.request (ScanType=0x01, ScanChannels==0x07fff800(for 2.4GHz O-QPSK PHY) ScanChannels==0x000007ff(for 868/915MHz BPSK PHY) ScanChannels==0x00000003fff(for 920MHz FSK PHY), ScanDuration=0x05, ChannelPage=0x00, SecurityLevel=0x00, KeyIDMode=xx, KeySource=NULL, KeyIndex=xx) Active Scan, all channels, duration=[*aBaseSuperframeDuration* * (2*n* + 1)] symbols per channel	Pass: It is possible to verify by a PAN analyzer that a beacon request command frame was issued from the DUT. Frame Length: 10 bytes IEEE 802.15.4 Frame Control: 0x0803 .... .011 = Frame Type: Command (0x0003) .... 0... = Security Enabled: Disabled ....0 .... = Frame Pending: No more data ....0. .... = Acknowledgment Request: Ack not required .... .0.. .... = Intra-PAN: Not within the PAN .... ..00 0... .... = Reserved .... 10.. .... = Destination Addressing Mode: Address field contains a 16-bit short address ..00 .... = Reserved for Zigbee PRO/RF4CE 00.. .... = Source Addressing Mode: PAN identifier and address field are not present Sequence Number: xx Destination PAN Identifier: 0xffff Destination Address: 0xffff MAC Payload Command Frame Identifier = Beacon Request: (0x07) Frame Check Sequence: Correct The DUT shall return a Success result, such that: MLME-SCAN.confirm (Status=0xea="NO_BEACON", ScanType=0x01=Active Scan, ChannelPage=0,

		UnscannedChannels=0x00 00 00 00, ResultListSize=0, EnergyDetectList=NULL, PANDescriptorList=NULL) Fail: DUT does not respond, or DUT responds with SUCCESS

Notes:

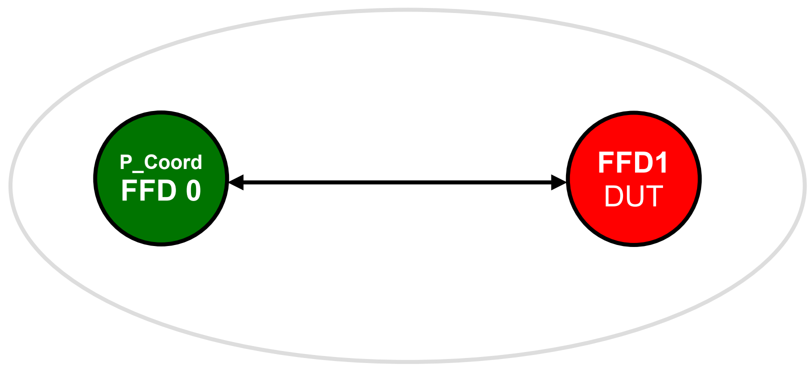
None.

TP/154/MAC/SCANNING-03

Perform an active scan on several channels to check positive response when coordinator(s) present (short address without beacon payload)

Setup Conditions:

Diagram



The Diagram is the image that represents the configuration deployed for the Test Procedure.

Initial Conditions:

Device name	Description
All the devices	*Security: no for all the devices* *Channel: a single channel TBD by, or at the discretion of, the test house for all devices.* *PAN id: 0x1AAA for all the devices*
P_Coord FFD0	*PAN Coordinator, Bo = 15, SO = 15* *MAC: 0xACDE480000000001* *Short Address: 0xbb00*
FFD1	*MAC: 0xACDE480000000002*

Test Procedure:

Step No.	Step description	Expected outcome
1	Set to DUT: MLME-RESET.request(TRUE).	Pass: FFD returns SUCCESS. Fail: FFD does not respond.
2	Set to DUT: MLME-SCAN.request (ScanType=0x01, ScanChannels==0x07fff800(for 2.4GHz O-QPSK PHY) ScanChannels==0x000007ff(for 868/915MHz BPSK PHY) ScanChannels==0x00000003fff(for 920MHz FSK PHY), ScanDuration=0x05, ChannelPage=0x00, SecurityLevel=0x00,	Pass: It is possible to verify by a PAN analyzer that a beacon request command frame was issued from the DUT. Frame Length: 10 bytes IEEE 802.15.4 Frame Control: 0x0803 .... .011 = Frame Type: Command (0x0003) .... 0... = Security Enabled: Disabled ....0 .... = Frame Pending: No more data ....0. .... = Acknowledgment Request: Ack not required .... .0.. .... = Intra PAN: Not within the PAN .... ..00 0... .... = Reserved

	KeyIDMode=xx, KeySource=NULL, KeyIndex=xx) Active Scan, all channels, duration=[*aBaseSuperframeDuration* * (2*n* + 1)] symbols per channel	.... 10.. .... = Destination Addressing Mode: Address field contains a 16-bit short address ..00 .... = Reserved for Zigbee PRO/RF4CE 00.. .... = Source Addressing Mode: PAN identifier and address field are not present Sequence Number: xx Destination PAN Identifier: 0xffff Destination Address: 0xffff MAC Payload Command Frame Identifier = Beacon Request: (0x07) Frame Check Sequence: Correct The DUT shall return a Success result, such that: In the case where macAutoRequest = True MLME-SCAN.confirm (Status=0x00="Success", ScanType=0x01=Active Scan, ChannelPage=0, UnscannedChannels=0x00 00 00 00, ResultListSize=01, EnergyDetectList=NULL, PANDescriptorList) PANDescriptorList: CoordAddrMode=0x2 CoordPanId=0x1aaa CoordAddr=0xbb00 LogicalChannel=0x14 SuperframeSpec = 0x4FFF GTSPermit=FALSE LinkQuality=0x00-0xff Timestamp SecurityFailure=SUCCESS SecurityLevel=0 [KeyIdMode = NULL KeySource=NULL KeyIndex=0x00] In the case where macAutoRequest = False MLME-BEACON-NOTIFY.indication(BSN = xx, PANDescriptor (see above), PendAddrSpec = 0x00 AddrList = NULL, sduLength = 0x00, sdu = NULL) MLME-SCAN.confirm (Status=0x00="Success", ScanType=0x01=Active Scan, ChannelPage=0, UnscannedChannels=0x00 00 00 00, ResultListSize=0, EnergyDetectList=NULL, PANDescriptorList=0) Fail: DUT does not respond, or DUT responds with NO_BEACON, or DUT responds with invalid pan descriptor list, or in the case where macAutoRequest = False DUT responds with a MLME-BEACON-NOTIFY.indication not followed by a MLME-SCAN.confirm

808 ***Notes:***

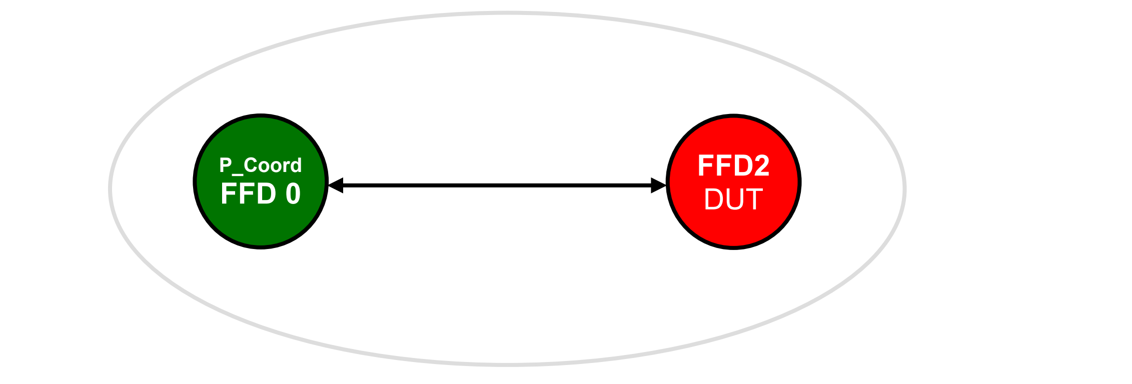
809 *None.*

TP/154/MAC/SCANNING-04

Perform an active scan on several channels to check positive response when coordinator(s) present (short address with beacon payload)

Setup Conditions:

Diagram



The Diagram is the image that represents the configuration deployed for the Test Procedure.

Initial Conditions:

Device name	**Description**
All the devices	*Security: no for all the devices* *Channel: 0x14 for all the devices* *PAN id: 0x1AAA for all the devices*
P_Coord FFD0	*PAN Coordinator, Bo = 15, SO = 15* *MAC: 0xACDE480000000001* *Short Address: 0xbb00* *Beacon Payload: 0x12 0x34*
FFD1	*MAC: 0xACDE480000000002*

Test Procedure:

Step No.	**Step description**	**Expected outcome**
1	Set to DUT: MLME-RESET.request(TRUE).	Pass: FFD returns SUCCESS. Fail: FFD does not respond.
2	Set to DUT: MLME-SCAN.request (ScanType=0x01, ScanChannels==0x07fff800(for 2.4GHz O-QPSK PHY) ScanChannels==0x000007ff(for 868/915MHz BPSK PHY) ScanChannels==0x00000003fff(for 920MHz FSK PHY), ScanDuration=0x05, ChannelPage=0x00, SecurityLevel=0x00, KeyIDMode=xx, KeySource=NULL,	Pass: It is possible to verify by a PAN analyzer that a beacon request command frame was issued from the DUT. Frame Length: 10 bytes IEEE 802.15.4 Frame Control: 0x0803 .... .011 = Frame Type: Command (0x0003) .... 0... = Security Enabled: Disabled ....0 .... = Frame Pending: No more data ....0. .... = Acknowledgment Request: Ack not required .... .0.. .... = Intra-PAN: Not within the PAN .... ..00 0... .... = Reserved .... 10.. .... = Destination Addressing Mode: Address field contains a 16-bit short address

	KeyIndex=xx) Active Scan, all channels, duration=[*aBaseSuperframeDuration* * (2*n* + 1)] symbols per channel	..00 .... = Reserved for Zigbee PRO/RF4CE 00.. .... = Source Addressing Mode: PAN identifier and address field are not present Sequence Number: xx Destination PAN Identifier: 0xffff Destination Address: 0xffff MAC Payload Command Frame Identifier = Beacon Request: (0x07) Frame Check Sequence: Correct The DUT shall return MLME-BEACON-NOTIFY.indication(BSN = xx, PANDescriptor, PendAddrSpec = 0x00 AddrList = NULL, sduLength = 0x02, sdu = 0x12 0x34) The DUT shall return a Success result, such that: In the case where macAutoRequest = True MLME-SCAN.confirm (Status=0x00=”Success”, ScanType=0x01=Active Scan, ChannelPage=0, UnscannedChannels=0x00 00 00 00, ResultListSize=01, EnergyDetectList=NULL, PANDescriptorList) PANDescriptorList: CoordAddrMode=0x2 CoordPanId=0x1aaa CoordAddr=0xbb00 LogicalChannel=0x14 ChannelPage=0 SuperframeSpec=0x4FFF GTSPermit=FALSE LinkQuality=0x00-0xff Timestamp SecurityFailure=SUCCESS SecurityLevel=0 [KeyIdMode = NULL KeySource=NULL KeyIndex=0x00] In the case where macAutoRequest = False MLME-BEACON-NOTIFY.indication(BSN = xx, PANDescriptor (see above), PendAddrSpec = 0x00 AddrList = NULL, sduLength = 0x00, sdu = NULL) MLME-SCAN.confirm (Status=0x00=”Success”, ScanType=0x01=Active Scan, ChannelPage=0, UnscannedChannels=0x00 00 00 00, ResultListSize=0, EnergyDetectList=NULL, PANDescriptorList=0) Fail:

		DUT does not respond, or DUT responds with NO_BEACON, or DUT responds with invalid pan descriptor list, or in the case where macAutoRequest = False DUT responds with a MLME-BEACON-NOTIFY.indication not followed by a MLME-SCAN.confirm

Notes:

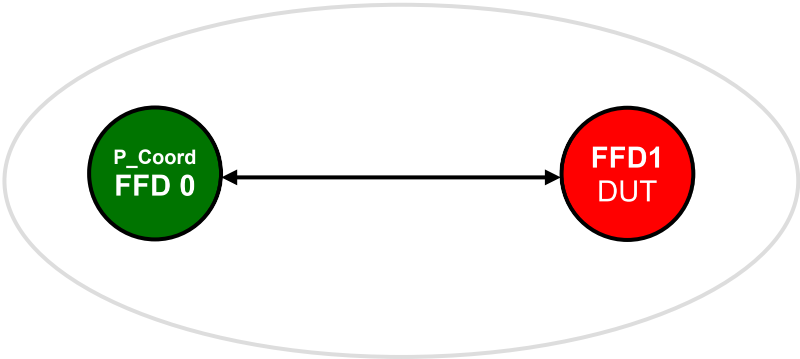
None.

TP/154/MAC/SCANNING-05

Perform an active scan on several channels to check positive response when coordinator(s) present (extended address without beacon payload)

Setup Conditions:

Diagram



The Diagram is the image that represents the configuration deployed for the Test Procedure.

Initial Conditions:

Device name	Description
All the devices	*Security: no for all the devices* *Channel: 0x14 for all the devices* *PAN id: 0x1AAA for all the devices*
P_Coord FFD0	*PAN Coordinator, Bo = 15, SO = 15* *MAC: 0xACDE480000000001* *Short Address: 0xfffe (use extended address)* *Beacon Payload: none*
FFD1	*MAC: 0xACDE480000000002*

Test Procedure:

Step No.	Step description	Expected outcome
1	Set to DUT: MLME-RESET.request(TRUE).	Pass: FFD returns SUCCESS. Fail: FFD does not respond.
2	Set to DUT: MLME-SCAN.request (ScanType=0x01, ScanChannels==0x07fff800(for 2.4GHz O-QPSK PHY) ScanChannels==0x000007ff(for 868/915MHz BPSK PHY) ScanChannels==0x00000003fff(for 920MHz FSK PHY), ScanDuration=0x05, ChannelPage=0x00, SecurityLevel=0x00, KeyIDMode=xx, KeySource=NULL,	Pass: It is possible to verify by a PAN analyzer that a beacon request command frame was issued from the DUT. Frame Length: 10 bytes IEEE 802.15.4 Frame Control: 0x0803 .... .011 = Frame Type: Command (0x0003) .... 0... = Security Enabled: Disabled ....0 .... = Frame Pending: No more data ....0. .... = Acknowledgment Request: Ack not required .... .0.. .... = Intra-PAN: Not within the PAN .... ..00 0... .... = Reserved .... 10.. .... = Destination Addressing Mode: Address field contains a 16-bit short address

	KeyIndex=xx) Active Scan, all channels, duration=[*aBaseSuperframeDuration* * (2*n* + 1)] symbols per channel	..00 .... = Reserved for Zigbee PRO/RF4CE 00.. .... = Source Addressing Mode: PAN identifier and address field are not present Sequence Number: xx Destination PAN Identifier: 0xffff Destination Address: 0xffff MAC Payload Command Frame Identifier = Beacon Request: (0x07) Frame Check Sequence: Correct The DUT shall return a Success result, such that: In the case where macAutoRequest = True MLME-SCAN.confirm (Status=0x00="Success", ScanType=0x01=Active Scan, ChannelPage=0, UnscannedChannels=0x00 00 00 00, ResultListSize=01, EnergyDetectList=NULL, PANDescriptorList) PANDescriptorList: CoordAddrMode=0x3 CoordPanId=0x1aaa CoordAddr=0xACDE480000000001 LogicalChannel=0x14 ChannelPage=0 SuperframeSpec=0x4FFF GTSPermit=FALSE LinkQuality=0x00-0xff Timestamp SecurityFailure=SUCCESS SecurityLevel=0 [KeyIdMode = NULL KeySource=NULL KeyIndex=0x00] In the case where macAutoRequest = False MLME-BEACON-NOTIFY.indication(BSN = xx, PANDescriptor (see above), PendAddrSpec = 0x00 AddrList = NULL, sduLength = 0x00, sdu = NULL) MLME-SCAN.confirm (Status=0x00="Success", ScanType=0x01=Active Scan, ChannelPage=0, UnscannedChannels=0x00 00 00 00, ResultListSize=0, EnergyDetectList=NULL, PANDescriptorList=0) Fail: DUT does not respond, or DUT responds with NO_BEACON, or DUT responds with invalid pan descriptor list, or in the case where macAutoRequest = False DUT responds with a MLME-BEACON-NOTIFY.indication not followed by a MLME-SCAN.confirm

Notes:

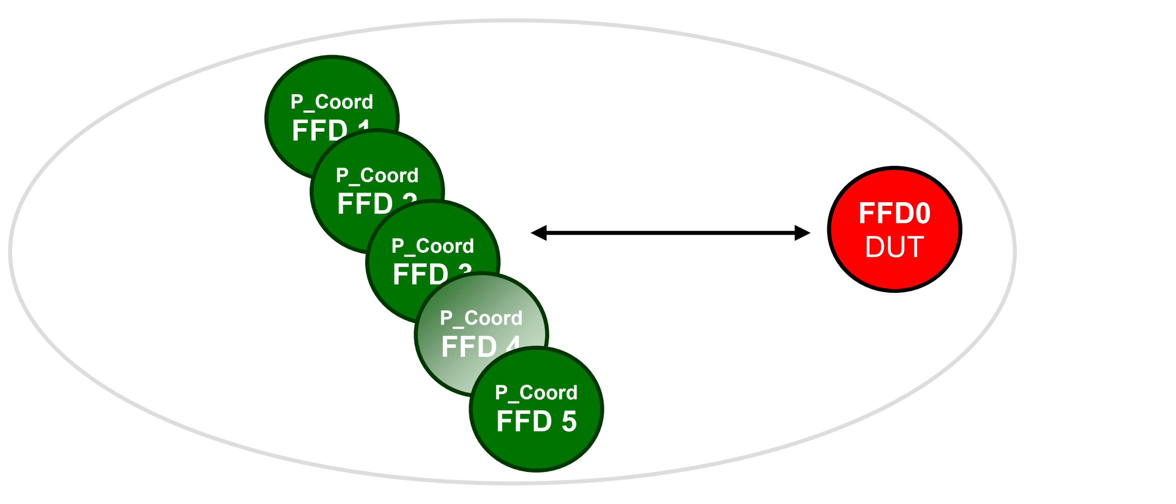
None.

TP/154/MAC/SCANNING-06

Perform an active scan on several channels to check positive response when multiple coordinators present (test for memory / communications overflow) See note below.

***Setup Conditions*:**

Diagram



The Diagram is the image that represents the configuration deployed for the Test Procedure.

Initial Conditions:

Device name	Description
All the devices	*Security: no for all the devices* *Channel: 0x14 for all the devices*
P_Coord FFDn (n = 1... 5)	*PAN Coordinator, Bo = 15, SO = 15* *PAN id: 0x000n* *MAC: 0xnnnnnnnn* *Short Address: 0x0000* *Beacon Payload: value ??*
FFD0	*MAC: 0xACDE480000000001*

Test Procedure:

Step No.	Step description	Expected outcome
1	Set to DUT: MLME-RESET.request(TRUE).	Pass: FFD returns SUCCESS. Fail: FFD does not respond.
2	Set to DUT: MLME-SCAN.request (ScanType=0x01, ScanChannels==0x07fff800(for 2.4GHz O-QPSK PHY) ScanChannels==0x000007ff(for 868/915MHz BPSK PHY) ScanChannels==0x00000003fff(for 920MHz FSK PHY),	Pass: It is possible to verify by a PAN analyzer that a beacon request command frame was issued from the DUT. Frame Length: 10 bytes IEEE 802.15.4 Frame Control: 0x0803 .... .011 = Frame Type: Command (0x0003) .... 0... = Security Enabled: Disabled ....0 .... = Frame Pending: No more data

	ScanDuration=0x05, ChannelPage=0x00, SecurityLevel=0x00, KeyIDMode=xx, KeySource=NULL, KeyIndex=xx) Active Scan, all channels, duration=[*aBaseSuperframeDuration* * (2*n* + 1)] symbols per channel	....0. .... = Acknowledgment Request: Ack not required .... .0.. .... = Intra-PAN: Not within the PAN .... ..00 0... .... = Reserved .... 10.. .... = Destination Addressing Mode: Address field contains a 16-bit short address ..00 .... = Reserved for Zigbee PRO/RF4CE 00.. .... = Source Addressing Mode: PAN identifier and address field are not present Sequence Number: xx Destination PAN Identifier: 0xffff Destination Address: 0xffff MAC Payload Command Frame Identifier = Beacon Request: (0x07) Frame Check Sequence: Correct The DUT shall return a Success result, such that: In the case where macAutoRequest = True MLME-SCAN.confirm (Status=0x00="Success", ScanType=0x01=Active Scan, ChannelPage=0, UnscannedChannels=0x00 00 00 00, ResultListSize=5 EnergyDetectList=NULL, PANDescriptorList) PANDescriptorList (multiple PAN descriptors): CoordAddrMode=0x2 CoordPanId=0x000n CoordAddr=0x0000 (short address of detected FFD) LogicalChannel=0x14 ChannelPage=0 SuperframeSpec GTSPermit=FALSE LinkQuality=0x00-0xff Timestamp SecurityFailure=SUCCESS SecurityLevel = 0x00 [KeyIdMode = NULL KeySource=NULL KeyIndex=0x00] In the case where macAutoRequest = True The below MLME-BEACON-NOTIFY.indication will be received 5 times, once for each of the 5 PA coordinators. MLME-BEACON-NOTIFY.indication(BSN = xx, PANDescriptor (see above), PendAddrSpec = 0x00 AddrList = NULL, sduLength = 0x00, sdu = NULL) MLME-SCAN.confirm (Status=0x00="Success", ScanType=0x01=Active Scan, ChannelPage=0, UnscannedChannels=0x00 00 00 00, ResultListSize=00, EnergyDetectList=NULL, PANDescriptorList=0) Fail: DUT does not respond, or

		DUT responds with NO_BEACON, or DUT responds with invalid pan descriptor list, or in the case where macAutoRequest = False DUT does not respond with 5 MLME-BEACON-NOTIFY.indication, or DUT responds with a MLME-BEACON-NOTIFY.indication not followed by a MLME-SCAN.confirm

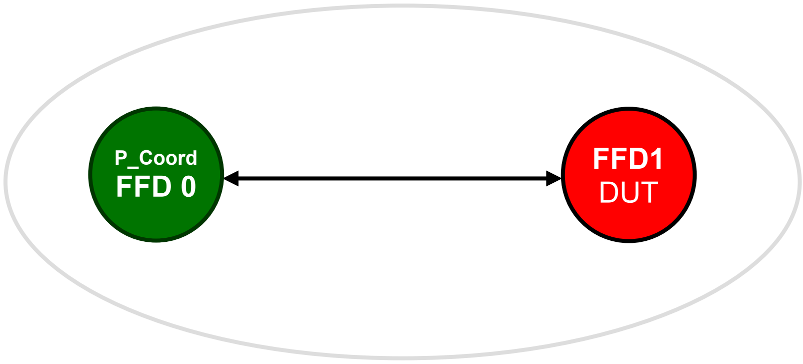
Notes:

*Test should be run with coordinators all on a single channel, or distributed across multiple channels. Full PAN descriptor list may not return, depending on number of coordinators that respond due to CSMA***.**

TP/154/MAC/SCANNING-07[1]

Perform an orphan scan to get on a network

Setup Conditions:

Diagram


The Diagram is the image that represents the configuration deployed for the Test Procedure.

Initial Conditions:

Device name	Description
All the devices	*Security: no for all the devices* *Channel: 0x14 for all the devices* *PAN id: 0x1AAA for all the devices*
P_Coord FFD0	*PAN Coordinator, Bo = 15, SO = 15* *MAC: 0xACDE480000000001* *Short Address: 0x0001* *Beacon Payload: none*
FFD1	*MAC: 0xACDE480000000002*

Test Procedure:

Step No.	Step description	Expected outcome
1	Set to DUT: MLME-SCAN.request (ScanType=0x03, ScanChannels==0x07fff800(for 2.4GHz O-QPSK PHY) ScanChannels==0x000007ff(for 868/915MHz BPSK PHY) ScanChannels==0x00000003fff(for 920MHz FSK PHY), ScanDuration=0x05, ChannelPage=0x00, SecurityLevel=0x00, KeyIDMode=xx, KeySource=NULL, KeyIndex=xx)	Pass: It is possible to verify by a PAN analyzer that a Orphan Scan command frame was issued from the DUT. Frame Length: 16 bytes IEEE 802.15.4 Frame Control: 0xC843 .... .011 = Frame Type: Command (0x0003) .... 0... = Security Enabled: Disabled ....0 .... = Frame Pending: No more data ....0. .... = Acknowledgment Request: Ack not required .... .1.. .... = Intra-PAN: Within the PAN .... ..00 0... .... = Reserved .... 10.. .... = Destination Addressing Mode: Address field contains a 16-bit short address ..00 .... = Reserved 11.. .... = Source Addressing Mode: Address field contains a 64-bit ieee address Sequence Number: xx Destination PAN Identifier: 0xffff

	Orphan Scan, all channels, duration=[*aBaseSuperframeDuration* * (2*n* + 1)] symbols per channel	Destination Address: 0xffff Source Address: 0xACDE480000000002 MAC Payload Command Frame Identifier = Orphan Notification: (0x06) Frame Check Sequence: Correct On the tester: MLME-ORPHAN.indication (OrphanAddress=0xACDE480000000002, SecurityLevel=0x00, KeyIdMode=xx, KeySource=NULL, KeyIndex=xx) Fail: DUT does not respond, or DUT responds with NO_BEACON, or DUT responds with invalid pan descriptor list
2	Set to Tester(FFD0) MLME-ORPHAN.response (OrphanAddress=0xACDE480000000002 ShortAddress=0x0002 AssociatedMember=TRUE SecurityLevel=0x00, KeyIDMode=xx, KeySource=NULL, KeyIndex=xx)	Pass: It is possible to verify by a PAN analyzer that a Coordinator Realignment command frame was issued from the Tester to the DUT. Frame Length: 31 bytes IEEE 802.15.4 Frame Control: 0xCC23 .... .011 = Frame Type: Command (0x0003) .... 0... = Security Enabled: Disabled ....0 .... = Frame Pending: No more data ....1. .... = Acknowledgment Request: Ack required .... .0.. .... = Intra-PAN: Not within the PAN .... ..00 0... .... = Reserved .... 11.. .... = Destination Addressing Mode: Address field contains a 64-bit long address ..00 .... = Reserved 11.. .... = Source Addressing Mode: Address field contains a 64-bit long address Sequence Number: xx Source PAN Identifier: 0x1AAA Destination PAN Identifier: 0xffff Source Address: 0xACDE480000000001 Destination Address: 0xACDE480000000002 MAC Payload Command Frame Identifier = Coordinator Realignment: (0x08) PAN ID = 0x1AAA Coordinator Short Address = 0x0001 Logical Channel = 0x14 Short Address = 0x0002 Frame Check Sequence: Correct The DUT shall return a Success result, such that: MLME-SCAN.confirm (Status=0x00, ScanType=0x03, ChannelPage=0, UnscannedChannels=0x00 00 00, ResultListSize=0x00, EnergyDetectList=NULL, PANDescriptorList=NULL) Fail: DUT does not ACK the coordinator Realignment DUT does not get the correct indication

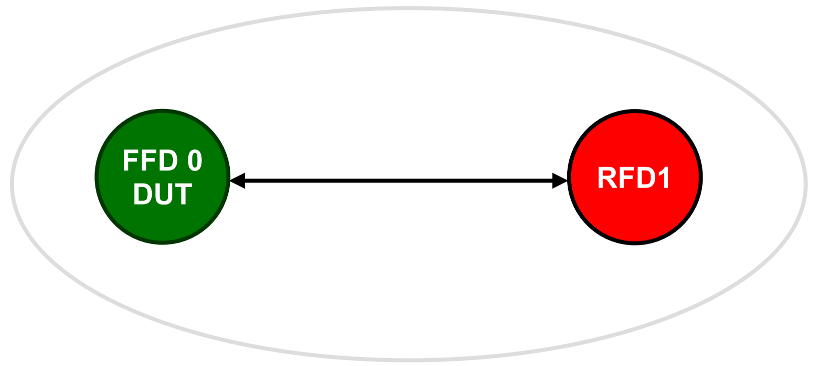
Notes:

None.

TP/154/MAC/SCANNING-08[1]

Receive an Orpan Scan and responding with a Coordinator Realignment

Setup Conditions:

Diagram


The Diagram is the image that represents the configuration deployed for the Test Procedure.

Initial Conditions:

Device name	Description
All the devices	*Security: no for all the devices* *Channel: 0x14 for all the devices* *PAN id: 0x1AAA for all the devices*
FFD0	*PAN Coordinator, Bo = 15, SO = 15* *MAC: 0xACDE480000000001* *Short Address: 0x0001* *Beacon Payload: none*
RFD1	*MAC: 0xACDE480000000002* Tester associates to DUT as described in TP/154/MAC/ASSOCIATION-01

Test Procedure:

Step No.	Step description	Expected outcome
1	Set to DUT: MLME-RESET.request(TRUE).	Pass: FFD returns SUCCESS. Fail: FFD does not respond.
2	Set to Tester(RFD1): MLME-SCAN.request (ScanType=0x03, ScanChannels==0x07fff800(for 2.4GHz O-QPSK PHY) ScanChannels==0x000007ff(for 868/915MHz BPSK PHY) ScanChannels==0x00000003fff(for 920MHz FSK PHY), ScanDuration=0x05, ChannelPage=0x00, SecurityLevel=0x00, KeyIDMode=xx,	Pass: It is possible to verify by a PAN analyzer that a Orphan Scan command frame was issued from the tester. Frame Length: 16 bytes IEEE 802.15.4 Frame Control: 0xC843 .... .011 = Frame Type: Command (0x0003) .... 0... = Security Enabled: Disabled ....0 .... = Frame Pending: No more data ....0. .... = Acknowledgment Request: Ack not required .... .1.. .... = Intra-PAN: Within the PAN .... ..00 0... .... = Reserved

	KeySource=NULL, KeyIndex=xx) Orphan Scan, all channels, duration=[*aBaseSuperframeDuration* * (2*n* + 1)] symbols per channel	.... 10.. .... = Destination Addressing Mode: Address field contains a 16-bit short address ..00 .... = Reserved 11.. .... = Source Addressing Mode: Address field contains a 64-bit ieee address Sequence Number: xx Destination PAN Identifier: 0xffff Destination Address: 0xffff Source Address: 0xACDE480000000002 MAC Payload Command Frame Identifier = Orphan Notification: (0x06) Frame Check Sequence: Correct On the DUT: MLME-ORPHAN.indication (OrphanAddress=0xACDE480000000002, SecurityLevel=0x00, KeyIdMode=xx, KeySource=NULL, KeyIndex=xx) Fail: DUT does not show the correct indication
3	Set to DUT MLME-ORPHAN.response (OrphanAddress=0xACDE480000000002 ShortAddress=0x0002 AssociatedMember=TRUE SecurityLevel=0x00, KeyIDMode=xx, KeySource=NULL, KeyIndex=xx)	Pass: The DUT shall return a Success result, such that: DUT returns MAC primitive MLME-COMM-STATUS.indication(PANid = 0x1aaa, SrcAddrMode = 0x03 = ExtendedAddress, SrcAddr = 0x ACDE480000000001, DstAddrMode = 0x03= 64-bit extended address, DstAddr = 0x ACDE480000000002, Status = 0x00 = SUCCESS, SecurityLevel = 0x00, KeyIdMode = xx, KeySource = xx, KeyIndex = xx) Pass: It is possible to verify by a PAN analyzer that a Coordinator Realignment command frame was issued from the DUT to the Tester. Frame Length: 31 bytes IEEE 802.15.4 Frame Control: 0xCC23 .... .011 = Frame Type: Command (0x0003) .... 0... = Security Enabled: Disabled ....0 .... = Frame Pending: No more data ....1. .... = Acknowledgment Request: Ack required .... .0.. .... = Intra-PAN: Not within the PAN .... ..00 0... .... = Reserved .... 11.. .... = Destination Addressing Mode: Address field contains a 64-bit long address ..00 .... = Reserved 11.. .... = Source Addressing Mode: Address field contains a 64-bit long address Sequence Number: xx Source PAN Identifier: 0x1AAA Destination PAN Identifier: 0xffff Source Address: 0xACDE480000000001 Destination Address: 0xACDE480000000002 MAC Payload

		Command Frame Identifier = Coordinator Realignment: (0x08) PAN ID = 0x1AAA Coordinator Short Address = 0x0001 Logical Channel = 0x14 Short Address = 0x0002 Frame Check Sequence: Correct The Tester shall return a Success result, such that: MLME-SCAN.confirm (Status=0x00, ScanType=0x03, ChannelPage=0, UnscannedChannels=0x00 00 00, ResultListSize=0x00, EnergyDetectList=NULL, PANDescriptorList=NULL) Fail: DUT does not return MLME-COMM-STATUS.indication primitive; Status is not SUCCESS DUT does not send the correct Coord Realignment packet

Notes:

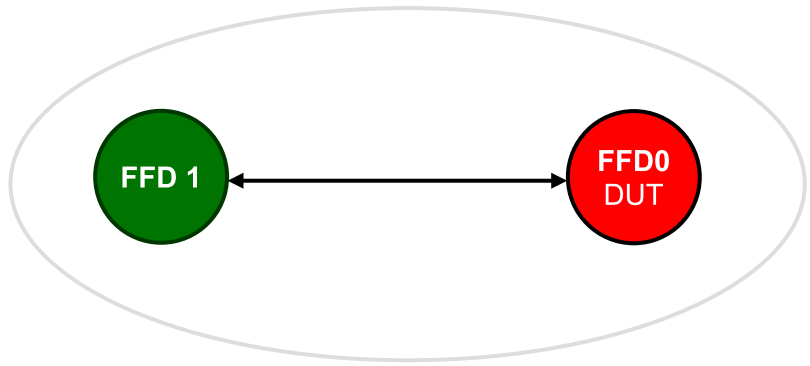
None.

TP/154/MAC/BEACON-MANAGEMENT-01

Check D.U.T. can generate beacons with short address, with beacon payload

Setup Conditions:

Diagram



The Diagram is the image that represents the configuration deployed for the Test Procedure.

Initial Conditions:

Device name	Description
All the devices	*Security: no for all the devices* *Channel: 0x14 for all the devices* *PAN id: 0x1AAA for all the devices*
FFD0	*MAC: 0xACDE480000000001*
FFD1	*MAC: 0xACDE480000000002*

Test Procedure:

Step No.	Step description	Expected outcome
1	Set to DUT: MLME-RESET.request(SetDefaultPIB=TRUE).	Pass: DUT returns primitive MLME-RESET.confirm(Status= SUCCESS). Fail: FFD does not respond.
2	MLME-SET.request for the DUT, such that: MLME-SET.request (PIBAttribute=0x53=macShortAddress, PIBAttributeValue=0x1122) Verify the PIB is correctly set MLME-GET.request (PIBAttribute=0x53=macShortAddress, PIBAttributeIndex = 0x00- ignore, Only used for MAC 2006 security PIBs)	Wait for MLME-Set_confirm MLME-SET.confirm (Status=SUCCESS, PIBAttribute=0x53=macShortAddress,) Wait for MLME-GET.confirm and confirm the parameters MLME-GET.confirm (Status = SUCCESS PIBAttribute=0x53=macShortAddress, PIBAttributeIndex = ignore. Only for security in MAC2006 PIBAttributeValue = 0x1122

		)
3	MLME-SET.request for the DUT, such that: MLME-SET.request (PIBAttribute=0x46=macBeaconPayload- LengthPIB, PIBAttributeValue=0x02) Verify the PIB is correctly set MLME-GET.request (PIBAttribute=0x46=macBeaconPayload, PIBAttributeIndex = 0x00 - ignore, Only used for MAC 2006 security PIBs)	Wait for MLME-Set_confirm MLME-SET.confirm (Status=SUCCESS, PIBAttribute=0x46=macBeaconPayload- LengthPIB,) Wait for MLME-GET.confirm and confirm the parameters MLME-GET.confirm (Status = SUCCESS PIBAttribute=0x46=macBeaconPayload, PIBAttributeIndex = ignore. Only for security in MAC2006 PIBAttributeValue =0x02)
4	MLME-SET.request for the DUT such that: MLME-SET.request(PIBAttribute=0x45=MacBeaconPayload, PIBAttributeValue=0x12 0x34 (to avoid confusion with short address)) Verify the PIB is correctly set MLME-GET.request (PIBAttribute=0x45=MacBeaconPayload, PIBAttributeIndex = 0x00 (ignore, Only used for MAC 2006 security PIBs)	Wait for MLME-Set_confirm MLME-SET.confirm (Status=SUCCESS, PIBAttribute=0x45=MacBeaconPayload,) Wait for MLME-GET.confirm and confirm the parameters MLME-GET.confirm (Status = SUCCESS PIBAttribute=0x45=MacBeaconPayload, PIBAttributeIndex = ignore. Only for security in MAC2006 PIBAttributeValue =0x12 0x34)
5	Finally, issue MLME-START.request command with the following parameters: MLME-START.request (PANId=0x1AAA, LogicalChannel=20=0x14, BeaconOrder=15, SuperframeOrder=15, PANCoordinator=TRUE, BatteryLifeExtension=FALSE, CoordRealignment=FALSE, CoordRealignmentSecurityLevel=0, [CoordRealignmentKeyIdMode =x, CoordRealignmentKeySource = x, CoordRealignmentKeyIndex = x,] BeaconSecurityLevel=0, [BeaconKeyIdMode =x, BeaconKeySource = x, BeaconKeyIndex = x,]).	Pass: DUT returns primitive MLME-START.confirm (Status=SUCCESS)
6	Tester(FFD1) sends Beacon request MAC command frame	Pass: It is possible to observe a beacon from the DUT after the Tester implements an active scan, with correct parameters in the beacon Frame Length: 15 bytes IEEE 802.15.4 Frame Control: 0x8000 .... .000 = Frame Type: Beacon(0x0000) .... 0... = Security Enabled: Disabled ....0 .... = Frame Pending: No more data ....0. .... = Acknowledgment Request: Ack not required .... .0.. .... = Intra-PAN: Not within the PAN .... ..00 0... .... = Reserved .... 00.. .... = Destination Addressing Mode:

		..00 .... = Reserved for Zigbee PRO/RF4CE 10.. .... = Source Addressing Mode: PAN identifier and Address field contains a 16-bit short address Sequence Number: xx Source PAN Identifier: 0x1aaa Source Address: 0x1122 Superframe Specification .... 1111 = BeaconOrder = 0x0f .... 1111.... = SuperframeOrder = 0x0f .... 1111 …. .... = FinalCapSlot = 0x0f ...0 .... = BLE = 0 ..0. .... = Reserved = 0 .1.. …. .... = PanCoordinator = 1 0... .... = AssociationPermit = 0 GTS Specification = 0x00 Pending Address Field = 0x00 Beacon Payload = 0x12 0x34 Frame Check Sequence: Correct Fail: FFD0 does not respond.

862 ***Notes:***

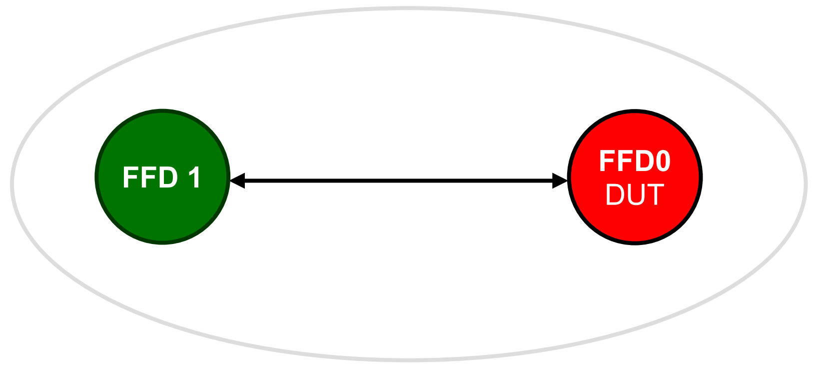
863 *None.*

TP/154/MAC/BEACON-MANAGEMENT-02[1] (optional)

Check D.U.T. can generate beacons with short address, without beacon payload

Setup Conditions:

Diagram



The Diagram is the image that represents the configuration deployed for the Test Procedure.

Initial Conditions:

Device name	**Description**
All the devices	*Security: no for all the devices* *Channel: 0x14 for all the devices* *PAN id: 0x1AAA for all the devices*
FFD0	*MAC: 0xACDE480000000001*
FFD1	*MAC: 0xACDE480000000002*

Test Procedure:

Step No.	**Step description**	**Expected outcome**
1	Set to DUT: MLME-RESET.request(SetDefaultPIB=TRUE).	Pass: DUT returns primitive MLME-RESET.confirm(Status= SUCCESS). Fail: FFD does not respond.
2	MLME-SET.request for the DUT, such that: MLME-SET.request (PIBAttribute=0x53=macShortAddress, PIBAttributeValue=0x1122) Verify the PIB is correctly set MLME-GET.request (PIBAttribute=0x53=macShortAddress, PIBAttributeIndex = 0x00- ignore, Only used for MAC 2006 security PIBs)	Wait for MLME-Set_confirm MLME-SET.confirm (Status=SUCCESS, PIBAttribute=0x53=macShortAddress,) Wait for MLME-GET.confirm and confirm the parameters MLME-GET.confirm (Status = SUCCESS PIBAttribute=0x53=macShortAddress, PIBAttributeIndex = ignore. Only for security in MAC2006 PIBAttributeValue = 0x1122)

3	Finally, issue MLME-START.request command with the following parameters: MLME-START.request (PANId=0x1AAA, LogicalChannel=20=0x14, BeaconOrder=15, SuperframeOrder=15, PANCoordinator=TRUE, BatteryLifeExtension=FALSE, CoordRealignment=FALSE, CoordRealignmentSecurityLevel=0, [CoordRealignmentKeyIdMode =x, CoordRealignmentKeySource = x, CoordRealignmentKeyIndex = x,] BeaconSecurityLevel=0, [BeaconKeyIdMode =x, BeaconKeySource = x, BeaconKeyIndex = x,]).	Pass: DUT returns primitive MLME-START.confirm (Status=SUCCESS)
4	Tester(FFD1) sends Beacon request MAC command frame	Pass: It is possible to observe a beacon from the DUT after the Tester implements an active scan, with correct parameters in the beacon Frame Length: 13 bytes IEEE 802.15.4 Frame Control: 0x8000 .... .000 = Frame Type: Beacon(0x0000) .... 0... = Security Enabled: Disabled ....0 .... = Frame Pending: No more data ....0. .... = Acknowledgment Request: Ack not required .... .0.. .... = Intra-PAN: Not within the PAN .... ..00 0... .... = Reserved .... 00.. .... = Destination Addressing Mode: ..00 .... = Reserved 10.. .... = Source Addressing Mode: PAN identifier and Address field contains a 16-bit short address Sequence Number: xx Source PAN Identifier: 0x1aaa Source Address: 0x1122 Superframe Specification .... 1111 = BeaconOrder = 0x0f .... 1111.... = SuperframeOrder = 0x0f .... 1111 .... = FinalCapSlot = 0x0f ...0 .... = BLE = 0 ..0. .... = Reserved = 0 .1.. .... = PanCoordinator = 1 0... .... = AssociationPermit = 0 GTS Specification = 0x00 Pending Address Field = 0x00 Frame Check Sequence: Correct Fail: FFD0 does not respond.

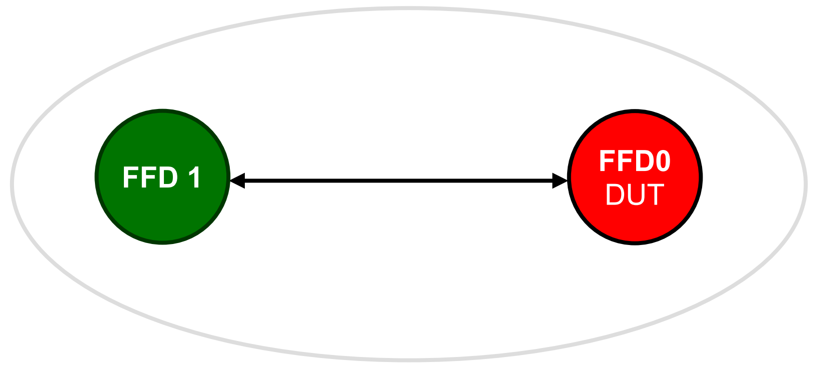
870 ***Notes:***

871 *None.*

TP/154/MAC/BEACON-MANAGEMENT-03[1] (optional)

Check D.U.T. can generate beacons with extended address, without beacon payload

Setup Conditions:

Diagram


The Diagram is the image that represents the configuration deployed for the Test Procedure.

Initial Conditions:

Device name	Description
All the devices	*Security: no for all the devices* *Channel: 0x14 for all the devices* *PAN id: 0x1AAA for all the devices*
FFD0	*MAC: 0xACDE480000000001*
FFD1	*MAC: 0xACDE480000000002*

Test Procedure:

Step No.	Step description	Expected outcome
1	Set to DUT: MLME-RESET.request(SetDefaultPIB=TRUE).	Pass: DUT returns primitive MLME-RESET.confirm(Status= SUCCESS). Fail: FFD does not respond.
2	MLME-SET.request for the DUT, such that: MLME-SET.request (PIBAttribute=0x53=macShortAddress, PIBAttributeValue=0xfffe) Verify the PIB is correctly set MLME-GET.request (PIBAttribute=0x53=macShortAddress, PIBAttributeIndex = 0x00- ignore, Only used for MAC 2006 security PIBs)	Wait for MLME-Set_confirm MLME-SET.confirm (Status=SUCCESS, PIBAttribute=0x53=macShortAddress,) Wait for MLME-GET.confirm and confirm the parameters MLME-GET.confirm (Status = SUCCESS PIBAttribute=0x53=macShortAddress, PIBAttributeIndex = ignore. Only for security in MAC2006 PIBAttributeValue = 0xfffe)

3	Finally, issue MLME-START.request command with the following parameters: MLME-START.request (PANId=0x1AAA, LogicalChannel=20=0x14, BeaconOrder=15, SuperframeOrder=15, PANCoordinator=TRUE, BatteryLifeExtension=FALSE, CoordRealignment=FALSE, CoordRealignmentSecurityLevel=0, [CoordRealignmentKeyIdMode =x, CoordRealignmentKeySource = x, CoordRealignmentKeyIndex = x,] BeaconSecurityLevel=0, [BeaconKeyIdMode =x, BeaconKeySource = x, BeaconKeyIndex = x,]).	Pass: DUT returns primitive MLME-START.confirm (Status=SUCCESS)
4	FFD1 sends Beacon request MAC command frame	Pass: It is possible to observe a beacon from the DUT after the Tester implements an active scan, with correct parameters in the beacon Frame Length: 19 bytes IEEE 802.15.4 Frame Control: 0xc000 .... .000 = Frame Type: Beacon(0x0000) .... 0... = Security Enabled: Disabled ....0 .... = Frame Pending: No more data ....0. .... = Acknowledgment Request: Ack not required .... .0.. .... = Intra-PAN: Not within the PAN .... ..00 0... .... = Reserved .... 00.. .... = Destination Addressing Mode: ..00 .... = Reserved 11.. .... = Source Addressing Mode: PAN identifier and Address field contains a 64-bit extended address Sequence Number: xx Source PAN Identifier: 0x1aaa Source Address: 0xACDE480000000001 Superframe Specification .... 1111 = BeaconOrder = 0x0f .... 1111.... = SuperframeOrder = 0x0f .... 1111 .... = FinalCapSlot = 0x0f ...0 .... = BLE = 0 ..0. .... = Reserved = 0 .1.. .... = PanCoordinator = 1 0... .... = AssociationPermit = 0 GTS Specification = 0x00 Pending Address Field = 0x00 Frame Check Sequence: Correct Fail: FFD0 does not respond.

878 ***Notes:***

879 *None.*

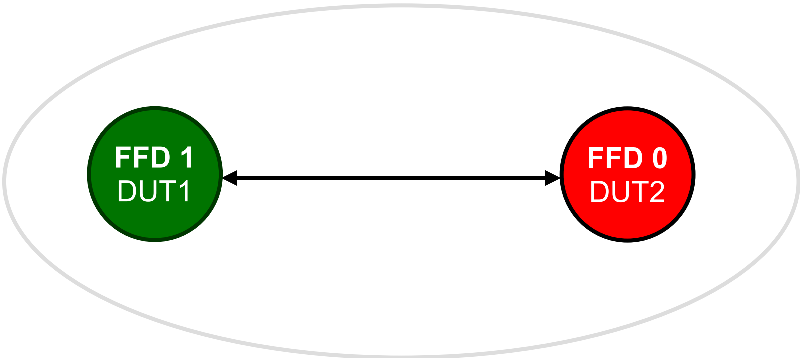
TP/154/MAC/BEACON-MANAGEMENT-04[1] (optional)

This test checks that the sequence of Enhanced Beacon Request, Enhanced Beacon, Association request, and Association response, used for inintal joining are propery executed. Specifically it checks:

- that an Enhanced Beacon Request (EBR), with an Enhanced Beacon (EB) Filter IE and a TX Power IE can be sent, per the requirements as described in Annex D of the Zigbee Specification [R2].
- that the EB Filter IE and TX Power IE are processed properly, along with additional filtering, and that an Enhanced Beacon (EB) with a EB Payload IE and TX Power IE are sent in response to the EBR, per the requirements as described in Annex D of the Zigbee Specification [R2].
- that the MAC Association request/response, occurring following a successful EBR/EB exchange for joining, are successful.
- and that the updates to the Power Control Information Table are evident in the EB and the following Association request/response.

The specific frame formats to be used for this test are described in Annex D of the Zigbee Specification [R2].

Setup Conditions:

Diagram


The Diagram is the image that represents the configuration deployed for the Test Procedure.

Initial Conditions:

Device name	**Description**
All the devices	*Security: no for all the devices* *Channel: 0x14 for all the devices* *PAN id: 0x1AAA for all the devices*
DUT1	*MAC: 0xACDE480000000001* *Prepare DUT1 to Transmit a packet on the channel to be tested*
DUT2	*MAC: 0xACDE480000000002* *Place DUT2 in Receive mode on the channel to be tested*

Test Procedure:

Step No.	**Step description**	**Expected outcome**
***mibJoiningPolicy* is set to NOJOIN**		

Step No.	Step description	Expected outcome
1	In DUT2 Set *mibJoiningPolicy* to NOJOIN, indicating that no device may join that coordinator/router	N/A
2	In DUT1 Set the transmit power and attenuators such that the target power is within the range of the receivers capability Transmit an EBR from DUT1 to DUT2, with an EB Filter IE and TX Power IE for initial joining, as described in Annex D of the Zigbee Specification [R2]	Observe the packet received by DUT2. Verify the following: - An EBR was received which included an EB Filter IE and TX Power IE, where the packet was set as described in Annex D of the Zigbee Specification [R2], for initial joining Pass: That the EBR, EB Filter IE, and TX Power IE are received properly for initial joining Fail: That any of the EBR, EB Filter IE, and TX Power IE are not received properly for initial joining Verify the following: - The EB Filter IE and TX Power IE and additional filtering are performed as described in Annex D of the Zigbee Specification [R2] and that an EB is not sent as described in Annex D of the Zigbee Specification [R2], for initial joining Pass: EB Filter IE, TX Power IE, and additional filter are processed and an EB is not sent Fail: Any of the EB Filter IE, TX Power IE, or additional filter are not processed and/or an EB with or without a EB Payload IE and/or TX Power IE is sent
3	Swap roles of DUT1 and DUT2 and Repeat steps 1-2	
***mibJoiningPolicy* is set to ALLJOIN**		
4	In DUT2 Set *mibJoiningPolicy* to ALLJOIN, indicating that any device may join that coordinator/router Clear the *mibJoiningIeeeList* in DUT2	N/A
5	In DUT1 Set the transmit power and attenuators such that the target power is within the range of the receivers capability Transmit an EBR from DUT1 to DUT2, with an EB Filter IE and TX Power IE for initial joining, as described in Annex D of the Zigbee Specification [R2]	Observe the packet received by DUT2. Verify the following: - An EBR was received which included an EB Filter IE and TX Power IE, where the packet was set as described in Annex D of the Zigbee Specification [R2], for initial joining Pass: That the EBR, EB Filter IE, and TX Power IE are received properly for initial joining Fail:

Step No.	Step description	Expected outcome
		That any of the EBR, EB Filter IE, and TX Power IE are not received properly for initial joining Verify the following: - The EB Filter IE and TX Power IE and additional filtering are performed as described in Annex D of the Zigbee Specification [R2] and that an EB is sent as described in Annex D of the Zigbee Specification [R2], for initial joining Pass: EB Filter IE, TX Power IE, and additional filter are processed and an EB with a EB Payload IE and TX Power IE is sent with a TX power setting reflecting evaluation of the received TX Power IE Fail: Any of the EB Filter IE, TX Power IE, or additional filter are not processed and/or an EB is not sent or an EB is sent without either or both of the EB Payload IE and TX Power IE
6	Swap roles of DUT1 and DUT2 and Repeat steps 4-5	
mibJoiningPolicy* is set to IEEELISTJOIN and DUT1 is on DUT2s *mibJoiningIeeeList		
7	In DUT2 Set *mibJoiningPolicy* to IEEELISTJOIN, indicating that only devices with their address in the *mibJoiningIeeeList* may join that coordinator/router Add the *macExtendedAddress* of DUT1 to the *mibJoiningIeeeList* of DUT2	N/A
8	In DUT1 Set the transmit power and attenuators such that the target power is within the range of the receivers capability Transmit an EBR from DUT1 to DUT2, with an EB Filter IE and TX Power IE for initial joining, as described in Annex D of the Zigbee Specification [R2]	Observe the packet received by DUT2. Verify the following: - An EBR was received which included an EB Filter IE and TX Power IE, where the packet was set as described in Annex D of the Zigbee Specification [R2], for initial joining Pass: That the EBR, EB Filter IE, and TX Power IE are received properly for initial joining Fail: That any of the EBR, EB Filter IE, and TX Power IE are not received properly for initial joining Verify the following: - The EB Filter IE and TX Power IE and additional filtering are performed as described Annex D of the Zigbee Specification [R2] and that an EB is sent as described in Annex D of the Zigbee Specification [R2], for initial joining Pass: EB Filter IE, TX Power IE, and additional filter are processed and an EB with a EB Payload IE and TX Power IE is sent with a TX power setting reflecting evaluation of the received TX Power IE

Step No.	Step description	Expected outcome
		Fail: Any of the EB Filter IE, TX Power IE, or additional filter are not processed and/or an EB is not sent, or an EB is sent without either or both of the EB Payload IE and TX Power IE
9	Swap roles of DUT1 and DUT2 and Repeat steps 7-8	
mibJoiningPolicy* is set to IEEELISTJOIN and DUT1 is not on DUT2s *mibJoiningIeeeList		
10	In DUT2 Set *mibJoiningPolicy* to IEEELISTJOIN, indicating that only devices with their address in the *mibJoiningIeeeList* may join that coordinator/router Remove the *macExtendedAddress* of DUT1 from the *mibJoiningIeeeList* of DUT2	N/A
11	In DUT1 Set the transmit power and attenuators such that the target power is within the range of the receivers capability Transmit an EBR from DUT1 to DUT2, with an EB Filter IE and TX Power IE for initial joining, as described in Annex D of the Zigbee Specification [R2]	Observe the packet received by DUT2. Verify the following: - An EBR was received which included an EB Filter IE and TX Power IE, where the packet was set as described in Annex D of the Zigbee Specification [R2], for initial joining Pass: That the EBR, EB Filter IE, and TX Power IE are received properly for initial joining Fail: That any of the EBR, EB Filter IE, and TX Power IE are not received properly for initial joining Verify the following: - The EB Filter IE and TX Power IE and additional filtering are performed as described in Annex D of the Zigbee Specification [R2] and that an EB is not sent as described in Annex D of the Zigbee Specification [R2], for initial joining Pass: EB Filter IE, TX Power IE, and additional filter are processed and an EB is not sent Fail: Any of the EB Filter IE, TX Power IE, or additional filter are not processed and/or an EB is sent with or without either or both of the EB Payload IE and TX Power IE
12	Swap roles of DUT1 and DUT2 and Repeat steps 10-11	
For all Cases Above Where an EB is Correctly Sent in Response to an EBR		
13	After DUT1 receives the EB for initial joining	Observe the packet received by DUT1. Verify the following: - An EB was received which included a EB Payload IE and TX Power IE, where the packet was set as described in Annex D of the Zigbee Specification [R2], for initial joining Pass:

Step No.	Step description	Expected outcome
		That the EB, EB Payload IE, and TX Power IE are received properly for initial joining Fail: That any of the EB, EB Payload IE, and TX Power IE are not received properly for initial joining Verify the following: - The TX Power of the device receiving the EB is adjusted based on the information received in the EB as described in Annex D of the Zigbee Specification [R2], for initial joining Pass: The next packet from DUT1 is sent with a TX power setting reflecting evaluation of the TX Power information in the EB Fail: The next packet from DUT1 is not sent with a TX power setting reflecting evaluation of the TX Power information in the EB
14	After a successful EBR/EB exchange for initial joining DUT1 sends an Association Request to DUT2, followed by DUT2 responding by sending a Association Response to DUT1 *** Rem: When the Association request/response follows an EBR/EB exchange the Association request/response MAY use the extended addressing mode. This is different from the short addressing mode Zigbee uses when the Association request/response follows a non-enhanced BR/B exchange.	Observe the packet received by DUT2. Verify the following: - An Association Request is received Pass: That the Association Request is received properly for initial joining Fail: That the Association Request is not received properly for initial joining In response to DUT2 properly receiving an Association Request form DUT1, DUT2 sends a Association Response to DUT1. Observe the packet received by DUT1. Verify the following: - A Association Response is received Pass: That the Association Response is received properly for initial joining Fail: That the Association Response is not received properly for initial joining

900 ***Notes:***

901 *Data (typically sent by Zigbee in a non-Enhanced Beacon Frame) to be sent in an Enhanced Beacon*
902 *Frame that is intended to be passed up from the MAC layer SHALL be sent in the Zigbee EB IE.*

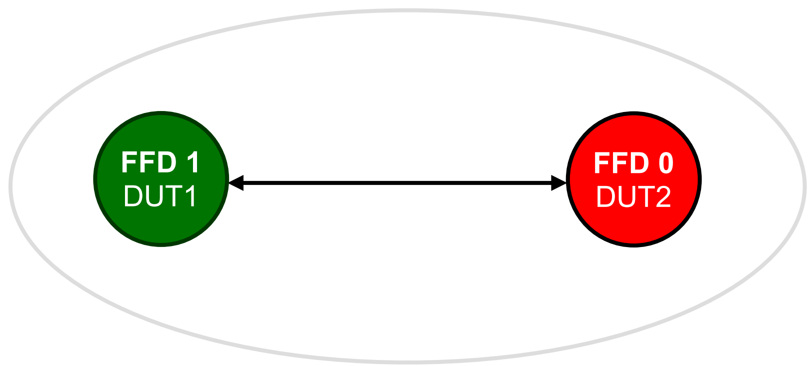
TP/154/MAC/BEACON-MANAGEMENT-05[1] (optional)

This test checks that the sequence of Enhanced Beacon Request, Enhanced Beacon, NWK Rejoin request, and NWK Rejoin response, used for rejoining are propery executed. Specifically it checks:

- that an EBR, with a Rejoin IE and TX Power IE can be sent, per the requirements as described in Annex D of the Zigbee Specification [R2].
- that the Rejoin IE and TX Power IE are processed properly, and that an EB with a EB Payload and TX Power IE are sent in response to an EBR, per the requirements as described in Annex D of the Zigbee Specification [R2].
- that the NWK Rejoin request/response, occurring following a successful EBR/EB exchange for rejoining, are successful.
- and that the updates to the Power Control Information Table are evident in the EB and the following NWK Rejoin request/response.

The specific frame formats to be used for this test are described in Annex D of the Zigbee Specification [R2].

Setup Conditions:

Diagram


The Diagram is the image that represents the configuration deployed for the Test Procedure.

Initial Conditions:

Device name	**Description**
All the devices	*Security: no for all the devices* *Channel: 0x14 for all the devices* *PAN id: 0x1AAA for all the devices*
DUT1	*MAC: 0xACDE480000000001* *Prepare DUT1 to Transmit a packet on the channel to be tested*
DUT2	*MAC: 0xACDE480000000002* *Place DUT2 in Receive mode on the channel to be tested*

Test Procedure:

Step No.	**Step description**	**Expected outcome**
nwkExtendedPANId field of the Rejoin IE in DUT1 is the same as the NIB of DUT2		

Step No.	Step description	Expected outcome
1	In DUT1 Set the *nwkExtendedPANId* field of the Rejoin IE to the NIB of DUT2 Set the transmit power and attenuators such that the target power is within the range of the receivers capability Transmit an EBR from DUT1 to DUT2, with an EB Filter IE and TX Power IE for initial joining, as described in Annex D of the Zigbee Specification [R2]	Observe the packet received by DUT2. Verify the following: - An EBR was received which included an Rejoin IE and TX Power IE, where the packet was set as described in Annex D of the Zigbee Specification [R2], for rejoining Pass: That the EBR, Rejoin IE, and TX Power IE are received properly for rejoining Fail: That any of the EBR, Rejoin IE, and TX Power IE are not received properly for rejoining Verify the following: - The Rejoin IE and TX Power IE and additional filtering are performed as described in Annex D of the Zigbee Specification [R2], and that an EB is sent as described in Annex D of the Zigbee Specification [R2], for re joining Pass: Rejoin IE, TX Power IE, and additional filter are processed and an EB with a EB Payload IE and TX Power IE is sent with a TX power setting reflecting evaluation of the received TX Power IE Fail: Any of the Rejoin IE, TX Power IE, or additional filter are not processed and/or an EB is not sent or an EB is sent without either or both of the EB Payload IE and TX Power IE
2	Swap roles of DUT1 and DUT2 and Repeat step 1	
nwkExtendedPANId field of the Rejoin IE in DUT1 is not the same as the NIB of DUT2		
3	Repeat steps 1-2, where the *nwkExtendedPANId* field of the Rejoin IE of DUT1 is cleared	Observe the packet received by DUT2. Verify the following: - An EBR was received which included an Rejoin IE and TX Power IE, where the packet was set as described in Annex D of the Zigbee Specification [R2], for rejoining Pass: That the EBR, Rejoin IE, and TX Power IE are received properly for rejoining Fail: That any of the EBR, Rejoin IE, and TX Power IE are not received properly for rejoining Verify the following: - The Rejoin IE and TX Power IE and additional filtering are performed as described in Annex D of the Zigbee Specification [R2] and that an EB is not sent as described in Annex D of the Zigbee Specification [R2], for rejoining Pass:

Step No.	Step description	Expected outcome
		Rejoin IE, TX Power IE, and additional filter are processed and an EB is not sent Fail: Any of the Rejoin IE, TX Power IE, or additional filter are not processed and/or an EB is sent
For the Case Above Where an EB is Correctly Sent in Response to an EBR		
4	After DUT1 receives the EB for rejoining	Observe the packet received by DUT1. Verify the following: - An EB was received which included a EB Payload IE and TX Power IE, where the packet was set as described in Annex D of the Zigbee Specification [R2], for rejoining Pass: That the EB, EB Payload IE, and TX Power IE are received properly for rejoining Fail: That any of the EB, EB Payload IE, and TX Power IE are not received properly for rejoining Verify the following: - The TX Power of the device receiving the EB is adjusted based on the information received in the EB as described in Annex D of the Zigbee Specification [R2], for rejoining Pass: The next packet from DUT1 is sent with a TX power setting reflecting evaluation of the TX Power information in the EB Fail: The next packet from DUT1 is not sent with a TX power setting reflecting evaluation of the TX Power information in the EB
5	After a successful EBR/EB exchange for rejoining DUT1 sends a NWK rejoin request to DUT2, followed by DUT2 responding by sending a NWK rejoin response to DUT1 *** Rem: When the NWK Rejoin request/response follows an EBR/EB exchange the NWK Rejoin request/response MAY use the extended addressing mode. This is different from the short addressing mode Zigbee uses when the NWK Rejoin request/response follows a non-enhanced BR/B exchange.	Observe the packet received by DUT2. Verify the following: - An NWK rejoin request is received Pass: That the NWK rejoin request is received properly for rejoining Fail: That the NWK rejoin request is not received properly for rejoining In response to DUT2 properly receiving a NWK rejoin request form DUT1, DUT2 sends a NWK rejoin response to DUT1. Observe the packet received by DUT1. Verify the following: - A NWK rejoin response is received Pass: That the NWK rejoin response is received properly for rejoining

Step No.	Step description	Expected outcome
		Fail: That the NWK rejoin response is not received properly for rejoining

Notes:

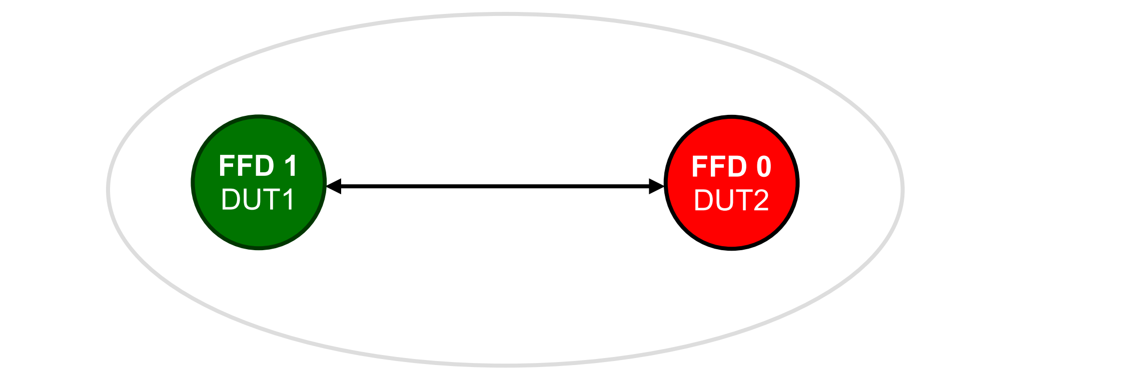
Data (typically sent by Zigbee in a non-Enhanced Beacon Frame) to be sent in an Enhanced Beacon Frame that is intended to be passed up from the MAC layer SHALL be sent in the Zigbee EB IE.

TP/154/MAC/BEACON-MANAGEMENT-06[1] (optional)

This test checks that unrecognized IEs, when received in an EB or EBR frame, are ignored and the remainder of the frame is processed normally. This test partially reuses the TP/154/MAC/BEACON-MANAGEMENT-04[1] (optional) test case.

Setup Conditions:

Diagram



The Diagram is the image that represents the configuration deployed for the Test Procedure.

Initial Conditions:

Device name	Description
All the devices	*Security: no for all the devices* *Channel: 0x14 for all the devices* *PAN id: 0x1AAA for all the devices*
DUT1	*MAC: 0xACDE480000000001* *Prepare DUT1 to Transmit a packet on the channel to be tested*
DUT2	*MAC: 0xACDE480000000002* *Place DUT2 in Receive mode on the channel to be tested*

Test Procedure:

Step No.	Step description	Expected outcome
***mibJoiningPolicy* is set to ALLJOIN**		
1	In DUT2 Set *mibJoiningPolicy* to ALLJOIN, indicating that any device may join that coordinator/router Clear the *mibJoiningIeeeList* in DUT2	N/A
2	In DUT1 Set the transmit power and attenuators such that the target power is within the range of the receivers capability Transmit an EBR from DUT1 to DUT2, with an EB Filter IE and TX Power IE for initial joining, as described in Annex D of the Zigbee Specification [R2] Additionally include:	Observe the packet received by DUT2. Verify the following: - An EBR was received which included an EB Filter IE and TX Power IE, where the packet was set as described in Annex D of the Zigbee Specification [R2], for initial joining Pass:

	- Any other 15.4 payload IE (such as another Vendor Specific IE with a different Vendor OUI than the one assigned to Zigbee), before the EB Filter IE A different Zigbee Payload (Nested) IE, whose Zigbee sub–ID is one of the reserved ones, with a random length, is within the Zigbee Payload IE and before the Zigbee Tx Power sub-ID IE	That the additional unrecognized 15.4 payload IE and additional unrecognized Zigbee Payload (Nested) IE are discarded, And That the EBR, EB Filter IE, and TX Power IE are received properly for initial joining Fail: That either of additional unrecognized 15.4 payload IE and additional unrecognized Zigbee Payload (Nested) IE are not discarded, And that any of the EBR, EB Filter IE, and TX Power IE are not received properly for initial joining Verify the following: - The EB Filter IE and TX Power IE and additional filtering are performed as described in Annex D of the Zigbee Specification [R2] and that an EB is sent as described in Annex D of the Zigbee Specification [R2], for initial joining Pass: EB Filter IE, TX Power IE, and additional filter are processed and an EB with a EB Payload IE and TX Power IE is sent with a TX power setting reflecting evaluation of the received TX Power IE Fail: Any of the EB Filter IE, TX Power IE, or additional filter are not processed and/or an EB is not sent or an EB is sent without either or both of the EB Payload IE and TX Power IE
3	Swap roles of DUT1 and DUT2 and Repeat steps 1-2	
For all Cases Above Where an EB is Correctly Sent in Response to an EBR		
4	After DUT1 receives the EB for initial joining	Observe the packet received by DUT1. Verify the following: - An EB was received which included a EB Payload IE and TX Power IE, where the packet was set as described in Annex D of the Zigbee Specification [R2], for initial joining Pass: That the EB, EB Payload IE, and TX Power IE are received properly for initial joining Fail: That any of the EB, EB Payload IE, and TX Power IE are not received properly for initial joining Verify the following: - The TX Power of the device receiving the EB is adjusted based on the information received in the EB as described in Annex D of the Zigbee Specification [R2], for initial joining Pass: The next packet from DUT1 is sent with a TX power setting reflecting evaluation of the TX Power information in the EB

		Fail: The next packet from DUT1 is not sent with a TX power setting reflecting evaluation of the TX Power information in the EB
5	After a successful EBR/EB exchange for initial joining DUT1 sends an Association Request to DUT2, followed by DUT2 responding by sending a Association Response to DUT1 *** Rem: When the Association request/response follows an EBR/EB exchange the Association request/response MAY use the extended addressing mode. This is different from the short addressing mode Zigbee uses when the Association request/response follows a non-enhanced BR/B exchange.	Observe the packet received by DUT2. Verify the following: - An Association Request is received Pass: That the Association Request is received properly for initial joining Fail: That the Association Request is not received properly for initial joining In response to DUT2 properly receiving an Association Request form DUT1, DUT2 sends a Association Response to DUT1. Observe the packet received by DUT1. Verify the following: - A Association Response is received Pass: That the Association Response is received properly for initial joining Fail: That the Association Response is not received properly for initial joining

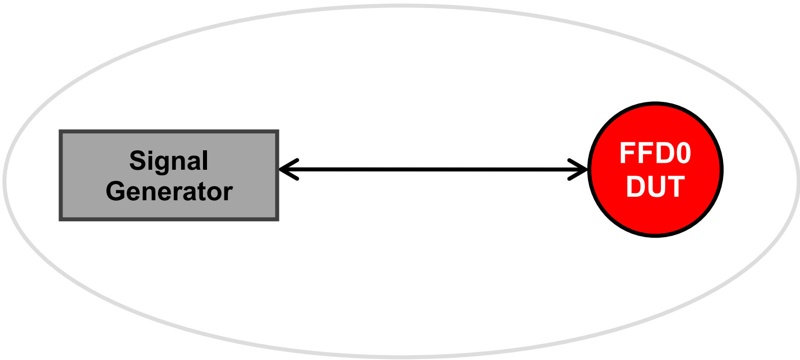
932 ***Notes:***

933 *None.*

TP/154/MAC/CHANNEL-ACCESS-01

Check D.U.T. does not transmit when channel is busy.

Setup Conditions:

Diagram


The Diagram is the image that represents the configuration deployed for the Test Procedure.

Initial Conditions:

Device name	Description
All the devices	*Security: no for all the devices* *Channel: 0x14 for all the devices* *PAN id: 0x1AAA for all the devices*
FFD0/DUT	*MAC: 0xACDE480000000001*

Test Procedure:

Step No.	Step description	Expected outcome
1	Set to DUT: MLME-RESET.request(SetDefaultPIB=TRUE).	Pass: DUT returns primitive MLME-RESET.confirm(Status= SUCCESS). Fail: FFD does not respond.
2	MLME-SET.request for the DUT, such that: MLME-SET.request (PIBAttribute=0x53=macShortAddress, PIBAttributeValue=0x1122) Verify the PIB is correctly set MLME-GET.request (PIBAttribute=0x53=macShortAddress, PIBAttributeIndex = 0x00- ignore, Only used for MAC 2006 security PIBs)	Wait for MLME-Set_confirm MLME-SET.confirm (Status=SUCCESS, PIBAttribute=0x53=macShortAddress,) Wait for MLME-GET.confirm and confirm the parameters MLME-GET.confirm (Status = SUCCESS PIBAttribute=0x53=macShortAddress, PIBAttributeIndex = ignore. Only for security in MAC2006 PIBAttributeValue = 0x1122)

3	Finally, issue MLME-START.request command with the following parameters: MLME-START.request (PANId=0x1AAA, LogicalChannel=20=0x14, BeaconOrder=15, SuperframeOrder=15, PANCoordinator=TRUE, BatteryLifeExtension=FALSE, CoordRealignment=FALSE, CoordRealignmentSecurityLevel=0, [CoordRealignmentKeyIdMode =x, CoordRealignmentKeySource = x, CoordRealignmentKeyIndex = x,] BeaconSecurityLevel=0, [BeaconKeyIdMode =x, BeaconKeySource = x, BeaconKeyIndex = x,]).	Pass: DUT returns primitive MLME-START.confirm (Status=SUCCESS)
4	Set Tester(Signal Generator) to generate constant carrier above RSSI threshold of DUT	
5	Set to DUT MCPS-DATA.request (SrcAddrMode = 0x02 = ShortAddress, DstAddrMode=0x02 = ShortAddress, DstPANId = 0x1aaa, DstAddr = 0x3344, msduLength=5, msdu = 0x00 0x01 0x02 0x03 0x04, msduHandle = 0x00, TxOptions = 0x00, SecurityLevel = 0x00, KeyIdMode = xx, KeySource = xx, KeyIndex = xx)	Pass: DUT returns: MCPS-DATA.confirm(msduHandle = 0x00, Status = CHANNEL_ACCESS_FAILURE) Fail: DUT returns: MCPS-DATA.confirm(msduHandle = 0x00, Status != CHANNEL_ACCESS_FAILURE)

940 ***Notes:***

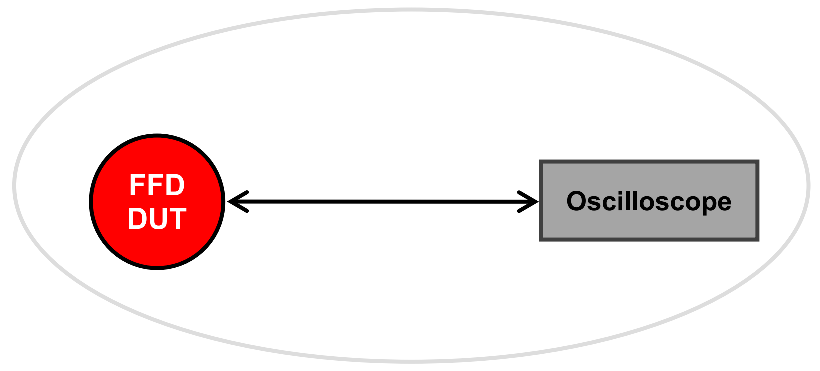
941 *None.*

TP/154/MAC/CHANNEL-ACCESS-02

Check D.U.T. correctly uses the channel access algorithm and does actually have a random backoff for sending packets.

Setup Conditions:

Diagram



The Diagram is the image that represents the configuration deployed for the Test Procedure.

Initial Conditions:

Device name	Description
All the devices	*Security: no for all the devices* *Channel: 0x14 for all the devices*
FFD DUT	PAN Coordinator, Bo = 15, SO = 15 MAC: 0xACDE480000000001 Short Address: 0x0000 Beacon Payload: none Coordinator shall set up a non-beacon enabled PAN.
Oscilloscope	*Waiting to trigger on DUT signal*

Test Procedure:

Step No.	Step description	Expected outcome
1	From DUT Coordinator send: MLME-Data.request (SrcAddrMode = 0x02 (short addr) DstAddrMode = 0x02 (short addr) DstPANId = 0x1aaa DstAddr = 0x1234 (non-existant addr) msduLength = 0x0a msdu = 0x0102030405060708090A TxOptions = 0x01 (Acknowledged Tx) SecurityLevel = 0x00 (no security))	Pass: DUT sends out first data packet, does not receive any acknowledgements, and sends the 3 other retries
2	Oscilloscope captures enough data to see all 4 retries. Operator measures the time between each 2 retries.	
3	Repeat steps 1 and 2 thirty (30) times.	Pass:

		All 90 values must be random enough as not to show a consistent and/or preset backoff. *Note: The time between any 2 retries will be largely affected by the software handling time (receiving the indication, deciding on retry, sending the request for retry, etc.). This being true, the software/firmware delays will be substantially larger than the backoff itself. This effectively has the same effect if the sum of both is still random enough to prevent collisions.*

Notes:

None.

TP/154/MAC/CHANNEL-ACCESS-03[1] (optional)

Check that the DUT executes Listen Before Talk (LBT), as defined in Annex D of the Zigbee Specification [R2] and that the rules as described by ETSI in EN 300-220 [R3] and EN 304-204 [R4] are properly followed to gain access to the channel. Full testing of the LBT functionality requires the upper layers, which are not covered in this layer of testing. The MAC layer test needs to check that *macLBTTDurationMs* is properly capturing the total transmit time over the last hour. The calculations for *macLBTTDurationMs* are a small part of the overall LBT functionality.

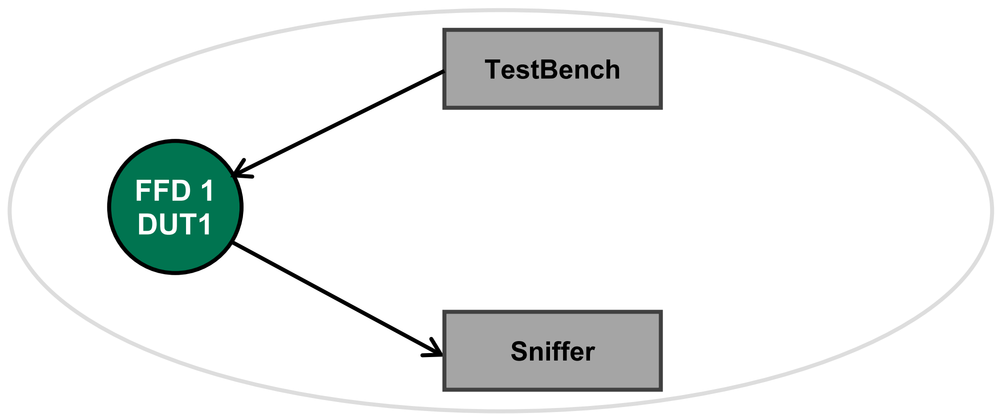
Assumptions:

- During 1st passes of LBT testing parameters will be changed to accelerate and simplify initial test.
- A symbol is defined to be 10µS.
- The test is of the MAC & PHY layer and not ZCL / application layers
- The rampUP and rampDOWN (ie the time before and after transmitting a packet where the transmitter is actively transmitting, eg carrier signal) is part of the transmitters radiated power and MUST be included in the duty cycle management

Setup Conditions:

In order to determine if the LBT is operating correctly a test harness uses one device to send out requests and the DUT responds accordingly.

Diagram



The Diagram is the image that represents the configuration deployed for the Test Procedure.

Criteria to Evaluate:

For LBT the following criteria need to be evaluated:

- A data transmission will NEVER occur less than 100mS after the last bit transmitted by the device
- The CCA will be at least 5mS and if not clear at the start it will require an additional random period to be added to the CCA
- The DUT will only transmit if the channel is clear for the CCA period (which will include additional backoff if channel is busy during CCA period).

Constants:

The following constants are used within the test harness and the DUT for LBT testing:

Parameter	Value	In Symbols	Description
SymbolDuration	0.01mS	1	Time duration of a symbol, ie 1mS = 100symbols
SymbosForl1mS	1mS	100	No of symbols for 1mS
ResponseStart	2mS	200	This is the point at which the DUT is expected to reply to the first packet. After performing LBT.
MaxInterference	30mS	3000	This is the max length of the interference packet
FrameOverhead	7 bytes	56	The 802.15.4 start and crc

For LBT the following criteria need to be evaluated:

Stack Requirements:

<u>Test Harness</u>

For the test harness packet transmissions do not require:

- LBT adherence, i.e. just output on demand
- Duty cycle limiting - the test is about testing LBT not duty cycle

<u>DUT</u>

For the DUT packet transmissions do not require:

- Duty cycle limiting - the test is about testing LBT not duty cycle

Test Harness Behaviour:

The test harness will:

- Output a sequence of packet pairs
  - The first packet will contain a time stamp (in uS) and information about the second packet of the pair, ie the start time relative to the end of transmission and the duration of the packet.
- The test harness will send the packet pairs up to 150mS between.
- The second packet will be of varying size and offset from the last pair it sent.

DUT Expected Behaviour:

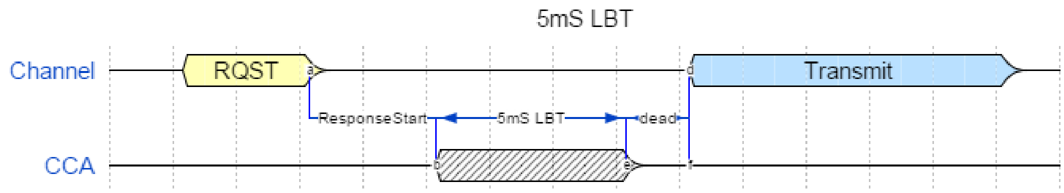
The DUT will attempt to:

- Send a reply to the first packet of the packet pair exactly ResponseStart after it receives the packet.

Perfect Test Sequence:

Test harness sends out request:

- DUT enters LBT exactly ResponseStart after receiving RQST
- DUT determines channel clear and monitors channel for 5mS
- On determining channel is free DUT changes from Rx to Tx mode
- DUT transmit packet

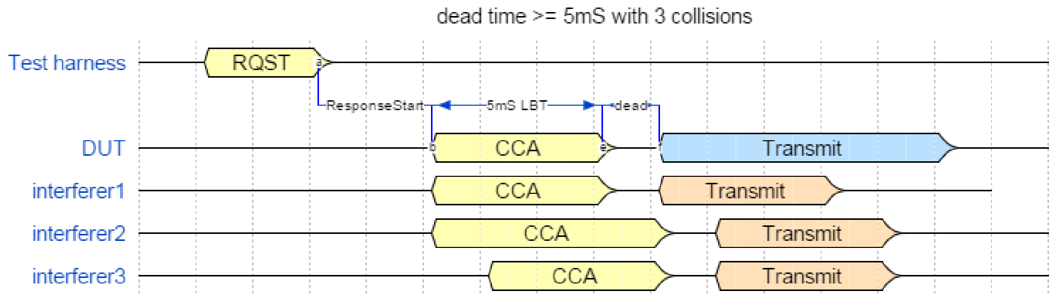


Potential Issues:

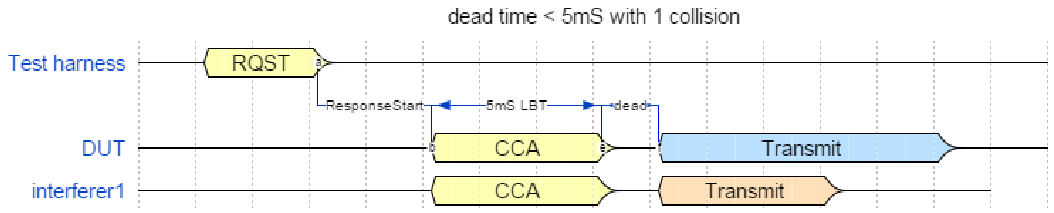
Dead time should be less than 0.5mS otherwise there is the potential for another device to assume after 5.5mS that the channel is clear and switches to transmit and transmits packet.

Potential Collisions Due to Rx/Tx Turnaround Time being ≥ 5mS:

<u>Dead Time ≥ 5mS with 3 Collisions</u>



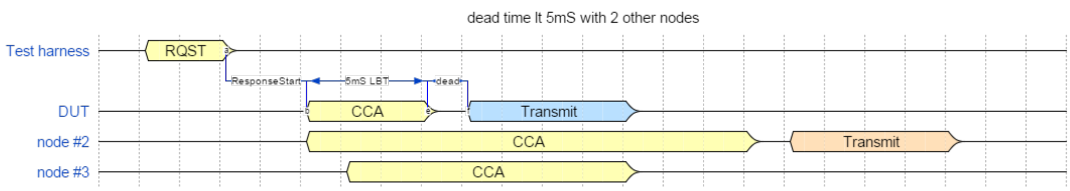
<u>Dead Time < 5mS with 1 Collision</u>



Correct operation with other node(s) other than DUT:

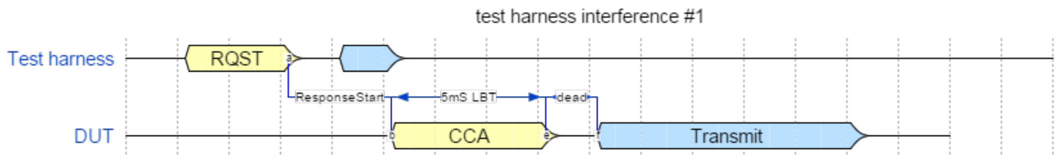
<u>Dead Time < 5mS with 2 Other Nodes</u>

The following diagram illustrates the correct behavior where two other nodes attempt to transmit, node #2 starts to monitor the channel at the same time as the DUT but it starts with a CCA ≥ 5.5mS and thus the DUT who attempts a CCA with 5mS sees the channel as free and transmits. Another node #3 has a CCA time of 5mS, the same as the DUT but starts accessing the channel after the DUT and sees the DUT start to transmit just before it would of attempted to transmit.



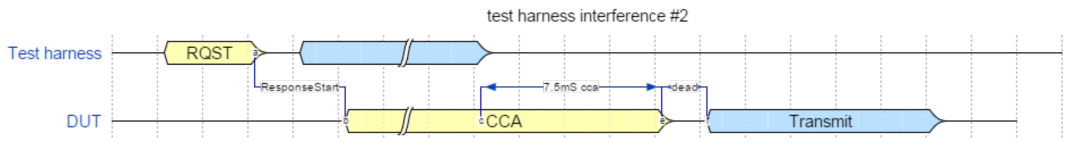
<u>Test Harness Interference #1</u>

The following diagram illustrates the correct behavior where the channel is busy before the DUT starts a 5mS CCA assessment of the channel. As the DUT does not detect any traffic on the channel during the 5mS CCA period it transmits.



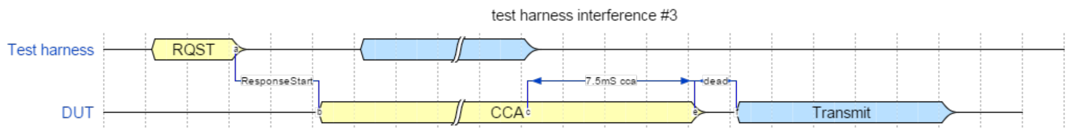
<u>Test Harness Interference #2</u>

The following diagram illustrates the correct behavior where the DUT sees the channel is busy at the beginning of the CCA assessment period and waits till the channel becomes free before waiting a further 7.5mS CCA where the channel is free and then transmitting.



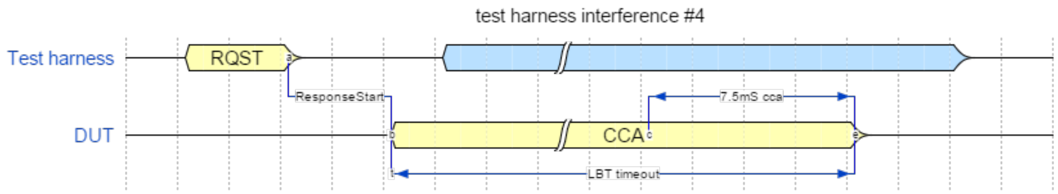
<u>Test Harness Interference #3</u>

The following diagram illustrates the correct behavior where the DUT sees the channel becomes busy during its initial 5mS CCA and waits till the channel becomes free before waiting a further 7.5mS CCA where the channel is free and then transmitting.



<u>Test Harness Interference #4</u>

The following diagram illustrates the correct behavior where the DUT times out as the channel does not become clear.



Test Procedure:

Packet format containing lead-in and trailing format as defined in 802.15.4 packet format.

<u>The 1st Test Harness Packet</u>

MAC payload SHOULD contain:

- 1st byte containing ascii character '?'

- 2 byte packet number, starting at 0
- 4 byte time stamp in symbols of start of transmission
- 2 byte Offset from end of 1st packet to start of second packet in symbols
- 2 byte Length of second packets in symbols
- 2-byte rampUP length (in symbols)
- 2-byte rampDOWN length (in symbols)
- Packet padded with ascii '5' to the required length (~ 112 bytes)

Offset	Length (octets)	Field	Value	Notes
0	1		'?'	Lead-in character
1	2	NN		Packet number
3	4	TTTT		Timestamp
7	2	OO		Offset
9	2	LL		Length
11	2	UU		rampUP length
13	2	DD		rampDOWN length

Test Harness 1st frame:

	Test Harness 1st frame																
	SHR	PHR	MHR					'?'	NN	TTTT	OO	LL	UU	DD	'5'*	MFR	
			FC	SNr	PANID	DST	SRC										total
bytes	10	2	2	1	2	2	2	1	2	4	2	2	2	2	101	2	139
symbols	80	16	16	8	16	16	16	8	16	32	16	16	16	16	808	16	1112

Test Harness 2nd frame (Interferer) – 1st fragment:

	Test Harness 2nd frame (Interferer) – 1st fragment													
	SHR	PHR	'#'	NN	OO	LL	F	GG	TTTT	UU	DD	'5'*	MFR	total
bytes	10	2	1	2	2	2	1	2	4	2	2	107	2	139
symbols	80	16	8	16	16	16	8	16	32	16	16	856	16	1112

Test Harness 2nd frame (Interferer) – Last fragment:

	Test Harness 2nd frame (Interferer) – Last fragment				
	SHR	PHR	'5'*	MFR	total
bytes	10	2	125	2	139
symbols	80	16	1000	16	1112

DUT response:

	DUT response																	
	SHR	PHR	MHR					'@'	NN	TTTT	OO	LL	UU	DD	TTTT	TTTT	'5'*	MFR
			FC	SNr	PANID	DST	SRC											
bytes	10	2	2	1	2	2	2	1	2	4	2	2	2	2	4	4	93	2
symbols	80	16	16	8	16	16	16	8	16	32	16	16	16	16	32	32	744	16

Max frame size	139	bytes	1112	symbols	symbols per mS	100	Calculated values
2nd frame overhead	17	bytes	136	symbols			

2nd frame(s)																	
test scen #	offset		Desired (approx) interference		fragment gap		Leading fragmaent inc trailing gap		Fragments			Last fragment		Actual interferer			
	mS	sym	mS	sym	mS (0-4.9)	sym	mS	sym	leading	size of last (% of max)	no of fragments	mS	sym	mS	sym	error
1	0	0	0	0	0	0	0.00	0	0	0.000	0	0.00	0	0.00	0	0.00%
2	1	100	10	1000	0	0	10.00	1000	1	0.899	1	0.00	0	10.00	1000	0.00%
3	3	300	10	1000	0	0	10.00	1000	1	0.899	1	0.00	0	10.00	1000	0.00%
4	6.5	650	4.5	450	0	0	4.50	450	1	0.405	1	0.00	0	4.56	456	1.33%
5	3	300	20	2000	1	100	12.12	1212	1	1.733	2	7.92	792	20.04	2004	0.20%
6	1	100	20	2000	4.5	450	15.62	1562	1	1.569	2	4.40	440	20.02	2002	0.10%
7	3	300	20	2000	4.5	450	15.62	1562	1	1.569	2	4.40	440	20.02	2002	0.10%
8	3	300	30	3000	1	100	12.12	1212	2	2.558	3	5.76	576	30.00	3000	0.00%
9	3	300	30	3000	4.5	450	15.62	1562	2	2.209	3	1.60	160	32.84	3284	9.47%
10	3	300	32	3200	1	100	12.12	1212	2	2.723	3	7.76	776	32.00	3200	0.00%
11	3	300	32	3200	4.5	450	15.62	1562	2	2.337	3	1.60	160	32.84	3284	2.63%
12	3	300	35	3500	1	100	12.12	1212	2	2.970	3	10.80	1080	35.04	3504	0.11%
13	3	300	35	3500	2	200	13.12	1312	2	2.820	3	8.80	880	35.04	3504	0.11%
14	1	100	50	5000	4.9	490	16.02	1602	3	3.427	4	2.00	200	50.06	5006	0.12%
15	4	400	55	5500	4.9	490	16.02	1602	3	3.739	4	6.96	696	55.02	5502	0.04%

The 2nd Test Harness Packet

of the pair in the MAC payload SHOULD contain:

- 1st byte containing ascii character '#'
- 2 byte packet number - which should be the same as the first packet
- 2 byte Offset from end of 1st packet to start of second packet in symbols
- 2 byte Length of second packets in symbols
- 1 byte - fragment count
- 2 byte - gap in symbols of between 0.5mS and 4.5ms (50 .. 450 symbols) if packet is fragmented
- 4 byte time stamp in symbols on entry into LBT (filled in by DUT)
- 2-byte rampUP length (in symbols)
- 2-byte rampDOWN length (in symbols)

- Packet padded with ascii '5' to the required length (~ 109 bytes)

The 2nd packet is at least 21 bytes (7 for 802.15.4 frame and 14 for required information fields), ie 139 symbols (1.39mS).

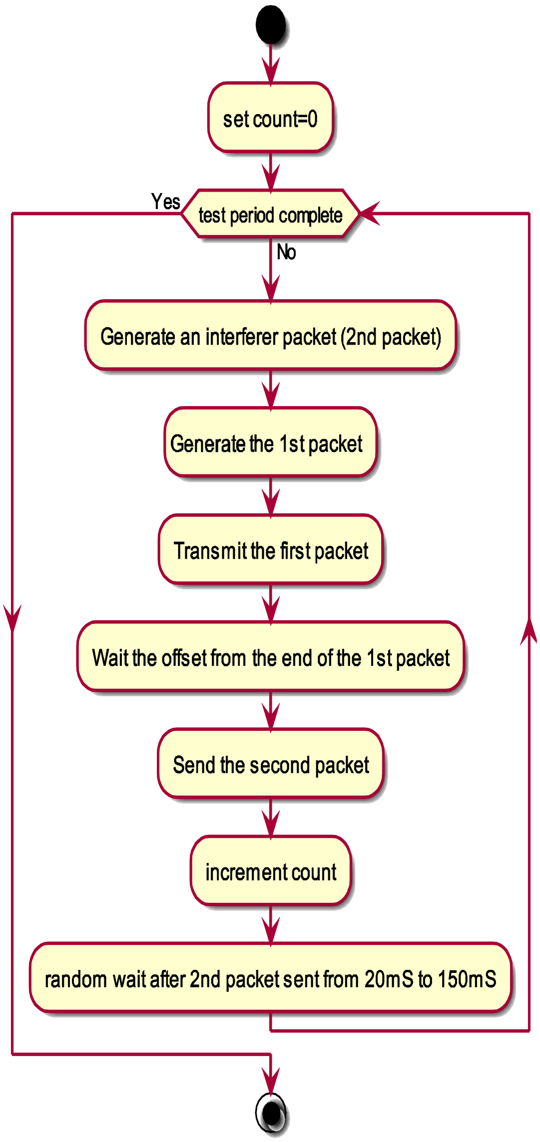
Fragmenting the 2nd Packet

If the 2nd packet is required to be more than one packet it is necessary to fragment it. This is required to enable long periods of interference testing and check transmission timeout and that LBT code transitions through periods of intermittent interference with quiet periods which are less than the LBTMinFree.

NOTE: This is a NICE to have feature but not essential for initial testing.

Test Harness

The test harness sends out a sequence of packet pairs, the first one is to stimulate the DUT to attempt to return a copy of the 1st pack adhering to the LBT rules.



Interferer packets:

The following table illustrates how the 2nd packet could be created:

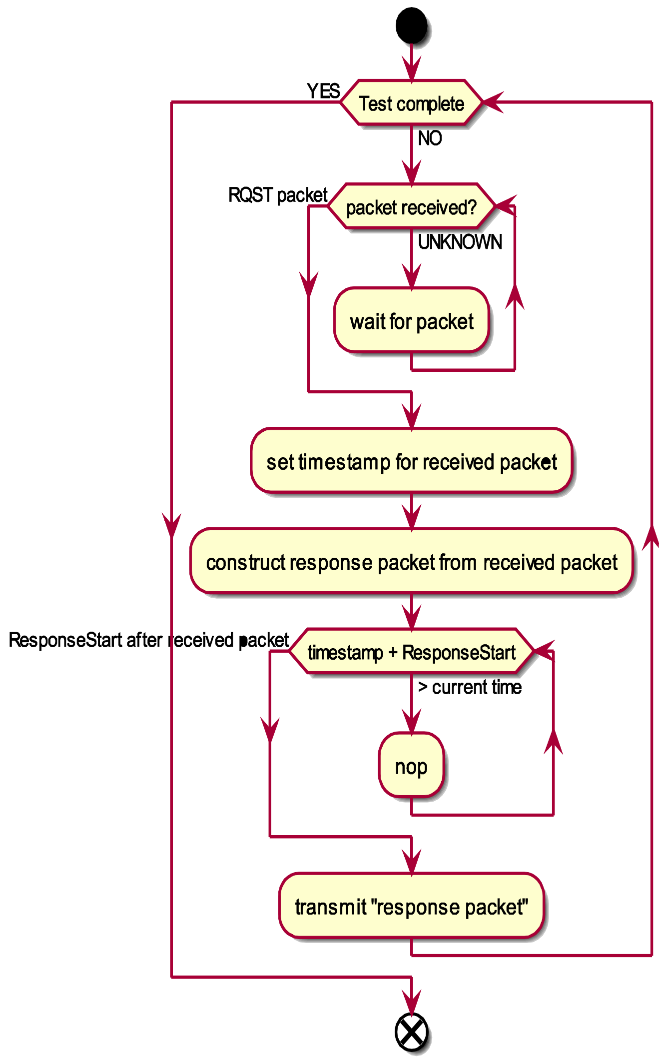
DUT Response

The DUT should respond to the 1st packet by echoing it back with the 1st byte changed from '?' to '@' starting at ResponseStart after receiving the 1st packet and using LBT to determine when to send the packet back. And it should include the time stamp of when the 1st packet is received and a timestamp of the start point the packet is expected to be transmitted. The packet format is based on the 1st packet sent from the test harness.

MAC payload SHOULD contain:

- 1st byte containing ascii character '@' [DUT]
- 2 byte packet number [Copied from received packet]
- 4 byte time stamp in symbols of start of transmission [Copied from received packet]
- 2 byte Offset from end of 1st packet to start of second packet in symbols [Copied from received packet]
- 2 byte Length of second packets in symbols [Copied from received packet]
- 2-byte rampUP length (in symbols) [DUT]
- 2-byte rampDOWN length (in symbols) [DUT]
- 4 byte time stamp in symbols of 1st packet received [DUT]
- 4 byte time stamp in symbols of start of transmission [DUT]
- Packet padded with ascii '5' to the required length (~ 101 bytes)

Offset	Length (octets)	Value	Notes
0	1	'@'	New Lead-in character
1	2	Packet number	Copied from incoming packet
3	4	Timestamp	Copied from incoming packet
7	2	Offset	Copied from incoming packet
9	2	Length	Copied from incoming packet
11	2	rampUP length	DUT's ramp up
13	2	rampDOWN length	DUT's ramp down
15	4	Received timestamp	DUT's timestamp of when 1st packet received
19	4	Transmit timestamp	DUT's timestamp when packet is to be transmitted
23	-	FILL	Fill character to pad packet to max length



Expected DUT behavior:

The DUT should NEVER:

- Transmit before (5mS + ResponseStart) AFTER the 1st packet is sent
- If 2nd packet starts transmission before ResponseStart after the 1st packet is sent the DUT should not start transmitting until at least LBTMinFree after the second packet stops transmission.
- If the DUT last transmitted < 100mS ago it MUST wait until 100mS has elapsed BEFORE attempting to transmit (ie entering LBT channel assessment).

Verification of LBT Tests

Sniffer logs will be generated and captured for these tests. The sniffer logs will then be processed and compared to the tables in Zigbee document 16-04012-002 LBT Test Parameters [R6] to determine if the device is behaving properly.

TP/154/MAC/CHANNEL-ACCESS-04[1] (optional)

Checks that the MAC properly supports Duty Cycle Monitoring, as defined in Annex D of the Zigbee Specification [R2]. Full testing of the Duty Cycle functionality requires the upper layers, which are not covered in this layer of testing. Specifically it checks:

- that the DUT executes the Duty Cycle NORMAL Operation Mode when the duty cycle is beneath the Normal Operation Duty Cycle. The MAC layer test needs to check that during NORMAL mode messages are transmitted at the MAC layer and that the MAC queue is enabled and existing messages already in the MAC queue are transmitted normally.
- that the DUT executes the Duty Cycle LIMITED Operation Mode when the Normal Operation Duty Cycle is exceeded. The MAC layer test needs to check that when LIMITED mode is entered the MAC queue is enabled, and existing messages already in the MAC queue are transmitted normally. However, the duty cycle may be reduced by the upper layers in order to prevent or delay from entering the CRITICAL mode.
- that the DUT executes the Duty Cycle CRITICAL Operation Mode when the Limited Operation Duty Cycle is exceeded. The MAC layer test needs to check that when CRITICAL mode is entered the MAC queue is enabled, and existing messages already in the MAC queue are transmitted normally. However, the duty cycle may be reduced by the upper layers in order to prevent or delay from entering the SUSPENDED mode.
- that the DUT executes the Duty Cycle SUSPENDED Operation Mode when the Critical Operation Duty Cycle is exceeded. The MAC layer the test needs to check that when SUSPENDED mode is entered the transmitter is disabled, all messages in the MAC queue are returned to the higher layer, and no messages are transmitted at the MAC layer until the device re-enters either the NORMAL, LIMITED, or CRITCAL mode.

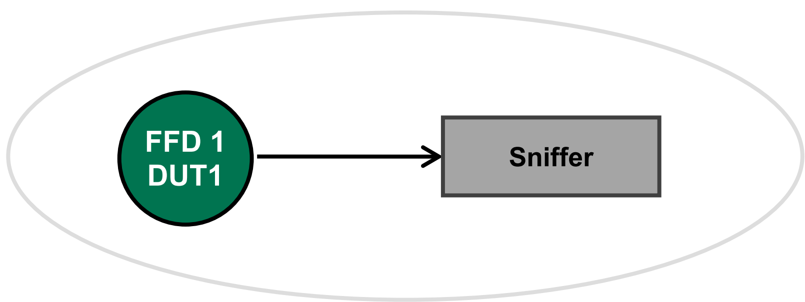
Assumptions:

- During 1st passes of Duty Cycle testing parameters will be changed to accelerate and simplify initial test.
- A symbol is defined to be 10µS.
- The test is of the MAC & PHY layer and not ZCL / application layers
- During the Duty Cycle test use of LBT during transmission is optional
- The rampUP and rampDOWN (ie the time before and after transmitting a packet where the transmitter is actively transmitting, eg carrier signal) is part of the transmitters radiated power and MUST be included in the duty cycle management

Setup Conditions:

For duty cycle testing a test harness is used to call the MAC/PHY stack as required in the relevant test. A sniffer is then used to collect the packets and time stamp them, allowing the packets to be automatically checked for conformance to the specification.

Diagram



The Diagram is the image that represents the configuration deployed for the Test Procedure.

Test Procedure 1: (Max Single Transmit is No MoreThan 1 Second):

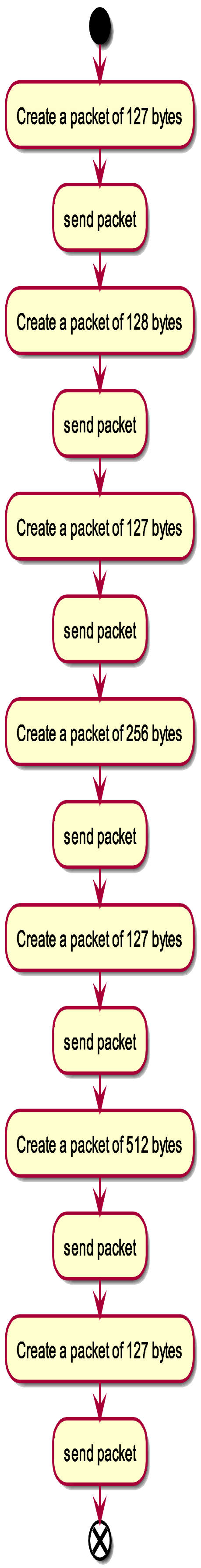
Initial Conditions / Assumptions

- Reset duty cycle counter, i.e. *mibDUTYCYCLECurrentMeas*
- MAX frame size is 127
- Set pauseinterval to 2mS, i.e. 200 symbols

Test Steps

Step No.	Step description	Expected outcome
1	Request transmission of 127 bytes	Packet is transmitted Transmitter active for < 1s Transmitter active for ~ 139 bytes: (112 symbols) + rampUP and rampDOWN time
2	Request transmission of 128 (or more) bytes	No transmission detected -OR- Multiple transmissions occur due to fragmentation (See note #1)
3	Try sending 150, 175 and largest size byte count defined by data type for frame length	No transmission detected -OR- Multiple transmissions occur due to fragmentation (See note #1)

Note #1 - In the event of fragmentation there should be at least 100mS between packets. When/if Dialog mode is defined, if it is used the total time to send the fragments MUST be less than 4s.


Create a packet of 127 bytes
send packet
Create a packet of 128 bytes
send packet
Create a packet of 127 bytes
send packet
Create a packet of 256 bytes
send packet
Create a packet of 127 bytes
send packet
Create a packet of 512 bytes
send packet
Create a packet of 127 bytes
send packet

1172

Verification of Max Transmission Test

Sniffer logs will be generated and captured for these tests. The sniffer logs will then be processed and compared to the tables in Zigbee document 16-04012-002 LBT Test Parameters [R6] to determine if the device is behaving properly.

Test Procedure 2: (Trigger Duty Cycle Limit(s) & Adherence to Regulatory Transmission Limit):

Initial Conditions / Assumptions

- Reset duty cycle counter
- Payload is 127 bytes

To accelerate testing the thresholds can be adjusted as described in the following table:

	No of *aDUTYCYCLEBuckets* buckets	13	User entry				
		(bytes)	(seconds)	(symbols)	(%)	(frames)	
NORMAL OPERATION	*aDUTYCYCLEBuckets* bucket duration		300	30,000,000			
	Total Frame size (bytes, inc 12 MAC)	139	0.0111	1,112			
	aDUTYCYCLEMeasurementPeriod		3,600	360,000,000	100.00%	323,741	
	aDUTYCYCLERegulated		100	10,000,000	2.78%	8,993	
						difference	1,799
	mibDUTYCYCLECriticalThresh		80	8,000,000	2.22%	7,194	
						difference	1,799
	mibDUTYCYCLELimitedThresh		60	6,000,000	1.67%	5,396	
	aLBTTxMinOff		0.1	10,000			
TEST MODE	*aDUTYCYCLEBuckets* bucket duration		20	2,000,000			
	Total Frame size (bytes, inc 12 MAC)	139	0.01112	1,112			
	aDUTYCYCLEMeasurementPeriod		240	24,000,000	100%	21,583	
	aDUTYCYCLERegulated		100	10,000,000	41.67%	8,993	
	difference						1,799
	mibDUTYCYCLECriticalThresh		80	8,000,000	33.33%	7,194	
	difference						1,799
	mibDUTYCYCLELimitedThresh		60	6,000,000	25.00%	5,396	
	aLBTTxMinOff		0.1	10,000			

Payload for Duty Cycle Testing

The payload of the transmitted frame should be padded to ensure it is 127 bytes and includes the state of the Duty Cycle Mode.

MLME-DUTY-CYCLE-MODE.indication	Payload to contain (in ascii)
NORMAL	###NORMAL###
LIMITED	###LIMITED###
CRITICAL	###CRITICAL###
SUSPENDED	###SUSPENDED###

Packet structure containing lead-in and trailing format as defined in 802.15.4 packet format.

The packet SHOULD contain:

- 1st byte containing ascii character '@'
- [NN] 2-byte packet number, starting at 0
- [TTTT] 4-byte time stamp in symbols of start of transmission
- [UU] 2-byte rampUP length (in symbols)
- [DD] 2-byte rampDOWN length (in symbols)
- [STEP] 1-byte value indicating state # of code currently being executed
- [MSG] 104 byte Message, either STARTOFTEST, NORMAL, LIMITED, CRITICAL, SUSPENDED or ENDOFTEST with 3 leading hashes and trailing padded with hashes to fill packet

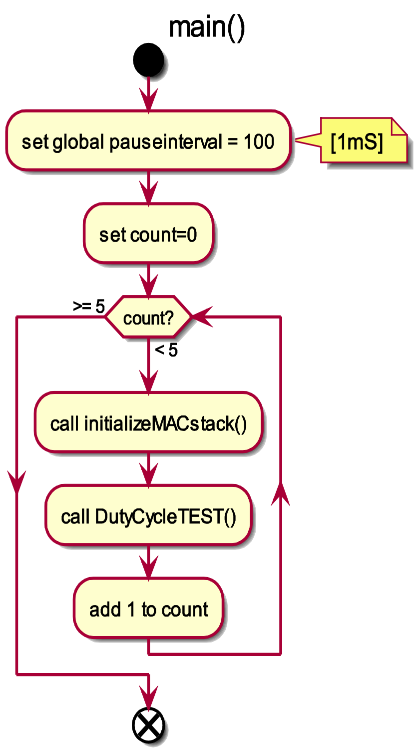
	duty cycle output frame															
	SHR	PHR	MHR					'@'	NN	TTTT	UU	DD	STEP	MSG	MFR	
			FC	SNr	PANID	DST	SRC									total
bytes	10	2	2	1	2	2	2	1	2	4	2	2	1	104	2	139
symbols	80	16	16	8	16	16	16	8	16	32	16	16	8	832	16	1112

Test Harness for Duty Cycle Testing

The following function runs above the MAC / PHY stack and cause the stack to generate packets that are sent over the air which should adhere to the duty cycle requirements in accordance with LBT use.

main():

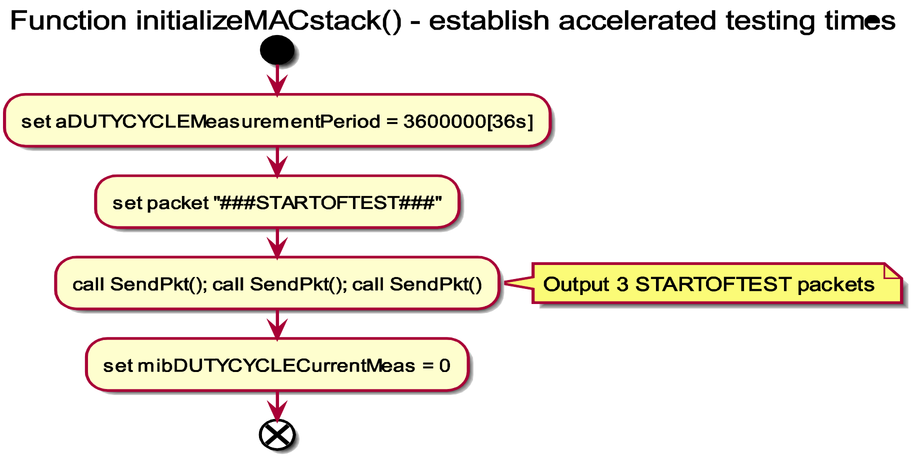
This is the main test code; it calls the DutyCycleTEST 5 times.



initializeMACstack():

This function changes a couple of the parameters in the MAC stack to accelerate testing, no other changes to the stack are made or allowed.

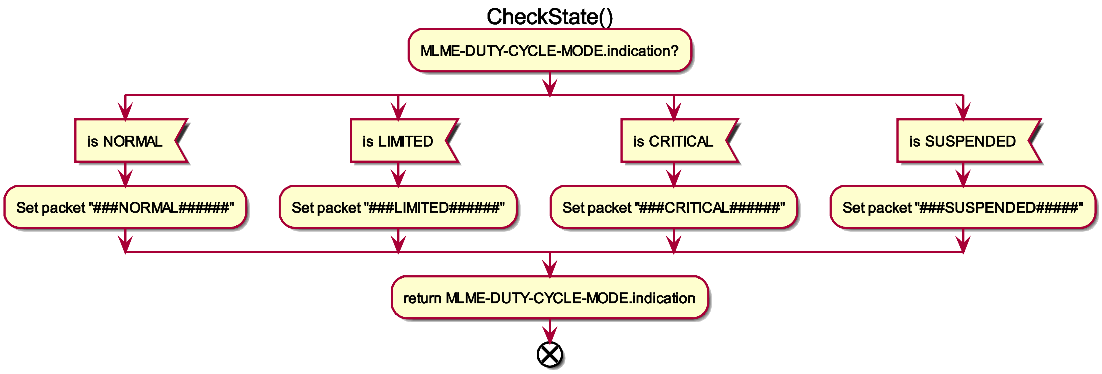
This function is called by main().



CheckState():

The purpose of this function is to check the state that the MAC stack is operating in and select the relevant output packet depending upon the state.

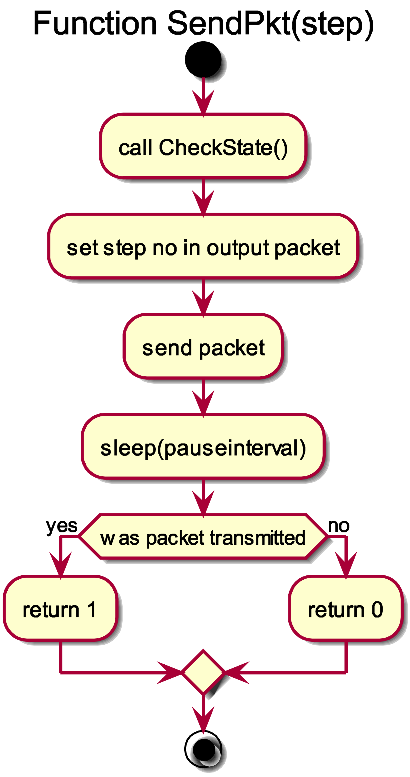
This function is called by DutyCycleTest().



SendPkt():

This function transmits the selected frame. It is not required to utilize stack's LBT but MUST adhere to the Duty Cycle checks.

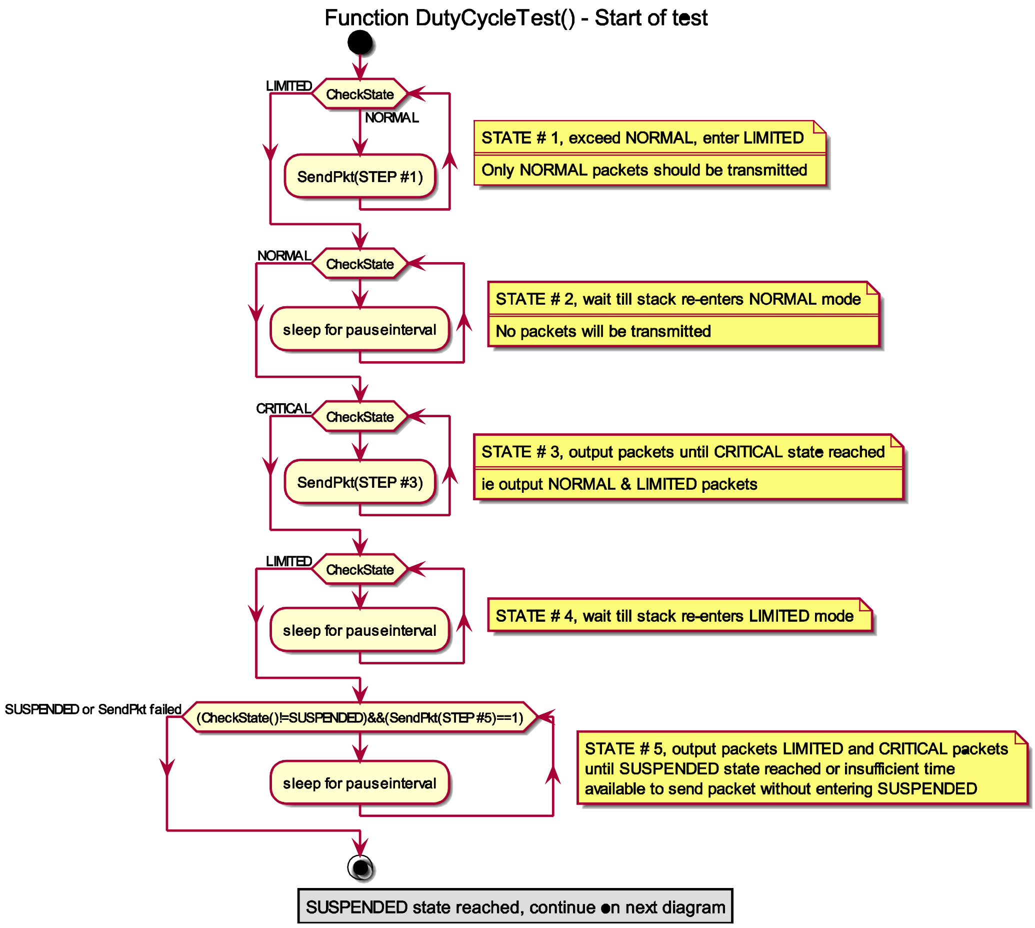
The function is called by DutyCycleTest().


Function SendPkt(step)
call CheckState()
set step no in output packet
send packet
sleep(pauseinterval)
yes
w as packet transmitted
no
return 1
return 0

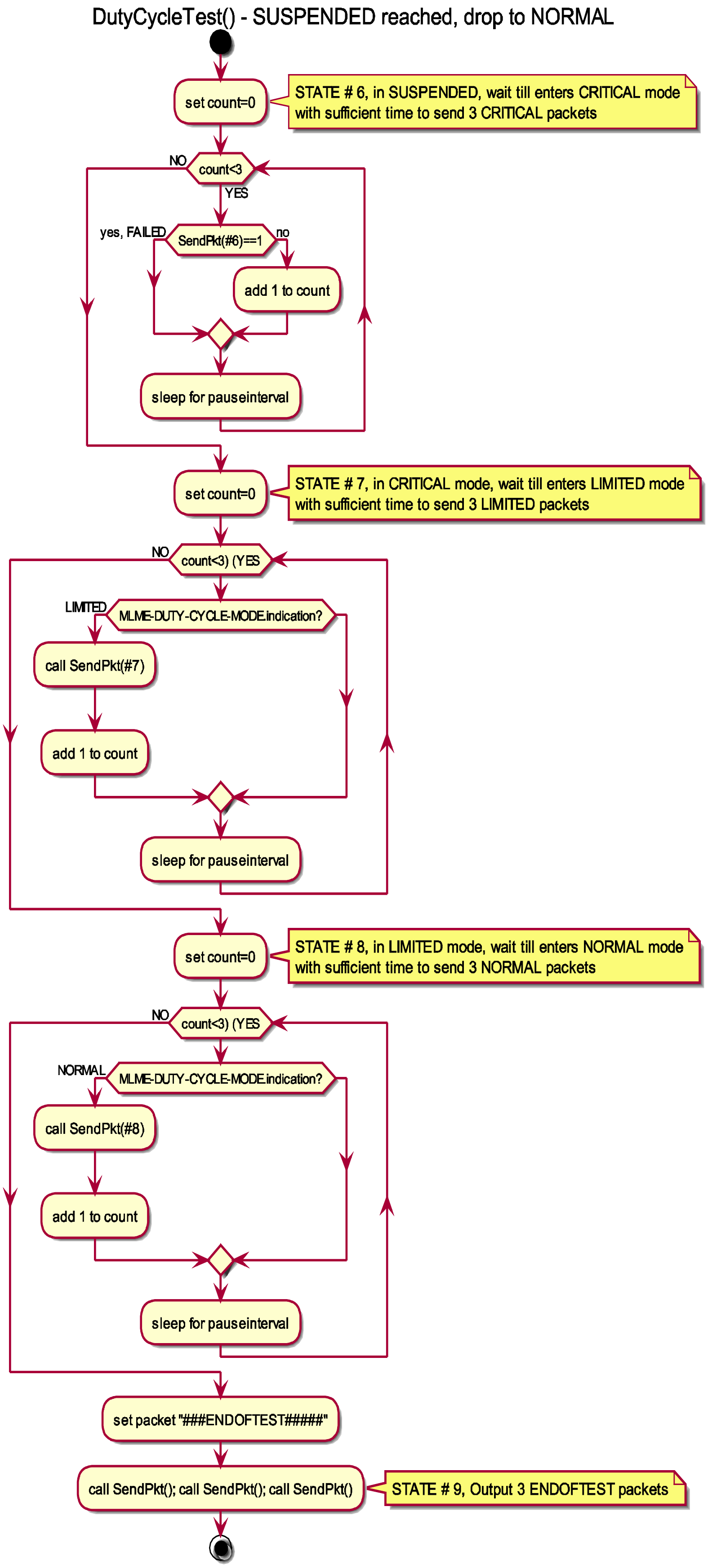
1218

1219

1220 ***DutyCycleTest():***



1221


DutyCycleTest() - SUSPENDED reached, drop to NORMAL
set count=0
STATE # 6, in SUSPENDED, wait till enters CRITICAL mode with sufficient time to send 3 CRITICAL packets
count<3
NO
YES
SendPkt(#6)==1
yes, FAILED
no
add 1 to count
sleep for pauseinterval
set count=0
STATE # 7, in CRITICAL mode, wait till enters LIMITED mode with sufficient time to send 3 LIMITED packets
count<3) (YES
NO
MLME-DUTY-CYCLE-MODE.indication?
LIMITED
call SendPkt(#7)
add 1 to count
sleep for pauseinterval
set count=0
STATE # 8, in LIMITED mode, wait till enters NORMAL mode with sufficient time to send 3 NORMAL packets
count<3) (YES
NO
MLME-DUTY-CYCLE-MODE.indication?
NORMAL
call SendPkt(#8)
add 1 to count
sleep for pauseinterval
set packet "###ENDOFTEST#####"
call SendPkt(); call SendPkt(); call SendPkt()
STATE # 9, Output 3 ENDOFTEST packets

1222

Monitoring Test

Using a packet sniffer monitor the packets and verify that the duty cycle in the modes meets the desired modes and that no packets are transmitted when the duty cycle limit for the measurement period has been reached.

Expected Packets:

Quiet period (based on resolution of rolling accumulated transmission period) is approximately:

$$\frac{aDUTYCYCLEMeasurementPeriod}{aDUTYCYCLEBuckets}$$

FrameSize = 139 bytes

Symbols per Frame = 139 * 8 = 1112

State	Expected Packets	No Expected Frames or Time Period	Notes
-	###STARTOFTEST##	At start of duty cycle testing 3 ###STARTOFTEST## packets will be transmitted	
1	###NORMAL###	$CEIL\left(\frac{aDUTYCYCLEWarningThresh}{FrameSize * 8}\right) packets$	These packets will be detected for the period from 0 to the LIMITED Threshold
2	No traffic	$< (aDUTYCYCLEMeasurementPeriod - time\ for\ step\ \#1)$	This time will be related to the no of measurement periods
3	###NORMAL### & ###LIMITED###	mibDUTYCYCLECriticalThresh - mibDUTYCYCLELimitedThresh	These packets will be detected for the period from the time period to the CRITICAL Threshold
4	No traffic	Quiet period	
5	###LIMITED### & ###CRITICAL###		Moving into SUSPENDED mode
6	###CRITICAL###	Quiet period and then 3 CRITICAL packets	May be more than 1mS between packets
7	###LIMITED###	Quiet period and then 3 LIMITED packets	
8	###NORMAL###	Quiet period and then 3 NORMAL packets	
9	###ENDOFTEST#####	End of test reached	

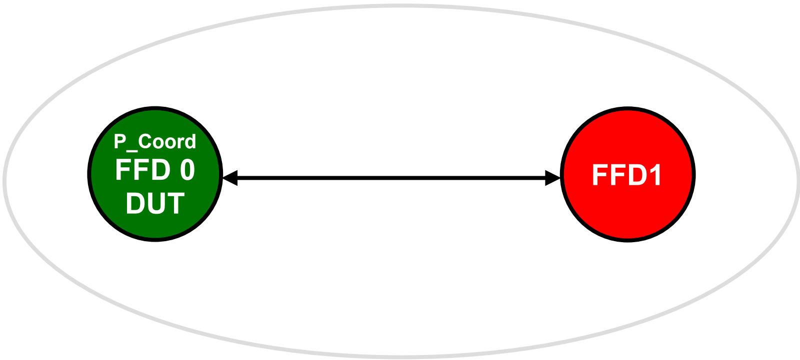
Verification of Duty Cycle Test

Sniffer logs will be generated and captured for these tests. The sniffer logs will then be processed and compared to the tables in Zigbee document 16-04012-002 LBT Test Parameters [R6] to determine if the device is behaving properly.

TP/154/MAC/FRAME-VALIDATION-01

Check D.U.T. correctly discards frame with invalid FCS.

***Setup Conditions*:**

Diagram


The Diagram is the image that represents the configuration deployed for the Test Procedure.

Initial Conditions:

Device name	**Description**
All the devices	*Security: no for all the devices* *Channel: 0x14 for all the devices* *PAN id: 0x1AAA for all the devices*
P_Coord FFD0	*PAN Coordinator, Bo = 15, SO = 15* *MAC: 0xACDE480000000001* *Short Address: 0x1122* *Beacon Payload: none* DUT starts PAN as described in TP/154/MAC/BEACON-MANAGEMENT-01
FFD1	*MAC: 0xACDE480000000002* *Short Address: 0x3344* Tester associates to DUT as described in TP/154/MAC/ASSOCIATION-01

Test Procedure:

Step No.	**Step description**	**Expected outcome**
1	Set DUT RxOnWhenIdle Set to DUT: MLME-SET.request (PIBAttribute=0x52=82=macRxOnWhenIdle, PIBAttributeValue=TRUE)	Obtain MLME-Set.confirm MLME-SET.confirm (Status=SUCCESS, PIBAttribute=0x52=82= macRxOnWhenIdle)
2	Set to Tester MCPS-DATA.request (SrcAddrMode = 0x02 = ShortAddress,	Tester transmits MAC Data frame with an **INVALID FCS value of bad1** Frame Length: 16 bytes IEEE 802.15.4 Frame Control: 0x8864

	DstAddrMode=0x02 = ShortAddress, DstPANId = 0x1aaa, DstAddr = 0x1122, msduLength=5, msdu = 0x00 0x01 0x02 0x03 0x04, msduHandle = 0x00, TxOptions = 0x01 = AckRequired, SecurityLevel = 0x00, KeyIdMode = xx, KeySource = xx, KeyIndex = xx)	.... .100 = Frame Type: Reserved (0x04) .... 0... = Security Enabled: Disabled ....0 .... = Frame Pending: No more data ....1. .... = Acknowledgment Request: Ack required .... .1.. .... = Intra-PAN: Within the PAN .... ..00 0... .... = Reserved .... 10.. .... = Destination Addressing Mode: Address field contains a 16-bit short address ..00 .... = Reserved for Zigbee PRO/RF4CE 10.. .... = Source Addressing Mode: Address field contains a 16-bit short address Sequence Number: xx Destination PAN Identifier: 0x1aaa Destination Address: 0x1122 Source Address: 0x3344 MAC Payload 0x00 0x01 0x02 0x03 0x04 Frame Check Sequence: 0x00 0x00 Pass: DUT does not acknowledge the frame DUT does not return MCPS-DATA.indication primitive Fail: DUT does acknowledge the frame DUT returns MAC Primitive: MCPS-DATA.indication(SrcAddrMode = 0x02 = ShortAddress, SrcPANId =0x1aaa, SrcAddr= 0x3344, DstAddrMode=0x02 = ShortAddress, DstPANId = 0x1aaa, DstAddr = 0x1122, msduLength=5, msdu = 0x00 0x01 0x02 0x03 0x04, mpduLinkQuality = xx, DSN = xx, Timestamp = xxxxxx SecurityLevel = 0x00, KeyIdMode = xx, KeySource = xx, KeyIndex = xx)
3	Tester returns MCPS-DATA.confirm(msduHandle = 0x00, Status = NO_ACK)	

1242 ***Notes:***

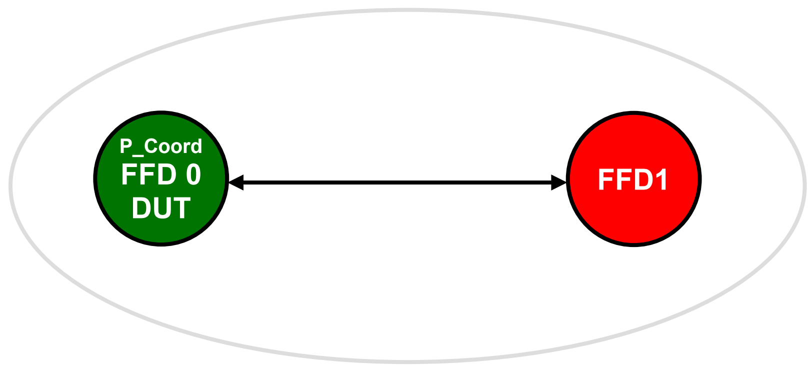
1243 *Requires special tester capable of generating an invalid frame.*

TP/154/MAC/FRAME-VALIDATION-02

Check that the MAC sublayer will discard a received frame that has a reserved value in the Frame Type subfield.

Setup Conditions:

Diagram



The Diagram is the image that represents the configuration deployed for the Test Procedure.

Initial Conditions:

Device name	Description
All the devices	*Security: no for all the devices* *Channel: 0x14 for all the devices* *PAN id: 0x1AAA for all the devices*
P_Coord FFD0	*PAN Coordinator, Bo = 15, SO = 15* *MAC: 0xACDE480000000001* *Short Address: 0x1122* *Beacon Payload: none* DUT starts PAN as described in TP/154/MAC/BEACON-MANAGEMENT-01
FFD1	*MAC: 0xACDE480000000002* *Short Address: 0x3344* Tester associates to DUT as described in TP/154/MAC/ASSOCIATION-01

Test Procedure:

Step No.	Step description	Expected outcome
1	Set DUT RxOnWhenIdle Set to DUT: MLME-SET.request (PIBAttribute=0x52=82=macRxOnWhenIdle, PIBAttributeValue=TRUE)	Obtain MLME-Set.confirm MLME-SET.confirm (Status=SUCCESS, PIBAttribute=0x52=82= macRxOnWhenIdle)
2	Set to Tester MCPS-DATA.request (	Tester transmits MAC Data frame with **RESERVED FRAME TYPE SUB FIELD** Frame Length: 16 bytes

	SrcAddrMode = 0x02 = ShortAddress, DstAddrMode=0x02 = ShortAddress, DstPANId = 0x1aaa, DstAddr = 0x1122, msduLength=5, msdu = 0x00 0x01 0x02 0x03 0x04, msduHandle = 0x00, TxOptions = 0x01, SecurityLevel = 0x00, KeyIdMode = xx, KeySource = xx, KeyIndex = xx)	IEEE 802.15.4 Frame Control: 0x8864 .... .100 = Frame Type: Reserved (0x04) .... 0... = Security Enabled: Disabled ....0 .... = Frame Pending: No more data ....1. .... = Acknowledgment Request: Ack required .... .1.. .... = Intra-PAN: Within the PAN .... ..00 0... .... = Reserved .... 10.. .... = Destination Addressing Mode: Address field contains a 16-bit short address ..00 .... = Reserved for Zigbee PRO/RF4CE 10.. .... = Source Addressing Mode: Address field contains a 16-bit short address Sequence Number: xx Destination PAN Identifier: 0x1aaa Destination Address: 0x1122 Source Address: 0x3344 MAC Payload 0x00 0x01 0x02 0x03 0x04 Frame Check Sequence: Correct Pass: DUT does not acknowledge the frame AND DUT does not return MCPS-DATA.indication primitive Fail: DUT acknowledges the frame DUT returns MAC Primitive: MCPS-DATA.indication(SrcAddrMode = 0x02 = ShortAddress, SrcPANId =0x1aaa, SrcAddr= 0x3344, DstAddrMode=0x02 = ShortAddress, DstPANId = 0x1aaa, DstAddr = 0x1122, msduLength=5, msdu = 0x00 0x01 0x02 0x03 0x04, mpduLinkQuality = xx, DSN = xx, Timestamp = xx xx xx, SecurityLevel = 0x00, KeyIdMode = xx, KeySource = xx, KeyIndex = xx)
3	Tester returns MCPS-DATA.confirm(msduHandle = 0x00, Status = NO_ACK)	

Notes:

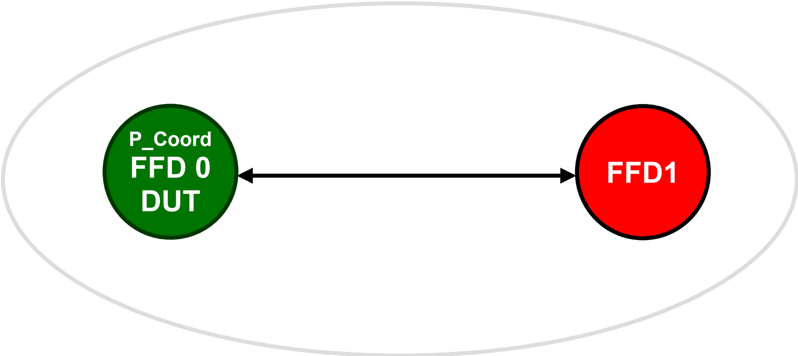
This test requires a special tester capable of generating an invalid frame.

1253 ## TP/154/MAC/FRAME-VALIDATION-03

1254 Check that the MAC sublayer will discard a received frame that implements MAC 2003 security.

1255 ***Setup Conditions***:

Diagram



1256 *The Diagram is the image that represents the configuration deployed for the Test Procedure.*

1257 ***Initial Conditions:***

Device name	Description
All the devices	*Security: no for all the devices* *Channel: 0x14 for all the devices* *PAN id: 0x1AAA for all the devices*
P_Coord FFD0	*PAN Coordinator, Bo = 15, SO = 15* *MAC: 0xACDE480000000001* *Short Address: 0x1122* *Beacon Payload: none* DUT starts PAN as described in TP/154/MAC/BEACON-MANAGEMENT-01
FFD1	*MAC: 0xACDE480000000002* *Short Address: 0x3344* Tester associates to DUT as described in TP/154/MAC/ASSOCIATION-01

1258 ***Test Procedure:***

Step No.	Step description	Expected outcome
1	Set DUT RxOnWhenIdle Set to DUT: MLME-SET.request (PIBAttribute=0x52=82=macRxOnWhenIdle, PIBAttributeValue=TRUE)	Obtain MLME-Set.confirm MLME-SET.confirm (Status=SUCCESS, PIBAttribute=0x52=82= macRxOnWhenIdle)
2	Set to Tester MCPS-DATA.request (SrcAddrMode = 0x02 = ShortAddress,	Tester transmits MAC Data frame with **2003 Security enabled** Frame Length: 16 bytes IEEE 802.15.4 Frame Control: 0x8869

	DstAddrMode=0x02 = ShortAddress, DstPANId = 0x1aaa, DstAddr = 0x1122, msduLength=5, msdu = 0x00 0x01 0x02 0x03 0x04, msduHandle = 0x00, TxOptions = 0x01, SecurityLevel = 0x00, KeyIdMode = xx, KeySource = xx, KeyIndex = xx)	.... .001 = Frame Type: Data (0x0001) .... 1... = Security Enabled: Enabled ....0 .... = Frame Pending: No more data ....1. .... = Acknowledgment Request: Ack required .... .1.. .... = Intra-PAN: Within the PAN .... ..00 0... .... = Reserved .... 10.. .... = Destination Addressing Mode: Address field contains a 16-bit short address ..00 .... = Reserved for Zigbee PRO/RF4CE 10.. .... = Source Addressing Mode: Address field contains a 16-bit short address Sequence Number: xx Destination PAN Identifier: 0x1aaa Destination Address: 0x1122 Source Address: 0x3344 frame counter: 0x00000004 key sequence counter: 0x05 MAC Payload 0x00 0x01 0x02 0x03 0x04 Frame Check Sequence: Correct Pass: DUT does not crash up on reception of frame and DUT acknowledges the frame DUT does not return MCPS-DATA.indication primitive DUT does receive a commstatusindication MLME-COMM-STATUS.indication(Panid =0x1AAA, srcAddrMode = 0x02 = ShortAddress, SrcAddr = 0x0100, DstAddrMode = 0x02=ShortAddress, DstAddr=0x1122, *The following fields only apply for ZIP and RF4CE* Status=0xDE= UnsupportedLegacy SecurityLevel = 0x00, KeyIdMode = xx, KeySource = xx, KeyIndex = xx) Fail: DUT crashes up on reception of frame or DUT does not acknowledges the frame or DUT returns MCPS-DATA.indication primitive or DUT does not receive a commstatusindication
3	Tester returns MCPS-DATA.confirm(msduHandle = 0x00, Status = SUCCESS)	

1259 ***Notes:***

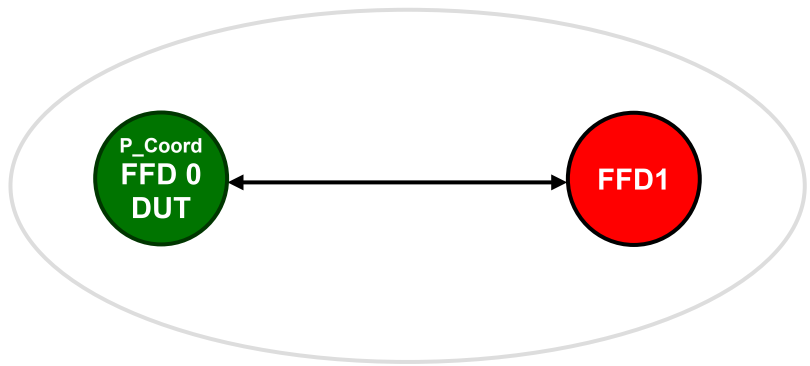
1260 *Requires special tester capable of generating a 802.15.4-2003 Secured frame!*

TP/154/MAC/FRAME-VALIDATION-04

Check that the MAC sublayer will discard a received frame if the destination pan id is not equal to the macPANId.

Setup Conditions:

Diagram



The Diagram is the image that represents the configuration deployed for the Test Procedure.

Initial Conditions:

Device name	Description
All the devices	*Security: no for all the devices* *Channel: 0x14 for all the devices* *PAN id: 0x1AAA for all the devices*
P_Coord FFD0	*PAN Coordinator, Bo = 15, SO = 15* *MAC: 0xACDE480000000001* *Short Address: 0x1122* *Beacon Payload: none* DUT starts PAN as described in TP/154/MAC/BEACON-MANAGEMENT-01
FFD1	*MAC: 0xACDE480000000002* *Short Address: 0x3344* Tester associates to DUT as described in TP/154/MAC/ASSOCIATION-01

Test Procedure:

Step No.	Step description	Expected outcome
1	Set DUT RxOnWhenIdle Set to DUT: MLME-SET.request (PIBAttribute=0x52=82=macRxOnWhenIdle, PIBAttributeValue=TRUE)	Obtain MLME-Set.confirm MLME-SET.confirm (Status=SUCCESS, PIBAttribute=0x52=82= macRxOnWhenIdle)
2	Set to Tester MCPS-DATA.request (	Tester transmits MAC Data frame **to different PAN** Pass:

	SrcAddrMode = 0x02 = ShortAddress, DstAddrMode=0x02 = ShortAddress, DstPANId = 0x1aa0, DstAddr = 0x1122, msduLength=5, msdu = 0x00 0x01 0x02 0x03 0x04, msduHandle = 0x00, TxOptions = 0x01, SecurityLevel = 0x00, KeyIdMode = xx, KeySource = xx, KeyIndex = xx)	DUT does not transmit ACK frame DUT does not return MCPS-DATA.indication primitive Fail: DUT transmits ACK frame DUT returns MAC Primitive: MCPS-DATA.indication(SrcAddrMode = 0x02 = ShortAddress, SrcPANId =0x1aaa, SrcAddr= 0x3344, DstAddrMode=0x02 = ShortAddress, DstPANId = 0x1aaa, DstAddr = 0x1122, msduLength=5, msdu = 0x00 0x01 0x02 0x03 0x04, mpduLinkQuality = xx, DSN = xx, Timestamp = xx xx xx, SecurityLevel = 0x00, KeyIdMode = xx, KeySource = xx, KeyIndex = xx)
3	Tester returns MCPS-DATA.confirm(msduHandle = 0x00, Status = NO_ACK)	

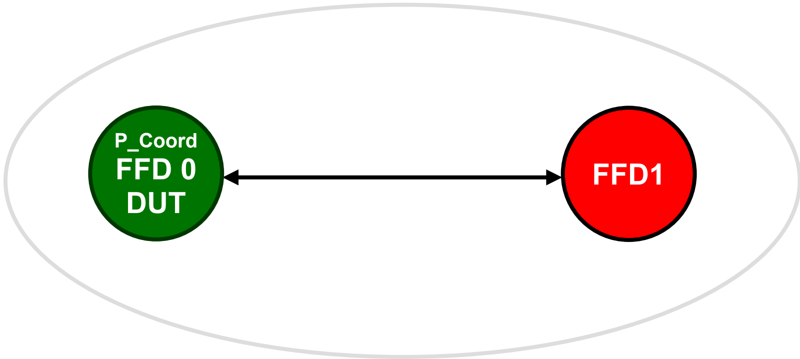
1268 ***Notes:***

1269 *None.*

TP/154/MAC/FRAME-VALIDATION-05

Check that the MAC sublayer will discard a received frame if the destination address is not equal to macShortAddress.

Setup Conditions:

Diagram


The Diagram is the image that represents the configuration deployed for the Test Procedure.

Initial Conditions:

Device name	Description
All the devices	*Security: no for all the devices* *Channel: 0x14 for all the devices* *PAN id: 0x1AAA for all the devices*
P_Coord FFD0	*PAN Coordinator, Bo = 15, SO = 15* *MAC: 0xACDE480000000001* *Short Address: 0x1122* *Beacon Payload: none* DUT starts PAN as described in TP/154/MAC/BEACON-MANAGEMENT-01
FFD1	*MAC: 0xACDE480000000002* *Short Address: 0x3344* Tester associates to DUT as described in TP/154/MAC/ASSOCIATION-01

Test Procedure:

Step No.	Step description	Expected outcome
1	Set DUT RxOnWhenIdle Set to DUT: MLME-SET.request (PIBAttribute=0x52=82=macRxOnWhenIdle, PIBAttributeValue=TRUE)	Obtain MLME-Set.confirm MLME-SET.confirm (Status=SUCCESS, PIBAttribute=0x52=82= macRxOnWhenIdle)
2	Set to Tester MCPS-DATA.request (	Tester transmits MAC Data frame **to different short Address** Pass: DUT does not transmit ACK frame

	SrcAddrMode = 0x02 = ShortAddress, DstAddrMode=0x02 = ShortAddress, DstPANId = 0x1aaa, DstAddr = 0x1000, msduLength=5, msdu = 0x00 0x01 0x02 0x03 0x04, msduHandle = 0x00, TxOptions = 0x00, SecurityLevel = 0x00, KeyIdMode = xx, KeySource = xx, KeyIndex = xx)	DUT does not return MCPS-DATA.indication primitive Fail: DUT transmits ACK frame DUT returns MAC Primitive: MCPS-DATA.indication(SrcAddrMode = 0x02 = ShortAddress, SrcPANId =0x1aaa, SrcAddr= 0x3344, DstAddrMode=0x02 = ShortAddress, DstPANId = 0x1aaa, DstAddr = 0x1122, msduLength=5, msdu = 0x00 0x01 0x02 0x03 0x04, mpduLinkQuality = xx, DSN = xx, Timestamp = xx xx xx, SecurityLevel = 0x00, KeyIdMode = xx, KeySource = xx, KeyIndex = xx)
3	Tester returns MCPS-DATA.confirm(msduHandle = 0x00, Status = NO_ACK)	

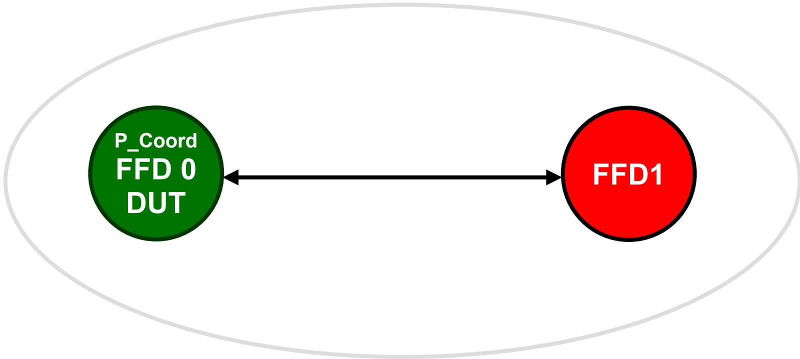
Notes:

None.

TP/154/MAC/FRAME-VALIDATION-06

Check that the MAC sublayer will discard a received frame if the destination address is not equal to device extended address.

Setup Conditions:

Diagram


The Diagram is the image that represents the configuration deployed for the Test Procedure.

Initial Conditions:

Device name	Description
All the devices	*Security: no for all the devices* *Channel: 0x14 for all the devices* *PAN id: 0x1AAA for all the devices*
P_Coord FFD0	*PAN Coordinator, Bo = 15, SO = 15* *MAC: 0xACDE480000000001* *Short Address: 0x1122* *Beacon Payload: none* DUT starts PAN as described in TP/154/MAC/BEACON-MANAGEMENT-01
FFD1	*MAC: 0xACDE480000000002* *Short Address: 0x3344* Tester associates to DUT as described in TP/154/MAC/ASSOCIATION-01

Test Procedure:

Step No.	Step description	Expected outcome
1	Set DUT RxOnWhenIdle Set to DUT: MLME-SET.request (PIBAttribute=0x52=82=macRxOnWhenIdle, PIBAttributeValue=TRUE)	Obtain MLME-Set.confirm MLME-SET.confirm (Status=SUCCESS, PIBAttribute=0x52=82= macRxOnWhenIdle)
2	Set to Tester MCPS-DATA.request (	Tester transmits MAC Data frame **to different extended Address** Frame Length: 16 bytes

	SrcAddrMode = 0x02 = ShortAddress, DstAddrMode=0x03= ExtendedAddress, DstPANId = 0x1aaa, DstAddr = *0xACDE480000000003,* msduLength=5, msdu = 0x00 0x01 0x02 0x03 0x04, msduHandle = 0x00, TxOptions = 0x01, SecurityLevel = 0x00, KeyIdMode = xx, KeySource = xx, KeyIndex = xx)	IEEE 802.15.4 Frame Control: 0x8C61 .... .001 = Frame Type: Data (0x0001) .... 0... = Security Enabled: Disabled ....0 .... = Frame Pending: No more data ....1. .... = Acknowledgment Request: Ack required .... .1.. .... = Intra-PAN: Within the PAN .... ..00 0... .... = Reserved .... 11.. .... = Destination Addressing Mode: Address field contains a 64 bit extended address ..00 .... = Reserved for Zigbee PRO/RF4CE 10.. .... = Source Addressing Mode: Address field contains a 16-bit short address Sequence Number: xx Destination PAN Identifier: 0x1aaa Destination Address: 0xACDE480000000003 Source Address: 0x3344 MAC Payload 0x00 0x01 0x02 0x03 0x04 Frame Check Sequence: Correct Pass: DUT does not transmit ACK frame DUT does not return MCPS-DATA.indication primitive Fail: DUT transmits ACK frame DUT returns MAC Primitive: MCPS-DATA.indication(SrcAddrMode = 0x02 = ShortAddress, SrcPANId =0x1aaa, SrcAddr= 0x3344, DstAddrMode=0x03 = ExtendedAddress, DstPANId = 0x1aaa, DstAddr = *0xACDE480000000003*, msduLength=5, msdu = 0x00 0x01 0x02 0x03 0x04, mpduLinkQuality = xx, DSN = xx, Timestamp = xx xx xx, SecurityLevel = 0x00, KeyIdMode = xx, KeySource = xx, KeyIndex = xx)
3	Tester returns MCPS-DATA.confirm(msduHandle = 0x00, Status = NO_ACK)	

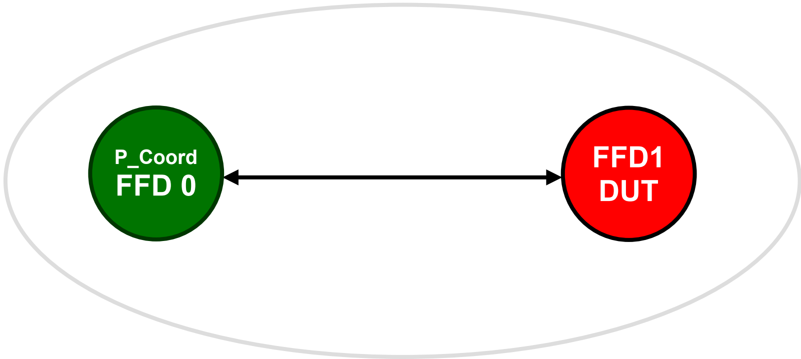
Notes:

None.

TP/154/MAC/FRAME-VALIDATION-07

Check that the MAC sublayer will discard a received frame if the destination address mode is 0 and the receiving device it not the pan coordinator.

Setup Conditions:

Diagram


The Diagram is the image that represents the configuration deployed for the Test Procedure.

Initial Conditions:

Device name	**Description**
All the devices	*Security: no for all the devices* *Channel: 0x14 for all the devices* *PAN id: 0x1AAA for all the devices*
P_Coord FFD0	*PAN Coordinator, Bo = 15, SO = 15* *MAC: 0xACDE480000000001* *Short Address: 0x1122* *Beacon Payload: none* Tester starts PAN as described in TP/154/MAC/BEACON-MANAGEMENT-01
FFD1	*MAC: 0xACDE480000000002* *Short Address: 0x3344* DUT associates to Tester as described in TP/154/MAC/ASSOCIATION-03

Test Procedure:

Step No.	**Step description**	**Expected outcome**
1	Set DUT RxOnWhenIdle Set to DUT: MLME-SET.request (PIBAttribute=0x52=82=macRxOnWhenIdle, PIBAttributeValue=TRUE)	Obtain MLME-Set.confirm MLME-SET.confirm (Status=SUCCESS, PIBAttribute=0x52=82= macRxOnWhenIdle)
2	Set to Tester MCPS-DATA.request (SrcAddrMode = 0x02 = ShortAddress, DstAddrMode=0x00= ImpliedAddress,	Tester transmits MAC Data frame **to impliedAddress** Frame Length: 14 bytes IEEE 802.15.4 Frame Control: 0x8021

	DstPANId = xx, DstAddr = *xx,* msduLength=5, msdu = 0x00 0x01 0x02 0x03 0x04, msduHandle = 0x00, TxOptions = 0x01=AckRequired, SecurityLevel = 0x00, KeyIdMode = xx, KeySource = xx, KeyIndex = xx)	.... .001 = Frame Type: Data (0x0001) .... 0... = Security Enabled: Disabled ....0 .... = Frame Pending: No more data ....1. .... = Acknowledgment Request: Ack required .... .0.. .... = Intra-PAN: Not within the PAN .... ..00 0... .... = Reserved .... 00.. .... = Destination Addressing Mode: Address field not present ..00 .... = Reserved for Zigbee PRO/RF4CE 10.. .... = Source Addressing Mode: Address field contains a 16-bit short address Sequence Number: xx Destination PAN Identifier: not present Destination Address: not present Source PAN Identifier: 0x1aaa Source Address: 0x3344 MAC Payload 0x00 0x01 0x02 0x03 0x04 Frame Check Sequence: Correct Pass: DUT does not transmit ACK frame DUT does not return MCPS-DATA.indication primitive Fail: DUT transmits ACK frame DUT returns MAC Primitive: MCPS-DATA.indication(SrcAddrMode = 0x02 = ShortAddress, SrcPANId =0x1aaa, SrcAddr= 0x1122, DstAddrMode=0x00= ImpliedAddress, DstPANId = 0x1aaa, DstAddr = *xx*, msduLength=5, msdu = 0x00 0x01 0x02 0x03 0x04, mpduLinkQuality = xx, DSN = xx, Timestamp = xx xx xx, SecurityLevel = 0x00, KeyIdMode = xx, KeySource = xx, KeyIndex = xx)
3	Tester returns MCPS-DATA.confirm(msduHandle = 0x00, Status = NO_ACK)	

1295 ***Notes:***

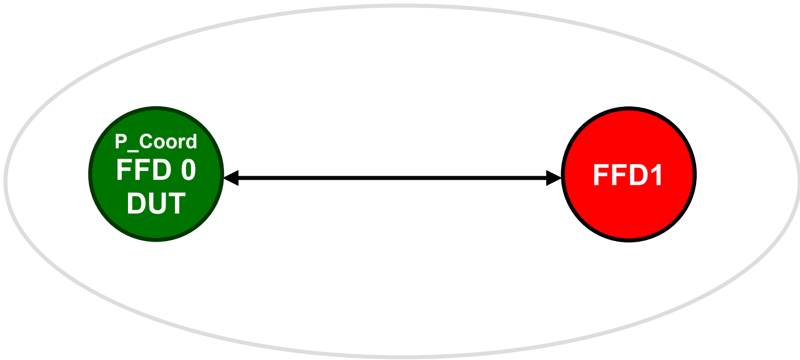
1296 *None.*

1297 ### TP/154/MAC/ACK-FRAME-DELIVERY-01

1298 Check D.U.T. correctly receives ACK frame

1299 ***Setup Conditions***:

Diagram



1300 *The Diagram is the image that represents the configuration deployed for the Test Procedure.*

1301 ***Initial Conditions:***

Device name	Description
All the devices	*Security: no for all the devices* *Channel: 0x14 for all the devices* *PAN id: 0x1AAA for all the devices*
P_Coord FFD0	*PAN Coordinator, Bo = 15, SO = 15* *MAC: 0xACDE480000000001* *Short Address: 0x1122* *Beacon Payload: none* DUT starts PAN as described in TP/154/MAC/BEACON-MANAGEMENT-01
FFD1	*MAC: 0xACDE480000000002* *Short Address: 0x3344* Tester associates to DUT as described in TP/154/MAC/ASSOCIATION-01

1302 ***Test Procedure:***

Step No.	Step description	Expected outcome
1	Set DUT RxOnWhenIdle Set to DUT: MLME-SET.request (PIBAttribute=0x52=82=macRxOnWhenIdle, PIBAttributeValue=TRUE) Set Tester RxOnWhenIdle to OFF Set to Tester: MLME-SET.request (	Obtain MLME-Set.confirm MLME-SET.confirm (Status=SUCCESS, PIBAttribute=0x52=82= macRxOnWhenIdle)

	PIBAttribute=0x52=82=macRxOnWhenIdle, PIBAttributeValue=FALSE)	
2	Set to DUT: MCPS-DATA.request (SrcAddrMode = 0x02 = ShortAddress, DstAddrMode=0x02= ShortAddress, DstPANId = 0x1aaa, DstAddr = *0x3344,* msduLength=5, msdu = 0x00 0x01 0x02 0x03 0x04, msduHandle = 0x00, TxOptions = 0x01= AckRequired, SecurityLevel = 0x00, KeyIdMode = xx, KeySource = xx, KeyIndex = xx)	Frame Length: 16 bytes IEEE 802.15.4 Frame Control: 0x8861 .... .001 = Frame Type: Data (0x0001) .... 0... = Security Enabled: Disabled ....0 .... = Frame Pending: No more data ....1. .... = Acknowledgment Request: Ack required .... .1.. .... = Intra-PAN: Within the PAN .... ..00 0... .... = Reserved .... 10.. .... = Destination Addressing Mode: Address field contains a 16-bit short address ..00 .... = Reserved for Zigbee PRO/RF4CE 10.. .... = Source Addressing Mode: Address field contains a 16-bit short address Sequence Number: [value] Destination PAN Identifier: 0x1aaa Destination Address: 0x3344 Source Address: 0x1122 MAC Payload 0x00 0x01 0x02 0x03 0x04 Frame Check Sequence: Correct Expected Outcome: DUT transmits MAC data frame 4 times to Tester – DSN (sequence number) does not change DUT returns MAC primitive: MCPS-DATA.confirm(msduHandle = 0x00, Status = 0xe9 = NO_ACK) Fail: DUT does not transmit MAC data frame 4 times to Tester - DSN does change DUT does not return MCPS-DATA.confirm primitive DUT returns MAC primitive: MCPS-DATA.confirm(msduHandle = 0x00, Status = 0x00 = SUCCESS)
3	Enable Receiver on Tester Set to Tester: MLME-SET.request (PIBAttribute=0x52=82=macRxOnWhenIdle, PIBAttributeValue=TRUE)	Obtain MLME-Set.confirm on Tester MLME-SET.confirm (Status=SUCCESS, PIBAttribute=0x52=82= macRxOnWhenIdle)
4	Set to DUT: MCPS-DATA.request (SrcAddrMode = 0x02 = ShortAddress, DstAddrMode=0x02= ShortAddress, DstPANId = 0x1aaa, DstAddr = *0x3344,* msduLength=5, msdu = 0x00 0x01 0x02 0x03 0x04, msduHandle = 0x00, TxOptions = 0x01= AckRequired,	Expected Outcome: DUT transmits MAC data frame to Tester Tester responds with an Ack Frame at 12 symbols from end of received frame DUT returns MAC primitive: MCPS-DATA.confirm(msduHandle = 0x00, Status = 0x00 = SUCCESS

	SecurityLevel = 0x00, KeyIdMode = xx, KeySource = xx, KeyIndex = xx)	) Tester returns MAC Primitive: MCPS-DATA.indication(SrcAddrMode = 0x02 = ShortAddress, SrcPANId =0x1aaa, SrcAddr= 0x1122, DstAddrMode=0x02 = ShortAddress, DstPANId = 0x1aaa, DstAddr = *0x3344*, msduLength=5, msdu = 0x00 0x01 0x02 0x03 0x04, mpduLinkQuality = xx, DSN = xx, Timestamp = xx SecurityLevel = 0x00, KeyIdMode = xx, KeySource = xx, KeyIndex = xx) Fail: DUT transmits more than one MAC data frame to Tester DUT does not return MCPS-DATA.confirm primitive DUT returns MAC primitive: MCPS-DATA.confirm(msduHandle = 0x00, Status = 0xe9 = NO_ACK)
5	Set Tester RxOnWhenIdle to OFF Set to Tester: MLME-SET.request (PIBAttribute=0x52=82=macRxOnWhenIdle, PIBAttributeValue=FALSE)	
6	Set to DUT: MCPS-DATA.request (SrcAddrMode = 0x02 = ShortAddress, DstAddrMode=0x02= ShortAddress, DstPANId = 0x1aaa, DstAddr = *0x3344,* msduLength=5, msdu = 0x00 0x01 0x02 0x03 0x04, msduHandle = 0x00, TxOptions = 0x01= AckRequired, SecurityLevel = 0x00, KeyIdMode = xx, KeySource = xx, KeyIndex = xx)	Frame Length: 16 bytes IEEE 802.15.4 Frame Control: 0x8861 .... .001 = Frame Type: Data (0x0001) .... 0... = Security Enabled: Disabled ....0 .... = Frame Pending: No more data ....1. .... = Acknowledgment Request: Ack required .... .1.. .... = Intra-PAN: Within the PAN .... ..00 0... .... = Reserved .... 10.. .... = Destination Addressing Mode: Address field contains a 16-bit short address ..00 .... = Reserved for Zigbee PRO/RF4CE 10.. .... = Source Addressing Mode: Address field contains a 16-bit short address Sequence Number: [value] Destination PAN Identifier: 0x1aaa Destination Address: 0x3344 Source Address: 0x1122 MAC Payload 0x00 0x01 0x02 0x03 0x04 Frame Check Sequence: Correct Pass: DUT transmits MAC data frame 4 times to Tester - DSN does not change DUT returns MAC primitive:

		MCPS-DATA.confirm(msduHandle = 0x00, Status = 0xe9 = NO_ACK) Fail: DUT does not transmit MAC data frame 4 times to Tester - DSN doeschange DUT does not return MCPS-DATA.confirm primitive DUT returns MAC primitive: MCPS-DATA.confirm(msduHandle = 0x00, Status = 0x00 = SUCCESS)

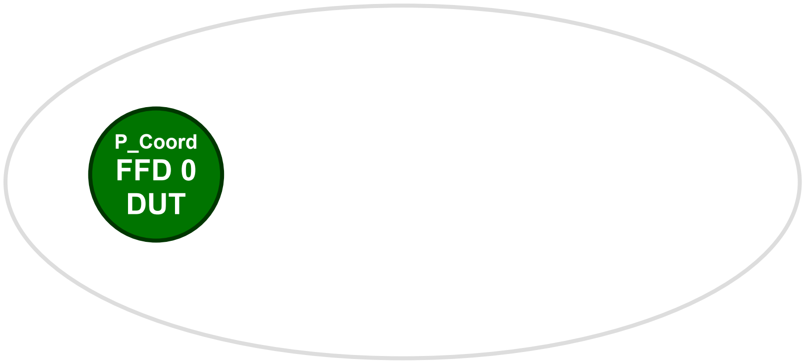
1303 ***Notes:***

1304 *None.*

1305 ## TP/154/MAC/RETRIES-01

1306 Check D.U.T. uses correct number of frame retries when transmission fails

1307 ***Setup Conditions***:

Diagram


1308 *The Diagram is the image that represents the configuration deployed for the Test Procedure.*

1309 ***Initial Conditions:***

Device name	**Description**
All the devices	*Security: no for all the devices* *Channel: 0x14 for all the devices* *PAN id: 0x1AAA for all the devices*
P_Coord FFD0	*PAN Coordinator, Bo = 15, SO = 15* *MAC: 0xACDE480000000001* *Short Address: 0x1122* *Beacon Payload: none* DUT starts PAN as described in TP/154/MAC/BEACON-MANAGEMENT-01

1310 ***Test Procedure:***

Step No.	**Step description**	**Expected outcome**
1	Set DUT RxOnWhenIdle Set to DUT: MLME-SET.request (PIBAttribute=0x52=82=macRxOnWhenIdle, PIBAttributeValue=TRUE) Verify the PIB is correctly set MLME-GET.request (PIBAttribute=0x52=macRxOnWhenIdle, PIBAttributeIndex = 0x00- ignore, Only used for MAC 2006 security PIBs)	Obtain MLME-Set.confirm MLME-SET.confirm (Status=SUCCESS, PIBAttribute=0x52=82= macRxOnWhenIdle) Wait for MLME-GET.confirm and confirm the parameters MLME-GET.confirm (Status = SUCCESS PIBAttribute=0x52=macRxOnWhenIdle, PIBAttributeIndex = ignore. Only for security in MAC2006 PIBAttributeValue = TRUE)
2	Set to DUT:	Frame Length: 16 bytes

	MCPS-DATA.request (SrcAddrMode = 0x02 = ShortAddress, DstAddrMode=0x02= ShortAddress, DstPANId = 0x1aaa, DstAddr = *0x3344,* msduLength=5, msdu = 0x00 0x01 0x02 0x03 0x04, msduHandle = 0x00, TxOptions = 0x01= AckRequired, SecurityLevel = 0x00, KeyIdMode = xx, KeySource = xx, KeyIndex = xx)	IEEE 802.15.4 Frame Control: 0x8861 .... .001 = Frame Type: Data (0x0001) .... 0... = Security Enabled: Disabled ....0 .... = Frame Pending: No more data ....1. .... = Acknowledgment Request: Ack required .... .1.. .... = Intra-PAN: Within the PAN .... ..00 0... .... = Reserved .... 10.. .... = Destination Addressing Mode: Address field contains a 16-bit short address ..00 .... = Reserved for Zigbee PRO/RF4CE 10.. .... = Source Addressing Mode: Address field contains a 16-bit short address Sequence Number: [value] Destination PAN Identifier: 0x1aaa Destination Address: 0x3344 Source Address: 0x1122 MAC Payload 0x00 0x01 0x02 0x03 0x04 Frame Check Sequence: Correct Expected Outcome: DUT transmits the same MAC data frame 4 times DUT returns MAC primitive: MCPS-DATA.confirm(msduHandle = 0x00, Status = 0xe9 = NO_ACK) Fail: DUT does not transmit the same MAC data frame 4 times to Tester DUT does not return MCPS-DATA.confirm primitive DUT returns MAC primitive: MCPS-DATA.confirm(msduHandle = 0x00, Status = 0x00 = SUCCESS)
3	Set DUT macMaxFrameRetries to 1 (non default value) Set to DUT: MLME-SET.request (PIBAttribute=0x59= macMaxFrameRetries, PIBAttributeValue=1) Verify the PIB is correctly set MLME-GET.request (PIBAttribute=0x59= macMaxFrameRetries, PIBAttributeIndex = 0x00- ignore, Only used for MAC 2006 security PIBs)	Obtain MLME-Set.confirm on DUT MLME-SET.confirm (Status=SUCCESS, PIBAttribute=0x59= macMaxFrameRetries) Wait for MLME-GET.confirm and confirm the parameters MLME-GET.confirm (Status = SUCCESS PIBAttribute=0x59= macMaxFrameRetries, PIBAttributeIndex = ignore. Only for security in MAC2006 PIBAttributeValue = 1)
4	Set to DUT: MCPS-DATA.request (SrcAddrMode = 0x02 = ShortAddress, DstAddrMode=0x02= ShortAddress, DstPANId = 0x1aaa, DstAddr = *0x3344,* msduLength=5, msdu = 0x00 0x01 0x02 0x03 0x04, msduHandle = 0x00, TxOptions = 0x01= AckRequired,	Frame Length: 16 bytes IEEE 802.15.4 Frame Control: 0x8861 .... .001 = Frame Type: Data (0x0001) .... 0... = Security Enabled: Disabled ....0 .... = Frame Pending: No more data ....1. .... = Acknowledgment Request: Ack required .... .1.. .... = Intra-PAN: Within the PAN .... ..00 0... .... = Reserved .... 10.. .... = Destination Addressing Mode: Address field contains a 16-bit short address

	SecurityLevel = 0x00, KeyIdMode = xx, KeySource = xx, KeyIndex = xx)	..00 .... = Reserved for Zigbee PRO/RF4CE 10.. .... = Source Addressing Mode: Address field contains a 16-bit short address Sequence Number: [value] Destination PAN Identifier: 0x1aaa Destination Address: 0x3344 Source Address: 0x1122 MAC Payload 0x00 0x01 0x02 0x03 0x04 Frame Check Sequence: Correct Pass: DUT transmits the same MAC data frame 2 times DUT returns MAC primitive: MCPS-DATA.confirm(msduHandle = 0x00, Status = 0xe9 = NO_ACK) Fail: DUT does not transmit the same MAC data frame 2 times . DUT does not return MCPS-DATA.confirm primitive DUT returns MAC primitive: MCPS-DATA.confirm(msduHandle = 0x00, Status = 0x00 = SUCCESS)

1311 ***Notes:***

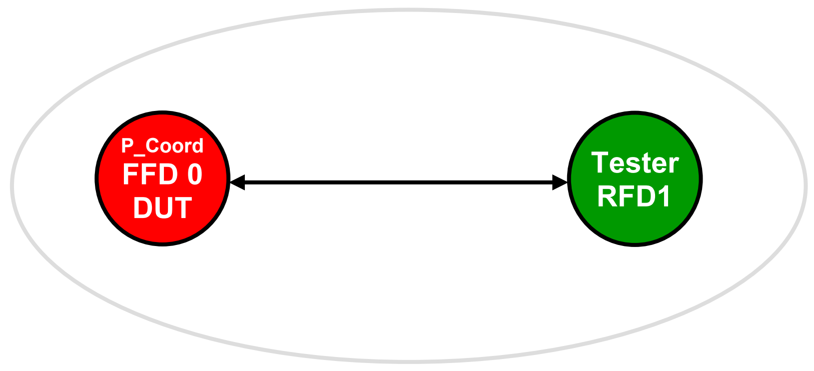
1312 *None.*

TP/154/MAC/ASSOCIATION-01[1]

Check D.U.T. (as coordinator) correctly receives association request frame and generates positive response

Setup Conditions:

Diagram



The Diagram is the image that represents the configuration deployed for the Test Procedure.

Initial Conditions:

Device name	Description
All the devices	*Security: no for all the devices* *Channel: 0x14 for all the devices* *PAN id: 0x1AAA for all the devices*
P_Coord FFD0	PAN Coordinator, Bo = 15, SO = 15 MAC: 0xACDE480000000001 Short Address: 0x1122 Beacon Payload: none DUT as coordinator shall set up a non-beacon enabled PAN.
RFD1	*MAC: 0xACDE480000000002* Short Address: None

Test Procedure:

Step No.	Step description	Expected outcome
1	Set DUT RxOnWhenIdle Set to DUT: MLME-SET.request (PIBAttribute=0x52=82=macRxOnWhenIdle, PIBAttributeValue=TRUE) Set to DUT: MLME-SET.request (PIBAttribute=0x41= macAssociationPermit, PIBAttributeValue=TRUE)	Pass: Obtain MLME-Set.confirm MLME-SET.confirm (Status=SUCCESS, PIBAttribute=0x52=82= macRxOnWhenIdle) Obtain MLME-Set.confirm MLME-SET.confirm (Status=SUCCESS, PIBAttribute=0x41= macAssociationPermit)

2	Set to Tester MLME-ASSOCIATE.request(LogicalChannel =0x14, ChannelPage=0, CoordAddrMode = 0x02 = Short Address, CoordPANId = 0x1aaa, CoordAddress = 0x1122, CapabilityInformation = 0x80, SecurityLevel = 0x00, KeyIdMode = xx, KeySource = xx, KeyIndex = xx)	Tester transmits MAC association request command frame . Pass: DUT returns MAC Primitive: MLME-ASSOCIATE.indication(DeviceAddress=0xACDE480000000002, CapabilityInformation=0x80, SecurityLevel = 0x00, KeyIdMode = xx, KeySource = xx, KeyIndex = xx) Fail: DUT does not return MLME-ASSOCIATE.indication primitive
3	Set to DUT MLME-ASSOCIATE.response(DeviceAddress =0xACDE480000000002, AssocShortAddress = 0x3344, Status = 0x00 = SUCCESS, SecurityLevel = 0x00, KeyIdMode = xx, KeySource = xx, KeyIndex = xx)	Tester will transmit MAC data request command frame to coordinator, DUT Pass: It is possible to verify by a PAN analyzer that an ACK frame was issued from the DUT with the frame pending field set to 1. Frame Length: 5 bytes IEEE 802.15.4 Frame Control: 0x0012 .... .010 = Frame Type: ack (0x0002) .... 0... = Security Enabled: Disabled ....1 .... = Frame Pending: data ....0. .... = Acknowledgment Request: Ack not required .... .0.. .... = Intra-PAN: Not within the PAN .... ..00 0... .... = Reserved .... 00.. .... = Destination Addressing Mode: None ..00 .... = Reserved 00.. .... = Source Addressing Mode: None Sequence Number: xx Frame Check Sequence: Correct Pass: It is possible to verify by a PAN analyzer that an association response MAC command frame was issued from the DUT. Frame Length: 27 bytes IEEE 802.15.4 Frame Control: 0xCC63 .... .011 = Frame Type: Command (0x0003) .... 0... = Security Enabled: Disabled ....0 .... = Frame Pending: No more data ....1. .... = Acknowledgment Request: Ack required .... .1.. .... = Intra-PAN: Within the PAN .... ..00 0... .... = Reserved .... 11.. .... = Destination Addressing Mode: Address field contains a 64-bit extended address ..00 .... = Reserved 11.. .... = Source Addressing Mode: Address field contains a 64-bit extended address Sequence Number: xx Destination PAN Identifier: 0x1aaa Destination Address: 0xACDE480000000002 Source Address: 0xACDE480000000001 MAC Payload MAC Command = 0x02 = Association Response ShortAddress = 0x3344, AssociationStatus = 0x00 = SUCCESS Frame Check Sequence: Correct Tester will transmit Ack frame in response to association response MAC command frame from DUT.

		The DUT shall return a Success result, such that: DUT returns MAC primitive MLME-COMM-STATUS.indication(PANid = 0x1aaa, SrcAddrMode = 0x03 = ExtendedAddress, SrcAddr = 0xACDE480000000001, DstAddrMode = 0x03= ExtendedAddress, DstAddr = 0xACDE480000000002, Status = 0x00 = SUCCESS, SecurityLevel = 0x00, KeyIdMode = xx, KeySource = xx, KeyIndex = xx) Fail: DUT does not send ACK in response to MAC data request command frame ACK does not have Frame pending field set DUT does not send association response frame DUT does not return MLME-COMM-STATUS.indication primitive Status is not SUCCESS

Notes:

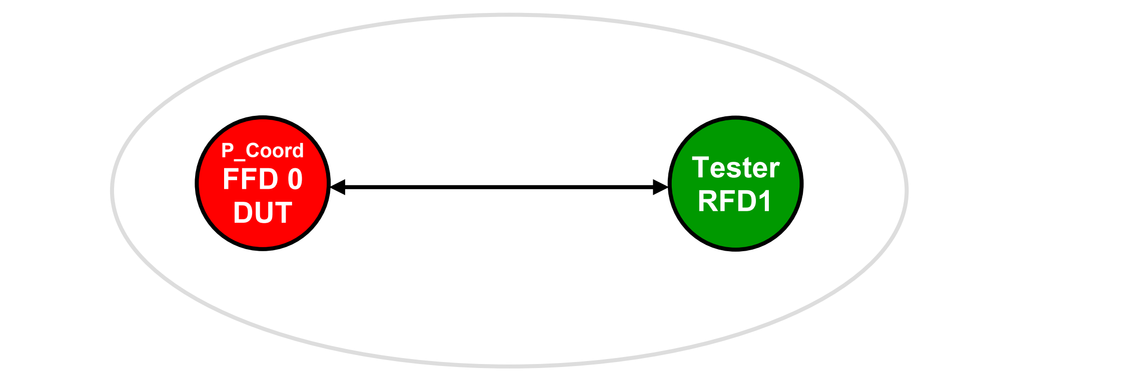
None.

TP/154/MAC/ASSOCIATION-02[1]

Check D.U.T. (as coordinator) correctly receives association request frame and generates negative response (PanAtCapacity, PanAccessDenied)

Setup Conditions:

Diagram



The Diagram is the image that represents the configuration deployed for the Test Procedure.

Initial Conditions:

Device name	Description
All the devices	*Security: no for all the devices* *Channel: 0x14 for all the devices* *PAN id: 0x1AAA for all the devices*
P_Coord FFD0	PAN Coordinator, Bo = 15, SO = 15 MAC: 0xACDE480000000001 Short Address: 0x1122 Beacon Payload: none DUT as coordinator shall set up a non-beacon enabled PAN.
RFD1	*MAC: 0xACDE480000000002* Short Address: 0x3344

Test Procedure:

Step No.	Step description	Expected outcome
1	Set DUT RxOnWhenIdle Set to DUT: MLME-SET.request (PIBAttribute=0x52=82=macRxOnWhenIdle, PIBAttributeValue=TRUE) Set to DUT: MLME-SET.request (PIBAttribute=0x41= macAssociationPermit, PIBAttributeValue=TRUE)	Pass: Obtain MLME-Set.confirm MLME-SET.confirm (Status=SUCCESS, PIBAttribute=0x52=82= macRxOnWhenIdle) Obtain MLME-Set.confirm MLME-SET.confirm (Status=SUCCESS, PIBAttribute=0x41= macAssociationPermit)
2	Set to Tester	Tester transmits MAC association request command frame .

	MLME-ASSOCIATE.request(LogicalChannel =0x14, ChannelPage=0, CoordAddrMode = 0x02 = Short Address, CoordPANId = 0x1aaa, CoordAddress = 0x1122, CapabilityInformation = 0x80, SecurityLevel = 0x00, KeyIdMode = xx, KeySource = xx, KeyIndex = xx)	Pass: DUT returns MAC Primitive: MLME-ASSOCIATE.indication(DeviceAddress=0xACDE480000000002, CapabilityInformation=0x80, SecurityLevel = 0x00, KeyIdMode = xx, KeySource = xx, KeyIndex = xx) Fail: DUT does not return MLME-ASSOCIATE.indication primitive
3	Set to DUT MLME-ASSOCIATE.response(DeviceAddress =0xACDE480000000002, AssocShortAddress = 0xffff, Status = 0x01 = Pan at capacity, SecurityLevel = 0x00, KeyIdMode = xx, KeySource = xx, KeyIndex = xx)	Tester will transmit MAC data request command frame to coordinator, DUT Pass: It is possible to verify by a PAN analyzer that an ACK frame was issued from the DUT with the frame pending field set to 1. Frame Length: 5 bytes IEEE 802.15.4 Frame Control: 0x0012 .... .010 = Frame Type: ack (0x0002) .... 0... = Security Enabled: Disabled ....1 .... = Frame Pending: data ....0. .... = Acknowledgment Request: Ack not required .... .0.. .... = Intra-PAN: Not within the PAN .... ..00 0... .... = Reserved .... 00.. .... = Destination Addressing Mode: None ..00 .... = Reserved 00.. .... = Source Addressing Mode: None Sequence Number: xx Frame Check Sequence: Correct Pass: It is possible to verify by a PAN analyzer that an association response MAC command frame was issued from the DUT. Frame Length: 27 bytes IEEE 802.15.4 Frame Control: 0xCC63 .... .011 = Frame Type: Command (0x0003) .... 0... = Security Enabled: Disabled ....0 .... = Frame Pending: No more data ....1. .... = Acknowledgment Request: Ack required .... .1.. .... = Intra-PAN: Within the PAN .... ..00 0... .... = Reserved .... 11.. .... = Destination Addressing Mode: Address field contains a 64-bit extended address ..00 .... = Reserved 11.. .... = Source Addressing Mode: Address field contains a 64-bit extended address Sequence Number: xx Destination PAN Identifier: 0x1aaa Destination Address: 0xACDE480000000002 Source Address: 0xACDE480000000001 MAC Payload MAC Command = 0x02 = Association Response ShortAddress = 0xffff, AssociationStatus = 0x0x1 = PAN at capacity Frame Check Sequence: Correct Tester will transmit Ack frame in response to association response MAC command frame from DUT The DUT shall return a Success result, such that:

		DUT returns MAC primitive MLME-COMM-STATUS.indication(PANid = 0x1aaa, SrcAddrMode = 0x03 = ExtendedAddress, SrcAddr = 0xACDE480000000001, DstAddrMode = 0x03= ExtendedAddress, DstAddr = 0xACDE480000000002, Status = 0x00 = SUCCESS, SecurityLevel = 0x00, KeyIdMode = xx, KeySource = xx, KeyIndex = xx) Fail: DUT does not send ACK in response to MAC data request command frame ACK does not have Frame pending field set DUT does not send association response frame DUT does not return MLME-COMM-STATUS.indication primitive Status is not SUCCESS

1329 ***Notes:***

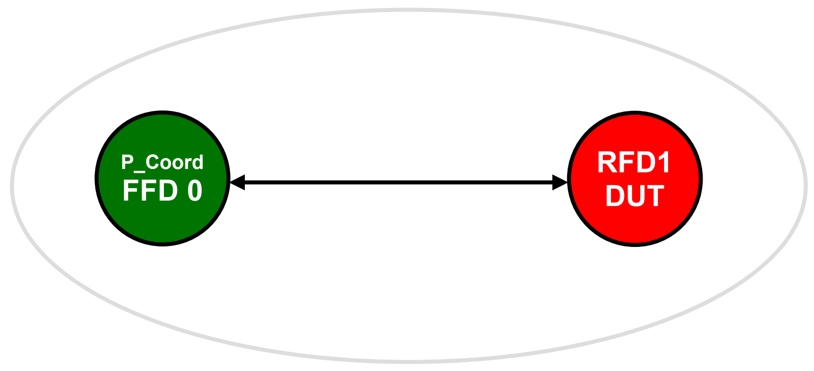
1330 *None.*

TP/154/MAC/ASSOCIATION-03[1]

Check D.U.T. (as device) correctly generates association request frame and receives association response frame with positive response (short address)

Setup Conditions:

Diagram



The Diagram is the image that represents the configuration deployed for the Test Procedure.

Initial Conditions:

Device name	Description
All the devices	*Security: no for all the devices* *Channel: 0x14 for all the devices* *PAN id: 0x1AAA for all the devices*
P_Coord FFD0	PAN Coordinator, Bo = 15, SO = 15 MAC: 0xACDE480000000001 Short Address: 0x1122 Beacon Payload: none Coordinator shall set up a non-beacon enabled PAN.
RFD1	*MAC: 0xACDE480000000002* Short Address: None

Test Procedure:

Step No.	Step description	Expected outcome
1	Set Coordinator RxOnWhenIdle Set to Tester: MLME-SET.request (PIBAttribute=0x52=82=macRxOnWhenIdle, PIBAttributeValue=TRUE) Set to tester: MLME-SET.request (PIBAttribute=0x41= macAssociationPermit, PIBAttributeValue=TRUE)	
2	Set to D.U.T. MLME-ASSOCIATE.request(	Pass:

	LogicalChannel =0x14, ChannelPage=0, CoordAddrMode = 0x02 = Short Address, CoordPANId = 0x1aaa, CoordAddress = 0x1122, CapabilityInformation = 0x80, SecurityLevel = 0x00, KeyIdMode = xx, KeySource = xx, KeyIndex = xx)	It is possible to verify by a PAN analyzer that an association request MAC command frame was issued from the DUT. Frame Length: 21 bytes IEEE 802.15.4 Frame Control: 0xC823 .... .011 = Frame Type: MAC Command (0x0003) .... 0... = Security Enabled: Disabled ....0 .... = Frame Pending: No more data ....1. .... = Acknowledgment Request: Ack required .... .0.. .... = Intra-PAN: Not within the PAN .... ..00 0... .... = Reserved .... 10.. .... = Destination Addressing Mode: Address field contains a 16-bit extended address ..00 .... = Reserved 11.. .... = Source Addressing Mode: Address field contains a 64-bit extended address Sequence Number: xx Destination PAN Identifier: 0x1aaa Destination Address: 0x1122 Source PAN Identifier: 0xffff Source Address: 0xACDE480000000002 MAC Payload MAC Command = 0x01 = Association Request CapabilityInformation = 0x80 = Allocate Address Frame Check Sequence: Correct Tester transmits ACK frame Tester sends MLME-ASSOCIATE.indication MAC primitive to NHL
3	Set to Tester MLME-ASSOCIATE.response(DeviceAddress =0xACDE480000000002, AssocShortAddress = 0x3344, Status = 0x00 = Success, SecurityLevel = 0x00, KeyIdMode = xx, KeySource = xx, KeyIndex = xx)	Pass: It is possible to verify by a PAN analyzer that the DUT transmits a MAC data request command frame to coordinator aResponseWaitTime after tester transmits ACK frame Frame Length: 12 bytes IEEE 802.15.4 Frame Control: 0xC863 .... .011 = Frame Type: Command (0x0003) .... 0... = Security Enabled: Disabled ....0 .... = Frame Pending: No more data ....1. .... = Acknowledgment Request: ACK required .... .1.. .... = Intra-PAN: Within the PAN .... ..00 0... .... = Reserved .... 10.. .... = Destination Addressing Mode: Address field contains a 16-bit short address ..00 .... = Reserved 11.. .... = Source Addressing Mode: Address field contains a 64-bit extended address Sequence Number: xx Destination PAN Identifier: 0x1aaa Destination Address: 0x1122 Source Address: 0xACDE480000000002 MAC Payload Command Frame Identifier = Data Request: (0x04) Frame Check Sequence: Correct Tester issues an ACK frame with the frame pending field set to 1. Frame Length: 5 bytes IEEE 802.15.4 Frame Control: 0x0012 .... .010 = Frame Type: ack (0x0002) .... 0... = Security Enabled: Disabled ....1 .... = Frame Pending: data ....0. .... = Acknowledgment Request: Ack not required .... .0.. .... = Intra-PAN: Not within the PAN .... ..00 0... .... = Reserved .... 00.. .... = Destination Addressing Mode: None

		..00 .... = Reserved 00.. .... = Source Addressing Mode: None Sequence Number: xx Frame Check Sequence: Correct Tester issues an association response MAC command frame. Frame Length: 27 bytes IEEE 802.15.4 Frame Control: 0xCC63 .... .011 = Frame Type: Command (0x0003) .... 0... = Security Enabled: Disabled ....0 .... = Frame Pending: No more data ....1. .... = Acknowledgment Request: Ack required .... .1.. .... = Intra-PAN: Within the PAN .... ..00 0... .... = Reserved .... 11.. .... = Destination Addressing Mode: Address field contains a 64-bit extended address ..00 .... = Reserved 11.. .... = Source Addressing Mode: Address field contains a 64-bit extended address Sequence Number: xx Destination PAN Identifier: 0x1aaa Destination Address: 0xACDE480000000002 Source Address: 0xACDE480000000001 MAC Payload MAC Command = 0x02 = Association Response ShortAddress = 0x3344, AssociationStatus = 0x0x0 = SUCCESS Frame Check Sequence: Correct It is possible to verify using a PAN Analyzer that the DUT issues an ACK frame with the frame pending field set to 0. Frame Length: 5 bytes IEEE 802.15.4 Frame Control: 0x0002 .... .010 = Frame Type: ack (0x0002) .... 0... = Security Enabled: Disabled ....0 .... = Frame Pending: data ....0. .... = Acknowledgment Request: Ack not required .... .0.. .... = Intra-PAN: Not within the PAN .... ..00 0... .... = Reserved .... 00.. .... = Destination Addressing Mode: None ..00 .... = Reserved 00.. .... = Source Addressing Mode: None Sequence Number: xx Frame Check Sequence: Correct The DUT shall return a Success result, such that: DUT returns MAC primitive MLME-ASSOCIATE.confirm (ShortAddress= 0x3344, Status = 0x00 = SUCCESS, SecurityLevel = 0x00, KeyIdMode = xx, KeySource = xx, KeyIndex = xx) Fail: DUT does not send MAC data request command frame DUT does not send ACK in response to Association response frame DUT does not return MLME-ASSOCIATE.confirm primitive ShortAddress is incorrect. Status is not SUCCESS

Notes:

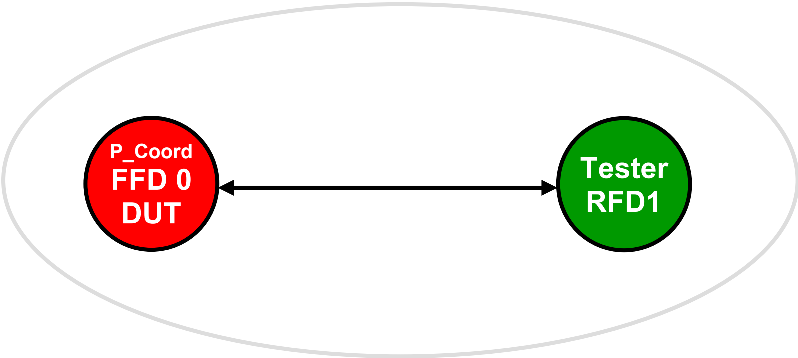
None.

TP/154/MAC/ASSOCIATION-04[1]

Check D.U.T. (as coordinator) correctly receives association request frame (sent to extended address) and generates positive response

Setup Conditions:

Diagram



The Diagram is the image that represents the configuration deployed for the Test Procedure.

Initial Conditions:

Device name	Description
All the devices	*Security: no for all the devices* *Channel: 0x14 for all the devices* *PAN id: 0x1AAA for all the devices*
P_Coord FFD0	PAN Coordinator, Bo = 15, SO = 15 MAC: 0xACDE480000000001 Short Address: 0xfffe (use extended address) Beacon Payload: none DUT as coordinator shall set up a non-beacon enabled PAN.
RFD1	*MAC: 0xACDE480000000002* Short Address: None

Test Procedure:

Step No.	Step description	Expected outcome
1	Set DUT RxOnWhenIdle Set to DUT: MLME-SET.request (PIBAttribute=0x52=82=macRxOnWhenIdle, PIBAttributeValue=TRUE) Set to DUT: MLME-SET.request (PIBAttribute=0x41= macAssociationPermit, PIBAttributeValue=TRUE)	Pass: Obtain MLME-Set.confirm MLME-SET.confirm (Status=SUCCESS, PIBAttribute=0x52=82= macRxOnWhenIdle) Obtain MLME-Set.confirm MLME-SET.confirm (Status=SUCCESS, PIBAttribute=0x41= macAssociationPermit)

2	Set to Tester MLME-ASSOCIATE.request(LogicalChannel =0x14, ChannelPage=0, CoordAddrMode = 0x03 = Extended Address, CoordPANId = 0x1aaa, CoordAddress = 0x0xACDE480000000001, CapabilityInformation = 0x80, SecurityLevel = 0x00, KeyIdMode = xx, KeySource = xx, KeyIndex = xx)	Tester transmits MAC association request command frame . Pass: the DUT generates an Ack in response to the Association Request frame from the Tester. DUT returns MAC Primitive: MLME-ASSOCIATE.indication(DeviceAddress=0xACDE480000000002, CapabilityInformation=0x80, SecurityLevel = 0x00, KeyIdMode = xx, KeySource = xx, KeyIndex = xx) Fail: DUT does not return MLME-ASSOCIATE.indication primitive
3	Set to DUT MLME-ASSOCIATE.response(DeviceAddress =0xACDE480000000002, AssocShortAddress = 0x3344, Status = 0x00 = SUCCESS, SecurityLevel = 0x00, KeyIdMode = xx, KeySource = xx, KeyIndex = xx)	Tester will transmit MAC data request command frame to coordinator, DUT after aResponseWaitTime of receiving the Ack from the DUT. Pass: It is possible to verify by a PAN analyzer that an ACK frame was issued from the DUT with the frame pending field set to 1. Frame Length: 5 bytes IEEE 802.15.4 Frame Control: 0x0012 .... .010 = Frame Type: ack (0x0002) .... 0... = Security Enabled: Disabled ....1 .... = Frame Pending: data ....0. .... = Acknowledgment Request: Ack not required .... .0.. .... = Intra-PAN: Not within the PAN .... ..00 0... .... = Reserved .... 00.. .... = Destination Addressing Mode: None ..00 .... = Reserved 00.. .... = Source Addressing Mode: None Sequence Number: xx Frame Check Sequence: Correct Pass: It is possible to verify by a PAN analyzer that an association response MAC command frame was issued from the DUT. Frame Length: 27 bytes IEEE 802.15.4 Frame Control: 0xCC63 .... .011 = Frame Type: Command (0x0003) .... 0... = Security Enabled: Disabled ....0 .... = Frame Pending: No more data ....1. .... = Acknowledgment Request: Ack required .... .1.. .... = Intra-PAN: Within the PAN .... ..00 0... .... = Reserved .... 11.. .... = Destination Addressing Mode: Address field contains a 64-bit extended address ..00 .... = Reserved 11.. .... = Source Addressing Mode: Address field contains a 64-bit extended address Sequence Number: xx Destination PAN Identifier: 0x1aaa Destination Address: 0xACDE480000000002 Source Address: 0xACDE480000000001 MAC Payload MAC Command = 0x02 = Association Response ShortAddress = 0x3344, AssociationStatus = 0x00 = SUCCESS

		Frame Check Sequence: Correct Tester will transmit Ack frame in response to association response MAC command frame from DUT The DUT shall return a Success result, such that: DUT returns MAC primitive MLME-COMM-STATUS.indication(PANid = 0x1aaa, SrcAddrMode = 0x03 = ExtendedAddress, SrcAddr = 0xACDE480000000001, DstAddrMode = 0x03= ExtendedAddress, DstAddr = 0xACDE480000000002, Status = 0x00 = SUCCESS, SecurityLevel = 0x00, KeyIdMode = xx, KeySource = xx, KeyIndex = xx) Fail: DUT does not send ACK in response to MAC data request command frame ACK does not have Frame pending field set DUT does not send association response frame DUT does not return MLME-COMM-STATUS.indication primitive Status is not SUCCESS

1347 ***Notes:***

1348 *None.*

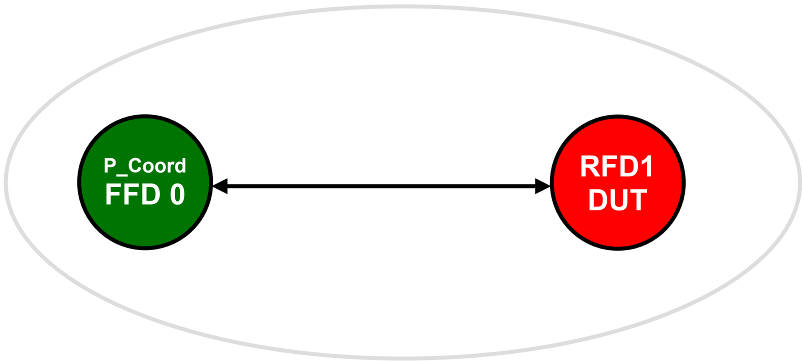
TP/154/MAC/DATA-01

Tests with Tester as PAN Coordinator and with D.U.T. associated as device. Perform the following tests:

a. D.U.T. to tester.: Short Address to Short Address, Unicast, no ACK
b. D.U.T. to tester.: Short Address to Short Address, Unicast, with ACK
c. D.U.T. to tester.: Short Address to Extended Address, Unicast with ACK
d. D.U.T. to tester.: Extended Address to Short Address , Unicast with ACK
e. D.U.T. to tester.: Extended Address to Extended Address , Unicast with ACK
f. D.U.T. to tester.: Extended Address to Broadcast Address
g. D.U.T. to tester.: Short Address to Broadcast Address
h. D.U.T. to tester: Short Address to Short Address, Unicast with ACK – maximum data size
i. Tester to D.U.T: Short Address to Short Address, Indirect, with ACK
j. Tester to D.U.T: Short Address to Extended Address, Indirect, with ACK
k. Tester to D.U.T: Extended Address to Short Address, Indirect, with ACK
l. Tester to D.U.T: Extended Address to Extended Address, Indirect, with ACK
m. Tester to D.U.T: Short Address to Short Address, Indirect, with ACK, maximum data size

Setup Conditions:

Diagram



The Diagram is the image that represents the configuration deployed for the Test Procedure.

Initial Conditions:

Device name	Description
All the devices	*Security: no for all the devices* *Channel: 0x14 for all the devices* *PAN id: 0x1AAA for all the devices*
P_Coord FFD0	PAN Coordinator, Bo = 15, SO = 15 MAC: 0xACDE480000000001 Short Address: 0x1122 Beacon Payload: none Tester as coordinator shall set up a non-beacon enabled PAN, and DUT and Tester shall be associated. Follow the steps to association in TP/154/MAC/ASSOCIATION-03; DUT shall associate with capability 0x80. Data by direct transmission is to be sent from DUT to the Tester and by indirect transmission from the Tester to the DUT
FFD1	*MAC: 0xACDE480000000002* Short Address: 0x3344

Test Procedure:

<table>
<tr><th>Step No.</th><th>Step description</th><th>Expected outcome</th></tr>
<tr><td>1</td><td>Set Tester RxOnWhenIdle
Set to Tester
MLME-SET.request (
PIBAttribute=0x52=82=macRxOnWhenIdle,
PIBAttributeValue=TRUE
)</td><td></td></tr>
<tr><td colspan="3">a. D.U.T. to tester.: Short Address to Short Address, Unicast, no ACK</td></tr>
<tr><td>2</td><td>Set to DUT
MCPS-DATA.request (
SrcAddrMode = 0x02 = ShortAddress,
DstAddrMode=0x02= ShortAddress ,
DstPANId = 0x1aaa,
DstAddr = 0x1122,
msduLength=5,
msdu = 0x00 0x01 0x02 0x03 0x04,
msduHandle = 0x00,
TxOptions = 0x00,
SecurityLevel = 0x00,
KeyIdMode = xx,
KeySource = xx,
KeyIndex = xx
)</td><td>DUT transmits MAC Data frame.

Pass:
It is possible to verify by a PAN analyzer that a data frame was issued from the DUT.
Frame Length: 16 bytes
IEEE 802.15.4
Frame Control: 0x8841
.... .001 = Frame Type: Data (0x0001)
.... 0... = Security Enabled: Disabled
....0 .... = Frame Pending: No more data
....0. .... = Acknowledgment Request: Ack not required
.... .1.. .... = Intra-PAN: Within the PAN
.... ..00 0... .... = Reserved
.... 10.. .... = Destination Addressing Mode: Address field contains a 16-bit short address
..00 .... = Reserved for Zigbee PRO/RF4CE
10.. .... = Source Addressing Mode: Address field contains a 16-bit short address
Sequence Number: xx
Destination PAN Identifier: 0x1aaa
Destination Address: 0x1122
Source Address: 0x3344
MAC Payload
0x00 0x01 0x02 0x03 0x04
Frame Check Sequence: Correct
The DUT shall return a Success result, such that:
DUT returns MAC primitive
MPCS-DATA.confirm(
msduHandle = 0x00,
status = 0x00 = SUCCESS,
Timestamp = xx xx xx
)

Fail:
DUT does not return MCPS-DATA.confirm primitive</td></tr>
<tr><td colspan="3">b. D.U.T. to tester.: Short Address to Short Address, Unicast, with ACK</td></tr>
<tr><td>2</td><td>Set to DUT
MCPS-DATA.request (
SrcAddrMode = 0x02 = ShortAddress,
DstAddrMode=0x02= ShortAddress ,
DstPANId = 0x1aaa,
DstAddr = 0x1122,
msduLength=5,
msdu = 0x01 0x02 0x03 0x04 0x05,
msduHandle = 0x00,
TxOptions = 0x01,
SecurityLevel = 0x00,
KeyIdMode = xx,
KeySource = xx,
KeyIndex = xx
)</td><td>DUT transmits MAC Data frame.

Expected Outcome::
It is possible to verify by a PAN analyzer that a data frame was issued from the DUT.
Frame Length: 16 bytes
IEEE 802.15.4
Frame Control: 0x8861
.... .001 = Frame Type: Data (0x0001)
.... 0... = Security Enabled: Disabled
....0 .... = Frame Pending: No more data
....1. .... = Acknowledgment Request: Ack required
.... .1.. .... = Intra-PAN: Within the PAN
.... ..00 0... .... = Reserved
.... 10.. .... = Destination Addressing Mode: Address field contains a 16-bit short address</td></tr>
</table>

		..00 .... = Reserved for Zigbee PRO/RF4CE 10.. .... = Source Addressing Mode: Address field contains a 16-bit short address Sequence Number: xx Destination PAN Identifier: 0x1aaa Destination Address: 0x1122 Source Address: 0x3344 MAC Payload 0x01 0x02 0x03 0x04 0x05 Frame Check Sequence: Correct Tester will respond with ACK frame. The DUT shall return a Success result, such that: DUT returns MAC primitive MPCS-DATA.confirm(msduHandle = 0x00, status = 0x00 = SUCCESS, Timestamp = xx xx xx) Fail: DUT does not return MCPS-DATA.confirm primitive
c. D.U.T. to tester.: Short Address to Extended Address, Unicast with ACK		
2	Set to DUT MCPS-DATA.request (SrcAddrMode = 0x02 = ShortAddress, DstAddrMode=0x03= ExtendedAddress, DstPANId = 0x1aaa, DstAddr = 0xACDE480000000001, msduLength=5, msdu = 0x10 0x11 0x12 0x13 0x14, msduHandle = 0x09, TxOptions = 0x01, SecurityLevel = 0x00, KeyIdMode = xx, KeySource = xx, KeyIndex = xx)	DUT transmits MAC Data frame. Pass: It is possible to verify by a PAN analyzer that a data frame was issued from the DUT. Frame Length: 22 bytes IEEE 802.15.4 Frame Control: 0x8C61 .... .001 = Frame Type: Data (0x0001) .... 0... = Security Enabled: Disabled ....0 .... = Frame Pending: No more data ....1. .... = Acknowledgment Request: Ack required .... .1.. .... = Intra-PAN: Within the PAN .... ..00 0... .... = Reserved .... 11.. .... = Destination Addressing Mode: Address field contains a 64-bit extended address ..00 .... = Reserved for Zigbee PRO/RF4CE 10.. .... = Source Addressing Mode: Address field contains a 16-bit short address Sequence Number: xx Destination PAN Identifier: 0x1aaa Destination Address: 0xACDE480000000001 Source Address: 0x3344 MAC Payload 0x10 0x11 0x12 0x13 0x14 Frame Check Sequence: Correct The tester sends an ACK frame in response. The DUT shall return a Success result, such that: DUT returns MAC primitive MPCS-DATA.confirm(msduHandle = 0x09, status = 0x00 = SUCCESS, Timestamp = xx xx xx) Fail: DUT does not return MCPS-DATA.confirm primitive
d. D.U.T. to tester.: Extended Address to Short Address , Unicast with ACK		
2	Set to DUT	DUT transmits MAC Data frame.

	MCPS-DATA.request (SrcAddrMode = 0x03 = ExtendedAddress, DstAddrMode=0x02= ShortAddress , DstPANId = 0x1aaa, DstAddr = 0x1122, msduLength=5, msdu = 0x20 0x21 0x22 0x23 0x24, msduHandle = 0x00, TxOptions = 0x01, SecurityLevel = 0x00, KeyIdMode = xx, KeySource = xx, KeyIndex = xx)	Pass: It is possible to verify by a PAN analyzer that a data frame was issued from the DUT. Frame Length: 22 bytes IEEE 802.15.4 Frame Control: 0xC861 .... .001 = Frame Type: Data (0x0001) .... 0... = Security Enabled: Disabled ....0 .... = Frame Pending: No more data ....1. .... = Acknowledgment Request: Ack required .... .1.. .... = Intra-PAN: Within the PAN .... ..00 0... .... = Reserved .... 10.. .... = Destination Addressing Mode: Address field contains a 16-bit short address ..00 .... = Reserved for Zigbee PRO/RF4CE 11.. .... = Source Addressing Mode: Address field contains a 64-bit extended address Sequence Number: xx Destination PAN Identifier: 0x1aaa Destination Address: 0x1122 Source Address: 0xACDE480000000002 MAC Payload 0x20 0x21 0x22 0x23 0x24 Frame Check Sequence: Correct Tester will respond with ACK frame. The DUT shall return a Success result, such that: DUT returns MAC primitive MPCS-DATA.confirm(msduHandle = 0x00, status = 0x00 = SUCCESS, Timestamp = xx xx xx) Fail: DUT does not return MCPS-DATA.confirm primitive
e. D.U.T. to tester.: Extended Address to Extended Address , Unicast with ACK		
2	Set to DUT MCPS-DATA.request (SrcAddrMode = 0x03 = ExtendedAddress, DstAddrMode=0x03= ExtendedAddress, DstPANId = 0x1aaa, DstAddr = 0xACDE480000000001, msduLength=5, msdu = 0x17 0x28 0x39 0x4a 0x5b, msduHandle = 0x00, TxOptions = 0x01, SecurityLevel = 0x00, KeyIdMode = xx, KeySource = xx, KeyIndex = xx)	DUT transmits MAC Data frame. Pass: It is possible to verify by a PAN analyzer that a data frame was issued from the DUT. Frame Length: 28 bytes IEEE 802.15.4 Frame Control: 0xCC61 .... .001 = Frame Type: Data (0x0001) .... 0... = Security Enabled: Disabled ....0 .... = Frame Pending: No more data ....1. .... = Acknowledgment Request: Ack required .... .1.. .... = Intra-PAN: Within the PAN .... ..00 0... .... = Reserved .... 11.. .... = Destination Addressing Mode: Address field contains a a 64-bit extended address ..00 .... = Reserved for Zigbee PRO/RF4CE 11.. .... = Source Addressing Mode: Address field contains a 64-bit extended address Sequence Number: xx Destination PAN Identifier: 0x1aaa Destination Address: : 0xACDE480000000001 Source Address: 0xACDE480000000002 MAC Payload 0x17 0x28 0x39 0x4a 0x5b Frame Check Sequence: Correct

		Tester will respond with ACK frame. The DUT shall return a Success result, such that: DUT returns MAC primitive MPCS-DATA.confirm(msduHandle = 0x00, status = 0x00 = SUCCESS, Timestamp = xx xx xx) Fail: DUT does not return MCPS-DATA.confirm primitive
f. D.U.T. to tester.: Extended Address to Broadcast Address		
2	Set to DUT MCPS-DATA.request (SrcAddrMode = 0x03 = Extended address, DstAddrMode=0x02= ShortAddress , DstPANId = 0x1aaa, DstAddr = 0xffff, msduLength=5, msdu = 0x6c 0x7d 0x08e 0x9f 0xa0, msduHandle = 0x00, TxOptions = 0x00, SecurityLevel = 0x00, KeyIdMode = xx, KeySource = xx, KeyIndex = xx)	DUT transmits MAC Data frame. Pass: It is possible to verify by a PAN analyzer that a data frame was issued from the DUT. Frame Length: 22 bytes IEEE 802.15.4 Frame Control: 0xC841 .... .001 = Frame Type: Data (0x0001) .... 0... = Security Enabled: Disabled ....0 .... = Frame Pending: No more data ....0. .... = Acknowledgment Request: Ack not required .... .1.. .... = Intra-PAN: Within the PAN .... ..00 0... .... = Reserved .... 10.. .... = Destination Addressing Mode: Address field contains a 16-bit short address ..00 .... = Reserved for Zigbee PRO/RF4CE 11.. .... = Source Addressing Mode: Address field contains a 64-bit extended address Sequence Number: xx Destination PAN Identifier: 0x1aaa Destination Address: 0xffff Source Address: 0xACDE480000000002 MAC Payload 0x6c 0x7d 0x8e 0x09f 0xa0 Frame Check Sequence: Correct The DUT shall return a Success result, such that: DUT returns MAC primitive MPCS-DATA.confirm(msduHandle = 0x00, status = 0x00 = SUCCESS, Timestamp = xx xx xx) Fail: DUT does not return MCPS-DATA.confirm primitive
g. D.U.T. to tester.: Short Address to Broadcast Address		
2	Set to DUT MCPS-DATA.request (SrcAddrMode = 0x02 = ShortAddress, DstAddrMode=0x02= ShortAddress , DstPANId = 0x1aaa, DstAddr = 0xffff, msduLength=5, msdu = 0x00 0x01 0x02 0x03 0x04, msduHandle = 0x00, TxOptions = 0x00, SecurityLevel = 0x00, KeyIdMode = xx, KeySource = xx,	DUT transmits MAC Data frame. Pass: It is possible to verify by a PAN analyzer that a data frame was issued from the DUT. Frame Length: 16 bytes IEEE 802.15.4 Frame Control: 0x8841 .... .001 = Frame Type: Data (0x0001) .... 0... = Security Enabled: Disabled ....0 .... = Frame Pending: No more data ....0. .... = Acknowledgment Request: Ack not required .... .1.. .... = Intra-PAN: Within the PAN

	KeyIndex = xx)	.... ..00 0... .... = Reserved .... 10.. .... = Destination Addressing Mode: Address field contains a 16-bit short address ..00 .... = Reserved for Zigbee PRO/RF4CE 10.. .... = Source Addressing Mode: Address field contains a 16-bit short address Sequence Number: xx Destination PAN Identifier: 0x1aaa Destination Address: 0xffff Source Address: 0x3344 MAC Payload 0x00 0x01 0x02 0x03 0x04 Frame Check Sequence: Correct The DUT shall return a Success result, such that: DUT returns MAC primitive MPCS-DATA.confirm(msduHandle = 0x00, status = 0x00 = SUCCESS, Timestamp = xx xx xx) Fail: DUT does not return MCPS-DATA.confirm primitive
h. D.U.T. to tester: Short Address to Short Address, Unicast with ACK – maximum data size		
2	Set to DUT MCPS-DATA.request (SrcAddrMode = 0x02 = ShortAddress, DstAddrMode=0x02= ShortAddress , DstPANId = 0x1aaa, DstAddr = 0x1122, msduLength=0x74, (max length) msdu = 0x00 0x01 0x02 0x03 0x04 0x05 0x06 0x07 0x08 0x09 0x0a 0x0b 0x0c 0x0d 0x0e 0x0f 0x10 0x11 0x12 0x13 0x14 0x15 0x16 0x17 0x18 0x19 0x1a 0x1b 0x1c 0x1d 0x1e 0x1f 0x20 0x21 0x22 0x23 0x24 0x25 0x26 0x27 0x28 0x29 0x2a 0x2b 0x2c 0x2d 0x2e 0x2f 0x30 0x31 0x32 0x33 0x34 0x35 0x36 0x37 0x38 0x39 0x3a 0x3b 0x3c 0x3d 0x3e 0x3f 0x40 0x41 0x42 0x43 0x44 0x45 0x46 0x47 0x48 0x49 0x4a 0x4b 0x4c 0x4d 0x4e 0x4f 0x50 0x51 0x52 0x53 0x54 0x55 0x56 0x57 0x58 0x59 0x5a 0x5b 0x5c 0x5d 0x5e 0x5f 0x60 0x61 0x62 0x63 0x64 0x65 0x66 0x67 0x68 0x69 0x6a 0x6b 0x6c 0x6d 0x6e 0x6f 0x70 0x71 0x72 0x73 msduHandle = 0x00, TxOptions = 0x01, SecurityLevel = 0x00, KeyIdMode = xx, KeySource = xx, KeyIndex = xx)	DUT transmits MAC Data frame. Expected Outcome: It is possible to verify by a PAN analyzer that a data frame was issued from the DUT. Frame Length: 127 bytes IEEE 802.15.4 Frame Control: 0x8861 .... .001 = Frame Type: Data (0x0001) .... 0... = Security Enabled: Disabled ....0 .... = Frame Pending: No more data ....1. .... = Acknowledgment Request: Ack required .... .1.. .... = Intra-PAN: Within the PAN .... ..00 0... .... = Reserved .... 10.. .... = Destination Addressing Mode: Address field contains a 16-bit short address ..00 .... = Reserved for Zigbee PRO/RF4CE 10.. .... = Source Addressing Mode: Address field contains a 16-bit short address Sequence Number: xx Destination PAN Identifier: 0x1aaa Destination Address: 0x1122 Source Address: 0x3344 MAC Payload 0x00 0x01 0x02 0x03 0x04 0x05 0x06 0x07 0x08 0x09 0x0a 0x0b 0x0c 0x0d 0x0e 0x0f 0x10 0x11 0x12 0x13 0x14 0x15 0x16 0x17 0x18 0x19 0x1a 0x1b 0x1c 0x1d 0x1e 0x1f 0x20 0x21 0x22 0x23 0x24 0x25 0x26 0x27 0x28 0x29 0x2a 0x2b 0x2c 0x2d 0x2e 0x2f 0x30 0x31 0x32 0x33 0x34 0x35 0x36 0x37 0x38 0x39 0x3a 0x3b 0x3c 0x3d 0x3e 0x3f 0x40 0x41 0x42 0x43 0x44 0x45 0x46 0x47 0x48 0x49 0x4a 0x4b 0x4c 0x4d 0x4e 0x4f 0x50 0x51 0x52 0x53 0x54 0x55 0x56 0x57 0x58 0x59 0x5a 0x5b 0x5c 0x5d 0x5e 0x5f 0x60 0x61 0x62 0x63 0x64 0x65 0x66 0x67 0x68 0x69 0x6a 0x6b 0x6c 0x6d 0x6e 0x6f 0x70 0x71 0x72 0x73 Frame Check Sequence: Correct

		Tester will respond with ACK frame. The DUT shall return a Success result, such that: DUT returns MAC primitive MPCS-DATA.confirm(msduHandle = 0x00, status = 0x00 = SUCCESS, Timestamp = xx xx xx) Fail: DUT does not return MCPS-DATA.confirm primitive
i. Tester to D.U.T: Short Address to Short Address, Indirect, with ACK		
2	Set to Tester MCPS-DATA.request (SrcAddrMode = 0x02 = ShortAddress, DstAddrMode=0x02= ShortAddress, DstPANId = 0x1aaa, DstAddr = 0x3344, msduLength=5, msdu = 0x00 0x01 0x02 0x03 0x04, msduHandle = 0x0d, TxOptions = 0x05 = Indirect transmission, ACK SecurityLevel = 0x00, KeyIdMode = xx, KeySource = xx, KeyIndex = xx)	
3	Set to DUT MLME-POLL.request (CoordAddrMode = 0x02 = ShortAddress, CoordPANId = 0x1aaa, CoordAddress = 0x1122, SecurityLevel = 0x00, KeyIdMode = xx, KeySource = xx, KeyIndex = xx)	Pass: It is possible to verify by a PAN analyzer that the DUT transmits a MAC data request command frame to coordinator. Frame Length: 12 bytes IEEE 802.15.4 Frame Control: 0x8863 .... .011 = Frame Type: Command (0x0003) .... 0... = Security Enabled: Disabled ....0 .... = Frame Pending: No more data ....1. .... = Acknowledgment Request: ACK required .... .1.. .... = Intra-PAN: Within the PAN .... ..00 0... .... = Reserved .... 10.. .... = Destination Addressing Mode: Address field contains a 16-bit short address ..00 .... = Reserved for Zigbee PRO/RF4CE 10.. .... = Source Addressing Mode: Address field contains a 16-bit short address Sequence Number: xx Destination PAN Identifier: 0x1aaa Destination Address: 0x1122 Source Address: 0x3344 MAC Payload Command Frame Identifier = Data Request: (0x04) Frame Check Sequence: Correct
4	The Tester will respond with an Acknowledgement frame with the frame pending field set to 1. The tester will then issue a data frame to the DUT.	Pass: The DUT shall receive the data frame and send to the next higher layer such that: DUT returns MAC Data indication primitive: MCPS-DATA.indication(SrcAddrMode = 0x02 = ShortAddress, SrcPANId =0x1aaa, SrcAddr= 0x1122,

		DstAddrMode=0x02 = ShortAddress, DstPANId = 0x1aaa, DstAddr = 0x3344, msduLength=5, msdu = 0x00 0x01 0x02 0x03 0x04, mpduLinkQuality = xx, DSN = xx, Timestamp = xx xx xx, SecurityLevel = 0x00, KeyIdMode = xx, KeySource = xx, KeyIndex = xx) The DUT will transmit an ACK frame. Fail: DUT does not return MCPS-DATA.indication primitive
j. Tester to D.U.T: Short Address to Extended Address, Indirect, with ACK		
2	Set to Tester MCPS-DATA.request (SrcAddrMode = 0x02 = ShortAddress, DstAddrMode=0x03= ExtendedAddress, DstPANId = 0x1aaa, DstAddr = 0xACDE480000000002, msduLength=5, msdu = 0x00 0x01 0x02 0x03 0x04, msduHandle = 0x0d, TxOptions = 0x05 = Indirect transmission, ACK SecurityLevel = 0x00, KeyIdMode = xx, KeySource = xx, KeyIndex = xx)	
3	*Note: First set device to poll using extended address.* Set to DUT MLME-SET.request(ShortAddress = 0x53, Index = 0 Value=0xfffe) MLME-POLL.request (CoordAddrMode = 0x02 = ShortAddress, CoordPANId = 0x1aaa, CoordAddress = 0x1122, SecurityLevel = 0x00, KeyIdMode = xx, KeySource = xx, KeyIndex = xx)	Pass: It is possible to verify by a PAN analyzer that the DUT transmits a MAC data request command frame to coordinator. Frame Length: 18 bytes IEEE 802.15.4 Frame Control: 0xC863 .... .011 = Frame Type: Command (0x0003) .... 0... = Security Enabled: Disabled ....0 .... = Frame Pending: No more data ....1. .... = Acknowledgment Request: ACK required .... .1.. .... = Intra-PAN: Within the PAN .... ..00 0... .... = Reserved .... 10.. .... = Destination Addressing Mode: Address field contains a 16-bit short address ..00 .... = Reserved for Zigbee PRO/RF4CE 11.. .... = Source Addressing Mode: Address field contains a 64-bit extended address Sequence Number: xx Destination PAN Identifier: 0x1aaa Destination Address: 0xACDE480000000002 Source Address: 0x1122 MAC Payload Command Frame Identifier = Data Request: (0x04) Frame Check Sequence: Correct
4	The Tester will respond with an Acknowledgement frame with the frame pending field set to 1.	Pass: The DUT shall receive the data frame and send to the next higher layer such that:

	The tester will then issue a data frame to the DUT.	DUT returns MAC Data indication primitive: MCPS-DATA.indication(SrcAddrMode = 0x02 = ShortAddress, SrcPANId =0x1aaa, SrcAddr= 0x1122, DstAddrMode=0x03 = ExtendedAddress, DstPANId = 0x1aaa, DstAddr = 0xACDE480000000002, msduLength=5, msdu = 0x00 0x01 0x02 0x03 0x04, mpduLinkQuality = xx, DSN = xx, Timestamp = xx xx xx, SecurityLevel = 0x00, KeyIdMode = xx, KeySource = xx, KeyIndex = xx) The DUT will transmit an ACK frame. Fail: DUT does not return MCPS-DATA.indication primitive
k. Tester to D.U.T: Extended Address to Short Address, Indirect, with ACK		
2	Set to Tester MCPS-DATA.request (SrcAddrMode = 0x03 = ExtendedAddress, DstAddrMode=0x02= ShortAddress, DstPANId = 0x1aaa, DstAddr = 0x3344, msduLength=5, msdu = 0x00 0x01 0x02 0x03 0x04, msduHandle = 0x0d, TxOptions = 0x05 = Indirect transmission, ACK SecurityLevel = 0x00, KeyIdMode = xx, KeySource = xx, KeyIndex = xx)	
3	*Note: First set the DUT back to using Short Address* Set to DUT MLME-SET.request(ShortAddress = 0x53, Index = 0 Value=0x3344) MLME-POLL.request (CoordAddrMode = 0x03 = ShortAddress, CoordPANId = 0x1aaa, CoordAddress = 0xACDE480000000001, SecurityLevel = 0x00, KeyIdMode = xx, KeySource = xx, KeyIndex = xx)	Pass: It is possible to verify by a PAN analyzer that the DUT transmits a MAC data request command frame to coordinator. Frame Length: 12 bytes IEEE 802.15.4 Frame Control: 0x8C63 .... .011 = Frame Type: Command (0x0003) .... 0... = Security Enabled: Disabled ....0 .... = Frame Pending: No more data ....1. .... = Acknowledgment Request: ACK required .... .1.. .... = Intra-PAN: Within the PAN .... ..00 0... .... = Reserved .... 11.. .... = Destination Addressing Mode: Address field contains a 64 bit extended address ..00 .... = Reserved for Zigbee PRO/RF4CE 10.. .... = Source Addressing Mode: Address field contains a 16-bit short address Sequence Number: xx Destination PAN Identifier: 0x1aaa Destination Address: 0xACDE480000000001 Source Address: 0x3344 MAC Payload Command Frame Identifier = Data Request: (0x04) Frame Check Sequence: Correct

4	The Tester will respond with an Acknowledgement frame with the frame pending field set to 1. The tester will then issue a data frame to the DUT.	Pass: The DUT shall receive the data frame and send to the next higher layer such that: DUT returns MAC Data indication primitive: MCPS-DATA.indication(SrcAddrMode = 0x03 = ExtendedAddress, SrcPANId =0x1aaa, SrcAddr= : 0xACDE480000000001,, DstAddrMode=0x02 = ShortAddress, DstPANId = 0x1aaa, DstAddr = 0x3344, msduLength=5, msdu = 0x00 0x01 0x02 0x03 0x04, mpduLinkQuality = xx, DSN = xx, Timestamp = xx xx xx, SecurityLevel = 0x00, KeyIdMode = xx, KeySource = xx, KeyIndex = xx) The DUT will transmit an ACK frame. Fail: DUT does not return MCPS-DATA.indication primitive
l. Tester to D.U.T: Extended Address to Extended Address, Indirect, with ACK		
2	Set to Tester MCPS-DATA.request (SrcAddrMode = 0x03 = ExtendedAddress, DstAddrMode=0x03= ExtendedAddress, DstPANId = 0x1aaa, DstAddr = 0xACDE480000000002, msduLength=5, msdu = 0x00 0x01 0x02 0x03 0x04, msduHandle = 0x0d, TxOptions = 0x05 = Indirect transmission, ACK SecurityLevel = 0x00, KeyIdMode = xx, KeySource = xx, KeyIndex = xx)	
3	Set to DUT MLME-SET.request(ShortAddress = 0x53, Index = 0 Value=0xfffe) MLME-POLL.request (CoordAddrMode = 0x03 = ShortAddress, CoordPANId = 0x1aaa, CoordAddress = 0xACDE480000000001, SecurityLevel = 0x00, KeyIdMode = xx, KeySource = xx, KeyIndex = xx)	Pass: It is possible to verify by a PAN analyzer that the DUT transmits a MAC data request command frame to coordinator. Frame Length: 12 bytes IEEE 802.15.4 Frame Control: 0xCC63 .... .011 = Frame Type: Command (0x0003) .... 0... = Security Enabled: Disabled ....0 .... = Frame Pending: No more data ....1. .... = Acknowledgment Request: ACK required

		.... .1.. .... = Intra-PAN: Within the PAN .... ..00 0... .... = Reserved .... 11.. .... = Destination Addressing Mode: Address field contains a 64 bit extended address ..00 .... = Reserved for Zigbee PRO/RF4CE 11.. .... = Source Addressing Mode: Address field contains a 64-bit extended address Sequence Number: xx Destination PAN Identifier: 0x1aaa Destination Address: 0xACDE480000000001 Source Address: 0xACDE480000000002 MAC Payload Command Frame Identifier = Data Request: (0x04) Frame Check Sequence: Correct
4	The Tester will respond with an Acknowledgement frame with the frame pending field set to 1. The tester will then issue a data frame to the DUT.	Pass: The DUT shall receive the data frame and send to the next higher layer such that: DUT returns MAC Data indication primitive: MCPS-DATA.indication(SrcAddrMode = 0x03 = ExtendedAddress, SrcPANId =0x1aaa, SrcAddr= : 0xACDE480000000001,, DstAddrMode=0x02 = ExtendedAddress, DstPANId = 0x1aaa, DstAddr = 0xACDE480000000002, msduLength=5, msdu = 0x00 0x01 0x02 0x03 0x04, mpduLinkQuality = xx, DSN = xx, Timestamp = xx xx xx, SecurityLevel = 0x00, KeyIdMode = xx, KeySource = xx, KeyIndex = xx) The DUT will transmit an ACK frame. Fail: DUT does not return MCPS-DATA.indication primitive
m. Tester to D.U.T: Short Address to Short Address, Indirect, with ACK, maximum data size		
2	Set to Tester MCPS-DATA.request (SrcAddrMode = 0x02 = ShortAddress, DstAddrMode=0x02= ShortAddress, DstPANId = 0x1aaa, DstAddr = 0x3344, msduLength=0x74, (max length) msdu = 0x00 0x01 0x02 0x03 0x04 0x05 0x06 0x07 0x08 0x09 0x0a 0x0b 0x0c 0x0d 0x0e 0x0f 0x10 0x11 0x12 0x13 0x14 0x15 0x16 0x17 0x18 0x19 0x1a 0x1b 0x1c 0x1d 0x1e 0x1f 0x20 0x21 0x22 0x23 0x24 0x25 0x26 0x27 0x28 0x29 0x2a 0x2b 0x2c 0x2d 0x2e 0x2f 0x30 0x31 0x32 0x33 0x34 0x35 0x36 0x37 0x38 0x39 0x3a 0x3b 0x3c 0x3d 0x3e 0x3f 0x40 0x41 0x42 0x43 0x44 0x45 0x46 0x47 0x48 0x49 0x4a 0x4b 0x4c 0x4d 0x4e 0x4f 0x50 0x51 0x52 0x53 0x54 0x55 0x56 0x57 0x58 0x59 0x5a 0x5b 0x5c 0x5d 0x5e 0x5f 0x60 0x61 0x62 0x63 0x64 0x65 0x66 0x67 0x68 0x69 0x6a 0x6b 0x6c 0x6d 0x6e 0x6f	

	0x70 0x71 0x72 0x73, msduHandle = 0x0d, TxOptions = 0x05 = Indirect transmission, ACK SecurityLevel = 0x00, KeyIdMode = xx, KeySource = xx, KeyIndex = xx)	
3	Set to DUT MLME-POLL.request (CoordAddrMode = 0x02 = ShortAddress, CoordPANId = 0x1aaa, CoordAddress = 0x1122, SecurityLevel = 0x00, KeyIdMode = xx, KeySource = xx, KeyIndex = xx)	Pass: It is possible to verify by a PAN analyzer that the DUT transmits a MAC data request command frame to coordinator. Frame Length: 12 bytes IEEE 802.15.4 Frame Control: 0x8863 .... .011 = Frame Type: Command (0x0003) .... 0... = Security Enabled: Disabled ....0 .... = Frame Pending: No more data ....1. .... = Acknowledgment Request: ACK required .... .1.. .... = Intra-PAN: Within the PAN .... ..00 0... .... = Reserved .... 10.. .... = Destination Addressing Mode: Address field contains a 16-bit short address ..00 .... = Reserved for Zigbee PRO/RF4CE 10.. .... = Source Addressing Mode: Address field contains a 16-bit short address Sequence Number: xx Destination PAN Identifier: 0x1aaa Destination Address: 0x1122 Source Address: 0x3344 MAC Payload Command Frame Identifier = Data Request: (0x04) Frame Check Sequence: Correct
4	The Tester will respond with an Acknowledgement frame with the frame pending field set to 1. The tester will then issue a data frame to the DUT.	Pass: The DUT shall receive the data frame and send to the next higher layer such that: DUT returns MAC Data indication primitive: MCPS-DATA.indication(SrcAddrMode = 0x02 = ShortAddress, SrcPANId =0x1aaa, SrcAddr= 0x1122, DstAddrMode=0x02 = ShortAddress, DstPANId = 0x1aaa, DstAddr = 0x3344, msduLength=0x74, (max length) msdu = 0x00 0x01 0x02 0x03 0x04 0x05 0x06 0x07 0x08 0x09 0x0a 0x0b 0x0c 0x0d 0x0e 0x0f 0x10 0x11 0x12 0x13 0x14 0x15 0x16 0x17 0x18 0x19 0x1a 0x1b 0x1c 0x1d 0x1e 0x1f 0x20 0x21 0x22 0x23 0x24 0x25 0x26 0x27 0x28 0x29 0x2a 0x2b 0x2c 0x2d 0x2e 0x2f 0x30 0x31 0x32 0x33 0x34 0x35 0x36 0x37 0x38 0x39 0x3a 0x3b 0x3c 0x3d 0x3e 0x3f 0x40 0x41 0x42 0x43 0x44 0x45 0x46 0x47 0x48 0x49 0x4a 0x4b 0x4c 0x4d 0x4e 0x4f 0x50 0x51 0x52 0x53 0x54 0x55 0x56 0x57 0x58 0x59 0x5a 0x5b 0x5c 0x5d 0x5e 0x5f 0x60 0x61 0x62 0x63 0x64 0x65 0x66 0x67 0x68 0x69 0x6a 0x6b 0x6c 0x6d 0x6e 0x6f 0x70 0x71 0x72 0x73, mpduLinkQuality = xx, DSN = xx, Timestamp = xx xx xx, SecurityLevel = 0x00, KeyIdMode = xx,

		KeySource = xx, KeyIndex = xx) The DUT will transmit an ACK frame. Fail: DUT does not return MCPS-DATA.indication primitive

Notes:

None.

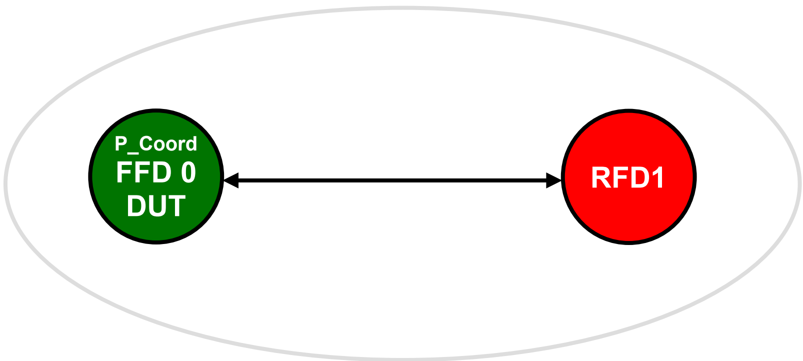
TP/154/MAC/DATA-02

Tests with DUT as coordinator for Intra-PAN, and Tester as device associated to DUT. Perform the following tests:

Tester to D.U.T.: Short Address to Short Address, Unicast, no ACK.
Tester to D.U.T.: Short Address to Short Address, Unicast, with ACK .
Tester to D.U.T.: Short Address to Extended Address, Unicast with ACK .
Tester to D.U.T.: Extended Address to Short Address , Unicast with ACK.
Tester to D.U.T.: Extended Address to Extended Address , Unicast with ACK.
Tester to D.U.T.: Extended Address to Broadcast Address.
Tester to D.U.T.: Short Address to Broadcast Address.
D.U.T. to tester: Short Address to Short Address, Indirect, with ACK.
D.U.T. to tester: Short Address to Extended Address, Indirect, with ACK.
D.U.T. to tester: Extended Address to Short Address, Indirect, with ACK.
D.U.T. to tester: Extended Address to Extended Address, Indirect, with ACK.
D.U.T. to tester: Short Address to Short Address, Indirect, with ACK (Tester does NOT poll, hence transaction expires).
D.U.T. to tester: Short Address to Short Address, Indirect, with ACK (Tester does NOT poll, DUT purges).

Setup Conditions:

Diagram



The Diagram is the image that represents the configuration deployed for the Test Procedure.

Initial Conditions:

Device name	Description
All the devices	*Security: no for all the devices* *Channel: 0x14 for all the devices* *PAN id: 0x1AAA for all the devices*
P_Coord FFD0	PAN Coordinator, Bo = 15, SO = 15 MAC: 0xACDE480000000001 Short Address: 0x1122 Beacon Payload: none DUT as coordinator shall set up a non-beacon enabled PAN, and DUT and Tester shall be associated. Follow the steps to association in TP/154/MAC/ASSOCIATION-01; Tester shall associate with capability 0x80
RFD1	*MAC: 0xACDE480000000002* Short Address: 0x3344

Test Procedure:

Step No.	Step description	Expected outcome
1	Set DUT RxOnWhenIdle Set to DUT: MLME-SET.request (PIBAttribute=0x52=82=macRxOnWhenIdle, PIBAttributeValue=TRUE)	Obtain MLME-Set.confirm MLME-SET.confirm (Status=SUCCESS, PIBAttribute=0x52=82= macRxOnWhenIdle)
Tester to D.U.T.: Short Address to Short Address, Unicast, no ACK		
2	Set to Tester MCPS-DATA.request (SrcAddrMode = 0x02 = ShortAddress, DstAddrMode=0x02 = ShortAddress, DstPANId = 0x1aaa, DstAddr = 0x1122, msduLength=5, msdu = 0x00 0x01 0x02 0x03 0x04, msduHandle = 0x00, TxOptions = 0x00, SecurityLevel = 0x00, KeyIdMode = xx, KeySource = xx, KeyIndex = xx)	Tester transmits MAC Data frame. Pass: DUT returns MAC Primitive: MCPS-DATA.indication(SrcAddrMode = 0x02 = ShortAddress, SrcPANId =0x1aaa, SrcAddr= 0x3344, DstAddrMode=0x02 = ShortAddress, DstPANId = 0x1aaa, DstAddr = 0x1122, msduLength=5, msdu = 0x00 0x01 0x02 0x03 0x04, mpduLinkQuality = xx, DSN = xx, Timestamp = xx xx xx, SecurityLevel = 0x00, KeyIdMode = xx, KeySource = xx, KeyIndex = xx) Fail: DUT transmits ACK frame DUT does not return MCPS-DATA.indication primitive
3	Tester returns MCPS-DATA.confirm(msduHandle = 0x00, Status = SUCCESS)	
Tester to D.U.T.: Short Address to Short Address, Unicast, with ACK		
2	Set to Tester MCPS-DATA.request (SrcAddrMode = 0x02 = ShortAddress, DstAddrMode=0x02 = ShortAddress, DstPANId = 0x1aaa, DstAddr = 0x1122, msduLength=5, msdu = 0x00 0x01 0x02 0x03 0x04, msduHandle = 0x00, TxOptions = 0x01, SecurityLevel = 0x00, KeyIdMode = xx, KeySource = xx, KeyIndex = xx)	Tester transmits MAC Data frame. Pass: DUT transmits ACK frame DUT returns MAC Primitive: MCPS-DATA.indication(SrcAddrMode = 0x02 = ShortAddress, SrcPANId =0x1aaa, SrcAddr= 0x3344, DstAddrMode=0x02 = ShortAddress, DstPANId = 0x1aaa, DstAddr = 0x1122, msduLength=5, msdu = 0x00 0x01 0x02 0x03 0x04, mpduLinkQuality = xx, DSN = xx, Timestamp = xx xx xx, SecurityLevel = 0x00, KeyIdMode = xx, KeySource = xx, KeyIndex = xx

		) Fail: DUT does not transmit ACK frame DUT transmits ACK frame at incorrect time DUT does not return MCPS-DATA.indication primitive
3	Tester returns MCPS-DATA.confirm(msduHandle = 0x00, Status = SUCCESS)	
Tester to D.U.T.: Short Address to Extended Address, Unicast with ACK		
2	Set to Tester MCPS-DATA.request (SrcAddrMode = 0x02 = ShortAddress, DstAddrMode=0x03= ExtendedAddress, DstPANId = 0x1aaa, DstAddr = 0xACDE480000000001, msduLength=5, msdu = 0x00 0x01 0x02 0x03 0x04, msduHandle = 0x00, TxOptions = 0x01, SecurityLevel = 0x00, KeyIdMode = xx, KeySource = xx, KeyIndex = xx)	Tester transmits MAC Data frame. Pass: DUT transmits ACK frame DUT returns MAC Primitive: MCPS-DATA.indication(SrcAddrMode = 0x02 = ShortAddress, SrcPANId =0x1aaa, SrcAddr= 0x3344, DstAddrMode=0x03 = ExtendedAddress, DstPANId = 0x1aaa, DstAddr = 0xACDE480000000001, msduLength=5, msdu = 0x00 0x01 0x02 0x03 0x04, mpduLinkQuality = xx, DSN = xx, Timestamp = xx xx xx, SecurityLevel = 0x00, KeyIdMode = xx, KeySource = xx, KeyIndex = xx) Fail: DUT does not return MCPS-DATA.indication primitive
3	Tester returns MCPS-DATA.confirm(msduHandle = 0x00, Status = SUCCESS)	
Tester to D.U.T.: Extended Address to Short Address , Unicast with ACK		
2	Set to Tester MCPS-DATA.request (SrcAddrMode = 0x03 = ExtendedAddress, DstAddrMode=0x02= ShortAddress, DstPANId = 0x1aaa, DstAddr = 0x1122, msduLength=5, msdu = 0x00 0x01 0x02 0x03 0x04, msduHandle = 0x00, TxOptions = 0x01, SecurityLevel = 0x00, KeyIdMode = xx, KeySource = xx, KeyIndex = xx)	Tester transmits MAC Data frame. Pass: DUT transmits ACK frame DUT returns MAC Primitive: MCPS-DATA.indication(SrcAddrMode = 0x03 = ExtendedAddress, SrcPANId =0x1aaa, SrcAddr= 0xACDE480000000002, DstAddrMode=0x02 = ShortAddress, DstPANId = 0x1aaa, DstAddr =0x1122, msduLength=5, msdu = 0x00 0x01 0x02 0x03 0x04, mpduLinkQuality = xx, DSN = xx, Timestamp = xx xx xx,

		SecurityLevel = 0x00, KeyIdMode = xx, KeySource = xx, KeyIndex = xx) Fail: DUT does not send ACK DUT does not send ACK at correct time DUT does not return MCPS-DATA.indication primitive
3	Tester returns MCPS-DATA.confirm(msduHandle = 0x00, Status = SUCCESS)	
Tester to D.U.T.: Extended Address to Extended Address , Unicast with ACK		
2	Set to Tester MCPS-DATA.request (SrcAddrMode = 0x03 = ExtendedAddress, DstAddrMode=0x03= ExtendedAddress, DstPANId = 0x1aaa, DstAddr = 0xACDE480000000001, msduLength=5, msdu = 0x00 0x01 0x02 0x03 0x04, msduHandle = 0x00, TxOptions = 0x01, SecurityLevel = 0x00, KeyIdMode = xx, KeySource = xx, KeyIndex = xx)	Tester transmits MAC Data frame. Pass: DUT transmits ACK frame DUT returns MAC Primitive: MCPS-DATA.indication(SrcAddrMode = 0x03 = ExtendedAddress, SrcPANId =0x1aaa, SrcAddr= 0xACDE480000000002, DstAddrMode=0x03 = ExtendedAddress, DstPANId = 0x1aaa, DstAddr =0xACDE480000000001, msduLength=5, msdu = 0x00 0x01 0x02 0x03 0x04, mpduLinkQuality = xx, DSN = xx, Timestamp = xx xx xx, SecurityLevel = 0x00, KeyIdMode = xx, KeySource = xx, KeyIndex = xx) Fail: DUT does not send ACK DUT does not send ACK at correct time DUT does not return MCPS-DATA.indication primitive
3	Tester returns MCPS-DATA.confirm(msduHandle = 0x00, Status = SUCCESS)	
Tester to D.U.T.: Extended Address to Broadcast Address		
2	Set to Tester MCPS-DATA.request (SrcAddrMode = 0x03 = ExtendedAddress, DstAddrMode=0x02= ShortAddress, DstPANId = 0x1aaa, DstAddr = 0xffff, msduLength=5, msdu = 0x00 0x01 0x02 0x03 0x04, msduHandle = 0x00, TxOptions = 0x00, SecurityLevel = 0x00, KeyIdMode = xx,	Tester transmits MAC Data frame. Pass: DUT returns MAC Primitive: MCPS-DATA.indication(SrcAddrMode = 0x03 = ExtendedAddress, SrcPANId =0x1aaa, SrcAddr= 0xACDE480000000002, DstAddrMode=0x02 = ShortAddress, DstPANId = 0x1aaa, DstAddr =0xffff,

	KeySource = xx, KeyIndex = xx)	msduLength=5, msdu = 0x00 0x01 0x02 0x03 0x04, mpduLinkQuality = xx, DSN = xx, Timestamp = xx xx xx, SecurityLevel = 0x00, KeyIdMode = xx, KeySource = xx, KeyIndex = xx) Fail: DUT sends ACK frame DUT does not return MCPS-DATA.indication primitive
3	Tester returns MCPS-DATA.confirm(msduHandle = 0x00, Status = SUCCESS)	
Tester to D.U.T.: Short Address to Broadcast Address		
2	Set to Tester MCPS-DATA.request (SrcAddrMode = 0x02 = ShortAddress, DstAddrMode=0x02= ShortAddress, DstPANId = 0x1aaa, DstAddr = 0xffff, msduLength=5, msdu = 0x00 0x01 0x02 0x03 0x04, msduHandle = 0x00, TxOptions = 0x00, SecurityLevel = 0x00, KeyIdMode = xx, KeySource = xx, KeyIndex = xx)	Tester transmits MAC Data frame. Pass: DUT transmits ACK frame DUT returns MAC Primitive: MCPS-DATA.indication(SrcAddrMode = 0x02 = ShortAddress, SrcPANId =0x1aaa, SrcAddr= 0x3344, DstAddrMode=0x02 = ShortAddress, DstPANId = 0x1aaa, DstAddr =0xffff, msduLength=5, msdu = 0x00 0x01 0x02 0x03 0x04, mpduLinkQuality = xx, DSN = xx, Timestamp = xx xx xx, SecurityLevel = 0x00, KeyIdMode = xx, KeySource = xx, KeyIndex = xx) Fail: DUT does not send ACK DUT does not send ACK at correct time DUT does not return MCPS-DATA.indication primitive
3	Tester returns MCPS-DATA.confirm(msduHandle = 0x00, Status = SUCCESS)	
D.U.T. to tester: Short Address to Short Address, Indirect, with ACK		
2	Set to DUT MCPS-DATA.request (SrcAddrMode = 0x02 = ShortAddress, DstAddrMode=0x02= ShortAddress, DstPANId = 0x1aaa, DstAddr = 0x3344, msduLength=5, msdu = 0x00 0x01 0x02 0x03 0x04,	Fail: DUT returns MCPS-DATA.confirm primitive

	msduHandle = 0x0c, TxOptions = 0x05 = Indirect transmission, ACK SecurityLevel = 0x00, KeyIdMode = xx, KeySource = xx, KeyIndex = xx)	
3	Set to tester MLME-POLL.request (CoordAddrMode = 0x02 = ShortAddress, CoordPANId = 0x1aaa, CoordAddress = 0x1122, SecurityLevel = 0x00, KeyIdMode = xx, KeySource = xx, KeyIndex = xx)	Tester will transmit MAC data request command frame to coordinator, DUT Pass: It is possible to verify by a PAN analyzer that an ACK frame was issued from the DUT with the frame pending field set to 1. Frame Length: 5 bytes IEEE 802.15.4 Frame Control: 0x0012 .... .010 = Frame Type: ack (0x0002) .... 0... = Security Enabled: Disabled ....1 .... = Frame Pending: data ....0. .... = Acknowledgment Request: Ack not required .... .0.. .... = Intra-PAN: Not within the PAN .... ..00 0... .... = Reserved .... 00.. .... = Destination Addressing Mode: None ..00 .... = Reserved for Zigbee PRO/RF4CE 00.. .... = Source Addressing Mode: None Sequence Number: xx Frame Check Sequence: Correct Pass: It is possible to verify by a PAN analyzer that a data frame was issued from the DUT. Frame Length: 16 bytes IEEE 802.15.4 Frame Control: 0x8861 .... .001 = Frame Type: Data (0x0001) .... 0... = Security Enabled: Disabled ....0 .... = Frame Pending: No more data ....1. .... = Acknowledgment Request: Ack required .... .1.. .... = Intra-PAN: Within the PAN .... ..00 0... .... = Reserved .... 10.. .... = Destination Addressing Mode: Address field contains a 16-bit short address ..00 .... = Reserved for Zigbee PRO/RF4CE 10.. .... = Source Addressing Mode: Address field contains a 16-bit short address Sequence Number: xx Destination PAN Identifier: 0x1aaa Destination Address: 0x3344 Source Address: 0x1122 MAC Payload 0x00 0x01 0x02 0x03 0x04 Frame Check Sequence: Correct The Tester will transmit an ACK frame The DUT shall return a Success result, such that: DUT returns MAC primitive MPCS-DATA.confirm(msduHandle = 0x0c, status = 0x00 = SUCCESS, Timestamp = xx xx xx) Fail: DUT does not send ACK in response to MAC data request command frame

		ACK does not have Frame pending field set DUT does not send data frame DUT does not return MCPS-DATA.confirm primitive
D.U.T. to tester: Short Address to Extended Address, Indirect, with ACK		
2	Set to DUT MCPS-DATA.request (SrcAddrMode = 0x02 = ShortAddress, DstAddrMode=0x03= ExtendedAddress, DstPANId = 0x1aaa, DstAddr = 0xACDE480000000002, msduLength=5, msdu = 0x00 0x01 0x02 0x03 0x04, msduHandle = 0x0c, TxOptions = 0x05 = Indirect transmission, ACK SecurityLevel = 0x00, KeyIdMode = xx, KeySource = xx, KeyIndex = xx)	Fail: DUT returns MCPS-DATA.confirm primitive
3	Set to tester MLME-SET.request (PIBAttribute=0x53=83=macShortAddress, PIBAttributeValue=0xfffe) Tester responds MLME-SET.confirm (Status=0=SUCCESS PIBAttribute=0x53=83=macShortAddress) MLME-POLL.request (CoordAddrMode = 0x02 = ShortAddress, CoordPANId = 0x1aaa, CoordAddress = 0x1122, SecurityLevel = 0x00, KeyIdMode = xx, KeySource = xx, KeyIndex = xx)	Tester will transmit MAC data request command frame to coordinator, DUT Pass: It is possible to verify by a PAN analyzer that an ACK frame was issued from the DUT with the frame pending field set to 1. Frame Length: 5 bytes IEEE 802.15.4 Frame Control: 0x0012 .... .010 = Frame Type: ack (0x0002) .... 0... = Security Enabled: Disabled ....1 .... = Frame Pending: data ....0. .... = Acknowledgment Request: Ack not required .... .0.. .... = Intra-PAN: Not within the PAN .... ..00 0... .... = Reserved .... 00.. .... = Destination Addressing Mode: None ..00 .... = Reserved for Zigbee PRO/RF4CE 00.. .... = Source Addressing Mode: None Sequence Number: xx Frame Check Sequence: Correct Pass: It is possible to verify by a PAN analyzer that a data frame was issued from the DUT. Frame Length: 22 bytes IEEE 802.15.4 Frame Control: 0x8C61 .... .001 = Frame Type: Data (0x0001) .... 0... = Security Enabled: Disabled ....0 .... = Frame Pending: No more data ....1. .... = Acknowledgment Request: Ack required .... .1.. .... = Intra-PAN: Within the PAN .... ..00 0... .... = Reserved .... 11.. .... = Destination Addressing Mode: Address field contains a 64-bit extended address ..00 .... = Reserved for Zigbee PRO/RF4CE 10.. .... = Source Addressing Mode: Address field contains a 16-bit short address Sequence Number: xx Destination PAN Identifier: 0x1aaa Destination Address: 0xACDE480000000002 Source Address: 0x1122 MAC Payload 0x00 0x01 0x02 0x03 0x04 Frame Check Sequence: Correct

		The Tester will transmit an ACK frame The DUT shall return a Success result, such that: DUT returns MAC primitive MPCS-DATA.confirm(msduHandle = 0x0c, status = 0x00 = SUCCESS, Timestamp = xx xx xx) Fail: DUT does not send ACK in response to MAC data request command frame ACK does not have Frame pending field set DUT does not send data frame DUT does not return MCPS-DATA.confirm primitive
D.U.T. to tester: Extended Address to Short Address, Indirect, with ACK		
2	Set to DUT MLME-SET.request (PIBAttribute=0x53=83=macShortAddress, PIBAttributeValue=0xfffe) DUT responds MLME-SET.confirm (Status=0=SUCCESS PIBAttribute=0x53=83=macShortAddress) MCPS-DATA.request (SrcAddrMode = 0x03 = ExtendedAddress, DstAddrMode=0x02 = ShortAddress, DstPANId = 0x1aaa, DstAddr = 0x3344, msduLength=5, msdu = 0x00 0x01 0x02 0x03 0x04, msduHandle = 0x0c, TxOptions = 0x05 = Indirect transmission, ACK SecurityLevel = 0x00, KeyIdMode = xx, KeySource = xx, KeyIndex = xx)	Fail: DUT returns MCPS-DATA.confirm primitive
3	Set to tester MLME-POLL.request (CoordAddrMode=0x03 =ExtendedAddress, CoordPANId = 0x1aaa, CoordAddress = 0xACDE480000000001, SecurityLevel = 0x00, KeyIdMode = xx, KeySource = xx, KeyIndex = xx)	Tester will transmit MAC data request command frame to coordinator, DUT Pass: It is possible to verify by a PAN analyzer that an ACK frame was issued from the DUT with the frame pending field set to 1. Frame Length: 5 bytes IEEE 802.15.4 Frame Control: 0x0012 .... .010 = Frame Type: ack (0x0002) .... 0... = Security Enabled: Disabled ....1 .... = Frame Pending: data ....0. .... = Acknowledgment Request: Ack not required .... .0.. .... = Intra-PAN: Not within the PAN .... ..00 0... .... = Reserved .... 00.. .... = Destination Addressing Mode: None ..00 .... = Reserved for Zigbee PRO/RF4CE 00.. .... = Source Addressing Mode: None Sequence Number: xx Frame Check Sequence: Correct Pass:

		It is possible to verify by a PAN analyzer that a data frame was issued from the DUT. Frame Length: 22 bytes IEEE 802.15.4 Frame Control: 0xC861 .... .001 = Frame Type: Data (0x0001) .... 0... = Security Enabled: Disabled ....0 .... = Frame Pending: No more data ....1. .... = Acknowledgment Request: Ack required .... .1.. .... = Intra-PAN: Within the PAN .... ..00 0... .... = Reserved .... 10.. .... = Destination Addressing Mode: Address field contains a 16 bit short address ..00 .... = Reserved for Zigbee PRO/RF4CE 11.. .... = Source Addressing Mode: Address field contains a 64-bit-extended address Sequence Number: xx Destination PAN Identifier: 0x1aaa Destination Address: 0xACDE480000000002 Source Address: 0xACDE480000000001 MAC Payload 0x00 0x01 0x02 0x03 0x04 Frame Check Sequence: Correct The Tester will transmit an ACK frame The DUT shall return a Success result, such that: DUT returns MAC primitive MPCS-DATA.confirm(msduHandle = 0x0c, status = 0x00 = SUCCESS, Timestamp = xx xx xx) Fail: DUT does not send ACK in response to MAC data request command frame ACK does not have Frame pending field set DUT does not send data frame DUT does not return MCPS-DATA.confirm primitive
D.U.T. to tester: Extended Address to Extended Address, Indirect, with ACK		
2	Set to DUT MLME-SET.request (PIBAttribute=0x53=83=macShortAddress, PIBAttributeValue=0xfffe) DUT responds MLME-SET.confirm (Status=0=SUCCESS PIBAttribute=0x53=83=macShortAddress) MCPS-DATA.request (SrcAddrMode = 0x03 = ExtendedAddress, DstAddrMode=0x03= ExtendedAddress, DstPANId = 0x1aaa, DstAddr = 0xACDE480000000002, msduLength=5, msdu = 0x00 0x01 0x02 0x03 0x04, msduHandle = 0x0c, TxOptions = 0x05 = Indirect transmission, ACK SecurityLevel = 0x00, KeyIdMode = xx,	Fail: DUT returns MCPS-DATA.confirm primitive

	KeySource = xx, KeyIndex = xx)	
3	Set to tester MLME-SET.request (PIBAttribute=0x53=83=macShortAddress, PIBAttributeValue=0xfffe) Tester responds MLME-SET.confirm (Status=0=SUCCESS PIBAttribute=0x53=83=macShortAddress) MLME-POLL.request (CoordAddrMode=0x03 =ExtendedAddress, CoordPANId = 0x1aaa, CoordAddress = 0xACDE480000000001, SecurityLevel = 0x00, KeyIdMode = xx, KeySource = xx, KeyIndex = xx)	Tester will transmit MAC data request command frame to coordinator, DUT Pass: It is possible to verify by a PAN analyzer that an ACK frame was issued from the DUT with the frame pending field set to 1. Frame Length: 5 bytes IEEE 802.15.4 Frame Control: 0x0012 .... .010 = Frame Type: ack (0x0002) .... 0... = Security Enabled: Disabled ....1 .... = Frame Pending: data ....0. .... = Acknowledgment Request: Ack not required .... .0.. .... = Intra-PAN: Not within the PAN .... ..00 0... .... = Reserved .... 00.. .... = Destination Addressing Mode: None ..00 .... = Reserved for Zigbee PRO/RF4CE 00.. .... = Source Addressing Mode: None Sequence Number: xx Frame Check Sequence: Correct Pass: It is possible to verify by a PAN analyzer that a data frame was issued from the DUT. Frame Length: 28 bytes IEEE 802.15.4 Frame Control: 0xCC61 .... .001 = Frame Type: Data (0x0001) .... 0... = Security Enabled: Disabled ....0 .... = Frame Pending: No more data ....1. .... = Acknowledgment Request: Ack required .... .1.. .... = Intra-PAN: Within the PAN .... ..00 0... .... = Reserved .... 11.. .... = Destination Addressing Mode: Address field contains a 64-bit extended address ..00 .... = Reserved for Zigbee PRO/RF4CE 11.. .... = Source Addressing Mode: Address field contains a 64-bit extended address Sequence Number: xx Destination PAN Identifier: 0x1aaa Destination Address: 0xACDE480000000002 Source Address: 0xACDE480000000001 MAC Payload 0x00 0x01 0x02 0x03 0x04 Frame Check Sequence: Correct The Tester will transmit an ACK frame The DUT shall return a Success result, such that: DUT returns MAC primitive MPCS-DATA.confirm(msduHandle = 0x0c, status = 0x00 = SUCCESS, Timestamp = xx xx xx) Fail: DUT does not send ACK in response to MAC data request command frame ACK does not have Frame pending field set DUT does not send data frame DUT does not return MCPS-DATA.confirm primitive
D.U.T. to tester: Short Address to Short Address, Indirect, with ACK (Tester does NOT poll, hence transaction expires)		

2	Set to DUT MCPS-DATA.request (SrcAddrMode = 0x02 = ShortAddress, DstAddrMode=0x02= ShortAddress, DstPANId = 0x1aaa, DstAddr = 0x3344, msduLength=5, msdu = 0x00 0x01 0x02 0x03 0x04, msduHandle = 0x0c, TxOptions = 0x05 = Indirect transmission, ACK SecurityLevel = 0x00, KeyIdMode = xx, KeySource = xx, KeyIndex = xx)	
3	Wait patiently	Fail: DUT returns MCPS-DATA.confirm primitive with Status = 0x00 = SUCCESS Pass: DUT returns MAC primitive MPCS-DATA.confirm(msduHandle = 0x0c, status = 0xf0 = TRANSACTION_EXPIRED Timestamp = xx xx xx)
4	Set to tester MLME-POLL.request (CoordAddrMode = 0x02 = ShortAddress, CoordPANId = 0x1aaa, CoordAddress = 0x1122, SecurityLevel = 0x00, KeyIdMode = xx, KeySource = xx, KeyIndex = xx)	Tester will transmit MAC data request command frame to coordinator, DUT Two possible pass conditions are valid: Pass 1: It is possible to verify by a PAN analyzer that an ACK frame was issued from the DUT with the frame pending field set to 0. Frame Length: 5 bytes IEEE 802.15.4 Frame Control: 0x0002 .... .010 = Frame Type: ack (0x0002) .... 0... = Security Enabled: Disabled ....0 .... = Frame Pending: no data ....0. .... = Acknowledgment Request: Ack not required .... .0.. .... = Intra-PAN: Not within the PAN .... ..00 0... .... = Reserved .... 00.. .... = Destination Addressing Mode: None ..00 .... = Reserved for Zigbee PRO/RF4CE 00.. .... = Source Addressing Mode: None Sequence Number: xx Frame Check Sequence: Correct Pass 2: It is possible to verify by a PAN analyzer that an ACK frame was issued from the DUT with the frame pending field set to 1, followed by a data frame with no payload. Frame Length: 5 bytes IEEE 802.15.4 Frame Control: 0x0012 .... .010 = Frame Type: ack (0x0002) .... 0... = Security Enabled: Disabled ....1 .... = Frame Pending: data ....0. .... = Acknowledgment Request: Ack not required .... .0.. .... = Intra-PAN: Not within the PAN .... ..00 0... .... = Reserved .... 00.. .... = Destination Addressing Mode: None

		..00 .... = Reserved for Zigbee PRO/RF4CE 00.. .... = Source Addressing Mode: None Sequence Number: xx Frame Check Sequence: Correct Frame Length: 12 bytes IEEE 802.15.4 Frame Control: 0x8861 .... .001 = Frame Type: Data (0x0001) .... 0... = Security Enabled: Disabled ....0 .... = Frame Pending: No more data ....1. .... = Acknowledgment Request: Ack required .... .1.. .... = Intra-PAN: Within the PAN .... ..00 0... .... = Reserved .... 10.. .... = Destination Addressing Mode: Address field contains a 16-bit short address ..00 .... = Reserved for Zigbee PRO/RF4CE 10.. .... = Source Addressing Mode: Address field contains a 16-bit short address Sequence Number: xx Destination PAN Identifier: 0x1aaa Destination Address: 0x3344 Source Address: 0x1122 MAC Payload Not present Frame Check Sequence: Correct Fail: DUT does not send ACK in response to MAC data request command frame ACK has Frame pending field set DUT sends data frame
D.U.T. to tester: Short Address to Short Address, Indirect, with ACK (Tester does NOT poll, DUT purges)		
2	Set to DUT MCPS-DATA.request (SrcAddrMode = 0x02 = ShortAddress, DstAddrMode=0x02= ShortAddress, DstPANId = 0x1aaa, DstAddr = 0x3344, msduLength=5, msdu = 0x00 0x01 0x02 0x03 0x04, msduHandle = 0x0c, TxOptions = 0x05 = Indirect transmission, ACK SecurityLevel = 0x00, KeyIdMode = xx, KeySource = xx, KeyIndex = xx)	
3	Set to DUT MLME-PURGE.request (msduHandle = 0x0c)	Fail: DUT returns MCPS-DATA.confirm primitive with Status = 0x00 = SUCCESS Pass: DUT returns MAC primitive MLME-PURGE.confirm(msduHandle = 0x0c, Staus = 0x00 - SUCCESS)
4	Set to tester MLME-POLL.request (CoordAddrMode = 0x02 = ShortAddress, CoordPANId = 0x1aaa, CoordAddress = 0x1122, SecurityLevel = 0x00,	Tester will transmit MAC data request command frame to coordinator, DUT One of two Pass conditions is possible as follows: Pass 1: It is possible to verify by a PAN analyzer that an ACK frame was issued from the DUT with the frame pending field set to 0.

	KeyIdMode = xx, KeySource = xx, KeyIndex = xx)	Frame Length: 5 bytes IEEE 802.15.4 Frame Control: 0x0012 .... .010 = Frame Type: ack (0x0002) .... 0... = Security Enabled: Disabled ....0 .... = Frame Pending: data ....0. .... = Acknowledgment Request: Ack not required .... .0.. .... = Intra-PAN: Not within the PAN .... ..00 0... .... = Reserved .... 00.. .... = Destination Addressing Mode: None ..00 .... = Reserved for Zigbee PRO/RF4CE 00.. .... = Source Addressing Mode: None Sequence Number: xx Frame Check Sequence: Correct Pass 2: It is possible to verify by a PAN analyzer that an ACK frame was issued from the DUT with the frame pending field set to 1, followed by a data frame with no payload. Frame Length: 5 bytes IEEE 802.15.4 Frame Control: 0x0012 .... .010 = Frame Type: ack (0x0002) .... 0... = Security Enabled: Disabled ....1 .... = Frame Pending: data ....0. .... = Acknowledgment Request: Ack not required .... .0.. .... = Intra-PAN: Not within the PAN .... ..00 0... .... = Reserved .... 00.. .... = Destination Addressing Mode: None ..00 .... = Reserved for Zigbee PRO/RF4CE 00.. .... = Source Addressing Mode: None Sequence Number: xx Frame Check Sequence: Correct Frame Length: 12 bytes IEEE 802.15.4 Frame Control: 0x8861 .... .001 = Frame Type: Data (0x0001) .... 0... = Security Enabled: Disabled ....0 .... = Frame Pending: No more data ....1. .... = Acknowledgment Request: Ack required .... .1.. .... = Intra-PAN: Within the PAN .... ..00 0... .... = Reserved .... 10.. .... = Destination Addressing Mode: Address field contains a 16-bit short address ..00 .... = Reserved for Zigbee PRO/RF4CE 10.. .... = Source Addressing Mode: Address field contains a 16-bit short address Sequence Number: xx Destination PAN Identifier: 0x1aaa Destination Address: 0x3344 Source Address: 0x1122 MAC Payload Not present Frame Check Sequence: Correct Fail: DUT does not send ACK in response to MAC data request command frame ACK has Frame pending field set DUT sends data frame

Notes:

None.

TP/154/MAC/DATA-03[1,2]

Check DUT correctly sends and receives data frames to/from a device on a different PAN.

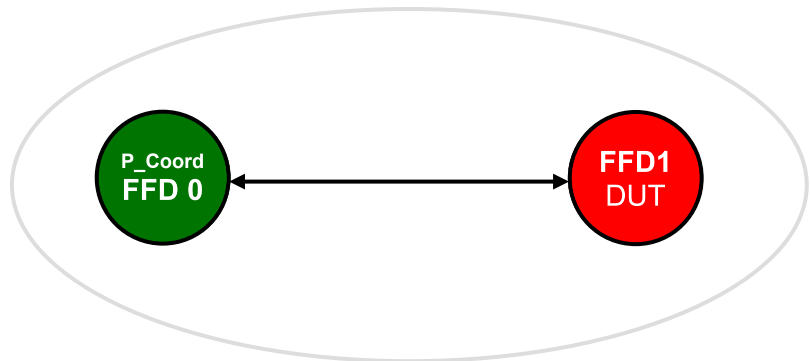
Tester is coordinator for PAN 1, DUT is coordinator for PAN2

D.U.T. to tester.: Short Address to Short Address, Unicast, no ACK
D.U.T. to tester.: Short Address to Short Address, Unicast, with ACK
D.U.T. to tester.: Short Address to Extended Address, Unicast with ACK
D.U.T. to tester.: Extended Address to Short Address , Unicast with ACK
D.U.T. to tester.: Extended Address to Extended Address , Unicast with ACK
D.U.T. to tester.: Extended Address to Broadcast Address
D.U.T. to tester.: Short Address to Broadcast Address

Tester to D.U.T.: Short Address to Short Address, Unicast, no ACK
Tester to D.U.T.: Short Address to Short Address, Unicast, with ACK
Tester to D.U.T.: Short Address to Extended Address, Unicast with ACK
Tester to D.U.T.: Extended Address to Short Address , Unicast with ACK
Tester to D.U.T.: Extended Address to Extended Address , Unicast with ACK
Tester to D.U.T.: Extended Address to Broadcast Address
Tester to D.U.T.: Short Address to Broadcast Address

Setup Conditions:

Diagram



The Diagram is the image that represents the configuration deployed for the Test Procedure.

Initial Conditions:

Device name	Description
All the devices	*Security: no for all the devices* *Channel: 0x14 for all the devices*
P_Coord FFD0	PAN Coordinator, Bo = 15, SO = 15 *PAN id: 0x1AAA* MAC: 0xACDE480000000001 Short Address: 0x1122 Beacon Payload: none Tester as coordinator shall set up a non-beacon enabled PAN, D.U.T. as coordinator shall set up a non-beacon enabled PAN as in test TP/154/MAC/BEACON-MANAGEMENT-01
FFD1	*PAN id: 0x1BBB* MAC: 0xACDE480000000002 Short Address: 0x3344

1417 ***Test Procedure:***

Step No.	Step description	Expected outcome
1	Set DUT RxOnWhenIdle Set to DUT: MLME-SET.request (PIBAttribute=0x52=82=macRxOnWhenIdle, PIBAttributeValue=TRUE)	Obtain MLME-Set.confirm MLME-SET.confirm (Status=SUCCESS, PIBAttribute=0x52=82= macRxOnWhenIdle)
D.U.T. to tester.: Short Address to Short Address, Unicast, no ACK		
2	Set to DUT MCPS-DATA.request (SrcAddrMode = 0x02 = ShortAddress, DstAddrMode=0x02= ExtendedAddress, DstPANId = 0x1aaa, DstAddr = 0x1122, msduLength=5, msdu = 0x00 0x01 0x02 0x03 0x04, msduHandle = 0x0a, TxOptions = 0x00, SecurityLevel = 0x00, KeyIdMode = xx, KeySource = xx, KeyIndex = xx)	DUT transmits MAC Data frame. Pass: It is possible to verify by a PAN analyzer that a data frame was issued from the DUT. Frame Length: 18 bytes IEEE 802.15.4 Frame Control: 0x8801 .... .001 = Frame Type: Data (0x0001) .... 0... = Security Enabled: Disabled ....0 .... = Frame Pending: No more data ....0. .... = Acknowledgment Request: Ack required .... .0.. .... = different PAN .... ..00 0... .... = Reserved .... 10.. .... = Destination Addressing Mode: Address field contains a 16-bit short address ..00 .... = Reserved 10.. .... = Source Addressing Mode: Address field contains a 16-bit short address Sequence Number: xx Destination PAN Identifier: 0x1aaa Destination Address: 0x1122 Source PAN Identifier: 0x1bbb Source Address: 0x3344 MAC Payload 0x00 0x01 0x02 0x03 0x04 Frame Check Sequence: Correct Tester will not transmit ACK frame The DUT shall return a Success result, such that: DUT returns MAC primitive MPCS-DATA.confirm(msduHandle = 0x0a, status = 0x00 = SUCCESS, Timestamp = xx xx xx) Fail: DUT does not return MCPS-DATA.confirm primitive
D.U.T. to tester.: Short Address to Short Address, Unicast, with ACK		
2	Set to DUT MCPS-DATA.request (SrcAddrMode = 0x02 = ShortAddress, DstAddrMode=0x02= ShortAddress, DstPANId = 0x1aaa, DstAddr = 0x1122, msduLength=5, msdu = 0x00 0x01 0x02 0x03 0x04, msduHandle = 0x0a, TxOptions = 0x01,	DUT transmits MAC Data frame. Pass: It is possible to verify by a PAN analyzer that a data frame was issued from the DUT. Frame Length: 18 bytes IEEE 802.15.4 Frame Control: 0x8821 .... .001 = Frame Type: Data (0x0001) .... 0... = Security Enabled: Disabled

	SecurityLevel = 0x00, KeyIdMode = xx, KeySource = xx, KeyIndex = xx)	....0 .... = Frame Pending: No more data ....1. .... = Acknowledgment Request: Ack required .... .0.. .... = different PAN .... ..00 0... .... = Reserved .... 10.. .... = Destination Addressing Mode: Address field contains a 16-bit short address ..00 .... = Reserved 10.. .... = Source Addressing Mode: Address field contains a 16-bit short address Sequence Number: xx Destination PAN Identifier: 0x1aaa Destination Address: 0x1122 Source PAN Identifier: 0x1bbb Source Address: 0x3344 MAC Payload 0x00 0x01 0x02 0x03 0x04 Frame Check Sequence: Correct Tester will transmit ACK frame The DUT shall return a Success result, such that: DUT returns MAC primitive MPCS-DATA.confirm(msduHandle = 0x0a, status = 0x00 = SUCCESS, Timestamp = xx xx xx) Fail: DUT does not return MCPS-DATA.confirm primitive
D.U.T. to tester.: Short Address to Extended Address, Unicast with ACK		
2	Set to DUT MCPS-DATA.request (SrcAddrMode = 0x02 = ShortAddress, DstAddrMode=0x03= ExtendedAddress, DstPANId = 0x1aaa, DstAddr = 0xACDE480000000001, msduLength=5, msdu = 0x00 0x01 0x02 0x03 0x04, msduHandle = 0x0c, TxOptions = 0x01, SecurityLevel = 0x00, KeyIdMode = xx, KeySource = xx, KeyIndex = xx)	DUT transmits MAC Data frame. Pass: It is possible to verify by a PAN analyzer that a data frame was issued from the DUT. Frame Length: 24 bytes IEEE 802.15.4 Frame Control: 0x8C21 .... .001 = Frame Type: Data (0x0001) .... 0... = Security Enabled: Disabled ....0 .... = Frame Pending: No more data ....1. .... = Acknowledgment Request: Ack required .... .0.. .... = different PAN .... ..00 0... .... = Reserved .... 11.. .... = Destination Addressing Mode: Address field contains a 64-bit extended address ..00 .... = Reserved 10.. .... = Source Addressing Mode: Address field contains a 16-bit short address Sequence Number: xx Destination PAN Identifier: 0x1aaa Destination Address0xACDE480000000001 Source PAN Identifier: 0x1bbb Source Address: 0x3344 MAC Payload 0x00 0x01 0x02 0x03 0x04 Frame Check Sequence: Correct The tester will transmit an ACK frame The DUT shall return a Success result, such that: DUT returns MAC primitive MPCS-DATA.confirm(msduHandle = 0x0c, status = 0x00 = SUCCESS,

<table>
<tr><td></td><td></td><td>Timestamp = xx xx xx
)

Fail:
DUT does not return MCPS-DATA.confirm primitive</td></tr>
<tr><td colspan="3">D.U.T. to tester.: Extended Address to Short Address , Unicast with ACK</td></tr>
<tr><td>2</td><td>Set to DUT

MCPS-DATA.request (
SrcAddrMode = 0x03 = ExtendedAddress,
DstAddrMode=0x02 = ShortAddress,
DstPANId = 0x1aaa,
DstAddr = 0x1122,
msduLength=5,
msdu = 0x00 0x01 0x02 0x03 0x04,
msduHandle = 0x0c,
TxOptions = 0x01,
SecurityLevel = 0x00,
KeyIdMode = xx,
KeySource = xx,
KeyIndex = xx
)</td><td>DUT transmits MAC Data frame.

Pass:

It is possible to verify by a PAN analyzer that a data frame was issued from the DUT.

Frame Length: 24 bytes
IEEE 802.15.4
Frame Control: 0xC821
.... .001 = Frame Type: Data (0x0001)
.... 0... = Security Enabled: Disabled
....0 .... = Frame Pending: No more data
....1. .... = Acknowledgment Request: Ack required
.... .0.. .... different PAN
.... ..00 0... .... = Reserved
.... 10.. .... = Destination Addressing Mode: Address field contains a 16-bit short address
..00 .... = Reserved
11.. .... = Source Addressing Mode: Address field contains a 64-bit extended address
Sequence Number: xx
Destination PAN Identifier: 0x1aaa
Destination Address: 0x1122
Source PAN Identifier: 0x1bbb
Source Address: 0xACDE480000000002
MAC Payload
0x00 0x01 0x02 0x03 0x04
Frame Check Sequence: Correct

The tester will transmit an ACK frame

The DUT shall return a Success result, such that:

DUT returns MAC primitive

MPCS-DATA.confirm(
msduHandle = 0x0c,
status = 0x00 = SUCCESS,
Timestamp = xx xx xx
)

Fail:
DUT does not return MCPS-DATA.confirm primitive</td></tr>
<tr><td colspan="3">D.U.T. to tester.: Extended Address to Extended Address , Unicast with ACK</td></tr>
<tr><td>2</td><td>Set to DUT

MCPS-DATA.request (
SrcAddrMode = 0x03 = ExtendedAddress,
DstAddrMode=0x03= ExtendedAddress,
DstPANId = 0x1aaa,
DstAddr = 0xACDE480000000001,
msduLength=5,
msdu = 0x00 0x01 0x02 0x03 0x04,
msduHandle = 0x0c,
TxOptions = 0x01,
SecurityLevel = 0x00,
KeyIdMode = xx,
KeySource = xx,
KeyIndex = xx
)</td><td>DUT transmits MAC Data frame.

Pass:

It is possible to verify by a PAN analyzer that a data frame was issued from the DUT.

Frame Length: 30 bytes
IEEE 802.15.4
Frame Control: 0xCC21
.... .001 = Frame Type: Data (0x0001)
.... 0... = Security Enabled: Disabled
....0 .... = Frame Pending: No more data
....1. .... = Acknowledgment Request: Ack required
.... .0.. .... = Different PAN
.... ..00 0... .... = Reserved
.... 11.. .... = Destination Addressing Mode: Address field contains a 64-bit extended address
..00 .... = Reserved</td></tr>
</table>

		11.. .... = Source Addressing Mode: Address field contains a 64-bit extended address Sequence Number: xx Destination PAN Identifier: 0x1aaa Destination Address: 0xACDE480000000001 Source PAN Identifier: 0x1bbb Source Address: 0xACDE480000000002 MAC Payload 0x00 0x01 0x02 0x03 0x04 Frame Check Sequence: Correct The tester will transmit an ACK frame The DUT shall return a Success result, such that: DUT returns MAC primitive MPCS-DATA.confirm(msduHandle = 0x0c, status = 0x00 = SUCCESS, Timestamp = xx xx xx) Fail: DUT does not return MCPS-DATA.confirm primitive
D.U.T. to tester.: Extended Address to Broadcast Address		
2	Set to DUT MCPS-DATA.request (SrcAddrMode = 0x03 = ExtendedAddress, DstAddrMode=0x02 = ShortAddress, DstPANId = 0x1aaa, DstAddr = 0xffff, msduLength=5, msdu = 0x00 0x01 0x02 0x03 0x04, msduHandle = 0x0c, TxOptions = 0x00, SecurityLevel = 0x00, KeyIdMode = xx, KeySource = xx, KeyIndex = xx)	DUT transmits MAC Data frame. Pass: It is possible to verify by a PAN analyzer that a data frame was issued from the DUT. Frame Length: 24 bytes IEEE 802.15.4 Frame Control: 0xC801 .... .001 = Frame Type: Data (0x0001) .... 0... = Security Enabled: Disabled ....0 .... = Frame Pending: No more data ....0. .... = Acknowledgment Request: Ack not required .... .0.. .... = different PAN .... ..00 0... .... = Reserved .... 10.. .... = Destination Addressing Mode: Address field contains a 16-bit short address ..00 .... = Reserved 11.. .... = Source Addressing Mode: Address field contains a 64-bit extended address Sequence Number: xx Destination PAN Identifier: 0x1aaa Destination Address: 0xffff Source PAN Identifier: 0x1bbb Source Address: 0xACDE480000000002 MAC Payload 0x00 0x01 0x02 0x03 0x04 Frame Check Sequence: Correct The tester will NOT transmit an ACK frame The DUT shall return a Success result, such that: DUT returns MAC primitive MPCS-DATA.confirm(msduHandle = 0x0c, status = 0x00 = SUCCESS, Timestamp = xx xx xx) Fail: DUT does not return MCPS-DATA.confirm primitive
D.U.T. to tester.: Short Address to Broadcast Address		

2	Set to DUT MCPS-DATA.request (SrcAddrMode = 0x02 = ShortAddress, DstAddrMode=0x02= ShortAddress, DstPANId = 0x1aaa, DstAddr = 0xffff, msduLength=5, msdu = 0x00 0x01 0x02 0x03 0x04, msduHandle = 0x0b, TxOptions = 0x00, SecurityLevel = 0x00, KeyIdMode = xx, KeySource = xx, KeyIndex = xx)	DUT transmits MAC Data frame. Pass: It is possible to verify by a PAN analyzer that a data frame was issued from the DUT. Frame Length: 18 bytes IEEE 802.15.4 Frame Control: 0x8801 .... .001 = Frame Type: Data (0x0001) .... 0... = Security Enabled: Disabled ....0 .... = Frame Pending: No more data ....0. .... = Acknowledgment Request: Ack not required .... .0.. .... = different PAN .... ..00 0... .... = Reserved .... 10.. .... = Destination Addressing Mode: Address field contains a 16-bit short address ..00 .... = Reserved 10.. .... = Source Addressing Mode: Address field contains a 16-bit short address Sequence Number: xx Destination PAN Identifier: 0x1aaa Destination Address: 0xffff Source PAN Identifier: 0x1bbb Source Address: 0x3344 MAC Payload 0x00 0x01 0x02 0x03 0x04 Frame Check Sequence: Correct The DUT shall return a Success result, such that: DUT returns MAC primitive MPCS-DATA.confirm(msduHandle = 0x0b, status = 0x00 = SUCCESS, Timestamp = xx xx xx) Fail: DUT does not return MCPS-DATA.confirm primitive
3	Set to DUT MCPS-DATA.request (SrcAddrMode = 0x02 = ShortAddress, DstAddrMode=0x02= ShortAddress, DstPANId = 0xffff, DstAddr = 0xffff, msduLength=5, msdu = 0x00 0x01 0x02 0x03 0x04, msduHandle = 0x0b, TxOptions = 0x00, SecurityLevel = 0x00, KeyIdMode = xx, KeySource = xx, KeyIndex = xx)	DUT transmits MAC Data frame. Pass: It is possible to verify by a PAN analyzer that a data frame was issued from the DUT. Frame Length: 18 bytes IEEE 802.15.4 Frame Control: 0x8801 .... .001 = Frame Type: Data (0x0001) .... 0... = Security Enabled: Disabled ....0 .... = Frame Pending: No more data ....0. .... = Acknowledgment Request: Ack not required .... .0.. .... = different PAN .... ..00 0... .... = Reserved .... 10.. .... = Destination Addressing Mode: Address field contains a 16-bit short address ..00 .... = Reserved 10.. .... = Source Addressing Mode: Address field contains a 16-bit short address Sequence Number: xx Destination PAN Identifier: 0xffff Destination Address: 0xffff Source PAN Identifier: 0x1bbb Source Address: 0x3344 MAC Payload 0x00 0x01 0x02 0x03 0x04 Frame Check Sequence: Correct

		The DUT shall return a Success result, such that: DUT returns MAC primitive MPCS-DATA.confirm(msduHandle = 0x0b, status = 0x00 = SUCCESS, Timestamp = xx xx xx) Fail: DUT does not return MCPS-DATA.confirm primitive
Tester to D.U.T.: Short Address to Short Address, Unicast, no ACK		
2	Set to Tester MCPS-DATA.request (SrcAddrMode = 0x02 = ShortAddress, DstAddrMode=0x02= ShortAddress, DstPANId = 0x1BBB, DstAddr = 0x3344, msduLength=5, msdu = 0x00 0x01 0x02 0x03 0x04, msduHandle = 0x00, TxOptions = 0x00, SecurityLevel = 0x00, KeyIdMode = xx, KeySource = xx, KeyIndex = xx)	Tester transmits MAC Data frame. Pass: DUT returns MAC Primitive: MCPS-DATA.indication(SrcAddrMode = 0x02 = ShortAddress, SrcPANId =0x1aaa, SrcAddr= 0x1122, DstAddrMode=0x02 = ShortAddress, DstPANId = 0x1bbb, DstAddr = 0x3344, msduLength=5, msdu = 0x00 0x01 0x02 0x03 0x04, mpduLinkQuality = xx, DSN = xx, Timestamp = xx xx xx, SecurityLevel = 0x00, KeyIdMode = xx, KeySource = xx, KeyIndex = xx) Fail: DUT does not return MCPS-DATA.indication primitive
Tester to D.U.T.: Short Address to Short Address, Unicast, with ACK		
2	Set to Tester MCPS-DATA.request (SrcAddrMode = 0x02 = ShortAddress, DstAddrMode=0x02= ShortAddress, DstPANId = 0x1BBB, DstAddr = 0x3344, msduLength=5, msdu = 0x00 0x01 0x02 0x03 0x04, msduHandle = 0x00, TxOptions = 0x01, SecurityLevel = 0x00, KeyIdMode = xx, KeySource = xx, KeyIndex = xx)	Tester transmits MAC Data frame. Pass: DUT sends ACK frame DUT returns MAC Primitive: MCPS-DATA.indication(SrcAddrMode = 0x02 = ShortAddress, SrcPANId =0x1aaa, SrcAddr= 0x1122, DstAddrMode=0x02 = ShortAddress, DstPANId = 0x1bbb, DstAddr = 0x3344, msduLength=5, msdu = 0x00 0x01 0x02 0x03 0x04, mpduLinkQuality = xx, DSN = xx, Timestamp = xx xx xx, SecurityLevel = 0x00, KeyIdMode = xx, KeySource = xx, KeyIndex = xx) Fail: DUT does not return MCPS-DATA.indication primitive or

		DUT does not send ACK frame
Tester to D.U.T.: Short Address to Extended Address, Unicast with ACK		
2	Set to Tester MCPS-DATA.request (SrcAddrMode = 0x02 = ShortAddress, DstAddrMode=0x03= ExtendedAddress, DstPANId = 0x1BBB, DstAddr = 0xACDE480000000002, msduLength=5, msdu = 0x00 0x01 0x02 0x03 0x04, msduHandle = 0x00, TxOptions = 0x01, SecurityLevel = 0x00, KeyIdMode = xx, KeySource = xx, KeyIndex = xx)	Tester transmits MAC Data frame. Pass: DUT transmits ACK frame DUT returns MAC Primitive: MCPS-DATA.indication(SrcAddrMode = 0x02 = ShortAddress, SrcPANId =0x1aaa, SrcAddr= 0x1122, DstAddrMode=0x03= ExtendedAddress, DstPANId = 0x1bbb, DstAddr = 0xACDE480000000002, msduLength=5, msdu = 0x00 0x01 0x02 0x03 0x04, mpduLinkQuality = xx, DSN = xx, Timestamp = xx xx xx, SecurityLevel = 0x00, KeyIdMode = xx, KeySource = xx, KeyIndex = xx) Fail: DUT does not return MCPS-DATA.indication primitive or DUT does not send ACK frame
Tester to D.U.T.: Extended Address to Short Address , Unicast with ACK		
2	Set to Tester MCPS-DATA.request (SrcAddrMode = 0x03 = ExtendedAddress, DstAddrMode=0x02= ShortAddress, DstPANId = 0x1BBB, DstAddr = 0x3344, msduLength=5, msdu = 0x00 0x01 0x02 0x03 0x04, msduHandle = 0x00, TxOptions = 0x01, SecurityLevel = 0x00, KeyIdMode = xx, KeySource = xx, KeyIndex = xx)	Tester transmits MAC Data frame. Pass: DUT transmits ACK frame DUT returns MAC Primitive: MCPS-DATA.indication(SrcAddrMode = 0x03 = ExtendedAddress, SrcPANId =0x1aaa, SrcAddr= 0xACDE480000000001, DstAddrMode=0x02= ShortAddress, DstPANId = 0x1bbb, DstAddr = 0x3344, msduLength=5, msdu = 0x00 0x01 0x02 0x03 0x04, mpduLinkQuality = xx, DSN = xx, Timestamp = xx xx xx, SecurityLevel = 0x00, KeyIdMode = xx, KeySource = xx, KeyIndex = xx) Fail: DUT does not return MCPS-DATA.indication primitive or DUT does not send ACK frame
Tester to D.U.T.: Extended Address to Extended Address , Unicast with ACK		
2	Set to Tester MCPS-DATA.request (SrcAddrMode = 0x03 = ExtendedAddress, DstAddrMode=0x03= ExtendedAddress, DstPANId = 0x1BBB, DstAddr = 0xACDE480000000002,	Tester transmits MAC Data frame. Pass: DUT transmits ACK frame DUT returns MAC Primitive:

	msduLength=5, msdu = 0x00 0x01 0x02 0x03 0x04, msduHandle = 0x00, TxOptions = 0x01, SecurityLevel = 0x00, KeyIdMode = xx, KeySource = xx, KeyIndex = xx)	MCPS-DATA.indication(SrcAddrMode = 0x03 = ExtendedAddress, SrcPANId =0x1aaa, SrcAddr= 0xACDE480000000001, DstAddrMode=0x03= ExtendedAddress, DstPANId = 0x1bbb, DstAddr = 0xACDE480000000002, msduLength=5, msdu = 0x00 0x01 0x02 0x03 0x04, mpduLinkQuality = xx, DSN = xx, Timestamp = xx xx xx, SecurityLevel = 0x00, KeyIdMode = xx, KeySource = xx, KeyIndex = xx) Fail: DUT does not return MCPS-DATA.indication primitive or DUT does not send ACK frame
Tester to D.U.T.: Extended Address to Broadcast Address		
2	Set to Tester MCPS-DATA.request (SrcAddrMode = 0x03 = ExtendedAddress, DstAddrMode=0x02= ShortAddress, DstPANId = 0x1BBB, DstAddr = 0xffff, msduLength=5, msdu = 0x00 0x01 0x02 0x03 0x04, msduHandle = 0x00, TxOptions = 0x00, SecurityLevel = 0x00, KeyIdMode = xx, KeySource = xx, KeyIndex = xx)	Tester transmits MAC Data frame. Pass: DUT does NOT transmit ACK frame DUT returns MAC Primitive: MCPS-DATA.indication(SrcAddrMode = 0x03 = ExtendedAddress, SrcPANId =0x1aaa, SrcAddr= 0xACDE480000000001, DstAddrMode=0x02= ShortAddress, DstPANId = 0x1bbb, DstAddr = 0xffff, msduLength=5, msdu = 0x00 0x01 0x02 0x03 0x04, mpduLinkQuality = xx, DSN = xx, Timestamp = xx xx xx, SecurityLevel = 0x00, KeyIdMode = xx, KeySource = xx, KeyIndex = xx) Fail: DUT transmits ACK frame DUT does not return MCPS-DATA.indication primitive
Tester to D.U.T.: Short Address to Broadcast Address		
2	Set to Tester MCPS-DATA.request (SrcAddrMode = 0x02 = ShortAddress, DstAddrMode=0x02= ShortAddress, DstPANId = 0x1BBB, DstAddr = 0xffff, msduLength=5, msdu = 0x00 0x01 0x02 0x03 0x04, msduHandle = 0x00, TxOptions = 0x00, SecurityLevel = 0x00, KeyIdMode = xx, KeySource = xx, KeyIndex = xx)	Tester transmits MAC Data frame. Pass: DUT does NOT send ACK frame DUT returns MAC Primitive: MCPS-DATA.indication(SrcAddrMode = 0x02 = ShortAddress, SrcPANId =0x1aaa, SrcAddr= 0x1122, DstAddrMode=0x02 = ShortAddress, DstPANId = 0x1bbb, DstAddr = 0xffff, msduLength=5, msdu = 0x00 0x01 0x02 0x03 0x04, mpduLinkQuality = xx,

		DSN = xx, Timestamp = xx xx xx, SecurityLevel = 0x00, KeyIdMode = xx, KeySource = xx, KeyIndex = xx) Fail: DUT sends ACK frame DUT does not return MCPS-DATA.indication primitive
3	Set to Tester MCPS-DATA.request (SrcAddrMode = 0x02 = ShortAddress, DstAddrMode=0x02= ShortAddress, DstPANId = 0xFFFF, DstAddr = 0xffff, msduLength=5, msdu = 0x00 0x01 0x02 0x03 0x04, msduHandle = 0x00, TxOptions = 0x00, SecurityLevel = 0x00, KeyIdMode = xx, KeySource = xx, KeyIndex = xx)	Tester transmits MAC Data frame. Pass: DUT sends ACK frame DUT returns MAC Primitive: MCPS-DATA.indication(SrcAddrMode = 0x02 = ShortAddress, SrcPANId =0x1aaa, SrcAddr= 0x1122, DstAddrMode=0x02 = ShortAddress, DstPANId = 0xffff, DstAddr = 0xffff, msduLength=5, msdu = 0x00 0x01 0x02 0x03 0x04, mpduLinkQuality = xx, DSN = xx, Timestamp = xx xx xx, SecurityLevel = 0x00, KeyIdMode = xx, KeySource = xx, KeyIndex = xx) Fail: DUT does not ACK frame DUT does not return MCPS-DATA.indication primitive

1418 ***Notes:***

1419 *None.*

1420 ## TP/154/MAC/DATA-04

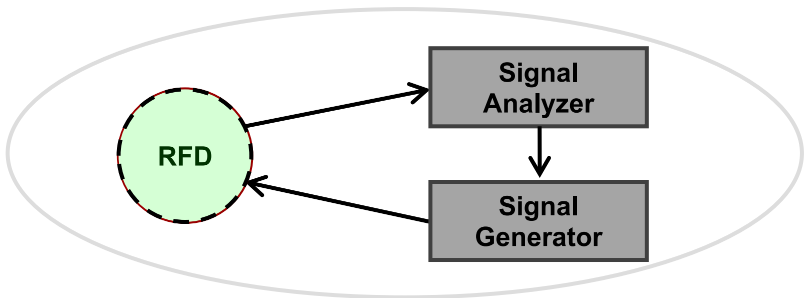
1421 Tests that the DUT (Sleepy Host) waits for a polled packet by the specified amount in the table below.

Band - PHY	Wait Time (symbols)	T* (msec)
2.4 GHz O-QPSK (250 kbps) 802.15.4-2003 (Zigbee PRO/RF4CE)	1220	20
868 MHz BPSK (20 kbps) 802.15.4-2003 (Zigbee PRO)	1220	61
915 MHz BPSK (40 kbps) 802.15.4-2003 (Zigbee PRO)	1220	31

1422 * The times above represent the time until the first bit of the preamble is received.

1423 ***Setup Conditions***:

Diagram



1424 *The Diagram is the image that represents the configuration deployed for the Test Procedure.*

1425 ***Initial Conditions:***

Device name	Description
DUT	*Channel: any channel in the band for the corresponding PHY being tested.*
Signal Analyzer	*Configure the Signal Analyzer to start acquisition on analog edge triggering*

Signal Generator	***DUT: RFD (Sleepy End Host)*** ***Signal generator: FFD (Parent Router)*** *Signal Generator generates a Acknowledgment frame with pending bit to respond data request from the DUT.* *Signal Generator generates a Data frame T (seconds) aftter it generates Ackowledgement frame.* *DUT shall transmit a polling signal (data request) and Signal Analyzer receives it and triggers Signal Generator to transmit Acknowledgement packet with pending data information .* *DUT shall receive Acknowledgement and sets its receiver on.* *The Signal Generator counts the time from its transmission end of Acknowledgement packet until T time reaches.* *The Signal Generator shall transmit the pended data packet to DUT.* *DUT shall report to Next Higher Layer on the reception of the expected data.* *The Signal Generator equipment shall be capable of providing tight synchronization with the Signal Analyzer.* *Signal Generator shall be configured to transmit a standards conforming signal and packet with an arbitrary payload and correct CRC with the following parameters:* Optionally, the signal generator may be replaced by a Test Harness. Observation: The Spectrum Analyzer is used to check that frames are transmitted at correct times. ***DUT: FFD (Parent Router)*** ***Signal Generator: RFD (Sleepy End Host)*** *DUT prepare the sending data which is transmitted by indirect transmission. Signal generator generates polling signal and send to the DUT.* *DUT shall send Acknowledgement with pending bit on.* *DUT shall send data frame after sending Ackknowledgment within T (symbol) time.* Optionally, the signal generator may be replaced by a Test Harness. Observation: The Spectrum Analyzer is used to check that frames are transmitted at correct times.

1426 ***Test Procedure:***

Step No.	Step description	Expected outcome
1	DUT and Test harness shall be associated before the steps.	
DUT: RFD, Testharness: Parent FFD		
2	Test harness sends data by indirect transmission to DUT.	Check if DUT surely waits data frame for *macMaxFrameTotalWaitTime*.

	DUT sends data request to the test harness. Test harness sends Acknowledgment frame to the DUT with pending bit. DUT waits T (sec)	Pass: DUT receives expected data frame and indication is reported to the upper layer. Fail: DUT does not receive data frame.
DUT: Parent FFD, Testharness: RFD		
2	DUT sends data by indirect transmission to the test harness. Test harness send data request to the parent FFD (DUT).	DUT shall send acknowledgement with pending bit. DUT shall send data frame within *macMaxFrameTotalWaitTime*. Pass: Test harness receives data frame within specified Wait Time Fail: Test harness does not receive data frame or receives it after specified Wait Time has passed

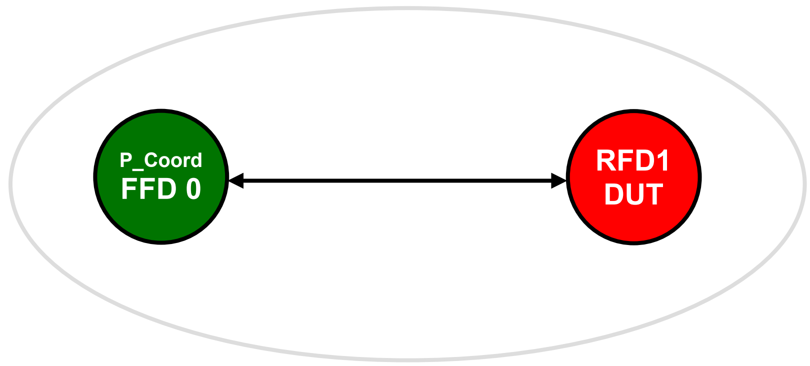
1427 ***Notes:***

1428 *None.*

TP/154/MAC/WARM-START-01

Check D.U.T. (as device) correctly starts up when PIB is restored using MLME-SET.request commands.

Setup Conditions:

Diagram


The Diagram is the image that represents the configuration deployed for the Test Procedure.

Initial Conditions:

Device name	Description
All the devices	*Security: no for all the devices* *Channel: 0x14 for all the devices* *PAN id: 0x1AAA for all the devices*
P_Coord FFD0	PAN Coordinator, Bo = 15, SO = 15 MAC: 0xACDE480000000001 Short Address: 0x1122 Beacon Payload: none Coordinator shall set up a non-beacon enabled PAN.
RFD1	*MAC: 0xACDE480000000002* Short Address: None

Test Procedure:

Step No.	Step description	Expected outcome
1	Set Coordinator RxOnWhenIdle Set to Tester: MLME-SET.request (PIBAttribute=0x52=82=macRxOnWhenIdle, PIBAttributeValue=TRUE)	
2	Set to D.U.T. MLME-SET.request (PIBAttribute=0x00=phyCurrentChannel, PIBAttributeValue=0x14) Verify the PIB is correctly set MLME-GET.request (PIBAttribute=0x00=phyCurrentChannel,	DUT responds: MLME-SET.confirm (Status = SUCCESS, PIBAttribute=0x00=phyCurrentChannel) Wait for MLME-GET.confirm and confirm the parameters MLME-GET.confirm (Status = SUCCESS PIBAttribute=0x00=phyCurrentChannel,

	PIBAttributeIndex = 0x00- ignore, Only used for MAC 2006 security PIBs) MLME-SET.request (PIBAttribute=0x50=macPANId, PIBAttributeValue=0x1AAA) Verify the PIB is correctly set MLME-GET.request (PIBAttribute=0x50=macPANId, PIBAttributeIndex = 0x00- ignore, Only used for MAC 2006 security PIBs) MLME-SET.request (PIBAttribute=0x53=macShortAddress, PIBAttributeValue=0x3344) Verify the PIB is correctly set MLME-GET.request (PIBAttribute=0x53=macShortAddress, PIBAttributeIndex = 0x00- ignore, Only used for MAC 2006 security PIBs) MLME-SET.request (PIBAttribute=0x4b=macCoordShortAddress, PIBAttributeValue=0x1122) Verify the PIB is correctly set MLME-GET.request (PIBAttribute=0x4b=macCoordShortAddress, PIBAttributeIndex = 0x00- ignore, Only used for MAC 2006 security PIBs) MLME-SET.request (PIBAttribute=0x4a=macCoordExtendedAddress, PIBAttributeValue=0xACDE480000000001) Verify the PIB is correctly set MLME-GET.request (PIBAttribute=0x4a=macCoordExtendedAddress, PIBAttributeIndex = 0x00- ignore, Only used for MAC 2006 security PIBs)	PIBAttributeIndex = ignore. Only for security in MAC2006 PIBAttributeValue = 0x14) MLME-SET.confirm (Status = SUCCESS, PIBAttribute=0x50=macPANId) Wait for MLME-GET.confirm and confirm the parameters MLME-GET.confirm (Status = SUCCESS PIBAttribute=0x50=macPANId, PIBAttributeIndex = ignore. Only for security in MAC2006 PIBAttributeValue = 0x1AAA) MLME-SET.confirm (Status = SUCCESS, PIBAttribute=0x53=macShortAddress) Wait for MLME-GET.confirm and confirm the parameters MLME-GET.confirm (Status = SUCCESS PIBAttribute=0x53=macShortAddress, PIBAttributeIndex = ignore. Only for security in MAC2006 PIBAttributeValue = 0x3344) MLME-SET.confirm (Status = SUCCESS, PIBAttribute=0x4b=macCoordShortAddress) Wait for MLME-GET.confirm and confirm the parameters MLME-GET.confirm (Status = SUCCESS PIBAttribute=0x4b=macCoordShortAddress, PIBAttributeIndex = ignore. Only for security in MAC2006 PIBAttributeValue = 0x1122) MLME-SET.confirm (Status = SUCCESS, PIBAttribute=0x4a=macCoordExtendedAddress) Wait for MLME-GET.confirm and confirm the parameters MLME-GET.confirm (Status = SUCCESS PIBAttribute=0x4a=macCoordExtendedAddress, PIBAttributeIndex = ignore. Only for security in MAC2006 PIBAttributeValue = 0xACDE480000000001)
3	Set to DUT MCPS-DATA.request (SrcAddrMode = 0x02 = ShortAddress, DstAddrMode=0x02= ShortAddress, DstPANId = 0x1aaa, DstAddr = 0x1122, msduLength=5, msdu = 0x00 0x01 0x02 0x03 0x04,	DUT transmits MAC Data frame. Pass: It is possible to verify by a PAN analyzer that a data frame was issued from the DUT. Frame Length: 18 bytes IEEE 802.15.4 Frame Control: 0x8861 .... .001 = Frame Type: Data (0x0001)

	msduHandle = 0x0a, TxOptions = 0x01, SecurityLevel = 0x00, KeyIdMode = xx, KeySource = xx, KeyIndex = xx)	.... 0... = Security Enabled: Disabled ....0 .... = Frame Pending: No more data ....1. .... = Acknowledgment Request: Ack required .... .1.. .... = Intra-PAN: Within the PAN .... ..00 0... .... = Reserved .... 10.. .... = Destination Addressing Mode: Address field contains a 16-bit short address ..00 .... = Reserved for Zigbee PRO/RF4CE 10.. .... = Source Addressing Mode: Address field contains a 16-bit short address Sequence Number: xx Destination PAN Identifier: 0x1aaa Destination Address: 0x1122 Source PAN Identifier: 0x1aaa Source Address: 0x3344 MAC Payload 0x00 0x01 0x02 0x03 0x04 Frame Check Sequence: Correct Tester will transmit ACK frame The DUT shall return a Success result, such that: DUT returns MAC primitive MPCS-DATA.confirm(msduHandle = 0x0a, status = 0x00 = SUCCESS, Timestamp = xx xx xx) Fail: DUT does not return MCPS-DATA.confirm primitive

1435 Check D.U.T. (as device) correctly starts up when PIB is restored using MLME-SET.request commands.

1436 ***Notes:***

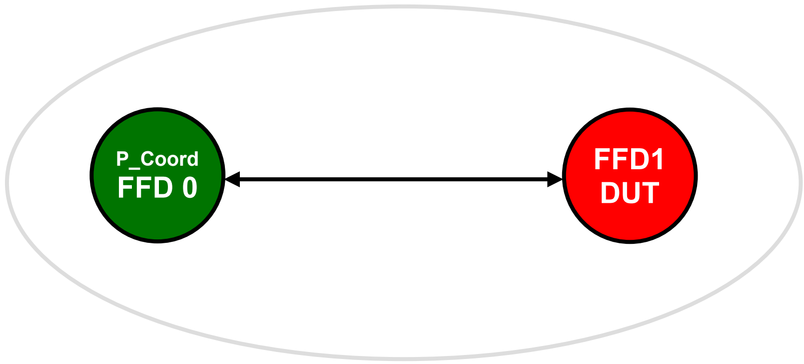
1437 *None.*

TP/154/MAC/WARM-START-02

Check D.U.T. (as coordinator) correctly starts up when PIB is restored using MLME-SET.request commands and MLME-START.request is issued. Note: that this test is similar to TP/154/MAC/WARM-START-01 with an additional step to “start” the device under test as a coordinator.

Setup Conditions:

Diagram



The Diagram is the image that represents the configuration deployed for the Test Procedure.

Initial Conditions:

Device name	Description
All the devices	*Security: no for all the devices* *Channel: 0x14 for all the devices* *PAN id: 0x1AAA for all the devices*
P_Coord FFD0	PAN Coordinator, Bo = 15, SO = 15 MAC: 0xACDE480000000001 Short Address: 0x1122 Beacon Payload: none Coordinator shall set up a non-beacon enabled PAN.
FFD1	*MAC: 0xACDE480000000002* Short Address: None

Test Procedure:

Step No.	Step description	Expected outcome
1	Set Coordinator RxOnWhenIdle Set to Tester: MLME-SET.request (PIBAttribute=0x52=82=macRxOnWhenIdle, PIBAttributeValue=TRUE)	
2	Set to D.U.T. MLME-SET.request (PIBAttribute=0x50=macPANId, PIBAttributeValue=0x1AAA) Verify the PIB is correctly set	DUT responds: MLME-SET.confirm (Status = SUCCESS, PIBAttribute=0x50=macPANId)Wait for MLME-GET.confirm and confirm the parameters MLME-GET.confirm (

	MLME-GET.request (PIBAttribute=0x50=macPANId, PIBAttributeIndex = 0x00- ignore, Only used for MAC 2006 security PIBs) MLME-SET.request (PIBAttribute=0x53=macShortAddress, PIBAttributeValue=0x3344) Verify the PIB is correctly set MLME-GET.request (PIBAttribute=0x53=macShortAddress, PIBAttributeIndex = 0x00- ignore, Only used for MAC 2006 security PIBs) MLME-SET.request (PIBAttribute=0x4b=macCoordShortAddress, PIBAttributeValue=0x1122) Verify the PIB is correctly set MLME-GET.request (PIBAttribute=0x4b=macCoordShortAddress, PIBAttributeIndex = 0x00- ignore, Only used for MAC 2006 security PIBs) MLME-SET.request (PIBAttribute=0x4a=macCoordExtendedAddress, PIBAttributeValue=0xACDE480000000001) Verify the PIB is correctly set MLME-GET.request (PIBAttribute=0x4a=macCoordExtendedAddress, PIBAttributeIndex = 0x00- ignore, Only used for MAC 2006 security PIBs)	Status = SUCCESS PIBAttribute=0x50=macPANId, PIBAttributeIndex = ignore. Only for security in MAC2006 PIBAttributeValue = 0x1AAA) MLME-SET.confirm (Status = SUCCESS, PIBAttribute=0x53=macShortAddress) Wait for MLME-GET.confirm and confirm the parameters MLME-GET.confirm (Status = SUCCESS PIBAttribute=0x53=macShortAddress, PIBAttributeIndex = ignore. Only for security in MAC2006 PIBAttributeValue = 0x3344) MLME-SET.confirm (Status = SUCCESS, PIBAttribute=0x4b=macCoordShortAddress) Wait for MLME-GET.confirm and confirm the parameters MLME-GET.confirm (Status = SUCCESS PIBAttribute=0x4b=macCoordShortAddress, PIBAttributeIndex = ignore. Only for security in MAC2006 PIBAttributeValue = 0x1122) MLME-SET.confirm (Status = SUCCESS, PIBAttribute=0x4a=macCoordExtendedAddress) Wait for MLME-GET.confirm and confirm the parameters MLME-GET.confirm (Status = SUCCESS PIBAttribute=0x4a=macCoordExtendedAddress, PIBAttributeIndex = ignore. Only for security in MAC2006 PIBAttributeValue = 0xACDE480000000001)
3	Set to D.U.T. Issue MLME-START.request command with the following parameters: MLME-START.request (PANId=0x1AAA, LogicalChannel=20=0x14, BeaconOrder=15, SuperframeOrder=15, PANCoordinator=TRUE, BatteryLifeExtension=FALSE, CoordRealignment=FALSE, CoordRealignmentSecurityLevel=0, [CoordRealignmentKeyIdMode =x, CoordRealignmentKeySource = x, CoordRealignmentKeyIndex = x,] BeaconSecurityLevel=0, [BeaconKeyIdMode =x, BeaconKeySource = x, BeaconKeyIndex = x,]).	DUT returns primitive MLME-START.confirm (Status=SUCCESS)

4	Set to DUT MCPS-DATA.request (SrcAddrMode = 0x02 = ShortAddress, DstAddrMode=0x02= ShortAddress, DstPANId = 0x1aaa, DstAddr = 0x1122, msduLength=5, msdu = 0x00 0x01 0x02 0x03 0x04, msduHandle = 0x0a, TxOptions = 0x01, SecurityLevel = 0x00, KeyIdMode = xx, KeySource = xx, KeyIndex = xx)	DUT transmits MAC Data frame. Pass: It is possible to verify by a PAN analyzer that a data frame was issued from the DUT. Frame Length: 16 bytes IEEE 802.15.4 Frame Control: 0x8861 .... .001 = Frame Type: Data (0x0001) .... 0... = Security Enabled: Disabled ....0 .... = Frame Pending: No more data ....1. .... = Acknowledgment Request: Ack required .... .1.. .... = Intra-PAN: Within the PAN .... ..00 0... .... = Reserved .... 10.. .... = Destination Addressing Mode: Address field contains a 16-bit short address ..00 .... = Reserved for Zigbee PRO/RF4CE 10.. .... = Source Addressing Mode: Address field contains a 16-bit short address Sequence Number: xx Destination PAN Identifier: 0x1aaa Destination Address: 0x1122 Source PAN Identifier: 0x1aaa Source Address: 0x3344 MAC Payload 0x00 0x01 0x02 0x03 0x04 Frame Check Sequence: Correct Tester will transmit ACK frame The DUT shall return a Success result, such that: DUT returns MAC primitive MPCS-DATA.confirm(msduHandle = 0x0a, status = 0x00 = SUCCESS, Timestamp = xx xx xx) Fail: DUT does not return MCPS-DATA.confirm primitive

1446 ***Notes:***

1447 *None.*

1448